Memoir — Approaches for Imitation Learning

Imitation by state-occupancy matching and by forward–backward representations

Proofs, approximations and a controlled comparison of f-policy gradients and forward–backward imitation

Ali Javani

Supervisor: Olivier Cappé

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Abstract. This memoir studies two reward-free approaches to imitation learning. The first matches state-occupancy measures with f-divergences: f-IRL learns a reward through a maximum-entropy inner loop, and f-policy gradients (f-PG) differentiate the divergence directly with respect to the policy. The second learns, without rewards, a forward–backward (FB) representation of successor measures once, and imitates by choosing a task vector: $\mathrm{ER}_{\mathrm{FB}}$ reads it off the demonstrations and $\mathrm{DM}_{\mathrm{FB}}$ fits it by matching successor measures. We write both families in one notation, prove their central results under explicit assumptions, and separate the exact statements from the approximations that the algorithms add: plug-in density estimates, finite rank, sampling, truncation and smoothing. We adapt f-PG to a fixed expert distribution and derive its gradient, estimator and baselines; we show that for softmax policies the FB optimality theorem is replaced by a temperature bound, and that the $\mathrm{DM}_{\mathrm{FB}}$ loss expansion holds without greedy policies. We then compare f-PG with FB imitation in two 7×7 finite mazes, where an exact oracle evaluates every policy, and in a continuous four-rooms maze, with three seeds and a protocol frozen before the final runs. In the finite mazes, all 18 f-PG runs reach the predeclared thresholds (held-out occupancy Jensen–Shannon divergence at most 0.10 nats and success at least 0.80), first at the 20,000- or 30,000-transition evaluation checkpoint (checkpoints every 10,000 transitions), whereas at the end of the budget $\mathrm{ER}_{\mathrm{FB}}$ meets them in 2 of 18 runs and $\mathrm{DM}_{\mathrm{FB}}$ in none. $\mathrm{DM}_{\mathrm{FB}}$'s model objective decreased in all 27 adaptations, while the held-out occupancy divergence increased in 20 of them (in 16 of the 18 finite ones, where it is computed exactly). In the continuous maze no method reaches the thresholds within its budget; f-PG ends closest (divergence 0.24–0.41 nats, success 0.92–0.99), but it reaches the goal without staying: only 6–14% of its rollouts end there, against 72–76% for the expert. In the finite mazes the FB models are provably approximate, because the rank of the transition kernel exceeds the representation rank, and their measured successor-measure errors are large. We report every seed, the cost of every phase, and the limits of three-seed evidence.

Note on tools and on the evidence. The source code, frozen configurations, and reproduction instructions for this memoir are publicly available at `https://github.com/AliJavanJ1/il-memoir-reproduction`. The code, the proof compendium and this text were drafted with the help of an AI assistant (Claude, Anthropic) under the author's direction. The tables of the results sections are generated from the saved run files by `report/verify/make_tables.py`. Every number of the text that is computed from the runs, from the environment and audit checks or from a closed-form expression is typeset through a macro that names a registered claim, and `report/verify/verify_results.py` checks that the printed literal is the registered one and that it equals the value recomputed from the saved files at the printed precision. Configuration settings quoted in the text are checked against the frozen configuration, but most of their occurrences are not tied to these checks (Section K). No theorem in this memoir is a statement about the convergence of neural-network training or about the optimality of an approximate model.

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1 Introduction

1.1 Imitation without rewards

In imitation learning an agent learns a behaviour from demonstrations of an expert instead of from a hand-crafted reward. Behavioural cloning treats imitation as supervised learning from states to actions [15]; it is simple, but its errors compound once the agent leaves the states the expert visited [16]. Inverse reinforcement learning (IRL) instead recovers a reward that explains the expert [1, 12], and the maximum-entropy principle resolves the ambiguity in the choice of a distribution over trajectories [20], although the reward itself is still not unique in general. A third view, which underlies both approaches studied here, describes a behaviour by the distribution of the states it visits. Two policies with the same discounted state-occupancy measure have the same return for every state reward (Theorem 3.3), so matching the expert's occupancy measure imitates the expert as far as any state reward can tell. Adversarial and divergence-minimization methods make this view operational [7–9].

The memoir compares two recent families built on this view, as asked by the assignment.

- **f-divergence occupancy matching.** f-IRL [13] learns a state reward whose maximum-entropy policy matches the expert's state marginal under an f-divergence, with an analytic reward gradient. f-policy gradients [2] differentiate an f-divergence between the agent's state visitation and a target directly with respect to the policy parameters.
- **Forward–backward (FB) representations.** Touati and Ollivier [18] learn, without any reward, a factorization $F(s, a, z)^\top B(s')$ of the successor measures of a family of policies indexed by a task vector z. A reward r is then optimized by the single vector $z_r = \mathbb{E}_\nu[rB]$, without planning. Pirotta et al. [14] use such a pretrained model for imitation: $\mathrm{ER}_{\mathrm{FB}}$ reads z off the demonstrations, and $\mathrm{DM}_{\mathrm{FB}}$ fits z by matching the successor measures of the expert.

Both families are reward-free during imitation and neither solves a Bellman optimality equation for the imitation task. They differ in what they learn, in what data they need, and in what their guarantees cover. f-PG needs fresh interaction with the environment for every task; FB imitation needs a task-blind pretraining phase once and then no interaction at all.

1.2 Research questions and where they are answered

The assignment asks five questions. Table 1 lists them with the sections that answer them; Section 16 collects the answers.

Table 1: The research questions of the assignment and where this memoir answers them.

	Question (assignment, pp. 1–2)	Answered in
Q1	Mathematical principles of f-divergence occupancy matching and of the FB representation; own proofs; exact arguments separated from the approximations used to build the algorithms.	Sections 3 to 7 (selected derivations); complete proofs in Appendices B–H
Q2	Similarities, differences and use cases of the two approaches.	Section 8
Q3	Role of the hyperparameters, in particular the choice of f in f-IRL and the algorithm used to learn the successor measure.	Section 9; ablations in Section 13
Q4	Performance, numerical stability and sample requirements in a common finite scenario where planning is exact.	Sections 10, 11 and 14
Q5	Behaviour in a continuous state space with a DQN-style approximation.	Section 12

1.3 Scope and contributions

What is implemented. The experiments compare f-PG, adapted to a fixed expert distribution, with a pretrained FB model and $\mathrm{DM}_{\mathrm{FB}}$ latent adaptation; $\mathrm{ER}_{\mathrm{FB}}$ on the same model is a cheap baseline. f-IRL is treated theoretically only: its reward gradient is reproved and related to f-PG, but it is not implemented. The environments are small mazes with five discrete actions: two 7×7 finite mazes (an open room and four rooms), for which an exact oracle computes occupancies, optimal values and the error of a trained FB model, and a continuous four-rooms maze with positions in $[0, 1]^2$. In the continuous maze the FB forward network $F(s, a, z)$ outputs one d-dimensional vector per action, as a DQN network outputs one value per action [11]; this is the DQN-style approximation of the state–action space asked for by Q5. The f-PG policy is a categorical network, not a DQN-style one.

State versus state–action matching. The assignment describes f-IRL and f-PG as matching the *state–action* occupancy measure. The supplied papers match *state* marginals: f-IRL's objective is a divergence between state marginals (FIRL Eq. 2, p. 3; Table 1, p. 2), f-PG's is a divergence between state visitations (FPG Eq. 4, p. 4), and FI24's successor measures are measures on next states (FI24 Eq. 1, p. 3). We therefore use state matching in the common experiments. We also define state–action occupancies, derive the state–action f-PG gradient (Theorem 4.5), and report the state–action divergence of every finite policy as a diagnostic. State-only matching does not identify the expert's actions (Theorem B.12), and we never claim that it does.

Interpretation of the assignment. The assignment names the first family after f-IRL and describes it as direct matching of the state–action occupancy measure using policy-gradient ideas, citing both FIRL and FPG; it asks for both approaches to be implemented in a finite scenario and in a continuous one, the latter with a DQN-style approximation of the state–action space (assignment, pp. 1–2). This memoir reads it in three ways, described above: (i) the experiments match *state* occupancies, as the supplied papers do, and report state–action divergences as diagnostics; (ii) f-PG is the implemented direct-matching method of the first family, while f-IRL is analysed theoretically and related to f-PG but not run; (iii) in the continuous experiment the DQN-style approximation is FB's forward network, and the f-PG policy is a categorical neural network, not a DQN-style one. These are the project's interpretations of the assignment; written confirmation of them by the supervisor is still outstanding.

Contributions.

(i) One notation for occupancies, successor measures, divergences and the three different expert distributions used by the methods (Section 2), with the source discrepancies that it resolves (Section J).

(ii) Complete proofs, under explicit assumptions, of the results that justify the algorithms, in Appendices B–H (the main text gives selected derivations): occupancy and successor identities, the fixed-target f-PG gradient with its discount weights, baselines and estimator, the f-IRL covariance gradient, FB's reward-to-task theorem, its softmax replacement, an approximation bound, the pretraining loss, and the $\mathrm{DM}_{\mathrm{FB}}$ expansion. Each result is labelled as a source result, our adaptation, an identity, or a comparison (Section A).

(iii) A derivation of every implemented update, with the point at which each exact guarantee stops applying: plug-in densities, cross-fitting, smoothing, finite rank, finite buffers, target networks, truncation and projection (Tables 3 and 4).

(iv) A controlled comparison with three seeds, an exact finite oracle, predeclared thresholds, every seed reported and a cost accounting that separates task-blind pretraining from per-task learning (Sections 10 to 14).

What is not claimed. No result here proves that neural-network training converges, that a trained FB model is optimal, or that an imitation method recovers the expert. Where a source theorem assumes exactness, we state where the assumption provably fails (the finite mazes) and that it is not established elsewhere. Three seeds support statements about these runs, not broad claims of superiority.

1.4 How to read this memoir

Sections 2 and 3 fix the notation and the exact occupancy identities. Sections 4 and 5 treat f-divergence matching, Sections 6 and 7 the FB representation and FB imitation. Sections 8 and 9 compare the approaches and discuss their hyperparameters. Sections 10 to 14 describe the experiments and their results, and Sections 15 and 16 discuss them and answer the research questions. The main text presents the methods and selected derivations; Appendices B–H contain the complete proofs of the central results, and the remaining appendices the map from each implemented update to its equation and code, the source audit, the reproduction record and the per-seed results (Section A).

Each result carries a tag such as [FPG-2 | D]. The type is S for a source result reproved under its own assumptions, D for our derived adaptation, I for an identity, and C for a source result cited for comparison only, from which no guarantee is drawn; S+D marks a result with both a source part and our part. Sources are cited as FIRL [13], FPG [2], FB21 [18] and FI24 [14], with the equation, theorem and page of the supplied PDFs. A reference such as [SA R1] points to an entry of the source audit in Section J.

2 Setting, conventions and notation

The sources use different symbols, timings and divergence orientations (Section J). This section fixes one convention for the whole memoir. It never identifies two objects that the sources treat differently: in particular the initial distribution, the replay reference measure, the expert distributions and the occupancies always have distinct symbols. Logarithms are natural and divergences are in nats.

2.1 Decision process, trajectories and timing

A Markov decision process is $(\mathcal{S}, \mathcal{A}, P, \mu_0, \gamma)$: a state space $\mathcal{S}$ (finite, or a Borel subset of $\mathbb{R}^2$), a finite action set $\mathcal{A}$ (here five actions: stay, right, up, left, down), a Markov kernel $P(\cdot \mid s, a)$, an initial law μ_0 and a discount $\gamma \in (0, 1)$; we use $\gamma = 0.95$. A trajectory is $(s_0, a_0, s_1, a_1, \dots)$: action a_t is chosen in s_t and leads to $s_{t+1} \sim P(\cdot \mid s_t, a_t)$. Rollouts and demonstrations have $H = 100$ transitions, and $Z_H := \sum_{t=0}^{H-1} \gamma^t = (1 - \gamma^H)/(1 - \gamma) \approx 19.88$. The environment is *continuing*: reaching a goal changes neither P nor the rollout, and a policy may stay at the goal. A policy is a stationary Markov kernel $\pi(\cdot \mid s)$ unless stated otherwise, and $P_\pi(s, \cdot) := \sum_a \pi(a \mid s) P(\cdot \mid s, a)$. $\mathbb{P}^\pi_{\mu_0}$ is the law of the trajectory started from $s_0 \sim \mu_0$, and $\mathbb{P}^\pi_{(s,a)}$ the law started from $(s_0, a_0) = (s, a)$; $\mathrm{Law}_t := \mathbb{P}^\pi_{\mu_0}(s_t \in \cdot)$. Rewards are functions of states, read at the *next* state: the return of π is

$$J_H(\pi) := \mathbb{E}^\pi_{\mu_0} \sum_{t=0}^{H-1} \gamma^t r(s_{t+1}), \qquad J_\infty(\pi) := \mathbb{E}^\pi_{\mu_0} \sum_{t \geq 0} \gamma^t r(s_{t+1}). \tag{2.1}$$

This timing is the one of FPG (pp. 2–3) and of FI24 Eq. (1); FI24 prints its Q-function with $r(s_t)$, which is inconsistent with its own successor measure [SA R1].

2.2 Occupancy and successor measures

Definition 2.1 (occupancies)**.** For a policy π and measurable $X \subseteq \mathcal{S}$, $Y \subseteq \mathcal{S} \times \mathcal{A}$,

$$d^+_\pi(X) := (1-\gamma)\sum_{t\geq 0}\gamma^t \mathrm{Law}_{t+1}(X), \qquad d^+_{\pi,H}(X) := Z_H^{-1}\sum_{t=0}^{H-1}\gamma^t \mathrm{Law}_{t+1}(X),$$
$$d^{\mathrm{SA}}_\pi(Y) := (1-\gamma)\sum_{t\geq 0}\gamma^t \mathbb{P}^\pi_{\mu_0}((s_t,a_t)\in Y), \qquad d^{\mathrm{SA}}_{\pi,H}(Y) := Z_H^{-1}\sum_{t=0}^{H-1}\gamma^t \mathbb{P}^\pi_{\mu_0}((s_t,a_t)\in Y),$$

and d^{pre}_π is the state marginal of d^{SA}_π. d^+_π is the *next-state occupancy*, the object matched by f-PG; $d^+_{\pi,H}$ is its H-step version, which is what rollouts of length H estimate; $d^{\mathrm{SA}}_{\pi,H}$ is the *pre-action state–action occupancy* used as a diagnostic.

Definition 2.2 (successor measures)**.** The next-state successor measure of π (FI24 Eq. 1, p. 3) and its action average (FI24 Eq. 3) are

$$M^{\pi,+}(s,a,X) := \sum_{t\geq 0}\gamma^t\, \mathbb{P}^\pi_{(s,a)}(s_{t+1}\in X), \qquad M^{\pi,+}(s,X) := \sum_a \pi(a\mid s) M^{\pi,+}(s,a,X).$$

Each has total mass $1/(1-\gamma) = 20$. FB21 (Eq. 1, p. 2) uses instead the state–action successor measure from $t=0$, $M^\pi_{21}(s,a,Y) := \sum_{t\geq 0}\gamma^t \mathbb{P}^\pi_{(s,a)}((s_t,a_t)\in Y)$, which contains the atom $\delta_{(s,a)}$; the two are related by $M^\pi_{21}(s,a,X\times\mathcal{A}) = \delta_s(X) + \gamma M^{\pi,+}(s,a,X)$ (Theorem 3.6).

2.3 Divergences

For probability vectors p, q on a finite set and a convex f with $f(1)=0$,

$$D_f(p\|q) := \sum_{x:\, q(x)>0} q(x)\, f\Big(\frac{p(x)}{q(x)}\Big) + f'(\infty)\, p(\{q=0\}), \qquad f'(\infty) := \lim_{u\to\infty} f(u)/u. \tag{2.2}$$

This is the convention of FPG Table 1 and its proofs and of FIRL Eq. (18); FPG Eq. (3) prints the opposite orientation [SA D1]. In our objectives the *learner is always the first argument.* The three generators used are

$$\mathrm{JS}(p,q):\ \ f_{\mathrm{JS}}(u) = \tfrac{1}{2}\big[u\log u - (1+u)\log\tfrac{1+u}{2}\big],$$
$$\mathrm{KL}(p\|q):\ \ f(u) = u\log u, \qquad \mathrm{KL}(q\|p):\ \ f(u) = -\log u.$$

JS is the standard Jensen–Shannon divergence with half weights, $0 \leq \mathrm{JS} \leq \log 2 \approx 0.693$; the generator printed in FPG Table 1 and FIRL Table 2 is $2f_{\mathrm{JS}}$ and gives 2JS [SA D3]. The names "forward" and "reverse" KL are avoided: FPG's "fkl" is KL(learner‖target), which FIRL calls "RKL" when the expert is the target [SA D2]; we always write the arguments. For a strictly positive reference law υ and $\varepsilon \in (0,1]$, the *smoothed* law is $p_\varepsilon := (1-\varepsilon)p + \varepsilon\upsilon$ (masses are mixed with masses and densities with densities); we use $\varepsilon = 0.01$.

2.4 Three expert distributions that must not be confused

The methods use three different summaries of the same demonstrations $(s^i_0,\dots,s^i_H)$, $i=1,\dots,N_E$, with $N_E = 4$.

- $q_E := d^+_{\pi_E,H}$ is the population next-state occupancy of the expert; $\hat{q}_E$ is its empirical version (weight $\gamma^t/(N_E Z_H)$ on s^i_{t+1}, $t=0,\dots,H-1$), and $q_{E,\varepsilon}$ its smoothed version, the *f-PG target.* The finite evaluation compares learners with the exact q_E.
- $\rho_e := \frac{1}{N_E}\sum_i \frac{1}{H}\sum_{t=1}^H \delta_{s^i_t}$ weights $s_1,\dots,s_H$ uniformly within each demonstration (FI24 Eq. 8 and footnote 3, pp. 4–5); it defines the *ER_{FB} latent.*
- $\bar{\rho}_{\mathcal{T}} := |\mathcal{T}|^{-1}\sum_{t\in\mathcal{T}} \mathrm{Law}^{\pi_E}_t$ with $\mathcal{T} = \{0,\dots,H-K\}$ and $K = 60$ is the *DM_{FB} start law*; it includes $t=0$ (decision D5, Theorem G.9).

For example, an expert that reaches an absorbing goal at step 4 puts mass 0.97 on the goal under ρ_e and 0.8565 under q_E ($\gamma = 0.95$, $H = 100$; proof of Theorem G.4(e)).

2.5 Objects of the forward–backward representation

- ν is the *replay reference measure*: the law of a replay next state. It is used for the target states of the successor loss, the Gram matrix, z_r and every density statement. $\hat{\nu}$ is the empirical law of the stored next states.
- $B : \mathcal{S} \to \mathbb{R}^d$ and $F : \mathcal{S} \times \mathcal{A} \times \mathbb{R}^d \to \mathbb{R}^d$ are the backward and forward networks, with rank $d = 16$; ξ, ω are their parameters and ξ^-, ω^- the target copies.
- $G_B := \mathbb{E}_\nu[BB^\top]$ is an *uncentered* Gram matrix (FB21 and FI24 call it "Cov B"), and $\hat{G}_B$ its empirical version over the buffer next states; nothing is mean-subtracted.
- $z_r := \mathbb{E}_\nu[r\, B] \in \mathbb{R}^d$ is the task vector of a state reward r.
- The policy attached to z is the softmax policy $\pi_{z,\tau}(a \mid s) := \mathrm{softmax}_a(F(s,a,z)^\top z/\tau)$ with temperature $\tau > 0$, and $F(s,z) := \sum_a \pi_{z,\tau}(a \mid s) F(s,a,z)$. The same family is used in pretraining targets, $\mathrm{ER}_{\mathrm{FB}}$, $\mathrm{DM}_{\mathrm{FB}}$ and evaluation. FB21's greedy policy is $\pi_z^{\mathrm{gr}}(s) \in \arg\max_a F(s,a,z)^\top z$.
- Latent vectors live on the sphere $\mathcal{Z}_R := \{\|z\|_2 = R\}$ with $R = \sqrt{d} = 4$; we use $\tau = 1$ (Section 10).

Table 2 summarizes the symbols; the map from each source's symbols to ours is in Section J.

Table 2: Main symbols. "Used by" names the method or analysis in which the object appears.

Symbol	Meaning	Used by
μ_0	initial law: the Dirac mass at the start cell $(1,1)$ (finite) or at its centre (continuous)	all
γ, H, Z_H	discount 0.95, horizon 100, $Z_H = \sum_{t<H} \gamma^t$	all
$d^+_{\pi,H}$, d_ϕ	H-step next-state occupancy; $d_\phi := d^+_{\pi_\phi,H}$	f-PG, evaluation
$d^{\mathrm{SA}}_{\pi,H}$	H-step pre-action state–action occupancy	diagnostic
$M^{\pi,+}$, M^π_{21}	next-state (2024) and state–action (2021) successor measures	FB theory
q_E, $\hat{q}_E$, $q_{E,\varepsilon}$	expert occupancy, its empirical version, its smoothed version	f-PG target, evaluation
ρ_e	empirical expert state law, uniform over $s_1, \dots, s_H$	$\mathrm{ER}_{\mathrm{FB}}$
$\bar{\rho}_{\mathcal{T}}$	expert state law averaged over start times $\mathcal{T} = \{0, \dots, 40\}$	$\mathrm{DM}_{\mathrm{FB}}$
π_ϕ, ϕ	f-PG policy and its parameters	f-PG
g_ϕ, G_k, χ_k	f-PG signal $f'(d_\phi/q)$, signal-to-go, score $\nabla \log \pi_\phi(a_k \mid s_k)$	f-PG
θ, r_θ, α	reward parameters, reward and MaxEnt temperature of f-IRL	f-IRL (theory)
ν, $\hat{\nu}$	replay reference measure and its empirical version	FB
F, B, d	forward and backward networks, rank 16	FB
G_B, z_r	uncentered Gram matrix $\mathbb{E}_\nu[BB^\top]$, task vector $\mathbb{E}_\nu[rB]$	FB
$\pi_{z,\tau}$, τ, R	softmax FB policy, temperature 1, latent radius 4	FB, $\mathrm{ER}_{\mathrm{FB}}$, $\mathrm{DM}_{\mathrm{FB}}$
z_{ER}, ℓ_{DM}	$\mathrm{ER}_{\mathrm{FB}}$ latent; $\mathrm{DM}_{\mathrm{FB}}$ objective	FB imitation
r_g	evaluation reward $\exp(-\lvert x - \bar{g}\rvert^2/(2w^2))$, $w = 1/14$	evaluation only

3 Occupancy measures: the exact identities

This section collects the identities on which both families rest. They are exact, they hold for every policy, and their complete proofs are in Section B. They involve no approximation; approximations enter only with the algorithms of Sections 4, 6 and 7.

Proposition 3.1 (normalization and time shift; Theorems B.4 and B.5)**.** [OCC-1, OCC-2 | I] *For every policy π: (a) d^+_π, $d^+_{\pi,H}$, d^{pre}_π are probability measures on $\mathcal{S}$, d^{SA}_π and $d^{\mathrm{SA}}_{\pi,H}$ on $\mathcal{S} \times \mathcal{A}$, and $M^{\pi,+}(s,a,\cdot)$ has mass $1/(1-\gamma)$; (b) $d^+_\pi = (1-\gamma) \int \mu_0(\mathrm{d}s)\, M^{\pi,+}(s,\cdot)$; (c) $d^{\mathrm{pre}}_\pi = (1-\gamma)\mu_0 + \gamma\, d^+_\pi$, and $Z_H\, d^{\mathrm{pre}}_{\pi,H} = \mu_0 + \gamma Z_H\, d^+_{\pi,H} - \gamma^H \mathrm{Law}_H$.*

Proof idea. Every object is a nonnegative series of probability measures with weights summing to 1 (or to $1/(1-\gamma)$), and Tonelli's theorem gives countable additivity. (b) disintegrates the

trajectory law along its first state–action pair. (c) separates the $t = 0$ term and shifts the index. □

The time shift (c) is the difference between the 2021 and 2024 conventions: the pre-action occupancy contains the uncontrolled start through the atom $(1-\gamma)\mu_0$, the next-state occupancy does not.

Proposition 3.2 (finite resolvent and Bellman identities; Theorems B.8 and B.9)**.** [OCC-3 | I] *If $\mathcal{S}$ is finite and $R_\pi := (I - \gamma P_\pi)^{-1} = \sum_{t\geq 0} \gamma^t P_\pi^t$, then, with measures written as row vectors,*

$$M^{\pi,+}(s,a,\cdot) = P(\cdot \mid s,a)\, R_\pi, \qquad d_\pi^+ = (1-\gamma)\, \mu_0^\top P_\pi R_\pi, \qquad d_{\pi,H}^+ = Z_H^{-1} \sum_{t=0}^{H-1} \gamma^t \mu_0^\top P_\pi^{t+1},$$

and, on any state space, $M^{\pi,+}(s,a,\cdot) = P(\cdot \mid s,a) + \gamma \int P(\,\mathrm{d}s_1 \mid s,a) \sum_{a_1} \pi(a_1 \mid s_1) M^{\pi,+}(s_1,a_1,\cdot)$.

The oracle of Section 10 computes $d_{\pi,H}^+$ both by the finite sum and by the resolvent form $Z_H^{-1}\mu_0^\top P_\pi (I - \gamma^H P_\pi^H) R_\pi$; the two agree up to round-off.

Proposition 3.3 (return identity; Theorem B.11)**.** [OCC-4 | D] *For every bounded state reward r read at the next state,*

$$J_H(\pi) = Z_H \int r \,\mathrm{d}d_{\pi,H}^+, \qquad J_\infty(\pi) = \frac{1}{1-\gamma} \int r \,\mathrm{d}d_\pi^+.$$

Hence $d_{\pi,H}^+ = d_{\pi',H}^+$ if and only if $J_H(\pi) = J_H(\pi')$ for every bounded r. If two policies induce the same state kernel, $\sum_a \pi(a \mid s) P(\cdot \mid s,a) = \sum_a \pi'(a \mid s) P(\cdot \mid s,a)$ for every s, all their state occupancies coincide even when $\pi \neq \pi'$.

Proof. By linearity, $J_H(\pi) = \sum_{t<H} \gamma^t \mathbb{E}\, r(s_{t+1}) = \sum_{t<H} \gamma^t \int r \,\mathrm{dLaw}_{t+1}$, which is $Z_H \int r \,\mathrm{d}d_{\pi,H}^+$ by the definition of $d_{\pi,H}^+$; integrating against a countable sum of measures is justified for indicators, then simple functions, then bounded functions by monotone convergence. The infinite-horizon identity follows in the same way after dominated convergence. Taking $r = \mathbf{1}_X$ gives the converse. For the last claim, $\mathrm{Law}_{t+1} = \mathrm{Law}_t P_\pi$ depends on π only through P_π, so induction from $\mathrm{Law}_0 = \mu_0$ gives equal marginals. □

The last statement is why state-occupancy matching cannot identify actions. In a corner cell, "move into the wall" and "stay" have the same next-state law; two policies that differ only there have the same state occupancies and the same return for every state reward, but different state–action occupancies (Theorem B.12).

Proposition 3.4 (finite versus infinite horizon; Theorem B.13)**.** [OCC-5 | D] $d_\pi^+ = (1-\gamma^H) d_{\pi,H}^+ + \gamma^H d_\pi^{+,\mathrm{tail}}$ *for a probability measure $d_\pi^{+,\mathrm{tail}}$. Hence* $\mathrm{TV}(d_\pi^+, d_{\pi,H}^+) \leq \gamma^H$ *and* $|J_\infty(\pi) - J_H(\pi)| \leq \gamma^H \|r\|_\infty/(1-\gamma)$. *For $\gamma = 0.95$ and $H = 100$, $\gamma^H \approx 5.92 \times 10^{-3}$ and $\gamma^H/(1-\gamma) \approx 0.118$.*

Learner and expert occupancies are both taken at the same horizon H, so the f-PG objective involves no truncation; d_π^+ serves only as a reference.

Proposition 3.5 (the finite oracle; Theorems B.15 to B.17)**.** [OCC-6 | I+D] *Let $\mathcal{S}$ be finite and r a state reward read at the next state. Value iteration with the next-state Bellman operator $(TV)(s) = \max_a \sum_{s'} P(s' \mid s,a)[r(s') + \gamma V(s')]$ converges to $V^\star$, and any greedy stationary $\pi^\star$ is optimal among all history-dependent policies; a residual $\eta_\pi = \|TV^\pi - V^\pi\|_\infty$ certifies $0 \leq V^\star - V^\pi \leq \eta_\pi/(1-\gamma)$. Backward induction $W_H = 0$, $W_k = TW_{k+1}$ gives the H-step optimum $V_H^\star = W_0$ over all policies, and for $r \in [0,1]$, $0 \leq V_H^\star - V_H^{\pi^\star} \leq \gamma^H/(1-\gamma) \approx 0.1184$.*

The finite experts of Section 10 are certified this way, and the "gap" of a learner is always measured against $V_H^\star$, the optimum for the exact reward, kernel and timing of the evaluation.

Proposition 3.6 (the 2021 and 2024 conventions; Theorem B.19)**.** [OCC-7 | D] *For a state reward r: $M^{\pi}_{21}(s,a,X\times\mathcal{A}) = \delta_s(X) + \gamma M^{\pi,+}(s,a,X)$; the 2021 action value $Q^{\pi}_{21} = \mathbb{E}\sum_t \gamma^t r(s_t)$ and the next-state value $Q^{\pi,+} = \mathbb{E}\sum_t \gamma^t r(s_{t+1})$ satisfy $Q^{\pi}_{21} = r(s) + \gamma Q^{\pi,+}$; and both conventions have the same greedy and optimal policies.*

FI24 Proposition 1's identity $Q^{\star}_r = F^{\top} z_r$ therefore holds with the reward read at s_{t+1}, as in its Eq. (1); with the Q printed on FI24 p. 3 it would read $Q^{\star}_{21,r} = r(s) + \gamma F^{\top} z_r$ (Theorem F.6(c)).

Densities. All identities above are identities between measures. Density statements need domination: if every $P(\cdot \mid s,a)$ has a density with respect to a reference measure, so do d^{+}_{π} and $M^{\pi,+}(s,a,\cdot)$, but d^{pre}_{π} and M^{π}_{21} keep the atoms μ_0 and δ_s (Theorem B.20). The continuous kernel of our maze has a bounded Lebesgue density and no atom (Theorem H.4); histogram estimates never prove absolute continuity.

4 f-divergence occupancy matching: f-PG with a fixed expert target

FPG minimizes $D_f(p_\theta \| p_g)$ between the agent's state visitation and a Dirac goal target $p_g = \delta_g$ (FPG Eq. 4, p. 4). For imitation we replace the goal by a *fixed expert distribution* and the visitation by the H-step next-state occupancy. This is an adaptation, not a reproduction of FPG's experiments: the gradient below is derived for this target, and FPG's Dirac-target optimality theorems do not transfer (Theorem 4.8). Complete proofs are in Sections C and D.

4.1 Divergences: the facts used

Proposition 4.1 (nonnegativity, orientation and smoothing; Theorems C.3 to C.5 and C.7)**.** [DIV-1–DIV-3 | I+D] *Let f be convex with $f(1) = 0$.*

(a) $D_f(p\|q) \geq 0$, with equality if and only if $p = q$ when f is strictly convex at 1; this holds for the three generators of Section 2. $0 \leq \mathrm{JS} \leq \log 2$; $\mathrm{KL}(p\|q) < \infty$ if and only if $\mathrm{supp} p \subseteq \mathrm{supp} q$.

(b) (orientation) With the perspective $f^{\circ}(u) := u f(1/u)$, $D_f(p\|q) = D_{f^{\circ}}(q\|p)$, including when p or q has zeros. $(u\log u)^{\circ} = -\log u$ and $f^{\circ}_{\mathrm{JS}} = f_{\mathrm{JS}}$.

(c) (smoothing) For a strictly positive reference law υ and $\varepsilon \in (0,1]$, every ratio $p_\varepsilon/q_\varepsilon$ lies in $[\varepsilon\upsilon_{\min}, 1/(\varepsilon\upsilon_{\min})]$, so $D_f(p_\varepsilon\|q_\varepsilon)$ is finite. If $p = d_\phi$ depends smoothly on ϕ and q is fixed, $\nabla_\phi D_f((d_\phi)_\varepsilon\|q_\varepsilon) = (1-\varepsilon)\sum_x f'((d_\phi)_\varepsilon(x)/q_\varepsilon(x))\nabla_\phi d_\phi(x)$.

Part (b) translates FIRL's expert-first objective into our learner-first convention: FIRL's "FKL" ($f = u\log u$, expert first) is our $\mathrm{KL}(q\|d)$, and its "RKL" is our $\mathrm{KL}(d\|q)$ (Theorem C.6). Smoothing changes the objective: the minimizers of the smoothed divergence need not be those of the unsmoothed one, and the evaluation always reports the unsmoothed divergence (Theorem C.8).

4.2 The objective and the exact gradient

Let $\{B_1,\dots,B_n\}$ be a finite partition of $\mathcal{S}$ into *cells*: the states of a finite maze, or the 356 histogram cells of positive free area of the continuous maze (Theorem H.6). For a parametric policy π_ϕ write $d_\phi(b) := d^{+}_{\pi_\phi,H}(B_b)$, and let q be a *fixed* probability vector on the cells. The objective is

$$J(\phi) := D_f(d_\phi\|q) = \sum_b q(b)\, f\big(d_\phi(b)/q(b)\big). \tag{4.1}$$

We assume (Theorem D.1): $\pi_\phi(a \mid s) > 0$, $\phi \mapsto \pi_\phi(a \mid s)$ is C^1, and the score $\chi_k := \nabla_\phi \log \pi_\phi(a_k \mid s_k)$ is bounded locally uniformly in ϕ; f is convex and C^1 on $(0,\infty)$; $q(b) > 0$ and $d_\phi(b) > 0$ on

every cell, or both arguments are smoothed. Tabular softmax policies satisfy the score condition with $\|\chi_k\|_2 \le \sqrt{2}$ (Theorem D.2); MLP policies satisfy it with C^1 activations on a bounded state space (Theorem D.3). Define the *f-PG signal* and the *signal-to-go*

$$g_\phi(b) := f'(d_\phi(b)/q(b)), \quad g_\phi(s) := g_\phi(b(s)), \qquad G_k := \sum_{t=k}^{H-1} \gamma^{t-k} g_\phi(s_{t+1}).$$

Theorem 4.2 (exact gradient, causal form; Theorems D.6 and D.7)**.** [FPG-1, FPG-2 | D] *Under these assumptions J is C^1 and*

$$\nabla d_\phi(b) = Z_H^{-1} \sum_{t=0}^{H-1} \gamma^t \, \mathbb{E}_\phi\Big[\mathbf{1}[s_{t+1} \in B_b] \sum_{k=0}^{t} \chi_k\Big], \tag{4.2}$$

$$\nabla J(\phi) = \sum_b g_\phi(b) \, \nabla d_\phi(b) = Z_H^{-1} \, \mathbb{E}_\phi\Big[\sum_{k=0}^{H-1} \gamma^k \, \chi_k \, G_k\Big]. \tag{4.3}$$

Proof. Step 1: likelihood ratio. Let h be a bounded measurable function of $(s_0, a_0, \dots, s_H)$ that does not depend on ϕ. Writing $\mathbb{E}_\phi h$ as an integral of $h \prod_{k<H} \pi_\phi(a_k \mid s_k)$ against the ϕ-free measure $\mu_0 \otimes \prod_k P(\,\mathrm{d}s_{k+1} \mid s_k, a_k)$ (times counting measure on actions), the integrand is C^1 in ϕ with gradient $h \prod_k \pi_\phi(a_k \mid s_k) \sum_k \chi_k$, bounded by $HC\|h\|_\infty$ on a ball around ϕ. The mean value theorem and dominated convergence allow differentiation under the integral:

$$\nabla_\phi \mathbb{E}_\phi h = \mathbb{E}_\phi\Big[h \sum_{k=0}^{H-1} \chi_k\Big].$$

Step 2: causality. Let $\mathcal{F}_k$ be generated by $(s_0, a_0, \dots, s_k)$. Given $\mathcal{F}_k$, $a_k \sim \pi_\phi(\cdot \mid s_k)$, so $\mathbb{E}[\chi_k \mid \mathcal{F}_k] = \sum_a \pi_\phi(a \mid s_k) \nabla \log \pi_\phi(a \mid s_k) = \nabla \sum_a \pi_\phi(a \mid s_k) = 0$. Hence $\mathbb{E}[Y \chi_k] = \mathbb{E}[Y \, \mathbb{E}[\chi_k \mid \mathcal{F}_k]] = 0$ for every bounded $\mathcal{F}_k$-measurable Y. *Step 3: occupancy gradient.* Apply Step 1 to $h = \mathbf{1}[s_{t+1} \in B_b]$. For $k \ge t+1$ the indicator is $\mathcal{F}_k$-measurable, so these terms vanish by Step 2; multiplying by γ^t / Z_H and summing over t gives (4.2). *Step 4: chain rule.* Since $d_\phi(b)/q(b) \in (0, \infty)$, $\nabla[q(b) f(d_\phi(b)/q(b))] = q(b) f'(d_\phi(b)/q(b)) \nabla d_\phi(b)/q(b) = g_\phi(b) \nabla d_\phi(b)$; sum over b. *Step 5: reordering.* Substitute (4.2) and use $\sum_b g_\phi(b) \mathbf{1}[s_{t+1} \in B_b] = g_\phi(s_{t+1})$: $\nabla J = Z_H^{-1} \sum_{t<H} \gamma^t \mathbb{E}[g_\phi(s_{t+1}) \sum_{k \le t} \chi_k]$. The index set $\{0 \le k \le t \le H-1\}$ can be summed over t first or over k first, and $\gamma^t = \gamma^k \gamma^{t-k}$; this gives the causal form. □

Signs and weights. The update is a *descent*, $\phi \leftarrow \phi - \eta \widehat{\nabla J}$: in REINFORCE terms [17, 19] the per-step "reward" is $-g_\phi(s_{t+1})$. Since f' is nondecreasing, g_ϕ is larger on cells that the learner over-visits relative to the target, and descent lowers the probability of actions whose signal-to-go is above the baseline (a first-order heuristic, not a theorem). The score at step k carries the weight γ^k and the signal at s_{t+1} the weight γ^{t-k} inside G_k. Dropping γ^k, common in discounted policy gradients, does *not* give ∇J; FPG's Eqs. (12)–(13) also differ from (4.3), while its on-policy Eq. (10), re-indexed, equals it [SA F5]. The signal g_ϕ is not a reward: it depends on ϕ through d_ϕ and is recomputed at every iteration.

Corollary 4.3 (baselines and Monte Carlo estimator; Theorems D.8 and D.9)**.** [FPG-3, FPG-4 | I] *(a) For every bounded $\mathcal{F}_k$-measurable β_k (a function of the history up to s_k, for example of (k, s_k)), $\nabla J = Z_H^{-1} \mathbb{E}_\phi[\sum_k \gamma^k \chi_k (G_k - \beta_k)]$; adding a constant to g_ϕ leaves ∇J unchanged; a "baseline" that depends on a_k changes it in general. (b) With the* exact *signal, the average $\hat{\nabla}_N$ of this expression over N independent trajectories is unbiased, has covariance $\mathrm{Cov}(\hat{\nabla}_1)/N$ and converges almost surely. If $\nabla J(\phi) \ne 0$, an exact step $\phi - \eta \nabla J(\phi)$ decreases J for small η.*

Nothing more is claimed about optimization: J is not convex in ϕ, and no statement is made about the stationary points that descent may reach.

4.3 From the exact gradient to the implemented estimator

The signal g_ϕ requires the unknown d_ϕ; the implementation replaces it by a *plug-in* estimate from sampled trajectories. The next result says exactly what this costs.

Proposition 4.4 (plug-in signals, smoothing and cross-fitting; Theorems D.10 and H.11)**.** [FPG-5, FPG-9 | D]

(a) If the signal $\hat{g}$ is a random function on cells independent *of the trajectories in the score sums, then $\mathbb{E}[\hat{\nabla}_N \mid \hat{g}] = \sum_b \hat{g}(b)\nabla d_\phi(b)$ and $\|\mathbb{E}[\hat{\nabla}_N \mid \hat{g}] - \nabla J\| \le \max_b|\hat{g}(b) - g_\phi(b)|\, C\kappa_H$ with $\kappa_H := Z_H^{-1}\sum_{t<H}(t+1)\gamma^t$.*

(b) The plug-in signal is biased in general, $\mathbb{E}f'(\hat{d}/q) \neq f'(d_\phi/q)$, because f' is nonlinear; the resulting gradient bias is $\sum_b(\mathbb{E}\hat{g}(b) - g_\phi(b))\nabla d_\phi(b)$. It vanishes if the signal bias is constant across cells, because $\sum_b \nabla d_\phi(b) = 0$, but constancy is not necessary; in general no unbiasedness of the gradient estimator is claimed. If $\hat{g}$ is computed from the same *trajectories as the scores, (a) fails: for a one-step example with $\mathrm{KL}(d\|q)$ the same-batch estimator has mean 0 while $\nabla J = 3\log 3/16 > 0$.*

(c) With both arguments smoothed, Theorem 4.2 and Theorem 4.3 hold verbatim for $J_\varepsilon(\phi) := D_f((d_\phi)_\varepsilon\|q_\varepsilon)$ with the signal $(1-\varepsilon)f'\big((d_\phi)_\varepsilon/q_\varepsilon\big)$, and no positivity of d_ϕ is needed.

(d) (cross-fitting) Split the N trajectories of an iteration into folds $\mathcal{I}_1, \dots, \mathcal{I}_{n_\mathrm{f}}$, $n_\mathrm{f} \ge 2$, and give the trajectories of fold j a signal $\hat{g}_{-j}$ and a time baseline β_{-j} computed from the other folds and the demonstrations only. Conditionally on the other folds, each term of fold j has mean $\Gamma_\phi(\hat{g}_{-j}) := \nabla_\phi \int \hat{g}_{-j}\,\mathrm{d}d^+_{\pi_\phi,H}$; the baseline contributes nothing; for cell signals $\Gamma_\phi(\hat{g}_{-j}) = \sum_b \hat{g}_{-j}(b)\nabla d_\phi(b)$ and (a) applies.

(e) For a kernel-density signal evaluated at points, no exact cell signal exists, and no bound between $\Gamma_\phi(\hat{g}_{-j})$ and a gradient of J is claimed (Theorem D.11).

Cross-fitting [in the spirit of 4] removes the same-batch dependence of (b), not the plug-in bias. Conditionally on the other fold, the sampled update is an unbiased but noisy estimate of the gradient of the surrogate $\int \hat{g}_{-j}\,\mathrm{d}d_\phi$, in which the fitted signal is held fixed. It is not an unbiased estimate of ∇J: the plug-in bias of (b) remains, and for the kernel-density signal no bound relates the surrogate gradient to ∇J (e).

Algorithm 1: implemented f-PG iteration (`il_memoir/fpg.py`, `FPGLearner.step`)

Fixed before learning. The target $\hat{q}_E$ from the demonstration *states* (weights $\gamma^t/(N_E Z_H)$ on s^i_{t+1}), smoothed with $\varepsilon = 0.01$: counts on the 40 or 49 states (finite), or a cut-and-normalize weighted Gaussian KDE with bandwidth $h = 0.05$ and the reference density $1/\lambda(\mathcal{S})$ (continuous; Theorem H.7). The learner never receives P, rewards, goals or expert actions.

Policy. Finite: tabular softmax logits, initialized at 0 (uniform policy). Continuous: categorical MLP on raw (x, y) with two tanh hidden layers of width 64, output layer scaled by 0.01 at initialization.

Each iteration.

1. Collect $N = 20$ trajectories of $H = 100$ transitions from μ_0 with π_ϕ; all 2,000 transitions are counted.
2. Assign trajectory i to fold $i \bmod 2$. For each fold j, estimate the learner occupancy $\hat{d}_{-j}$ from the other fold with the same weights and estimator as the target; set $\hat{g}_{-j}(s^i_{t+1}) = (1-\varepsilon)f'\big(\hat{d}_{-j,\varepsilon}(s^i_{t+1})/\hat{q}_{E,\varepsilon}(s^i_{t+1})\big)$ for $i \in \mathcal{I}_j$, the signal-to-go $\hat{G}^i_k = \sum_{t\ge k}\gamma^{t-k}\hat{g}_{-j}(s^i_{t+1})$, and the baseline $\beta_{-j}(k)$, the mean signal-to-go of the other fold at step k.
3. Minimize the surrogate $L(\phi) = \frac{1}{N}\sum_i \sum_{k<H} \frac{\gamma^k}{Z_H}(\hat{G}^i_k - \beta_{-j(i)}(k))\log \pi_\phi(a^i_k \mid s^i_k)$, whose gradient is the estimator of Theorem 4.4(d), by one Adam step [10] with learning rate 0.3 (finite) or 0.003 (continuous).

Budget: 25 iterations (50,000 transitions) in the finite mazes, 50 iterations (100,000) in the continuous maze. No PPO clipping and no entropy bonus are used.

Table 3: f-PG: exact objects and their implemented approximations. [I] implementation approximation, [C] change of objective.

Quantity	Exact object	Implemented	Consequence
objective	$D_f(d_\phi \| q_E)$	$D_f((d_\phi)_\varepsilon \| \hat{q}_{E,\varepsilon})$, 4 demonstrations [C]	the gradient formula of Theorem 4.2 holds for the smoothed objective (Theorem 4.4(c)); its minimizers need not be those of the unsmoothed one
signal	$f'(d_\phi/q)$	plug-in from the other fold [I]	biased (b); conditional mean $\Gamma_\phi(\hat{g}_{-j})$ (d); bound (a) for counts
continuous signal	cell signal	KDE at the sampled points [I]	no bound to a gradient of J (e); Euclidean truncation ignores walls (Theorem H.8)
gradient	expectation	20 trajectories per step [I]	unbiased for ∇J only with the exact signal (Theorem 4.3(b)); with the fitted signal, conditionally unbiased for the surrogate gradient only (d)
step	exact small step	Adam, fixed learning rate [I]	no descent or convergence guarantee
policy class	any C^1 family	tabular softmax; tanh MLP	a deterministic expert is not representable with finite logits (Theorem 4.7)

4.4 State–action occupancies

Proposition 4.5 (state–action extension; Theorems D.13 and D.14)**.** [FPG-6 | D] *In a finite MDP, for* $J^{\mathrm{SA}}(\phi) := D_f(d^{\mathrm{SA}}_{\pi_\phi,H} \| q^{\mathrm{SA}})$ *with* q^{SA} *fixed and both positive on the computation set,*

$$\nabla J^{\mathrm{SA}} = Z_H^{-1} \mathbb{E}_\phi\Big[\sum_{k=0}^{H-1} \gamma^k \chi_k \, G_k^{\mathrm{SA}}\Big], \qquad G_k^{\mathrm{SA}} := \sum_{t=k}^{H-1} \gamma^{t-k} f'\Big(\frac{d_\phi^{\mathrm{SA}}}{q^{\mathrm{SA}}}(s_t, a_t)\Big).$$

The current-step term $t = k$ *is* not *a baseline, because* $\mathbf{1}[a_t = a]$ *is not* $\mathcal{F}_t$*-measurable. Against an unsmoothed deterministic expert,* JS *and* $\mathrm{KL}(q_E^{\mathrm{SA}} \| d_\phi^{\mathrm{SA}})$ *are finite while* $\mathrm{KL}(d_\phi^{\mathrm{SA}} \| q_E^{\mathrm{SA}}) = +\infty$ *for every positive policy; smoothing both arguments makes both* KL *directions finite.*

We do not optimize J^{SA}: the learners use states only. The finite evaluation reports $\mathrm{JS}(d^{\mathrm{SA}}_{\pi,H}, q_E^{\mathrm{SA}})$ as a diagnostic of how far state matching is from action matching.

4.5 The state-entropy reading and the optimum

Proposition 4.6 (state-MaxEnt identity; FPG Lemma 5.1, p. 6; Theorem D.16)**.** [FPG-7 | S+D] *If* $\mathrm{supp} d \subseteq \mathrm{supp} q$*, then* $\mathrm{KL}(d\|q) = -\mathcal{H}(d) - \mathbb{E}_d[\log q]$*. Minimizing* $\mathrm{KL}(d^+_{\pi,H}\|q)$ *is maximizing* $\mathcal{H}(d^+_{\pi,H}) + \mathbb{E}_{d^+_{\pi,H}}[\log q]$*: the entropy of the* state occupancy *enters with a plus sign. It is not the entropy of the action distribution, and neither maximizes the other in general (Theorem D.17).*

Proof. $\mathrm{KL}(d\|q) = \sum_{d>0} d \log d - \sum_{d>0} d \log q$, and every $\log q$ is finite on $\mathrm{supp} d$. □

Proposition 4.7 (the fixed-target optimum; Theorem D.18)**.** [FPG-8 | D] *In a finite MDP with tabular softmax policies:* $J \geq 0$ *with equality if and only if* $d_\phi = q$*; if* $q = q_E$ *and* $J(\phi) = 0$*, the learner has the expert's* H*-step return for every state reward (Theorem 3.3); if the learner can reach a state outside* $\mathrm{supp} q_E$*,* $\mathrm{KL}(d_\phi\|q_E) = +\infty$ *for every* ϕ*; for* JS *and* $\mathrm{KL}(q_E\|d_\phi)$*,* $\inf_\phi J = 0$*, but the infimum need not be attained with finite logits (Theorem D.19).*

The third statement is why $\mathrm{KL}(d\|q)$ needs smoothing against a deterministic expert, which never visits most states.

Remark 4.8 (FPG's Dirac-target theorems; Theorem D.21). [FPG-C1 | C] FPG's optimality results (Theorem 5.1, p. 6; Theorem 4.2, p. 5) use $D_f(p\|\delta_g) = f(p(g)) + f'(\infty)(1-p(g))$, a monotone function of the goal visitation. For a non-degenerate expert target the divergence depends on the whole occupancy and the argument has no analogue. We draw no optimality guarantee from these theorems.

5 f-IRL: reward learning by marginal matching (theory only)

f-IRL [13] learns a state reward r_θ such that the maximum-entropy (MaxEnt) policy of r_θ matches the expert's state marginal under an f-divergence. It is not implemented here; we reprove its central result under its exact hypothesis and relate it to f-PG. The complete proofs are in Section E.

Setting. f-IRL uses a finite, undiscounted horizon T_{IRL}, a marginal that counts $s_1, \dots, s_{T_{\mathrm{IRL}}}$ (not s_0), an expert-first objective, and parameters θ of the *reward*:

$$L_f(\theta) := D_f(q_E^{\mathrm{IRL}}\|\rho_\theta), \qquad \rho_\theta(s) := \frac{1}{T_{\mathrm{IRL}}}\mathbb{E}_{\tau\sim\rho_\theta}\eta_\tau(s), \quad \eta_\tau(s) := \sum_{t=1}^{T_{\mathrm{IRL}}} \mathbf{1}[s_t = s]$$

(FIRL Eq. 2, p. 3; A.1, p. 11). The central hypothesis is that the agent's law over state sequences is the *Gibbs* law

$$\rho_\theta(\tau) = \bar{p}(\tau)\exp\Big(\frac{1}{\alpha}\sum_{t=1}^{T_{\mathrm{IRL}}} r_\theta(s_t)\Big)\Big/ Z(\theta) \tag{5.1}$$

for a base weight $\bar{p}$ that does not depend on θ and a temperature $\alpha > 0$ (Theorem E.1). FIRL writes $\bar{p}(\tau) = \rho_0(s_0)\prod_t p(s_{t+1} \mid s_t, a_t)$ although τ contains no actions; (5.1) is the property the proof actually uses. We do not claim that any trained MaxEnt or soft actor–critic policy realizes it [SA U-7].

Theorem 5.1 (FIRL Theorem 4.1, p. 3; Theorem E.4)**.** [FIRL-1 | S] *Under* (5.1)*, a finite state space, C^1 rewards in θ, visitability of every state, and f convex and C^1 on $(0,\infty)$ (with $f(0)$ finite if the expert marginal has zeros),*

$$\nabla_\theta L_f(\theta) = \frac{1}{\alpha T_{\mathrm{IRL}}}\mathrm{Cov}_{\tau\sim\rho_\theta}\Big(\sum_{t=1}^{T_{\mathrm{IRL}}} h_f(u_\theta(s_t)),\ \sum_{t=1}^{T_{\mathrm{IRL}}} \nabla_\theta r_\theta(s_t)\Big),$$

where $h_f(u) := f(u) - uf'(u)$ and $u_\theta := q_E^{\mathrm{IRL}}/\rho_\theta$.

Proof. Gibbs derivative. With $R_\theta(\tau) = \sum_t r_\theta(s_t)$, $\log\rho_\theta(\tau) = \log\bar{p}(\tau) + R_\theta(\tau)/\alpha - \log Z(\theta)$ and $\nabla\log Z = \mathbb{E}_{\rho_\theta}[\nabla R_\theta]/\alpha$, so $\nabla\rho_\theta(\tau) = \alpha^{-1}\rho_\theta(\tau)(\nabla R_\theta(\tau) - \mathbb{E}\nabla R_\theta)$. Summing against $\eta_\tau(s)/T_{\mathrm{IRL}}$ gives $\nabla\rho_\theta(s) = (\alpha T_{\mathrm{IRL}})^{-1}\mathrm{Cov}(\eta_\tau(s), \nabla R_\theta)$; all sums are finite. *Chain rule.* Visitability gives $\rho_\theta(s) > 0$, and for $q_E^{\mathrm{IRL}}(s) > 0$, $\nabla[\rho_\theta f(q/\rho_\theta)] = \nabla\rho_\theta\, f(u) - \rho_\theta f'(u)\, q\,\nabla\rho_\theta/\rho_\theta^2 = h_f(u)\nabla\rho_\theta$; for $q_E^{\mathrm{IRL}}(s) = 0$ the term is $\rho_\theta(s)f(0)$, with gradient $h_f(0)\nabla\rho_\theta(s)$ where $h_f(0) := f(0)$. *Covariance.* By bilinearity, $\nabla L_f = (\alpha T_{\mathrm{IRL}})^{-1}\mathrm{Cov}(\sum_s h_f(u(s))\eta_\tau(s), \nabla R_\theta)$, and $\sum_s h_f(u(s))\eta_\tau(s) = \sum_t h_f(u(s_t))$. □

Proposition 5.2 (the weight h_f and the orientation; Theorem E.6)**.** [FIRL-2 | S+D] $h_f'(u) = -uf''(u) \le 0$*, and $h_f(u) = (f^\circ)'(1/u)$: FIRL's weight is the derivative of the* learner-first *generator evaluated at the learner-to-expert ratio. For the three divergences:*

FIRL name	*f (expert first)*	$h_f(u)$, $u = q/\rho$	*our divergence*	$(f^\circ)'(v)$, $v = \rho/q$
FKL	$u \log u$	$-u$	$\mathrm{KL}(q\|d)$	$-1/v$
RKL	$-\log u$	$1 - \log u$	$\mathrm{KL}(d\|q)$	$1 + \log v$
JS	f_{JS}	$\frac{1}{2}\log\frac{2}{1+u}$	JS	$\frac{1}{2}\log\frac{2v}{1+v}$

Proposition 5.3 (relation with f-PG; Theorem E.7)**.** [FIRL-3 | D] *For any C^1 parameterization $\vartheta \mapsto m_\vartheta > 0$ of a learner marginal and a fixed expert marginal $q > 0$,*

$$\nabla_\vartheta D_f(q\|m_\vartheta) = \sum_x h_f\Big(\frac{q(x)}{m_\vartheta(x)}\Big)\nabla m_\vartheta(x) = \sum_x (f^\circ)'\Big(\frac{m_\vartheta(x)}{q(x)}\Big)\nabla m_\vartheta(x).$$

So f-IRL and f-PG weight each state by the *same* function of the occupancy ratio. They differ in three respects: (i) how ∇m is obtained, by the Gibbs covariance over reward parameters in f-IRL and by the likelihood ratio over policy parameters in f-PG; (ii) which marginal is matched, the undiscounted uniform marginal over $s_1, \ldots, s_{T_{\mathrm{IRL}}}$ in f-IRL and the discounted next-state occupancy $d^+_{\pi,H}$ here; (iii) what is learned, a reward that needs a MaxEnt inner loop after every reward update (FIRL Algorithm 1, p. 4) in f-IRL, and the policy itself in f-PG.

Sign of the reward update. FIRL descends L_f. With tabular rewards $r_\theta(s) = \theta_s$, the s'-component of the gradient is $(\alpha T_{\mathrm{IRL}})^{-1}\mathrm{Cov}(\sum_t h_f(u(s_t)), \eta_\tau(s'))$, so the reward of s' increases when its visit count is negatively correlated with the trajectory signal $\sum_t h_f(u(s_t))$. Since h_f is nonincreasing, trajectories through states where the expert/agent ratio is large have a small signal; the paper's reading that such states gain reward (p. 4) is a heuristic about the typical sign of this covariance, not a consequence of the formula (Theorem E.5). Adding a constant to h_f does not change a covariance, which is why FIRL's $h_{\mathrm{JS}} = -\log(1+u)$ may drop $\log 2$ [SA D3].

Exact versus approximate in f-IRL. The covariance formula is exact only for the Gibbs law (5.1). The practical algorithm departs from it in four ways (Theorem E.8): the agent is an approximately trained soft actor–critic policy, so its law is not exactly Gibbs; ρ_θ is estimated by kernel density estimation (FIRL App. C.1, p. 18); the covariance is estimated from samples; and the practical Eq. (6), p. 5, mixes expert trajectories into the covariance, which the paper itself notes biases the gradient. FIRL's claim that the learned state-only reward is recovered up to a constant (§4.3, p. 4; App. A.4, p. 13) rests on a theorem of AIRL [7], external to the supplied PDFs; we report it as the source's claim and do not reprove it.

6 Forward–backward representations

The FB representation [18] learns successor measures, an object introduced for value prediction by Dayan [6] and generalized to features by Barreto et al. [3], for a whole family of policies at once. We work in the next-state convention of FI24 and translate FB21's 2021 statements with Theorem 3.6. Complete proofs are in Section F; the setting is Theorem F.1: ν a finite reference measure, B bounded, $F(\cdot,\cdot,z)$ bounded for each z.

6.1 Successor measures and the FB model

Proposition 6.1 (successor Bellman identity; FB21 Eq. 4, p. 4; Theorem F.5)**.** [FB-1 | S] *For every stationary policy π,*

$$M^{\pi,+}(s,a,\cdot) = P(\cdot \mid s,a) + \gamma \int P(\,\mathrm{d}s_1 \mid s,a) \sum_{a_1} \pi(a_1 \mid s_1)\, M^{\pi,+}(s_1,a_1,\cdot),$$

and $M^{\pi,+}$ is the unique bounded solution. In the 2021 convention the first term is the atom $\delta_{(s,a)}$ instead of $P(\cdot \mid s,a)$.

The *FB model* of the successor measure is $\hat{M}_z(s,a,\mathrm{d}x) := F(s,a,z)^\top B(x)\,\nu(\mathrm{d}x)$. It is *exact at* z for a policy family (π_z) if $\hat{M}_z(s,a,\cdot) = M^{\pi_z,+}(s,a,\cdot)$ for every (s,a) (Theorem F.2); FI24 Eq. (2) states this with "$\approx$".

Theorem 6.2 (reward-to-task; FB21 Theorem 2, p. 3; FI24 Proposition 1, p. 4; Theorem F.6)**.** [FB-2 | S] *Let r be a bounded state reward and $z_r = \int rB\,\mathrm{d}\nu$.*

(a) (policy evaluation, any family) If the representation is exact at z, then $Q_r^{+,\pi_z}(s,a) = F(s,a,z)^\top z_r$ for all (s,a).

(b) (optimality, greedy family) If the greedy *representation is exact at z_r, then $\pi^{\mathrm{gr}}_{z_r}$ is optimal among all policies and $Q_r^\star = F(\cdot,\cdot,z_r)^\top z_r$.*

Proof. (a) $Q_r^{+,\pi_z}(s,a) = \mathbb{E}^{\pi_z}_{(s,a)} \sum_t \gamma^t r(s_{t+1}) = \int r\,\mathrm{d}M^{\pi_z,+}(s,a,\cdot)$ by dominated convergence and (B.3). Exactness replaces the measure by $F(s,a,z)^\top B\,\nu$, and linearity of the integral over the d bounded functions rB_i gives $F(s,a,z)^\top \int rB\,\mathrm{d}\nu = F(s,a,z)^\top z_r$. (b) Let $\pi := \pi^{\mathrm{gr}}_{z_r}$. By (a) at $z = z_r$, $F(s,\cdot,z_r)^\top z_r = Q_r^{+,\pi}(s,\cdot)$, and $\pi(s)$ maximizes the left-hand side, so π is greedy with respect to its own action values. A policy that is greedy with respect to its own Q-function is a fixed point of the Bellman optimality operator, hence optimal among all history-dependent policies (Theorem F.4(v), applied with the one-step reward $c_r(s,a) = \int P(\mathrm{d}x \mid s,a) r(x)$, Theorem F.3). Then $Q_r^\star = Q_r^{+,\pi} = F^\top z_r$. □

Exactness is needed only at the single latent z_r; FB21 assumes it for all z so that the conclusion holds for every reward. Greediness enters only in (b). Exactness also forces $M^{\pi_z,+}(s,a,\cdot) \ll \nu$, which is stronger than FB21's support condition (Remark 5, p. 16; Theorem F.25). In the 2021 convention the same argument gives $Q^\star_{21,r} = r(s) + \gamma F^\top z_r$.

6.2 When can the model be exact? The rank obstruction

Proposition 6.3 (finite realizability and its limit; FB21 Proposition 6, p. 17; Theorems F.8 and F.10)**.** [FB-3, FB-4 | S+D] *Let $\mathcal{S}$ and $\mathcal{A}$ be finite.*

(a) If $\nu(s) > 0$ for every s and $d \geq \#\mathcal{S}$, an exact greedy representation exists on $\mathbb{R}^d$, with B one-hot (our next-state analogue of FB21 Proposition 6, which needs $d \geq \#\mathcal{S} \cdot \#\mathcal{A}$). For every $\tau > 0$ an exact softmax representation also exists, because a softmax fixed point $\pi = \mathrm{softmax}(Q_r^{+,\pi}/\tau)$ exists for every reward (Brouwer's theorem; uniqueness is not claimed).

(b) For every stationary π, the matrix $[M^{\pi,+}(s,a,\{s'\})]$ has the rank k of the matrix $\tilde{P}$ with rows $P(\cdot \mid s,a)$, so an exact representation needs $d \geq k$. If some action has a nonsingular transition matrix, for example a stay action $(1-\epsilon)\delta_s + \epsilon u_s$ with $\epsilon < \frac{1}{2}$, then $k = \#\mathcal{S}$.

Proof of (b). $M^{\pi,+}(s,a,\cdot) = P(\cdot \mid s,a)R_\pi$ with R_π invertible (Theorem 3.2), so the successor matrix is $\tilde{P}R_\pi$ and has rank k; an exact model is a product $\mathbf{F}_z\mathbf{B}_\nu^\top$ through $\mathbb{R}^d$, of rank at most d. A matrix $(1-\epsilon)I + \epsilon U$ with U row-stochastic is strictly diagonally dominant when $\epsilon < \frac{1}{2}$, hence nonsingular, and its rows are rows of $\tilde{P}$. □

Consequence for this study. Our finite kernels have such a stay action ($\epsilon = 0.1$), and $\mathrm{rank}\,\tilde{P} = \#\mathcal{S}$ is 49 (open) or 40 (four rooms) (Theorem H.2); the oracle measures smallest singular values of about 2.01. Since $d = 16 < 40$, *no exact rank-16 FB model exists in either finite maze, for any policy family.* Every result that assumes exactness (Theorem 6.2, Theorem 6.4(c)–(d), Theorem 7.1, Theorem 7.4(a)–(b),(d) and their corollaries) is therefore an interpretation for the trained finite-maze models, not a guarantee; only the conditional bound of Theorem 6.5 applies, with an error that the oracle measures. In the continuous maze exactness is not disproved, but it is not established either, and the error cannot be computed.

6.3 Softmax policies

The implementation uses the softmax family $\pi_{z,\tau}$ in pretraining targets, imitation and evaluation (FB21 itself uses a softmax in its training targets, Algorithm 1, line 18, with $\tau = 200$). Theorem 6.2(b) does not apply to it.

Proposition 6.4 (what replaces optimality for softmax policies; Theorems F.12 and F.14)**.** [FB-5 | D] *Let c be a bounded one-step reward and $\tau > 0$.*

(a) A softmax fixed point $\pi = \mathrm{softmax}(Q_c^\pi/\tau)$ satisfies $0 \le V_c^\star - V_c^\pi \le \tau \log|\mathcal{A}|/(1-\gamma)$.
(b) It is strictly suboptimal at every state where $Q_c^\pi(s,\cdot)$ is not constant.
(c) Under an exact softmax representation at z_r, $\pi_{z_r,\tau}$ is such a fixed point for r.
(d) If instead the representation is exact at ζz_r, $\zeta > 0$, the bound holds with temperature τ/ζ; on the sphere $\mathcal{Z}_R$ the effective temperature *relative to r is $\tau\|z_r\|_2/R$. If $G_B = I$, $\|z_r\|_2$ is the $L^2(\nu)$ norm of the projection of r onto* span(B).

Proof of (a). By the log-sum-exp inequality, $\sum_a \pi(a \mid s) Q_c^\pi(s,a) \ge \max_a Q_c^\pi(s,a) - \tau \log|\mathcal{A}|$, that is $V_c^\pi \ge T_c V_c^\pi - \tau \log|\mathcal{A}|$, while $V_c^\pi \le T_c V_c^\pi$. The residual bound $\|V - V_c^\star\|_\infty \le \|T_c V - V\|_\infty/(1-\gamma)$ concludes. □

At the frozen $\tau = 1$ with five actions and $\gamma = 0.95$, $\tau \log 5/(1-\gamma) \approx 32.2$, which exceeds the largest possible return $1/(1-\gamma) = 20$ of a reward in $[0,1]$: at $\zeta = 1$ the bound is vacuous. The temperature was chosen in the pilot for keeping the FB policy stochastic (Section 10), not for this bound. Part (d) shows that τ and the scale of z are coupled under exactness [SA T2]; for a trained network $F(s,a,\zeta z)^\top \zeta z \ne \zeta F(s,a,z)^\top z$ in general, so the coupling is only empirical.

6.4 Approximate representations

Theorem 6.5 (approximation bound; FB21 Theorem 8, p. 19, Eqs. 100–103; Theorem F.17)**.** [FB-6 | S+D] *Without any exactness, let $\mathcal{E}_z(s,a,\cdot) := M^{\pi_z,+}(s,a,\cdot) - \hat{M}_z(s,a,\cdot)$ be the error kernel, and let $\|\cdot\|_A$, $\|\cdot\|_B$ be dual norms, $|\int f \,\mathrm{d}\mu| \le \|f\|_A \|\mu\|_B$ (for example the sup norm and total variation).*

(a) For every z and every policy family, $|Q_r^{+,\pi_z}(s,a) - F(s,a,z)^\top z_r| \le \|r\|_A \|\mathcal{E}_z(s,a,\cdot)\|_B$.
(b) With $\delta := \|r\|_A \sup_{s,a} \|\mathcal{E}_{z_r}(s,a,\cdot)\|_B$: the greedy policy satisfies $V_r^\star - V_r^\pi \le 2\delta/(1-\gamma)$, and the softmax policy $V_r^\star - V_r^\pi \le (\tau \log|\mathcal{A}| + 2\delta)/(1-\gamma)$. With a scaled latent ζz_r the same holds with τ/ζ and the error at ζz_r.

Part (a) needs no greediness and is the only FB guarantee that applies to our trained models. It is conditional on the error, which the pretraining loss does not bound (Theorem F.18). In the finite mazes the oracle computes the error exactly for evaluation only: at the latents actually used, the sup-norm total-variation error ranges from 36.2 to 79.7, against a true successor mass of 20, and $\max_{s,a}|Q_{r_g}^{+,\pi} - F^\top z_{r_g}|$ from 2.69 to 15.22 (Section 11). With such errors the value bounds of (b) are vacuous.

6.5 Learning the successor measure

The loss. Let replay transitions (s,a,s_1) be drawn from $\mathsf{D}(\,\mathrm{d}s,\,\mathrm{d}a,\,\mathrm{d}s_1) = \mathsf{D}_{\mathrm{SA}}(\,\mathrm{d}s,\,\mathrm{d}a) P(\,\mathrm{d}s_1 \mid s,a)$, let ν be the law of s_1, latents $z \sim \mathcal{D}_z$ and target states $x \sim \nu$ independent (Theorem F.21). With the target policy $\pi_z^- = \mathrm{softmax}(F^-(\cdot,\cdot,z)^\top z/\tau)$ and $\bar{F}^-(s_1,z) := \sum_a \pi_z^-(a \mid s_1) F^-(s_1,a,z)$ computed from the target copies, the per-sample loss is

$$\ell(\xi,\omega; s,a,s_1,z) := \tfrac{1}{2}\int \big(F_\xi(s,a,z)^\top B_\omega(x) - \gamma \bar{F}^-(s_1,z)^\top B^-(x)\big)^2 \nu(\,\mathrm{d}x) - F_\xi(s,a,z)^\top B_\omega(s_1), \tag{6.1}$$

estimated on a minibatch of b transitions and an *independent* batch of b' target states by

$$\hat{L} = \frac{1}{2bb'} \sum_{i,j} \big(F_i^\top B(x_j) - \gamma \bar{F}_i^{-\top} B^-(x_j)\big)^2 - \frac{1}{b} \sum_i F_i^\top B(s_{i+1}).$$

Proposition 6.6 (the pretraining loss; Theorem F.22)**.** [FB-8 | D]

(a) If $P(\cdot \mid s,a)$ has a density $p_\nu \in L^2(\nu)$, then $\int P(\mathrm{d}s_1 \mid s,a)\ell = \frac{1}{2}\|\hat{M}_z(s,a,\cdot) - (\mathcal{T}_z\hat{M}^-)(s,a,\cdot)\|^2_{L^2(\nu)^\star} - C_z(s,a)$, where $\mathcal{T}_z\hat{M}^- = P + \gamma P\,\bar{F}^{-\top}B^-\nu$ is the Bellman target of Theorem 6.1 built from the target model and C_z does not depend on (ξ,ω). Without such a density the loss is not a residual and can be unbounded below.

(b) $\mathbb{E}\hat{L} = L$ and $\mathbb{E}\nabla\hat{L} = \nabla L$ for independent draws; with a stored buffer this is unbiasedness for the empirical buffer loss*, not for the population loss. Target states taken from the same transitions bias the $i = j$ terms.*

(c) If $(\xi^-,\omega^-) = (\xi,\omega)$ and the softmax representation is exact on $\mathcal{S}\times\mathcal{A}$ for every latent in the support of $\mathcal{D}_z$, then $\nabla_{\xi,\omega}L = 0$. There is no converse and no convergence statement.

Proof of (a). Expand the square, integrate over $s_1 \sim P(\cdot \mid s,a)$ and exchange the integrals (all terms are bounded): with $\hat{m}(x) = F_\xi(s,a,z)^\top B_\omega(x)$ and $\bar{n}(x) = \gamma\int P(\mathrm{d}s_1 \mid s,a)\bar{F}^-(s_1,z)^\top B^-(x)$, the left-hand side is $\frac{1}{2}\int \hat{m}^2\,\mathrm{d}\nu - \int \hat{m}(p_\nu + \bar{n})\,\mathrm{d}\nu + C_1$. Completing the square gives $\frac{1}{2}\|\hat{m} - p_\nu - \bar{n}\|^2_{L^2(\nu)}$ minus a constant, and $\hat{m} - p_\nu - \bar{n}$ is the ν-density of $\hat{M}_z - \mathcal{T}_z\hat{M}^-$, whose $L^2(\nu)^\star$ norm is that $L^2(\nu)$ norm (Theorem F.15). □

The diagonal term $-F(s_i,a_i,z_i)^\top B(s_{i+1})$ is the integral of the model density against the first term $P(\cdot \mid s,a)$ of the next-state Bellman identity, sampled by the observed next state. FB21's loss (Eq. 164, p. 36) has $B(s_i,a_i)$ instead, the integral against the atom $\delta_{(s,a)}$ of the 2021 convention (Theorem F.23). Because the target moves, the update is a semi-gradient, as in temporal-difference learning; FB21 notes that "the loss itself is not observable".

Orthonormality. The FB factorization is invariant under $B \mapsto CB$, $F \mapsto C^{-\top}F(\cdot,\cdot,C^{-1}z)$ (FB21 Remark 7, p. 18), which also maps the softmax family to itself (Theorem F.20(d)). FB21 fixes this freedom softly with the penalty $\mathcal{R}(\omega) = \frac{1}{4}\|G_{B_\omega} - I\|^2_F$.

Proposition 6.7 (the orthonormality penalty and the Gram matrix; FB21 Eqs. 165–166; Theorems F.20 and F.24)**.** [FB-9, FB-10 | S, I+D] *(a) $\nabla_\omega\mathcal{R} = \mathbb{E}_{x,x'\sim\nu\otimes\nu}[\partial_\omega B(x)^\top(B(x')B(x')^\top B(x) - B(x))]$, and the stop-gradient loss L_{reg} evaluated on two independent ν-batches has this gradient in expectation; in our convention both batches must be next states (the transition next states and the target batch). (b) G_B is symmetric positive semidefinite, invertible if and only if $B_1,\dots,B_d$ are linearly independent in $L^2(\nu)$, and* uncentered*: $\mathrm{Cov}_\nu(B) = G_B/\nu(\mathcal{S}) - m_B m_B^\top$. Under exactness at any z, $F(s,a,z)^\top\int B\,\mathrm{d}\nu = 1/(1-\gamma)$, so B has nonzero mean and must not be constrained to mean zero. (c) The penalty is soft: $G_B \neq I$ in general.*

Domination. The model kernel $\hat{M}_z(s,a,\cdot)$ always has a density with respect to ν. A successor measure with a part singular to ν, such as the 2021 atom δ_s or a reachable ν-null set, therefore has no exact representation, and the total-variation error of any model is at least the singular mass (Theorem F.25). Our continuous kernel has a Lebesgue density bounded by $p_{\max} \approx 6366$ and no atom (Theorem H.4); the population replay law has a Lebesgue density at least $n_{\min} \approx 1.2\times10^{-3}$ on the free space, so every successor measure has a bounded ν-density (Theorem H.10). The trained model, however, sees only the empirical $\hat{\nu}$, which is atomic: no measure with a Lebesgue density is dominated by it, and the model relates to the population ν only through the generalization of a smooth network, about which nothing is claimed.

Algorithm 2: implemented FB pretraining (`il_memoir/fb.py`, `FBPretrainer.update`)

Task-blind replay. Segments of 50 transitions from resets uniform on the free space, with i.i.d. uniform actions; 50,000 transitions (finite, 1,000 resets) or 100,000 (continuous, 2,000 resets) per layout and seed; no reward, no expert data. The population reference law is uniform on the states in the finite mazes (Theorem H.10(a)); every state–action pair (finite) and every pair of a free maze-lattice cell of side 1/7 and an action (continuous) is covered; at the finer resolution of the evaluation histogram the

continuous coverage is incomplete near walls (Section 12).

Networks. $F_\xi(s,a,z)$: input $[\mathrm{enc}(s), z]$, output $|\mathcal{A}| \times d$ (one d-vector per action, the DQN-style head); $B_\omega(x)$: input $\mathrm{enc}(x)$, output $d = 16$. Two hidden layers of width 64, the first Linear–LayerNorm–tanh, the second SiLU; one-hot inputs (finite) or raw (x, y) (continuous); float32.

Each update. Sample $b = 128$ transitions, an independent batch of $b' = 128$ replay next states, and 128 latents uniform on $\mathcal{Z}_4$. Compute $\bar{F}^-$ from the target copies with $\pi^-_{z,\tau}$, $\tau = 1$. Take one Adam step (learning rate 3×10^{-4}) on $\hat{L}$ for ξ and on $\hat{L} + \lambda_{\mathrm{orth}} L_{\mathrm{reg}}$ for ω, $\lambda_{\mathrm{orth}} = 1$; then Polyak-average the target copies with momentum 0.99.

After 10,000 **(finite) or** 20,000 **(continuous) updates.** Freeze the *online* copy (decision D1); compute $\hat{G}_B$ over all buffer next states in float64. Snapshots are also frozen every 2,500 or 5,000 updates to trace imitation over pretraining.

7 Imitation with a pretrained FB model: $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$

After pretraining, the networks (F, B) and the Gram estimate $\hat{G}_B$ are *frozen*; imitation chooses only the latent z, from the demonstration states, and collects no new transition [14]. We use the online copy of the networks for everything after pretraining (decision D1), the softmax family $\pi_{z,\tau}$, and the expert as a stationary Markov policy [SA A4]. Complete proofs are in Section G.

Proposition 7.1 (successor features of B; FI24 Proposition 1, second part; Theorem F.19)**.** [FB-7 | S] *If the representation is exact at z, for any policy family, then $G_B F(s,a,z) = \mathbb{E}^{\pi_z}_{(s,a)} \sum_{t\geq 0} \gamma^t B(s_{t+1})$ and $G_B F(s,z) = \int B \,\mathrm{d}M^{\pi_z,+}(s,\cdot)$. Greediness is not used.*

7.1 $\mathrm{ER}_{\mathrm{FB}}$: a reward read off the demonstrations

$\mathrm{ER}_{\mathrm{FB}}$ sets (FI24 Eq. 8, p. 5)

$$z_{ER} := \int B \,\mathrm{d}\rho_e = \frac{1}{N_E} \sum_{i=1}^{N_E} \frac{1}{H} \sum_{t=1}^{H} B(s_t^i), \tag{7.1}$$

projects it onto the sphere $\mathcal{Z}_R$, and plays $\pi_{z,\tau}$ at the projected latent.

Proposition 7.2 (what z_{ER} is; Theorem G.4)**.** [FBIL-1 | S+D]

(a) If $\rho_e \ll \nu$, then $z_{ER} = z_{r_e}$ for the bounded reward $r_e := \mathrm{d}\rho_e / \mathrm{d}\nu$ (FI24 p. 5: "a reward at every state visited by the expert"). If G_B is invertible, $z_{ER} = z_{\tilde{r}_e}$ with $\tilde{r}_e := B^\top G_B^{-1} z_{ER}$, the reward of minimal $L^2(\nu)$ norm with this task vector, whether or not $\rho_e \ll \nu$; when ν is diffuse, as in the continuous maze, only this reading exists.

(b) Under an exact greedy representation at z_{ER}, $\pi^{\mathrm{gr}}_{z_{ER}}$ maximizes the return of r_e; under an exact softmax representation the gap is at most $\tau \log|\mathcal{A}|/(1-\gamma)$, and after projection the effective temperature is $\tau\|z_{ER}\|_2/R$.

(c) This is reward maximization, not distribution matching: optimality for r_e does not make the occupancy close to q_E (Theorem 7.3).

(d) z_{ER} is unbiased for $\int B \,\mathrm{d}\bar{\rho}_e$, with covariance of order $1/N_E$; ρ_e and the f-PG target $\hat{q}_E$ are different laws.

Example 7.3 ($\mathrm{ER}_{\mathrm{FB}}$ is not distribution matching; Theorem G.5)**.** Two states x, y; actions stay and switch; deterministic effects; start at x; the expert always switches. For even H, $\rho_e = (\frac{1}{2}, \frac{1}{2})$ while $q_E = (\frac{\gamma}{1+\gamma}, \frac{1}{1+\gamma})$. With $\nu = (\frac{2}{3}, \frac{1}{3})$ the reward is $r_e = (\frac{3}{4}, \frac{3}{2})$, whose unique optimal policy switches at x and stays at y; its occupancy is δ_y, at total variation $\gamma/(1+\gamma)$ from q_E, and the expert itself is suboptimal for r_e.

7.2 $\mathrm{DM}_{\mathrm{FB}}$: matching successor measures

$\mathrm{DM}_{\mathrm{FB}}$ compares, at expert states, the successor measure of $\pi_{z,\tau}$ with the expert's, in the norm $\|\mu\|_{L^2(\nu)^\star} = \|\,\mathrm{d}\mu/\,\mathrm{d}\nu\|_{L^2(\nu)}$ (FI24 Eqs. 30–31, p. 17). For a start law $\bar\rho$,

$$\mathcal{L}_{DM}(z;\bar\rho) := \int \bar\rho(\,\mathrm{d}x)\,\big\|M^{\pi_{z,\tau},+}(x,\cdot) - M^{\pi_E,+}(x,\cdot)\big\|^2_{L^2(\nu)^\star}, \tag{7.2}$$

$$\ell_{DM}(z;\bar\rho) := \int \bar\rho(\,\mathrm{d}x)\,[F(x,z)^\top G_B F(x,z) - 2F(x,z)^\top \bar\psi_E(x)], \tag{7.3}$$

where $\bar\psi_E(x) := \mathbb{E}^{\pi_E}_{\delta_x}\sum_{k\ge 0}\gamma^k B(s_{k+1})$. $\mathcal{L}_{DM}$ is the true objective; ℓ_{DM} is what can be computed from the model and the demonstrations.

Theorem 7.4 (the $\mathrm{DM}_{\mathrm{FB}}$ expansion; FI24 Theorem 3, p. 17; Theorem G.6)**.** [FBIL-2 | S+D]

(a) If the softmax *representation is exact at z, G_B is computed under the same ν, and for $\bar\rho$-almost every x the expert successor measure has a jointly measurable density $m_E(x,\cdot)$ with $C_E := \int \bar\rho(\,\mathrm{d}x)\|m_E(x,\cdot)\|^2_{L^2(\nu)} < \infty$, then $\mathcal{L}_{DM}(z;\bar\rho) = \ell_{DM}(z;\bar\rho) + C_E$, with C_E independent of z. Greediness is not needed.*

(b) Under exactness and invertible G_B, $\ell_{DM} = \int \bar\rho\,\|G_B F(x,z) - \bar\psi_E(x)\|^2_{G_B^{-1}} - \mathrm{const}$, and $\mathcal{L}_{DM} = \int \bar\rho\,[\|F^\top B - \Pi_B m_E\|^2_{L^2(\nu)} + \|m_E - \Pi_B m_E\|^2_{L^2(\nu)}]$: $\mathrm{DM}_{\mathrm{FB}}$ can match only the projection of the expert's successor density onto $\mathrm{span}(B)$.

(c) Without exactness and for invertible G_B, ℓ_{DM} is by definition a mismatch between the model's predicted features $G_B F(x,z) = \int B\,\mathrm{d}\hat M_z(x,\cdot)$ and the expert's; pointwise in x, its G_B^{-1}-norm differs from the feature-projected distance of the true successor measures by at most $\lambda_{\min}(G_B)^{-1/2}\beta_B\|\mathcal{E}_z(x,\cdot)\|_{\mathrm{TV}}$, with $\beta_B = \sup\|B\|_2$.

(d) Under the hypotheses of (a), $\int \bar\rho(\,\mathrm{d}x)\big(V_r^{\pi_{z,\tau}}(x) - V_r^{\pi_E}(x)\big)^2 \le \|r\|^2_{L^2(\nu)}\mathcal{L}_{DM}(z;\bar\rho)$ for every bounded reward.

(e) If $\bar\rho \ge c\,\mu_0$, then for every stationary π (with a measurability condition on the density of the difference), $\mathrm{TV}(d^+_{\pi,H}, q_E) \le 2\gamma^H + \frac{1-\gamma}{2c}\nu(\mathcal{S})^{1/2}\mathcal{L}(\pi;\bar\rho)^{1/2}$, where $\mathcal{L}(\pi;\bar\rho)$ is (7.2) *with π in place of $\pi_{z,\tau}$. With the implemented start law, $c = 1/41$.*

Proof of (a) and (b). (a) Under exactness, $M^{\pi_{z,\tau},+}(x,\cdot) = \sum_a \pi_{z,\tau}(a \mid x)F(x,a,z)^\top B\,\nu = F(x,z)^\top B\,\nu$, a measure with the bounded density $F(x,z)^\top B$. The difference of the two successor measures therefore has the density $\dot\mu_x := F(x,z)^\top B - m_E(x,\cdot) \in L^2(\nu)$, and its dual norm is $\|\dot\mu_x\|_{L^2(\nu)}$ (Theorem F.15). Expanding,

$$\|\dot\mu_x\|^2_{L^2(\nu)} = F(x,z)^\top\Big(\int BB^\top\,\mathrm{d}\nu\Big)F(x,z) - 2F(x,z)^\top\int Bm_E(x,\cdot)\,\mathrm{d}\nu + \|m_E(x,\cdot)\|^2_{L^2(\nu)},$$

and $\int Bm_E(x,\cdot)\,\mathrm{d}\nu = \int B\,\mathrm{d}M^{\pi_E,+}(x,\cdot) = \bar\psi_E(x)$ (Theorem F.3). Integrating over $\bar\rho$ (Tonelli; the first two terms are bounded in x) gives the claim. Only the identity $M^{\pi_{z,\tau},+}(x,\cdot) = F(x,z)^\top B\,\nu$ was used, so greediness is not needed. (b) By Theorem 7.1 averaged over actions, $\int B\,\mathrm{d}M^{\pi_{z,\tau},+}(x,\cdot) = G_B F(x,z)$; expanding $(G_BF - \bar\psi_E)^\top G_B^{-1}(G_BF - \bar\psi_E)$ reproduces the integrand of ℓ_{DM} plus a z-independent term. The projection of $m_E(x,\cdot)$ onto $\mathrm{span}(B)$ is $w^\top B$ with $G_B w = \bar\psi_E(x)$, and Pythagoras' theorem splits $\|F^\top B - m_E\|^2$. □

Part (b) also shows that under exactness every successor measure has a ν-density bounded by $\beta_B\sup_{s,a}\|F(s,a,z)\|_2/(1-\gamma)$, so the integrability hypothesis of (a) is then automatic up to measurability (Theorem G.6(b)). Part (e) is why the start law matters: if it excluded the start state, a policy could have zero true objective and yet the maximal occupancy divergence $\log 2$ (Theorem G.7). We therefore include $t = 0$ in the start times (decision D5). Part (e) concerns the *true* objective. Nothing transfers it to the model's estimate: $\hat\ell$ determines $\mathcal{L}_{DM}$ only up to the unknown constant C_E and the estimation, truncation and model errors.

7.3 The estimator and its biases

The demonstrations are finite, so the expert continuation $\bar{\psi}_E(s_t^i)$ is replaced by a truncated discounted sum along the demonstration. With $K = 60$ and the start times $\mathcal{T}_K = \{0, \dots, H-K\}$ (41 per demonstration),

$$\hat{\ell}^{(K)}(z; \hat{G}_B) := \frac{1}{N_E|\mathcal{T}_K|} \sum_{i=1}^{N_E} \sum_{t \in \mathcal{T}_K} \Big[F(s_t^i, z)^\top \hat{G}_B F(s_t^i, z) - 2F(s_t^i, z)^\top \psi_t^{(K),i} \Big], \tag{7.4}$$

with the truncated targets $\psi_t^{(K),i} := \sum_{k=0}^{K-1} \gamma^k B(s_{t+k+1}^i)$.

Proposition 7.5 (estimator, truncation and Gram estimate; Theorem G.8)**.** [FBIL-3 | D] *For a fixed z (not chosen from the demonstrations): (a) with the infinite continuation and G_B, the estimator is unbiased for $\ell_{DM}(z; \bar{\rho}_\mathcal{T})$ (Markov property at a fixed time); (b) $\|\psi_t^{(\infty),i} - \psi_t^{(K),i}\|_2 \le \gamma^K \beta_B/(1-\gamma)$, about $0.92\,\beta_B$ for $K = 60$, against $20\,\beta_B$ for the full sum; (c) conditionally on $\hat{G}_B$, $|\mathbb{E}\hat{\ell}^{(K)}(z; \hat{G}_B) - \ell_{DM}(z; \bar{\rho}_{\mathcal{T}_K})| \le 2\gamma^K \beta_B \bar{F}(z)/(1-\gamma) + \|\hat{G}_B - G_B\|_{\mathrm{op}} \bar{F}(z)^2$, with $\bar{F}(z) = \sup_x \|F(x,z)\|_2$; (d) changing the expert state in a future slot changes the target of every earlier start time within distance K.*

FI24's "unbiased" therefore holds for a fixed z, the population start law and the infinite continuation; it does not transfer unchanged to the truncated estimator with an estimated Gram matrix, nor to a z fitted on the same demonstrations.

7.4 Optimizing the latent

Proposition 7.6 (the latent gradient with frozen networks; Theorems G.10 and G.11)**.** [FBIL-4, FBIL-5 | D] *Write $\hat{Q}_a := F(x,a,z)^\top z$, $\mathcal{J}_a := \partial_z F(x,a,z)$ and $\pi_a := \pi_{z,\tau}(a \mid x)$. Then $\nabla_z \hat{Q}_a = F(x,a,z) + \mathcal{J}_a^\top z$, $\nabla_z \pi_a = \frac{\pi_a}{\tau}(\nabla_z \hat{Q}_a - \sum_b \pi_b \nabla_z \hat{Q}_b)$, $\mathcal{J}(x,z) := \partial_z F(x,z) = \sum_a [\pi_a \mathcal{J}_a + F(x,a,z)(\nabla_z \pi_a)^\top]$, and*

$$\nabla_z \hat{\ell}(z) = \frac{2}{n} \sum_{(i,t)} \mathcal{J}(s_t^i, z)^\top (\hat{G}_B F(s_t^i, z) - \psi_t^{(K),i}).$$

Treating the policy probabilities as constants gives a different vector. On the sphere $\mathcal{Z}_R$ the first-order condition is $(I - zz^\top/R^2)\nabla\hat{\ell}(z) = 0$, and an exact projected gradient step decreases $\hat{\ell}$ for a small step. Under exactness the temperature relative to the reward $r_z = B^\top G_B^{-1} z$ encoded by z is pinned at τ: rescaling cannot sharpen the policy for a fixed reward direction.

Algorithm 3: implemented $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ (`il_memoir/fb_il.py`, `er_fb`, `dm_fit`)

Input. A frozen FB snapshot (online networks, $\hat{G}_B$, $\tau = 1$, $R = 4$) and the demonstration states $s_0^i, \dots, s_H^i$, $i = 1, \dots, 4$. No simulator access.

$\mathrm{ER}_{\mathrm{FB}}$. z_{ER} by (7.1); play $\pi_{z,\tau}$ at $Rz_{ER}/\|z_{ER}\|_2$.

$\mathrm{DM}_{\mathrm{FB}}$. Build the targets $\psi_t^{(K),i}$, $K = 60$, $t \in \{0, \dots, 40\}$, with the frozen B. Start from $z_0 = Rz_{ER}/\|z_{ER}\|_2$ (our choice; FI24 does not state how $\mathrm{DM}_{\mathrm{FB}}$ is initialized). For 300 steps: compute $\hat{\ell}^{(K)}(z; \hat{G}_B)$ and its gradient through both $F(s,a,z)$ and $\pi_{z,\tau}$ (nothing detached, networks and $\hat{G}_B$ unchanged, which the code checks), take one Adam step with learning rate 0.05, and project onto $\mathcal{Z}_R$. Play $\pi_{z,\tau}$ at the final iterate.

Evaluated. $\mathrm{ER}_{\mathrm{FB}}$ and the final $\mathrm{DM}_{\mathrm{FB}}$ latent at every pretraining snapshot, and the $\mathrm{DM}_{\mathrm{FB}}$ iterates at steps $0, 50, \dots, 300$ at the final snapshot.

8 Comparison of the approaches

This section answers Q2 at the level of objectives, information and guarantees; Sections 11, 12 and 14 add the empirical side.

Table 4: FB imitation: exact objects and their implemented approximations (condensed from Section G.7). [I] implementation approximation, [C] change of objective.

Quantity	Exact object	Implemented	Consequence
FB model	$M^{\pi_z,+}$ for all z	rank-16 networks [I]	exactness impossible in the finite mazes (Theorem 6.3); Theorem 6.5 with an error we measure
reference law	population ν	buffer law $\hat{\nu}$ [I]	one buffer for loss, $\hat{G}_B$ and z_r; $\hat{\nu}$ atomic in the continuous maze
pretraining	minimizer of L	Adam, Polyak targets, 10^4 or 2×10^4 updates [I]	no convergence; loss value does not bound the error
policy	greedy	softmax, $\tau = 1$ [C]	Theorem 6.4 replaces optimality; the bound is vacuous at $\tau = 1$
latent	$z \in \mathbb{R}^d$	sphere $\|z\| = 4$ [C]	effective temperature $\tau\|z_r\|/R$ under exactness
$\mathrm{ER}_{\mathrm{FB}}$ latent	$\int B \, \mathrm{d}\bar{\rho}_e$	4 demonstrations [I]	covariance $O(1/N_E)$; reward reading, not matching
$\mathrm{DM}_{\mathrm{FB}}$ start law	ρ_e ($t \geq 1$, FI24)	$\bar{\rho}_{\mathcal{T}_K}$, $t = 0, \ldots, 40$ [C]	occupancy control through Theorem 7.4(e), for the true objective only
$\mathrm{DM}_{\mathrm{FB}}$ target	infinite continuation	$K = 60$ terms [I]	bias bound of Theorem 7.5(c)
$\mathrm{DM}_{\mathrm{FB}}$ optimization	minimizer	300 projected Adam steps [I]	no descent or convergence claim; data reuse biases an exact minimum optimistically

8.1 Four objectives that are not the same

Table 5 writes the four objectives in the common notation. They are different objects, and none reduces to another.

Table 5: The objectives of the four methods. d is a learner occupancy, q an expert one; all divergences are learner-first except where FIRL's expert-first form is shown.

Method	Objective	Learned object	What is compared
f-PG (ours)	$D_f\big((d^+_{\pi_\phi,H})_\varepsilon \| \hat{q}_{E,\varepsilon}\big)$	policy π_ϕ	discounted next-state *marginal* from μ_0 over H steps, on cells
f-IRL	$D_f(q_E^{\mathrm{IRL}} \| \rho_\theta)$, with ρ_θ the Gibbs marginal of r_θ	reward r_θ (then a MaxEnt policy)	undiscounted uniform *marginal* over $s_1, \ldots, s_T$
$\mathrm{DM}_{\mathrm{FB}}$	$\int \bar{\rho}(\mathrm{d}x) \|M^{\pi_{z,\tau},+}(x,\cdot) - M^{\pi_E,+}(x,\cdot)\|^2_{L^2(\nu)^\star}$, estimated through the model as $\hat{\ell}^{(K)}$	latent z of a frozen model	*conditional* successor measures at expert states, in a replay-weighted L^2 norm
$\mathrm{ER}_{\mathrm{FB}}$	none: $z_{ER} = \int B \, \mathrm{d}\rho_e = z_{r_e}$, then reward maximization	latent z of a frozen model	nothing is matched; ρ_e defines a reward

How they are related. f-PG and f-IRL minimize divergences between *marginals* of the same kind, and Theorem 5.3 shows that they weight each state by the same function $(f^\circ)'$ of the occupancy ratio; they differ in the parameterization (policy versus reward with an exact MaxEnt map), in the orientation, and in the discounting. $\mathrm{DM}_{\mathrm{FB}}$ compares *conditional* successor measures, one per expert state. The two kinds of object are linked by Theorem 3.1(b), $d^+_\pi = (1-\gamma) \int \mu_0(\mathrm{d}x) M^{\pi,+}(x,\cdot)$: the occupancy is the successor measure averaged over the start law. In one direction this gives Theorem 7.4(e): if the start law charges μ_0, a small *true* $\mathrm{DM}_{\mathrm{FB}}$ objective forces the occupancies to be close. The converse fails: $d^+_{\pi,H} = q_E$ does not imply equal

successor measures at every expert state, because a marginal averages over trajectories. And when the start law misses μ_0, the true $\mathrm{DM}_{\mathrm{FB}}$ objective can vanish while the occupancy divergence is maximal (Theorem G.7). $\mathrm{ER}_{\mathrm{FB}}$ is a different kind of method altogether: under exactness it maximizes the reward $r_e = \mathrm{d}\rho_e / \mathrm{d}\nu$, and Theorem 7.3 shows a case where the resulting occupancy is at total variation $\gamma/(1+\gamma)$ from the expert's. Every one of these objectives is blind to actions that do not change the next-state law (Theorem 3.3).

8.2 Similarities

- **Reward-free imitation of state behaviour.** All four use only expert states; none uses expert actions or a task reward.
- **Discounted next-state objects.** The f-PG target $d^{+}_{\pi,H}$ and the FB successor measure $M^{\pi,+}$ are both discounted laws of next states, related by Theorem 3.1(b), and both satisfy linear Bellman identities (Theorems 3.2 and 6.1).
- **No Bellman optimality equation for the imitation task.** f-PG follows a likelihood-ratio gradient; FB amortizes control into pretraining and turns a task into a latent vector. (The f-IRL inner loop does solve a MaxEnt control problem for every reward update.)
- **Approximate in practice.** The exact statements of both families hold for population quantities and exact models; both implementations rely on plug-in estimates (densities for f-PG, the model's successor measure for FB) whose errors are not controlled by the optimized quantity.

8.3 Differences

Table 6: What each method needs and what its theory guarantees in this study.

	f-PG (fixed target)	FB pretraining with $\mathrm{ER}_{\mathrm{FB}}$ / $\mathrm{DM}_{\mathrm{FB}}$
interaction	on-policy rollouts for every task	task-blind replay once per environment; none per task
expert information	a density estimate of the expert occupancy	demonstration states summarized by B
per-task optimization	policy gradient over the whole policy	$d = 16$ numbers (the latent)
exact guarantee	an exact gradient formula for the population objective (Theorem 4.2); local descent for exact steps only	optimality or matching only under exactness, which is impossible in the finite mazes (Theorem 6.3) and not established in the continuous one
what fails first	plug-in signal bias and variance; density estimation in continuous spaces	model error of the successor measure, which the latent optimization can exploit
temperature, scale	none (policy logits are free)	for an exact model, τ and $\|z\|$ act together as one inverse temperature (Theorem 6.4(d)); not guaranteed for the trained network, which also depends on z

The key structural difference is *where the approximation sits.* f-PG samples the real environment and approximates only the occupancy densities inside its signal. Conditionally on the fitted signal, its sampled update is an unbiased but noisy estimate of the gradient of a surrogate in which that signal is held fixed; it is not an unbiased estimate of the gradient of the divergence (Theorem 4.4). FB imitation keeps the imitation step cheap, but it never meets the real environment again: $\mathrm{ER}_{\mathrm{FB}}$ reads a latent off four demonstrations, and $\mathrm{DM}_{\mathrm{FB}}$ takes a finite number of projected Adam steps on an empirical, truncated estimate of a model-based

objective. Every statement about the real environment passes through the model's successor measures, which a rank-16 network cannot represent exactly in the finite mazes. $\mathrm{DM}_{\mathrm{FB}}$ then optimizes the model's estimate of its objective; when the model is wrong, decreasing the estimate need not decrease the true objective, and the results of Section 11 show exactly this.

8.4 Use cases

- **f-PG** fits a single task with a cheap simulator: it needs fresh rollouts for each new expert, but nothing else. In a finite state space its occupancy estimates are unbiased counts, so its signal errors come from sampling alone (they still bias the nonlinear signal, Theorem 4.4(b)). Its weak point is density estimation in high-dimensional or continuous spaces.
- **f-IRL** is the choice when a reward is wanted, to inspect or to transfer to new dynamics; its cost is a MaxEnt inner loop per reward update, and its guarantees rest on the Gibbs hypothesis. The learned reward is not unique in general, and FIRL's transfer claim rests on an external result that we do not reprove (Section 5).
- **FB imitation** pays once for task-blind pretraining and then imitates any number of experts without interaction, in milliseconds for $\mathrm{ER}_{\mathrm{FB}}$ and below a second here for $\mathrm{DM}_{\mathrm{FB}}$ (Section 14). It is attractive when many tasks share one environment and interaction is expensive, provided the model is accurate enough on the states that matter; the representation rank and the coverage of the replay data then decide what can be imitated.

9 The role of the hyperparameters

This section answers Q3. It separates what the theory says about each hyperparameter from what the experiments show; the empirical statements refer to Sections 11 to 13.

9.1 The choice of f

Where f enters. In f-PG the generator enters only through the signal $g = f'(v)$ evaluated at the learner-to-target ratio $v = d_\phi/q$ (Theorem 4.2); in f-IRL it enters through $h_f(u) = (f^\circ)'(1/u)$, which is the *same* function of $v = 1/u$ (Theorems 5.2 and 5.3). So the choice of f in both methods is the choice of how strongly each cell is pushed according to how much the learner over- or under-visits it. Adding a constant to the signal changes nothing (Theorem 4.3(a)), so only the *shape* of $v \mapsto f'(v)$ matters, together with the finiteness of the objective. In descent, a large signal makes a cell repulsive and a very negative one attractive.

divergence (learner first)	signal $f'(v)$	$v \to 0$ (under-visited)	$v \to \infty$ (over-visited)	finite against a deterministic expert?
$\mathrm{JS}(d, q)$	$\frac{1}{2}\log\frac{2v}{1+v}$	$-\infty$	$\frac{1}{2}\log 2$ (bounded)	always
$\mathrm{KL}(d\|q)$	$1 + \log v$	$-\infty$	$+\infty$	no, unless smoothed (Theorem 4.7)
$\mathrm{KL}(q\|d)$	$-1/v$	$-\infty$ (fast)	0 (bounded)	yes, for a positive policy

$\mathrm{KL}(d\|q)$ penalizes visits to cells the expert rarely visits without bound: it is "zero-forcing" and infinite against an unsmoothed target with zeros, which is why both arguments are smoothed. $\mathrm{KL}(q\|d)$ penalizes under-coverage most strongly ($-1/v$) and over-visits hardly at all: it is "mass-covering" and tolerates a spread-out learner. JS is bounded on the over-visit side and logarithmic on the under-visit side, and is the only one of the three that is bounded by $\log 2$. These are statements about the signal; which stationary point descent reaches is not determined by them. For f-IRL the table of h_f (Theorem 5.2) has the same content, and FIRL's names FKL and RKL correspond to our $\mathrm{KL}(q\|d)$ and $\mathrm{KL}(d\|q)$.

What the ablation shows. On the four-rooms goal $(5, 5)$, the three generators all reach the thresholds in all three seeds, but not equally fast or equally well (Section 13): $\mathrm{KL}(d\|q)$ behaves like JS, while $\mathrm{KL}(q\|d)$ is slower (first meeting the thresholds at the 40,000- to 50,000-transition checkpoints, against 20,000 to 30,000), ends at a larger held-out divergence and a lower return, and keeps a more stochastic policy. The last observation is consistent with the mass-covering shape of its signal. All three use the learning rate tuned for JS, so the comparison mixes the generator with a step size chosen for another one. f-IRL is not implemented, so its choice of f is discussed theoretically only.

9.2 The other f-PG hyperparameters

- **Smoothing** ε changes the objective to J_ε (Theorem 4.1(c)); it keeps the signal bounded because every ratio lies in $[\varepsilon\upsilon_{\min}, 1/(\varepsilon\upsilon_{\min})]$. We use $\varepsilon = 0.01$ and always evaluate the unsmoothed divergence.
- **Batch size and folds.** $N = 20$ trajectories per update with two folds: each fold's signal is estimated from 10 trajectories of the other fold. Cross-fitting removes the same-batch dependence, not the plug-in bias (Theorem 4.4(b),(d)), at the price of a noisier signal.
- **Learning rate.** Adam with 0.3 (tabular) and 0.003 (MLP), chosen on one pilot seed and one goal as the largest values of a three-point grid (Table 8). Both are at the edge of the tested grid, so a larger rate might have done better.
- **Density estimator.** In the continuous maze, a cut-and-normalize Gaussian KDE with $h = 0.05$; it ignores walls, with a hidden mass share of mean 1.1% and up to about 31.5% near doorway corners (Theorem H.8). At equal learning rate the histogram signal gave the same pilot divergence (0.488 against 0.486 nats).
- γ **and** H define the objective itself: with $\gamma = 0.95$ and $H = 100$ the matched occupancy weighs early steps more and is within $\gamma^H \approx 0.006$ in total variation of its infinite-horizon version.

9.3 The FB hyperparameters

- **Rank** d**.** In the finite mazes exactness needs $d \geq \operatorname{rank} \tilde{P}$, that is 40 or 49 (Theorem 6.3), so $d = 16$ makes the model approximate by construction there; for the continuous maze no such statement is proved. Larger d was an optional study that was not run.
- **Temperature** τ **and latent radius** R**.** Under exactness only τ/ζ matters, and on the sphere the effective temperature relative to a reward r is $\tau\|z_r\|/R$ (Theorem 6.4(d)). In the pilot, $\tau = 0.1$ and 0.3 made the $\mathrm{DM}_{\mathrm{FB}}$ policy nearly greedy (score spread divided by τ of 110 and 35 in the finite maze, policy entropy 0.05 and 0.06 nats), while $\tau = 1$ kept it stochastic (spread 11, entropy 0.62); $\tau = 1$ also gave the best $\mathrm{ER}_{\mathrm{FB}}$ results in the finite maze and the best $\mathrm{ER}_{\mathrm{FB}}$ success in the continuous one, and was frozen with $R = 4$ (Table 8); for $\mathrm{DM}_{\mathrm{FB}}$ alone, $\tau = 0.3$ had the lower finite divergence.
- **Orthonormality weight** λ_{orth}**.** It fixes the gauge $B \mapsto CB$ only softly. Without it, the Gram matrix drifts far from the identity and $\|z_{ER}\|$ grows from about 2.6–2.9 to 13.8–17.2 (Section 13); since the latent is projected to radius 4, this changes which direction is chosen, not the scale.
- **Target networks** (Polyak momentum 0.99 after every update), the **independent target batch** (without it the $i = j$ terms are biased, Theorem 6.6(b)), the **latent distribution** (uniform on $\mathcal{Z}_4$; the loss constrains the model only on its support), and the **inputs** (one-hot or raw (x, y); FB21 fed 21×21 radial-basis features to B in its continuous maze) are design choices without a theorem behind them. Raw inputs are one untested candidate cause of the nearly singular continuous Gram matrix of Section 12.

9.4 The algorithm used to learn the successor measure

The successor measure is not observed: it is the solution of the measure-valued Bellman equation of Theorem 6.1. In a finite MDP with known P it is the matrix $P(\cdot \mid s,a)(I-\gamma P_\pi)^{-1}$, which our oracle computes for evaluation; a learner without P must estimate it from transitions. FB21's algorithm, which we implement in the next-state convention, does so by temporal-difference learning on a parametric model:

(i) the model $F(s,a,z)^\top B(x)\nu(\mathrm{d}x)$ represents the measures of all policies π_z at once through the latent z;

(ii) the loss (6.1) is, up to a constant, the squared $L^2(\nu)^\star$ distance between the model and its one-step Bellman target built from a slowly moving target copy (Theorem 6.6(a));

(iii) the target uses the policy π_z^- itself, so the learned measure is that of π_z, whatever policy collected the data (off-policy learning from task-blind replay);

(iv) the orthonormality penalty, the latent distribution and the reference measure ν decide the gauge, the latents on which the model is trained, and the norm in which errors are weighed.

What this algorithm guarantees is limited: its minibatch gradient is unbiased for the buffer loss, and an exact representation is a stationary point (Theorem 6.6(b)–(c)). There is no converse, no convergence statement, and the loss value omits a constant that can be infinite, so it cannot certify the model error that Theorem 6.5 needs. In the runs, the imitation results moved substantially from one pretraining snapshot to the next: the held-out divergence of the same imitation method changed by up to 0.53 nats between consecutive snapshots in the finite mazes, and 12 of the 36 finite $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ curves reached success 0.8 at some earlier snapshot but not at the last one (Section 11). The successor-measure learner was not stationary at the budget.

9.5 The $\mathrm{DM}_{\mathrm{FB}}$ hyperparameters

- **Truncation** $K=60$ bounds the missing tail of each target by $\gamma^K\beta_B/(1-\gamma)\approx 0.92\,\beta_B$, against $20\,\beta_B$ for the full sum (Theorem 7.5(b)).
- **Start times** $\{0,\dots,40\}$ include $t=0$, which gives $\bar\rho\ge\mu_0/41$ and the occupancy control of Theorem 7.4(e) for the true objective.
- **Optimization**: 300 projected Adam steps with learning rate 0.05 from the projected z_{ER}. In every final adaptation, at least 97% of the total decrease of the model objective was reached within the first 100 steps (Section 11).

10 Experimental protocol

The protocol was fixed in a written specification before any code was run, and its numerical settings were frozen after a one-seed pilot and before any final seed ran (Section 10.4). Nothing was changed after the final runs started.

10.1 Environments

Finite mazes (Theorem H.1). A 7×7 lattice of cells (x,y), in two layouts: *open*, with 49 free cells, and *four rooms*, with walls at $x=3$ except $y\in\{1,5\}$ and at $y=3$ except $x\in\{1,5\}$, which leaves 40 free cells and four doorways. The five actions move by one cell or stay; a move into a wall or out of the grid stays. With probability $p_{\mathrm{slip}}=0.1$ the executed action is drawn uniformly from all five actions. The start is the cell $(1,1)$ and the goals are $(5,1)$, $(5,5)$ and $(1,5)$; from the start, $(5,1)$ and $(1,5)$ are 4 steps away and $(5,5)$ is 8 steps away in both layouts, so the layouts differ by their walls, not by their shortest paths. The transition matrix is used only by the oracle, the expert construction and the evaluation; its rank is 49 or 40 (Theorem H.2).

Continuous maze (Theorem H.3). The four-rooms layout rescaled to $[0,1]^2$: the free space is the open unit square minus the closed wall squares (area $40/49$), the step length is one cell, $L = 1/7$, and the start is the centre of cell $(1,1)$. An action moves the agent by L if the segment to the target is free, and otherwise stays; Gaussian noise of standard deviation 0.01 is then added, redrawn until the noise segment is free. This kernel has no clipping, no atom and no tunnelling, and a Lebesgue density bounded by about 6366 (Theorem H.4).

Evaluation reward and success. The reward of goal g is $r_g(x) = \exp(-|x - \bar{g}|^2/(2w^2))$ with $\bar{g}$ the centre of the goal cell and $w = 1/14$, read at the next state; in the finite maze x is the cell centre. Success (a "hit") means that some s_t with $1 \le t \le H$ lies in the goal region $\{|x - \bar{g}| \le 1/14\}$, which in the finite maze is the goal cell. Rewards and success are used for expert construction and evaluation only; no learner sees them.

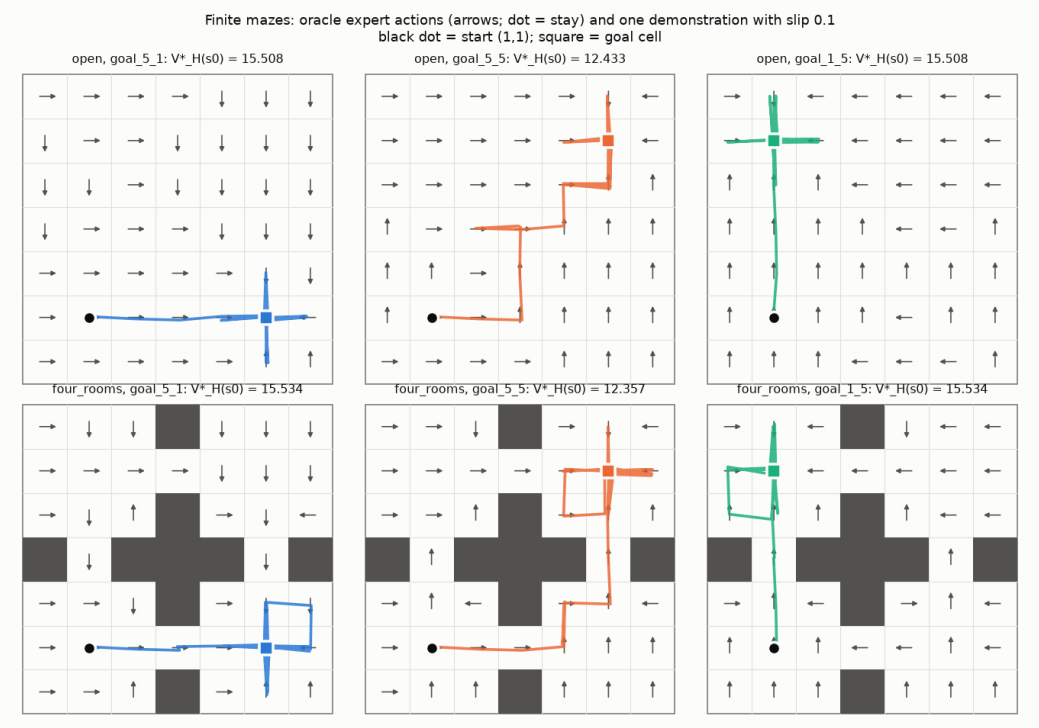


Figure 1: Finite mazes: actions of the certified oracle expert (arrows; a dot is "stay") and one demonstration with slip 0.1, for the three goals of each layout. Illustration produced by `scripts/check_environments.py` (laptop run, seed 20261005, same environment configuration as the final runs); the final demonstrations are other draws from the same experts.

10.2 Experts and data

Experts. In the finite mazes the expert is the stationary optimal policy for r_g, computed by value iteration with a deterministic tie-breaking order and certified by its Bellman residual (Theorem H.9(a)): over the six tasks the value-iteration residual is at most 9.7×10^{-13} and the policy residual $\eta_{\pi^\star}$ at most 1.1×10^{-14}, and the expert's H-step return equals the H-step optimum $V_H^\star$ at the start state. In the continuous maze the expert is a constructed route controller, $\pi_E(x) = \Gamma(\lfloor 7x \rfloor)$ for a fixed table Γ of shortest-path actions on free cells: a stationary state-feedback policy, as $\mathrm{DM}_{\mathrm{FB}}$ requires, whose planned moves are almost surely never blocked (Theorem H.9(b)–(c)). It is a reference controller, not a proven optimal policy. In the final evaluation it hits the goal in every rollout, and 72–76% of its final states lie in the goal region.

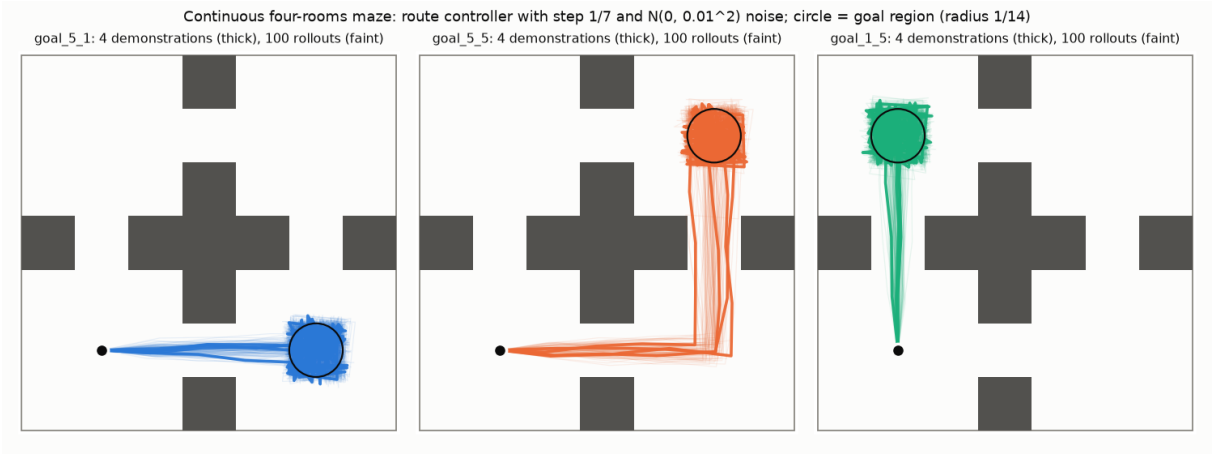


Figure 2: Continuous four-rooms maze: the route controller, with 4 demonstrations (thick) and 100 rollouts (faint) per goal; the circle is the goal region. Same source and caveat as Fig. 1.

Data. For every task and seed: $N_E = 4$ demonstrations of $H = 100$ transitions, of which learners receive the states only (actions are stored for the state–action diagnostic); for every layout and seed, a task-blind replay buffer for FB (Section 6.5), whose coverage of every state–action pair was checked (at least 137 visits per state–action pair in the finite buffers, and 423 per pair of a free maze-lattice cell of side 1/7 and an action in the continuous ones; at the finer resolution of the evaluation histogram the continuous coverage is incomplete near walls, Section 12); and in the continuous maze, 2,000 held-out expert trajectories as the evaluation reference. All data were generated once, with recorded seeds and content hashes, and shared by all methods.

10.3 Methods, settings and budgets

Table 7: Frozen settings of the final runs (`configs/core.yaml`, `configs/compact_ablations.yaml`).

	finite mazes	continuous maze
f-PG (Algorithm 1)		
policy	tabular softmax logits	MLP, 2 tanh layers of 64, raw (x, y)
signal	smoothed counts, $\varepsilon = 0.01$	KDE, $h = 0.05$, $\varepsilon = 0.01$
updates	25 Adam steps, lr 0.3, 20 trajectories each	50 Adam steps, lr 0.003, 20 trajectories each
budget	50,000 learner transitions per task	100,000 learner transitions per task
FB pretraining (Algorithm 2), one model per layout and seed, shared by its 3 goals		
networks	$d = 16$, width 64, one-hot inputs	$d = 16$, width 64, raw (x, y)
data	50,000 replay transitions	100,000 replay transitions
updates	10,000 (batch $128 + 128$), Adam 3×10^{-4}	20,000, same
policy	softmax, $\tau = 1$, latents on the sphere $R = 4$	same
ER_{FB} and DM_{FB} (Algorithm 3)		
DM_{FB}	$K = 60$, start times 0..40, 300 Adam steps, lr 0.05	same

Suites and seeds. The *core* suite has 144 jobs: per seed, 3 FB pretraining runs, 9 ER_{FB}/DM_{FB} adaptations, 9 f-PG runs with JS, their 18 evaluations and 9 expert evaluations. It covers both finite layouts and the continuous maze, three goals each, and the final seeds 0, 1 and 2: 27 f-PG runs, 9 FB models and 27 adaptations. The *compact ablations* add 21 jobs on the four-rooms goal

$(5, 5)$: f-PG with $\mathrm{KL}(d\|q)$ and $\mathrm{KL}(q\|d)$ (6 runs) and FB pretraining without the orthonormality penalty (3 models), with their adaptations and evaluations; the JS runs and the base FB models are the core's own jobs, reused by identifier. Each job derives its random streams from its seed and labels, and evaluation streams are distinct from every training stream. Some streams are shared on purpose, as common random numbers for paired comparisons: the KL runs share the initialization and the first batch of the JS run of the same seed; FB pretraining with and without the orthonormality penalty shares its initialization and minibatch streams; every checkpoint of a run, and every method of a goal and seed, is evaluated with the same evaluation random numbers, so continuous learning curves and paired differences carry correlated Monte Carlo noise (in the finite mazes the reported metrics are exact and only the Monte Carlo cross-checks use these streams); and the three seeds share one set of 2,000 continuous reference rollouts (reference seed 314159), so the standard deviations across seeds do not include the sampling variance of that reference. The finite FB-imitation runs are 18 adaptations on 6 pretrained FB models: the $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ runs of a goal and seed come from one adaptation job, and the three goals of a layout and seed share one model. The budgets are the ceilings of the specification; they were not chosen as sufficient for convergence.

10.4 Pilot and tuning scope

A single-seed learning pilot (seed 999, distinct from the final seeds, four-rooms goal $(5, 5)$ in both mazes) chose the settings that the specification left open, by rules declared before any final run: for f-PG the learning rate with the lowest last-checkpoint held-out divergence among three, and for FB the temperature with the best mean divergence and success over $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ in both mazes, together with the policy-entropy diagnostics. Table 8 shows the evidence.

Table 8: Learning pilot (seed 999, four rooms, goal $(5, 5)$): held-out divergence (nats) and success at the last checkpoint. Used only to choose the frozen settings; these are not results.

setting	value	finite: JS	finite: success	continuous: JS	continuous: success
f-PG learning rate	0.03 / 0.0003	0.361	0.708	0.508	0.602
f-PG learning rate	0.1 / 0.001	0.160	1.000	0.486	0.776
f-PG learning rate	0.3 / 0.003 (frozen)	0.047	1.000	0.342	0.968
f-PG histogram signal	lr 0.001	–	–	0.488	0.700
$\mathrm{ER}_{\mathrm{FB}}$, τ	0.1	0.296	0.806	0.604	0.002
$\mathrm{DM}_{\mathrm{FB}}$, τ	0.1	0.340	0.446	0.693	0.000
$\mathrm{ER}_{\mathrm{FB}}$, τ	0.3	0.299	0.803	0.577	0.040
$\mathrm{DM}_{\mathrm{FB}}$, τ	0.3	0.288	0.591	0.692	0.000
$\mathrm{ER}_{\mathrm{FB}}$, τ	1 (frozen)	0.179	0.983	0.581	0.128
$\mathrm{DM}_{\mathrm{FB}}$, τ	1 (frozen)	0.384	0.621	0.687	0.014

The resulting settings have a narrow basis, which the results must be read with:

- they were chosen on one goal of the four-rooms mazes with one seed; the open layout and the goals $(5, 1)$ and $(1, 5)$ inherit them untested;
- both frozen f-PG learning rates are the largest of the tested grid, so a larger rate might have done better;
- the KL ablations use the learning rate tuned for JS;
- in the pilot, $\mathrm{DM}_{\mathrm{FB}}$ ended below its own $\mathrm{ER}_{\mathrm{FB}}$ starting point at $\tau = 1$: in the finite maze its held-out divergence rose from 0.179 at the $\mathrm{ER}_{\mathrm{FB}}$ start to 0.384 after the 300 latent steps, while its model objective fell from -1456 to -1647;
- in the pilot the continuous FB Gram matrix was nearly singular ($\lambda_{\min}(\hat{G}_B) \approx 2 \times 10^{-5}$), and continuous $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ failed.

The $\mathrm{DM}_{\mathrm{FB}}$ settings, the network sizes and the budgets were not in the pilot grid and were kept at their initial values.

10.5 Evaluation and operational thresholds

Finite mazes: exact evaluation. For every checkpointed policy the oracle computes, from P: the occupancy divergence $\mathrm{JS}(d^{+}_{\pi,H}, q_E)$ against the exact expert occupancy, the state–action diagnostic $\mathrm{JS}(d^{\mathrm{SA}}_{\pi,H}, q^{\mathrm{SA}}_{E})$, the exact H-step return and its gap to $V^{\star}_{H}$, and the exact hit probability; 200 Monte Carlo rollouts cross-check them. For FB models it also computes the exact model error of Theorem 6.5.

Continuous maze: Monte Carlo evaluation. 500 fresh rollouts per checkpoint give the success rate, the return, the fraction of final states in the goal region and the occupancy divergence: a plug-in JS between the masses that the policy's rollouts and the 2,000 reference rollouts put on the 356 free-space histogram cells, with the discount weights $\gamma^{t}/(nZ_H)$. This plug-in estimate is biased upward; the expert's own row measures its floor, 0.0005 to 0.0013 nats.

Checkpoints and thresholds. f-PG policies are evaluated every 5 iterations, $\mathrm{ER}_{\mathrm{FB}}$ and the final $\mathrm{DM}_{\mathrm{FB}}$ latent at every FB snapshot, and the $\mathrm{DM}_{\mathrm{FB}}$ iterates every 50 steps at the final snapshot. All tables use the *last* checkpoint, i.e. the full budget. A run *meets the operational thresholds* at a checkpoint if JS ≤ 0.10 nats and success ≥ 0.80. These thresholds were declared in the specification and kept unchanged; they are study choices, not universal standards. "First reached" is the first assessed checkpoint at which both hold, in learner transitions for f-PG and in FB pretraining updates for $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$, where $\mathrm{DM}_{\mathrm{FB}}$ is assessed after its completed 300 latent steps at each snapshot; "not reached" means at no assessed checkpoint within the budget. With f-PG evaluated every 10,000 learner transitions, a first hit is an observed bound at that resolution, not a measured sample requirement.

10.6 Execution and provenance

All runs executed sequentially on the CPU of the author's laptop (4 PyTorch threads, NumPy and BLAS thread counts neither set nor recorded; Windows, Python 3.11.16, PyTorch 2.14.1), chosen because the pilot measured it faster than the laptop GPU for these small networks. The core suite ran on 5 October 2026 from 21:46:55 to 22:02:35 UTC, the ablations from 22:20:00 to 22:22:55 UTC; all 165 jobs completed at their first attempt, with no failure or interruption, and the peak process memory was 823 MiB. Every job records its configuration, code, data and dependency hashes, device, wall times and outputs; the provenance audit of the results (`scripts/audit_results.py`) reports no error and no warning, and a second implementation (`report/verify/verify_results.py`), which recomputes the results from the raw run files without the project's analysis code, reproduces every table (Section K).

11 Results in the finite mazes

This section answers the performance and stability parts of Q4; the sample and compute requirements are in Section 14. All quantities are exact oracle values for the policy at the last checkpoint, i.e. after the full budget, reported as mean and sample standard deviation over the seeds 0, 1, 2; every seed is listed in Section L. The Monte Carlo cross-checks agree with the exact values and are not repeated.

11.1 Experts and the demonstration target

The six experts are optimal for their evaluation rewards: their gap to $V^{\star}_{H}$ is 0, and $V^{\star}_{H}$ at the start is 15.51 (open) or 15.53 (four rooms) for the goals $(5, 1)$ and $(1, 5)$, and 12.43 (open) or 12.36 (four rooms) for $(5, 5)$. An expert reaches its goal in every episode and then stays there: its next-state occupancy q_E puts 0.61–0.77 of its mass on the goal cell, and the probability that

s_H lies in the goal is 0.916. To imitate it, a policy must therefore reach the goal quickly *and stay*; reaching it is what the success rate measures, staying is what the return and the divergence measure.

The learners never see q_E. f-PG matches the smoothed empirical occupancy $\hat{q}_E$ of the four demonstrations, $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ use the same demonstration states. That target is itself at a held-out divergence $\mathrm{JS}(\hat{q}_E, q_E)$ of 0.007 to 0.022 nats from the expert. All held-out divergences below are computed against the exact q_E.

11.2 Final performance

Table 9: Finite mazes, core suite, last checkpoint (mean ± sample SD over 3 seeds). JS is the held-out next-state occupancy divergence $\mathrm{JS}(d^{+}_{\pi,H}, q_E)$, success the exact probability of reaching the goal within H steps, the gap is $V^{\star}_H - J_H(\pi)$ for the evaluation reward, and $\mathrm{JS}^{\mathrm{SA}}$ the state–action diagnostic. Generated from the run files by `report/verify/make_tables.py`.

layout, goal	method	$V^{\star}_H$	JS (nats)	success	gap to $V^{\star}_H$	$\mathrm{JS}^{\mathrm{SA}}$ (nats)
open (5,1)	f-PG (JS)	15.51	0.010 ± 0.014	1.000 ± 0.000	0.63 ± 0.50	0.026 ± 0.020
	$\mathrm{DM}_{\mathrm{FB}}$		0.535 ± 0.082	0.218 ± 0.221	14.35 ± 0.89	0.572 ± 0.086
	$\mathrm{ER}_{\mathrm{FB}}$		0.437 ± 0.013	0.868 ± 0.128	13.49 ± 0.51	0.532 ± 0.003
open (5,5)	f-PG (JS)	12.43	0.019 ± 0.014	1.000 ± 0.000	1.36 ± 1.15	0.065 ± 0.045
	$\mathrm{DM}_{\mathrm{FB}}$		0.457 ± 0.051	0.450 ± 0.081	11.00 ± 0.31	0.507 ± 0.055
	$\mathrm{ER}_{\mathrm{FB}}$		0.292 ± 0.074	0.992 ± 0.012	8.65 ± 2.24	0.418 ± 0.067
open (1,5)	f-PG (JS)	15.51	0.002 ± 0.001	1.000 ± 0.000	0.35 ± 0.18	0.012 ± 0.004
	$\mathrm{DM}_{\mathrm{FB}}$		0.397 ± 0.252	0.441 ± 0.477	11.68 ± 4.39	0.441 ± 0.228
	$\mathrm{ER}_{\mathrm{FB}}$		0.339 ± 0.213	0.808 ± 0.293	11.03 ± 4.09	0.433 ± 0.220
four rooms (5,1)	f-PG (JS)	15.53	0.002 ± 0.000	1.000 ± 0.000	0.39 ± 0.21	0.009 ± 0.003
	$\mathrm{DM}_{\mathrm{FB}}$		0.589 ± 0.049	0.087 ± 0.042	15.09 ± 0.33	0.620 ± 0.047
	$\mathrm{ER}_{\mathrm{FB}}$		0.436 ± 0.011	0.944 ± 0.027	13.87 ± 0.21	0.530 ± 0.017
four rooms (5,5)	f-PG (JS)	12.36	0.015 ± 0.005	1.000 ± 0.000	1.44 ± 0.34	0.048 ± 0.019
	$\mathrm{DM}_{\mathrm{FB}}$		0.476 ± 0.204	0.327 ± 0.540	11.38 ± 1.64	0.550 ± 0.183
	$\mathrm{ER}_{\mathrm{FB}}$		0.348 ± 0.109	0.772 ± 0.196	10.25 ± 1.78	0.425 ± 0.085
four rooms (1,5)	f-PG (JS)	15.53	0.006 ± 0.006	0.999 ± 0.001	0.68 ± 0.34	0.016 ± 0.008
	$\mathrm{DM}_{\mathrm{FB}}$		0.365 ± 0.226	0.628 ± 0.505	11.50 ± 3.71	0.421 ± 0.219
	$\mathrm{ER}_{\mathrm{FB}}$		0.278 ± 0.181	0.936 ± 0.094	9.95 ± 4.99	0.364 ± 0.223

Table 9 and Fig. 3 summarize the final policies.

f-PG ends close to the expert in every run: the held-out divergence is 0.0011 to 0.032 nats, the success probability at least 0.998, the gap to $V^{\star}_H$ between 0.15 and 2.68, and the probability of ending in the goal 0.853 to 0.912 against the expert's 0.916. The final policies are nearly deterministic on the states they visit (mean policy entropy 0.03 to 0.17 nats, against $\log 5 \approx 1.61$ for a uniform policy).

$\mathrm{ER}_{\mathrm{FB}}$ reaches the goal in most runs but rarely stays: success is 0.47 to 1.00, and at least 0.80 in 14 of the 18 runs, but the probability of ending in the goal is 0.001 to 0.51 (at least 0.5 in a single run), the gap is 4.3 to 14.1, and the held-out divergence is 0.079 to 0.497 nats. Under exactness $\mathrm{ER}_{\mathrm{FB}}$ would maximize the reward $\mathrm{d}\rho_e / \mathrm{d}\nu$ (Theorem 7.2), which is largest at the goal where the expert spends most of its time; the trained model does not deliver this.

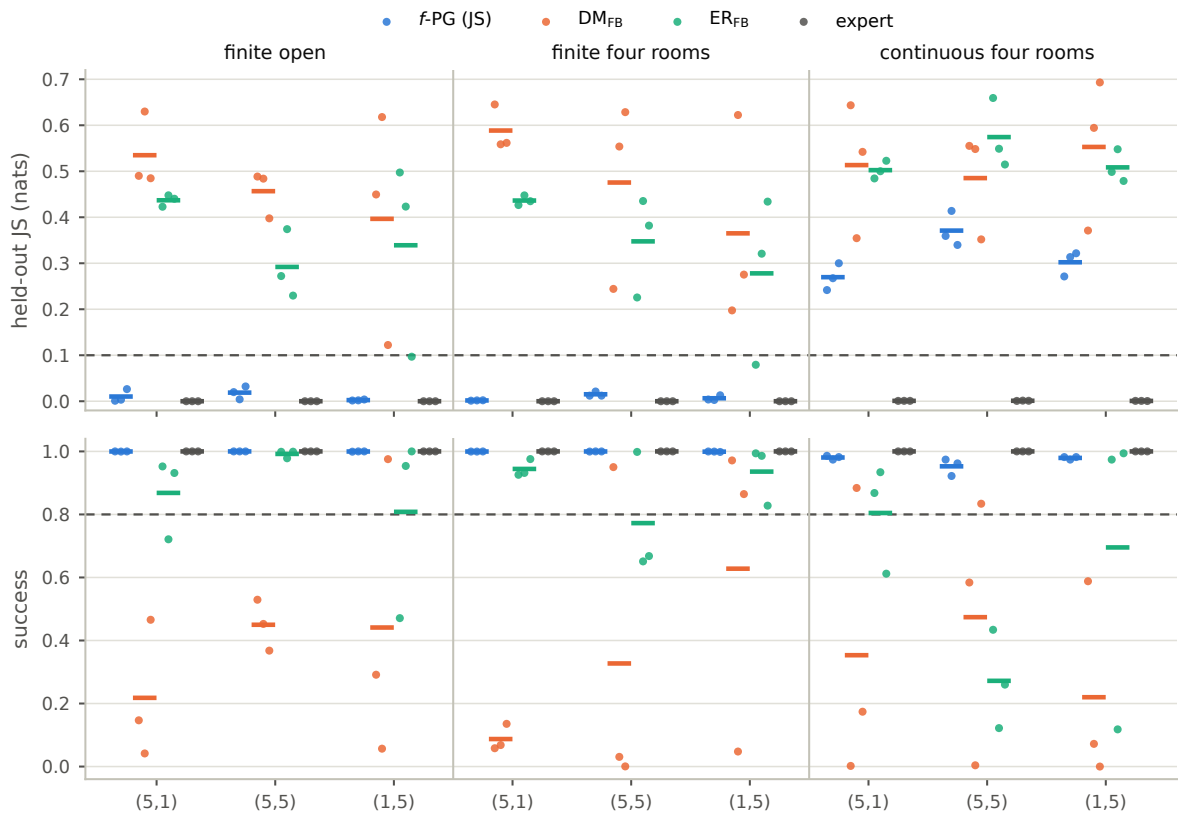


Figure 3: Every core run at the last checkpoint, finite and continuous: held-out divergence (top) and success (bottom), with the operational thresholds (dashed). Dots are seeds, bars the means over the three seeds; the expert is the reference.

$\mathrm{DM}_{\mathrm{FB}}$ is worse than $\mathrm{ER}_{\mathrm{FB}}$ in almost every run: its divergence is 0.122 to 0.645 nats, its success 0.0005 to 0.975 (at least 0.80 in 4 runs), its gap 6.7 to 15.4. Paired by layout, goal and seed, $\mathrm{DM}_{\mathrm{FB}}$ has a larger divergence than $\mathrm{ER}_{\mathrm{FB}}$ in 16 of the 18 runs and a lower success in 17. Since $\mathrm{DM}_{\mathrm{FB}}$ starts from the $\mathrm{ER}_{\mathrm{FB}}$ latent, this is a statement about what the 300 latent steps did; Section 11.4 examines it.

State–action diagnostic. For every policy $\mathrm{JS}^{\mathrm{SA}} \geq \mathrm{JS}$, as it must be: $d^{+}_{\pi,H}$ is the image of $d^{\mathrm{SA}}_{\pi,H}$ under the kernel P, and an f-divergence cannot increase under a Markov kernel [5]. For f-PG the state–action divergence is 0.006 to 0.111 nats. State matching does not determine the expert's actions (Theorem B.12), and we do not claim that f-PG recovers them.

11.3 Learning curves and the operational thresholds

f-PG (Fig. 4, left) starts from the uniform policy at a divergence of 0.47 to 0.52 nats and meets the thresholds in all 18 runs, first at the 20,000-transition checkpoint (10 policy updates) in 17 runs and at the 30,000-transition checkpoint in the remaining one (four rooms, goal $(5, 5)$, seed 0); all 18 still meet them at the end of the budget (Table 10). With checkpoints every 10,000 transitions, these first hits are observed bounds at that resolution: the thresholds were first met somewhere between the previous checkpoint and this one. The curves are smooth and every seed follows the mean closely.

$\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ (Fig. 5, left) behave differently. Their quality does not improve steadily with pretraining: between consecutive snapshots the held-out divergence of a run changes by up

Table 10: Operational thresholds (JS ≤ 0.10 nats and success ≥ 0.80), declared before the runs: number of seeds that meet them at the last checkpoint, and the first assessed checkpoint at which each seed met them ("–": at no assessed checkpoint within the budget; $\mathrm{DM}_{\mathrm{FB}}$ is assessed after its completed latent steps). f-PG is counted in learner transitions, $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ in FB pretraining updates (the latent adaptation itself uses no transition).

maze	goal	method	met at last	first reached (seeds 0, 1, 2)
finite open	(5,1)	f-PG (JS)	3/3	20k, 20k, 20k (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	–, –, 2.5k (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	0/3	2.5k, –, – (FB updates)
	(5,5)	f-PG (JS)	3/3	20k, 20k, 20k (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
	(1,5)	f-PG (JS)	3/3	20k, 20k, 20k (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	2.5k, –, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	1/3	–, –, 10k (FB updates)
finite four rooms	(5,1)	f-PG (JS)	3/3	20k, 20k, 20k (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
	(5,5)	f-PG (JS)	3/3	30k, 20k, 20k (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	–, 5k, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
	(1,5)	f-PG (JS)	3/3	20k, 20k, 20k (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	7.5k, 5k, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	1/3	10k, –, – (FB updates)
continuous	(5,1)	f-PG (JS)	0/3	–, –, – (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
	(5,5)	f-PG (JS)	0/3	–, –, – (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
	(1,5)	f-PG (JS)	0/3	–, –, – (transitions)
		$\mathrm{DM}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)
		$\mathrm{ER}_{\mathrm{FB}}$	0/3	–, –, – (FB updates)

to 0.53 nats, and 12 of the 36 finite curves reach a success of 0.80 at an earlier snapshot but not at the last one. $\mathrm{ER}_{\mathrm{FB}}$ meets the thresholds at the last snapshot in 2 runs (open, goal $(1, 5)$, seed 2, and four rooms, goal $(1, 5)$, seed 0), and in one more run only at the first snapshot (open, goal $(5, 1)$, seed 0, after 2,500 updates). $\mathrm{DM}_{\mathrm{FB}}$ meets them at some snapshot in 5 runs, after 2,500, 2,500, 5,000, 5,000 and 7,500 updates, and at the last snapshot in none. So 34 of the 36 finite FB-imitation runs do not meet the thresholds at the end of the budget: 6 of them met them at an earlier snapshot only, and 28 did not meet the thresholds at any assessed FB snapshot after completed adaptation (*not reached* in Table 10). Of these 28 runs, three are $\mathrm{DM}_{\mathrm{FB}}$ runs (open maze, goal $(1, 5)$, seeds 1 and 2; open maze, goal $(5, 1)$, seed 0) whose latent path at the final snapshot crossed both thresholds at an intermediate latent step and left the threshold region before the final step; counting every evaluated checkpoint, including these intermediate iterates, 25 of the 36 runs, rather than 28, never met the thresholds. The convergence rule, which assesses $\mathrm{DM}_{\mathrm{FB}}$ after its completed adaptation, is unchanged. Their full curves are in Section L.

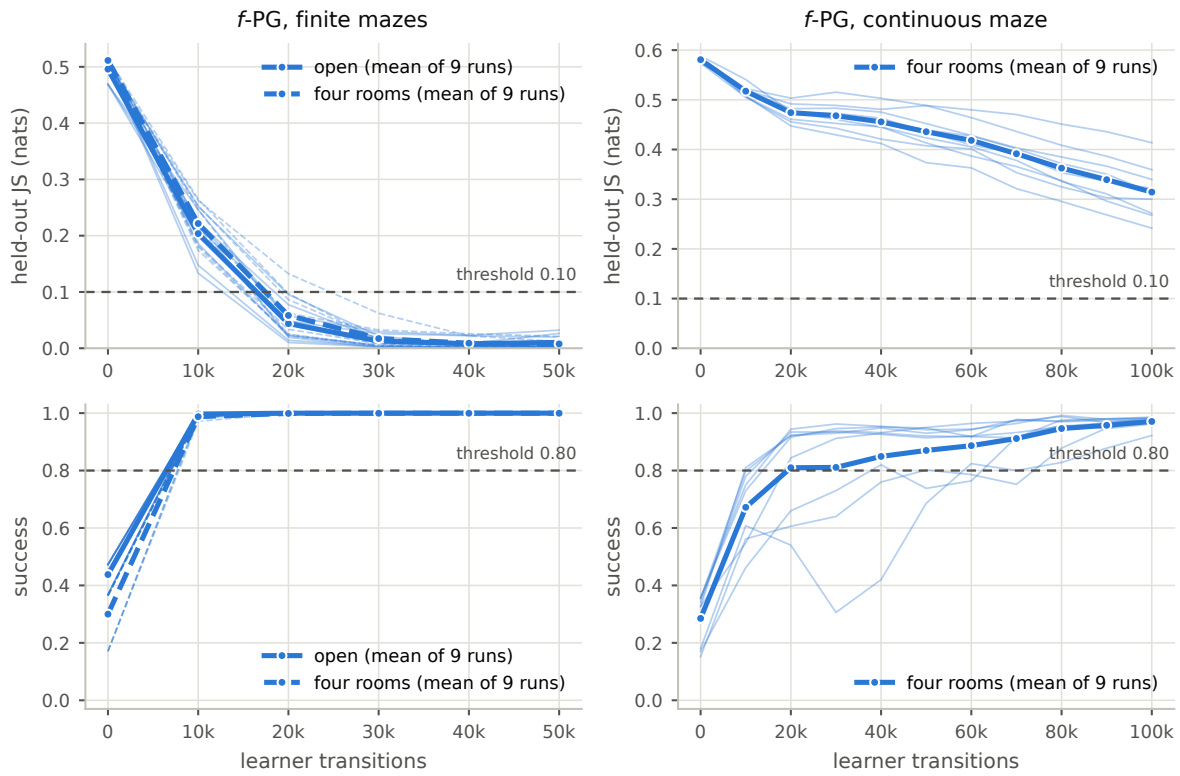


Figure 4: f-PG learning curves (core suite): held-out divergence and success against learner transitions. Thin lines are the individual runs, thick lines the mean over the 9 runs of a maze (finite: open solid, four rooms dashed). The dashed horizontal lines are the operational thresholds.

11.4 Why $\mathrm{DM}_{\mathrm{FB}}$ moves away from the expert

Figure 6 follows the latent iterates at the final snapshot. The model objective decreased in all 27 adaptations, at least 97% of its total decrease within the first 100 steps; the projected Adam iteration did what it was asked. Yet the held-out divergence at step 300 is larger than at step 0 in 16 of the 18 finite adaptations (and in 4 of the 9 continuous ones), and the success is lower in 17 of the 18 finite ones. The pilot had shown the same pattern on its single seed (Section 10.4).

The theory explains why this can happen without proving that it must. Theorem 7.4(a) identifies the *true* $\mathrm{DM}_{\mathrm{FB}}$ objective with $\ell_{DM} + C_E$ only for an exact model; the algorithm minimizes the model's estimate $\hat{\ell}^{(K)}$, in which the model's successor measure replaces the true one (Theorem 7.5). The oracle measures how far apart the two are (Table 11). At the latents actually used, the sup-norm total-variation error of the model's successor measure is 36.2 to 79.7, against a true successor mass of 20, and the largest action-value error $\max_{s,a}|Q_{r_g}^{+,\pi_z} - F^\top z_{r_g}|$ is 2.9 to 8.9 at the $\mathrm{ER}_{\mathrm{FB}}$ latents and 2.7 to 15.2 at the $\mathrm{DM}_{\mathrm{FB}}$ ones, for values of at most 20. The model is accurate in a weak sense only: its Gram matrix is well conditioned ($\lambda_{\min}(\hat{G}_B)$ between 0.93 and 0.96, $\lambda_{\max}$ between 2.26 and 2.65) and its total mass $F^\top \int B \, \mathrm{d}\hat{\nu}$ averages 18.2 to 19.2 instead of 20. With errors of this size, Theorem 6.5 gives no useful bound, and a latent that lowers $\hat{\ell}^{(K)}$ can do so by moving to policies whose successor measures the model represents worst. This is the measured fact; the mechanism (the optimizer exploiting model error) is our interpretation, which these runs are consistent with but do not prove.

11.5 Stability and failed seeds

No run failed numerically: all 165 jobs completed at their first attempt, every loss and gradient stayed finite (the code stops on a non-finite value), and the recorded $\mathrm{DM}_{\mathrm{FB}}$ iterates at the final

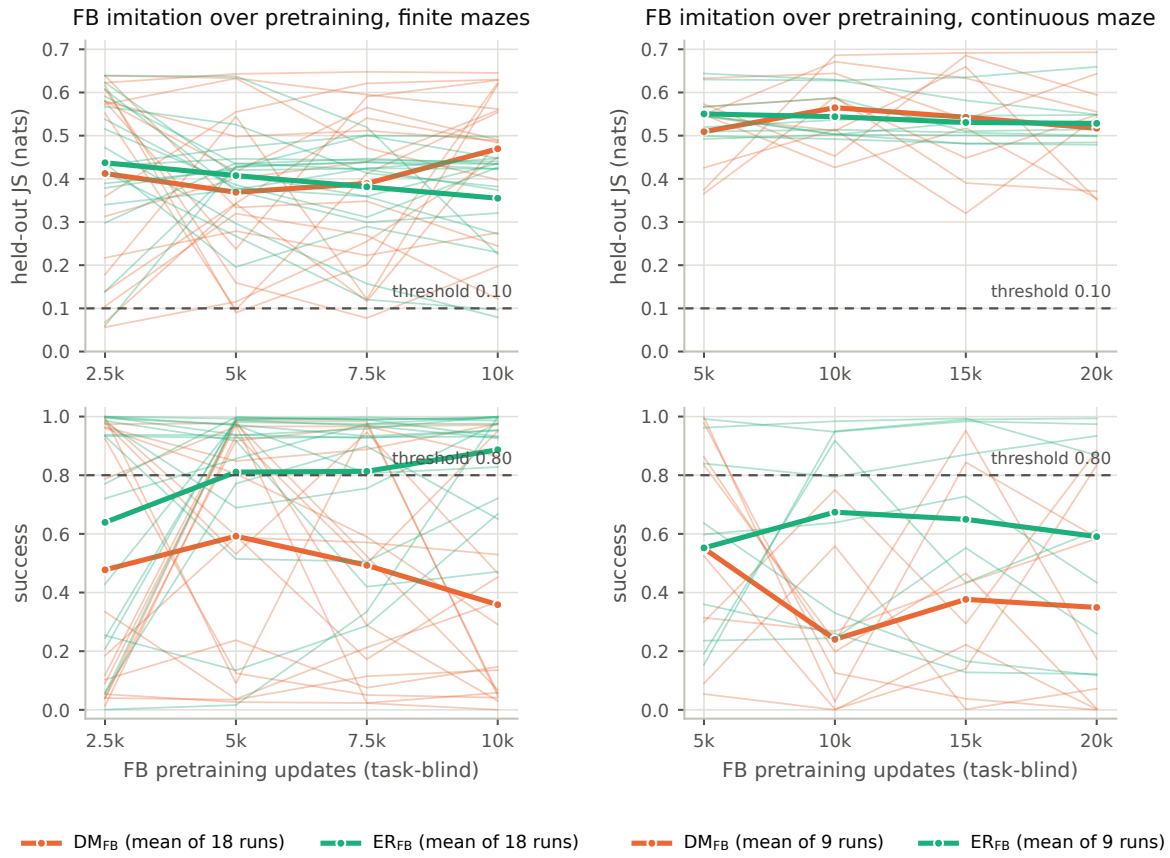


Figure 5: $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ (after its 300 latent steps) evaluated at every FB pretraining snapshot, against the number of task-blind pretraining updates. Thin lines are the runs, thick lines the means (finite: 18 runs per method; continuous: 9).

snapshot stayed on the sphere of radius 4 (to 10^{-5}). The instability is statistical, not numerical, and it is specific to FB imitation. Across seeds, the standard deviation of the final divergence is no more than about 0.014 nats for f-PG, but reaches 0.20 to 0.25 nats for $\mathrm{DM}_{\mathrm{FB}}$ and 0.21 for $\mathrm{ER}_{\mathrm{FB}}$ on some goals (Table 9). The clearest failure is $\mathrm{DM}_{\mathrm{FB}}$ on the four-rooms goal $(5, 5)$: seed 0 ends at a divergence of 0.244 nats with success 0.95, while seeds 1 and 2 end at 0.554 and 0.629 nats with success 0.031 and 0.0005. With three seeds, these spreads are descriptions of the runs, not estimates of a population variance.

12 Results in the continuous maze

This section answers Q5. In the continuous four-rooms maze the state is a position in $[0, 1]^2$ and the actions are the same five moves. FB uses the DQN-style approximation of the state–action space: the forward network maps a raw position and a latent to one d-dimensional vector per action, as a DQN maps a state to one value per action, and the policy is a softmax over the five scores $F(s, a, z)^\top z$. f-PG uses a categorical MLP policy and a kernel density estimate of the occupancies. No exact oracle exists here: every number is a Monte Carlo estimate from 500 fresh rollouts per checkpoint, and the occupancy divergence is the plug-in histogram estimate of Section 10.

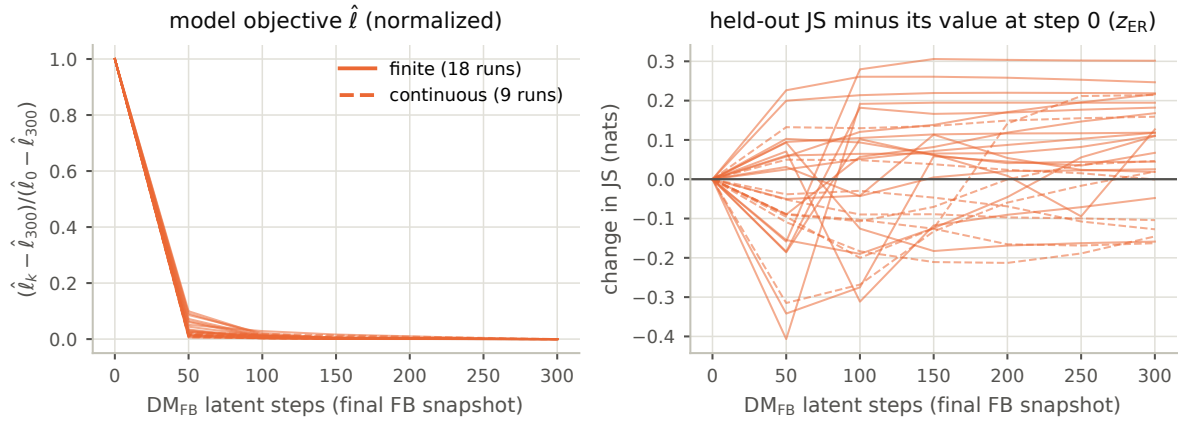


Figure 6: The 300 DM_{FB} latent steps at the final FB snapshot, for all 27 core adaptations. Left: the model objective $\hat{\ell}^{(K)}$, normalized to fall from 1 at the start (z_{ER}, projected) to 0 at step 300. Right: the held-out divergence minus its value at step 0; above the zero line, the latent steps moved the policy away from the expert.

Table 11: FB diagnostics at the final pretraining snapshot (evaluation only). $\hat{G}_B$ is the uncentred Gram matrix of B under the replay measure; $\overline{F^\top \int B \, d\hat{\nu}}$ the model's mean total successor mass (exact value 20); $\|z_{ER}\|$ the norm before projection to radius 4; the entropy is the mean policy entropy of the ER_{FB} and DM_{FB} policies at the DM_{FB} start states; the last two columns are the exact FB-6 errors of Theorem 6.5 at the ER_{FB} and DM_{FB} latents (finite mazes only). Ranges are over the three goals that share a model.

model	seed	$\lambda_{\min}(\hat{G}_B)$	$\lambda_{\max}$	$\|\hat{G}_B - I\|_F$	$\overline{F^\top \int B \, d\hat{\nu}}$	$\|z_{ER}\|$	entropy (nats)	sup TV error	max $\|Q - F^\top z_r\|$
finite open	0	0.943	2.53	2.46	18.24	3.79–3.92	0.07–0.60	38.9–71.6	4.0–15.2
	1	0.945	2.49	2.67	18.16	3.83–3.93	0.04–0.62	36.2–79.7	6.1–14.2
	2	0.929	2.26	2.09	18.21	3.12–3.74	0.08–0.59	37.4–72.3	5.5–13.9
finite four rooms	0	0.951	2.51	3.24	18.72	2.87–3.68	0.05–0.68	36.9–67.1	3.4–14.4
	1	0.959	2.65	3.01	19.20	2.79–3.30	0.07–0.81	36.3–64.7	2.9–12.8
	2	0.948	2.34	2.77	19.20	2.59–3.24	0.15–0.72	37.9–55.0	2.7–11.4
four rooms, no orth.	0	0.105	131.38	283.66	19.91	16.87	1.35–1.37	21.7–29.1	3.3–6.2
	1	0.190	145.69	266.36	19.45	13.79	1.30–1.44	29.6–36.0	1.6–4.0
	2	0.050	118.40	264.21	19.79	17.17	1.34–1.41	27.7–44.8	3.4–3.6
continuous	0	1.7×10^{-5}	2.16	3.39	19.42	2.28–2.52	0.05–0.89	n/a	n/a
	1	1.9×10^{-5}	2.01	3.40	20.54	2.30–2.44	0.09–1.01	n/a	n/a
	2	4.2×10^{-5}	2.06	3.43	18.18	2.24–2.60	0.04–0.61	n/a	n/a

12.1 Final performance

Table 12 and Fig. 3 give the final policies. **No method meets the operational thresholds in the continuous maze, at any checkpoint of any run** (Table 10). The expert rows show that this is not an artefact of the estimator: the expert's own divergence is 0.0005 to 0.0013 nats, two orders of magnitude below the learners'.

f-PG ends closest to the expert, with a divergence of 0.24 to 0.41 nats, and reaches the goal in 92 to 99% of the rollouts in all nine runs. But it does not imitate the expert's behaviour at the goal: only 6 to 14% of its rollouts end in the goal region, against 72 to 76% for the expert, and after the first hit it spends 10 to 18% of the remaining steps there, against 77 to 79%. Its policy remains very stochastic (mean entropy 1.04 to 1.47 nats on visited states, against $\log 5 \approx 1.61$ for the uniform policy), and its divergence was still decreasing at the end of the budget: over the last 10,000 learner transitions it fell by 0.003 to 0.039 nats (0.025 on average). Together with a learning rate at the edge of the tested grid (Section 10.4), this suggests that the budget, not an obstacle we can identify, limited f-PG here; the runs do not show where it would have stopped.

Table 12: Continuous four-rooms maze, core suite, last checkpoint (mean ± sample SD over 3 seeds; 500 evaluation rollouts each). JS is the plug-in histogram divergence against 2,000 reference expert rollouts; "final in goal" is the fraction of rollouts whose state s_H lies in the goal region. The expert rows measure the estimator's floor.

goal	method	JS (nats)	success	return	final in goal
(5,1)	f-PG (JS)	0.270 ± 0.029	0.981 ± 0.006	4.42 ± 0.64	0.103 ± 0.028
	$\mathrm{DM}_{\mathrm{FB}}$	0.513 ± 0.147	0.353 ± 0.468	1.09 ± 1.70	0.040 ± 0.034
	$\mathrm{ER}_{\mathrm{FB}}$	0.502 ± 0.019	0.805 ± 0.170	1.49 ± 0.25	0.001 ± 0.001
	expert	0.0008 ± 0.0001	1.000 ± 0.000	13.72 ± 0.09	0.732 ± 0.016
(5,5)	f-PG (JS)	0.371 ± 0.038	0.953 ± 0.027	1.13 ± 0.26	0.086 ± 0.014
	$\mathrm{DM}_{\mathrm{FB}}$	0.485 ± 0.115	0.474 ± 0.426	0.75 ± 0.87	0.043 ± 0.042
	$\mathrm{ER}_{\mathrm{FB}}$	0.574 ± 0.076	0.272 ± 0.156	0.21 ± 0.04	0.001 ± 0.002
	expert	0.0010 ± 0.0003	1.000 ± 0.000	10.80 ± 0.03	0.741 ± 0.009
(1,5)	f-PG (JS)	0.302 ± 0.027	0.979 ± 0.005	3.87 ± 0.46	0.080 ± 0.014
	$\mathrm{DM}_{\mathrm{FB}}$	0.553 ± 0.165	0.220 ± 0.321	0.95 ± 1.58	0.045 ± 0.077
	$\mathrm{ER}_{\mathrm{FB}}$	0.508 ± 0.036	0.695 ± 0.500	1.73 ± 0.66	0.000 ± 0.000
	expert	0.0007 ± 0.0002	1.000 ± 0.000	13.72 ± 0.06	0.745 ± 0.014

$\mathrm{ER}_{\mathrm{FB}}$ reaches the goal in 12 to 99% of the rollouts (at least 80% in 4 of the 9 runs) but almost never stays: at most 0.4% of its rollouts end in the goal region. Its divergence is 0.48 to 0.66 nats.

$\mathrm{DM}_{\mathrm{FB}}$ has divergences of 0.35 to 0.69 nats and success 0 to 0.88 (at least 0.80 in 2 runs). Its largest divergence, 0.693 nats, is essentially the maximum $\log 2$ of JS, reached when the learner's rollouts and the expert's occupy disjoint histogram cells. Paired with $\mathrm{ER}_{\mathrm{FB}}$ by goal and seed, $\mathrm{DM}_{\mathrm{FB}}$ has a larger divergence in 4 of the 9 runs and a lower success in 7; its latent steps increased the divergence in 4 of the 9 adaptations (Fig. 6).

12.2 Learning curves

f-PG starts from a nearly uniform policy at 0.57–0.59 nats. Averaged over the nine runs, its success rises from 0.29 at the start to 0.81 after 20,000 transitions and 0.97 at the end, while its mean divergence decreases at every checkpoint (Fig. 4, right). $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ change little with pretraining: at every snapshot, their mean divergence over the nine runs lies between 0.51 and 0.56 nats (Fig. 5, right). Between consecutive snapshots the divergence of a run changes by up to 0.31 nats, and 5 of the 18 curves reach a success of 0.80 at an earlier snapshot but not at the last.

12.3 The FB model with a DQN-style forward network

The replay data contained every pair of a free maze-lattice cell (side 1/7) and an action at least 423 times. At the resolution of the 356 cells of the evaluation histogram, coverage is incomplete near walls: the smallest number of visits of a pair of a histogram cell and an action is 0, 1 and 2 in the buffers of seeds 0, 1 and 2, and 29 to 34 pairs per buffer have fewer than 10 visits, all in histogram cells of which walls leave only 14% free (counts from the stored buffers by `report/verify/replay_hist_coverage.py`). Neither count rules out coverage problems of the continuous state space at finer scales. The learned backward representation is nearly degenerate (Table 11): the smallest eigenvalue of its Gram matrix under the replay measure is 1.7×10^{-5} to 4.2×10^{-5}, against 0.93 to 0.96 in the finite mazes, while the largest is 2.0 to 2.2 and $\|\hat{G}_B - I\|_F$ is about 3.4. For the eigenvector u of $\lambda_{\min}$, $\int (u^\top B)^2 \, \mathrm{d}\hat{\nu} = \lambda_{\min}$: at least one direction of B is almost zero on the replay data, despite the orthonormality penalty that pushes $\hat{G}_B$ towards the identity.

The diagnostics record only the smallest and the largest eigenvalue, so this report does not say how many directions are poorly excited. The pilot had already shown this ($\lambda_{\min} \approx 2 \times 10^{-5}$ at all three temperatures); the protocol was frozen and the settings were not changed in response. We could not compute the model error exactly here, so we cannot say how much of the failure comes from the representation and how much from the latent step. Plausible causes that these runs do not test are the raw (x, y) inputs to B (FB21 used radial-basis features in its continuous maze), the budget of 20,000 updates, and the rank $d = 16$.

Answer for Q5, in short. With a DQN-style forward network, FB pretraining on 100,000 task-blind transitions produced a backward representation with a nearly singular Gram matrix, and neither $\mathrm{ER}_{\mathrm{FB}}$ nor $\mathrm{DM}_{\mathrm{FB}}$ imitated the expert: both sometimes reach the goal but neither stays. f-PG with a KDE signal and an MLP policy made steady progress but did not reach the thresholds within 100,000 learner transitions. These are statements about this maze, these budgets and three seeds.

13 Ablations: the choice of f and the orthonormality penalty

The compact ablations vary one hyperparameter of each family on the finite four-rooms maze with goal $(5, 5)$, the goal farthest from the start, with the seeds 0, 1, 2 (Table 13 and Fig. 7). The base runs are the core runs of that goal, reused by identifier; everything else in the protocol is unchanged. One goal, one layout and three seeds make these ablations descriptive: they show what happened in these runs, not a ranking that would hold elsewhere.

Table 13: Compact ablations, finite four rooms, goal $(5, 5)$, last checkpoint (mean ± SD over 3 seeds), with the first checkpoint at which each seed met the operational thresholds. The f-PG (JS), $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ rows are the core runs.

method	JS (nats)	success	gap to $V_H^\star$	$\mathrm{JS}^{\mathrm{SA}}$ (nats)	first reached (seeds 0, 1, 2)
f-PG (JS)	0.015 ± 0.005	1.000 ± 0.000	1.44 ± 0.34	0.048 ± 0.019	30k, 20k, 20k (transitions)
f-PG (KL($d\|q$))	0.012 ± 0.005	0.999 ± 0.001	1.72 ± 0.44	0.044 ± 0.013	30k, 30k, 30k (transitions)
f-PG (KL($q\|d$))	0.061 ± 0.009	0.998 ± 0.002	4.37 ± 1.09	0.146 ± 0.044	40k, 50k, 40k (transitions)
$\mathrm{ER}_{\mathrm{FB}}$	0.348 ± 0.109	0.772 ± 0.196	10.25 ± 1.78	0.425 ± 0.085	–, –, – (FB updates)
$\mathrm{ER}_{\mathrm{FB}}$, no orth.	0.358 ± 0.044	0.965 ± 0.017	11.10 ± 0.29	0.490 ± 0.039	–, –, – (FB updates)
$\mathrm{DM}_{\mathrm{FB}}$	0.476 ± 0.204	0.327 ± 0.540	11.38 ± 1.64	0.550 ± 0.183	–, 5k, – (FB updates)
$\mathrm{DM}_{\mathrm{FB}}$, no orth.	0.467 ± 0.067	0.476 ± 0.113	11.85 ± 0.26	0.568 ± 0.048	–, –, – (FB updates)

13.1 The generator f of f-PG

The three generators of Section 9.1 were run with the same smoothing $\varepsilon = 0.01$, the same initialization and first batch per seed, and the learning rate 0.3 tuned for JS; the KL runs were not tuned. All nine runs meet the thresholds at the end of the budget, and the held-out divergence is the same $\mathrm{JS}(d^+_{\pi,H}, q_E)$ for all of them, whatever f was optimized.

- KL($d\|q$) behaves like JS: final divergences 0.0085 to 0.0169 nats against 0.0121 to 0.0212, thresholds first met at the 30,000-transition checkpoint in all three seeds (against 20,000 to 30,000), and similar returns (10.64 against 10.92 on average).
- KL($q\|d$) is slower and ends further from the expert: thresholds first met at the 40,000- to 50,000-transition checkpoints, final divergences 0.055 to 0.071 nats, a mean return of 7.99 (gap 4.37 against 1.44 for JS), and a final state in the goal with probability 0.82 against 0.89. Its policies stay more stochastic (mean entropy 0.38 nats on visited states, against 0.12 for JS and 0.16 for KL($d\|q$)), and their state–action divergence is three times larger (0.146 against 0.048).

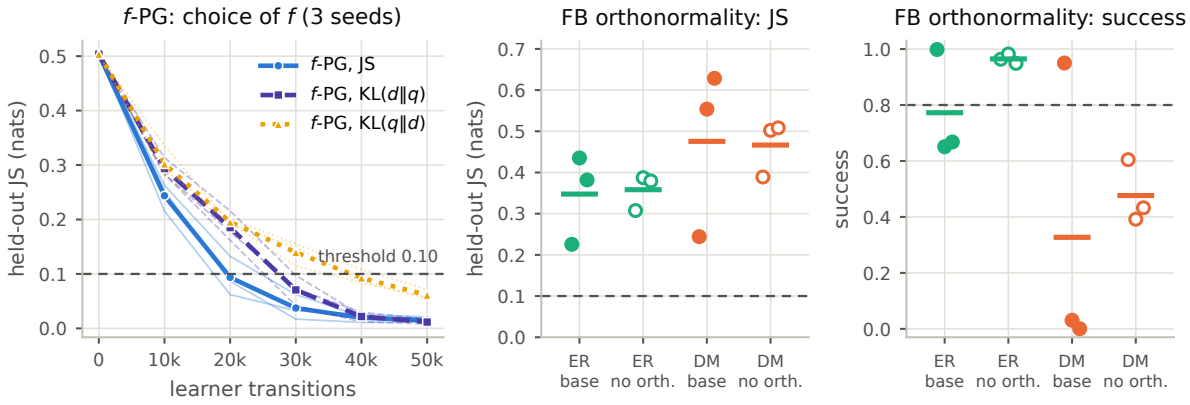


Figure 7: Ablations (four rooms, goal $(5, 5)$). Left: f-PG held-out divergence for three generators f (thin lines: seeds; thick: mean). Middle and right: final $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ divergence and success with and without the orthonormality penalty (dots: seeds, open dots without the penalty; bars: means).

The last observations agree with the shape of the signal (Section 9.1): $f'(v) = -1/v$ pushes hard towards under-visited expert cells and barely penalizes extra mass elsewhere, so a spread-out policy is cheap under $\mathrm{KL}(q\|d)$. They cannot be separated from the learning rate: Adam makes the update roughly insensitive to the overall scale of the signal, but not to its shape, and the rate was tuned for JS only. For f-IRL the same correspondence between f and the per-state weight holds (Theorem 5.3), but nothing was run.

13.2 The orthonormality penalty of FB

Without the penalty, the backward Gram matrix drifts far from the identity (Table 11): $\lambda_{\min}(\hat{G}_B)$ falls to 0.05–0.19 and $\lambda_{\max}$ rises to 118–146, against 0.95–0.96 and 2.3–2.7 with it, and $\|z_{ER}\|$ grows from 2.6–2.9 to 13.8–17.2 before projection. The model is not worse by the oracle's measures: its sup-norm total-variation error is 21.7 to 44.8 (against 36.9 to 53.9) and its largest action-value error 1.6 to 6.2 (against 2.7 to 12.3). But its policies are almost uniform: their entropy at the $\mathrm{DM}_{\mathrm{FB}}$ start states is 1.30 to 1.44 nats (against 0.25 to 0.72, and $\log 5 \approx 1.61$). This is consistent with the gauge freedom of the factorization: $(F, B) \mapsto (C^{-\top}F, CB)$ leaves the model's successor measures $F^\top B\nu$ unchanged but changes the scores $F^\top z$ that define π_z at a fixed latent radius (Theorem 6.4(d)); without the penalty nothing fixes this gauge.

The imitation outcome changes less than the model. No run, with or without the penalty, meets the thresholds at the end of the budget. $\mathrm{ER}_{\mathrm{FB}}$ without the penalty reaches the goal more often (success 0.95 to 0.98, against 0.65 to 1.00) at a similar divergence (0.358 against 0.348 nats on average); this is consistent with nearly uniform policies that diffuse to the goal without staying there (final state in the goal with probability 0.13 on average, against 0.12). $\mathrm{DM}_{\mathrm{FB}}$ has a similar mean divergence with and without the penalty (0.467 against 0.476 nats) but a much smaller spread across seeds. The penalty is therefore not what limits FB imitation in these runs. What differed with and without it is the conditioning of $\hat{G}_B$ and the entropy of the policies, as reported above: with the penalty, $\hat{G}_B$ was well conditioned and the policies were far from uniform. This is an observation on three seeds of one finite task, not a guarantee: the penalty is soft and only encourages conditioning and scale control (Theorem 6.7(c)), and with the penalty on, the continuous Gram matrices were nearly singular ($\lambda_{\min}$ between 1.7×10^{-5} and 4.2×10^{-5}, Section 12).

14 Cost accounting

This section answers the sample-requirement part of Q4 and gives the cost side of the comparison of Section 8.

14.1 Accounting rules

The rules were fixed in the specification before any run.

(i) *Training transitions* are the environment transitions that a learner receives: the on-policy rollouts of f-PG, and the task-blind replay buffer of FB. The four demonstrations (400 transitions, states only) are common to all methods and counted separately.

(ii) *Evaluation transitions* are fresh rollouts generated by the evaluator. They are never given to a learner and never counted as training data; they are reported separately.

(iii) FB pretraining is task-blind and serves the three goals of a layout and seed. The *one-task* cost charges the whole pretraining to a single task; the *amortized* cost divides it by the three goals that used it. The adaptation cost of $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ is added in both cases; it uses no new transition.

(iv) Wall times are measured on one laptop CPU with 4 PyTorch threads (NumPy and BLAS thread counts were neither set nor recorded), with jobs run one after another. They include the diagnostics that the training loops compute in passing (for example the logged objectives and entropies); this overhead was not timed separately and is not removed. Two conventions are used, never mixed within a comparison. *Per-task phase times* (Tables 14 and 15 and the comparisons below them) sum the phase timers in each job's record; for FB pretraining they add the recorded generation time of its replay buffer, which the data-preparation step performed once, before and outside the timed jobs. *Whole-job wall times* (the totals at the end of this section) run from the start to the end of each job: they include job set-up, checkpoints and output writing, but not the replay generation. (This description of the two conventions and of the diagnostics was added after the runs; it changes no measurement.)

Table 14: Training and evaluation cost per task (core suite; means over the 18 finite and 9 continuous runs of each method). Training transitions are those given to the learner; evaluation transitions are generated by the evaluator only. Wall times are per-task phase times in seconds on one laptop CPU (rule (iv)). $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ share one evaluation job per adaptation (4 $\mathrm{ER}_{\mathrm{FB}}$ and 10 $\mathrm{DM}_{\mathrm{FB}}$ checkpoints): its cost is shown once for the pair and counted once in the totals.

		one task		amortized over 3 goals		held-out evaluation	
maze	method	transitions	wall (s)	transitions	wall (s)	transitions	wall (s)
finite	f-PG (JS)	50,000	0.38	50,000	0.38	120,000	0.14
	$\mathrm{DM}_{\mathrm{FB}}$	50,000	51.78	16,667	17.62	280,000	1.28
	$\mathrm{ER}_{\mathrm{FB}}$	50,000	51.25	16,667	17.08		
continuous	f-PG (JS)	100,000	16.03	100,000	16.03	550,000	2.64
	$\mathrm{DM}_{\mathrm{FB}}$	100,000	100.32	33,333	33.80	700,000	3.79
	$\mathrm{ER}_{\mathrm{FB}}$	100,000	99.78	33,333	33.26		

14.2 What the runs cost

Tables 14 and 15 give the costs. By design the two families had the same transition budget for one task: 50,000 transitions in a finite maze and 100,000 in the continuous maze, collected on-policy by f-PG and task-blind by FB. Amortized over its three goals, FB used a third of that per task. The compute differs much more than the data. In the finite mazes, tabular f-PG trained in 0.38 s per task, while FB pretraining took 51.2 s; in the continuous maze, f-PG with

Table 15: Cost components per phase (core suite; per-task phase times; mean ± sample SD over runs where it varies, and over the distinct models for FB pretraining, each model counted once).

phase	finite	continuous
expert demonstrations (transitions, every method)	400	400
FB task-blind replay (transitions, once per layout and seed)	50,000	100,000
FB replay resets	1,000	2,000
FB updates (batch 128 + target batch 128)	10,000	20,000
FB pretraining wall time per model (s)	51.2 ± 1.0	99.8 ± 0.5
$\mathrm{ER}_{\mathrm{FB}}$ latent at the final snapshot (s)	0.0005	0.0004
$\mathrm{DM}_{\mathrm{FB}}$ 300 latent steps at the final snapshot (s)	0.54 ± 0.02	0.54 ± 0.01
new transitions during latent adaptation	0	0
f-PG learner transitions per task	50,000	100,000
f-PG policy updates per task	25	50
f-PG wall time per task (s)	0.38 ± 0.01	16.0 ± 0.1

an MLP policy and a KDE signal took 16.0 s and FB pretraining 99.8 s. Once a model is trained, imitation is almost free: $\mathrm{ER}_{\mathrm{FB}}$ needs one weighted average of B over the demonstrations (below 1 ms), and the 300 $\mathrm{DM}_{\mathrm{FB}}$ steps take 0.54 s.

Samples to reach the thresholds. In the finite mazes f-PG first met the operational thresholds at the 20,000- or 30,000-transition checkpoint (Table 10), i.e. within 40 to 60% of its budget; with checkpoints every 10,000 transitions, these are upper bounds at that resolution. FB imitation did not reach that quality with its whole budget, except for 2 $\mathrm{ER}_{\mathrm{FB}}$ runs, so the number of task-blind transitions that FB would need to match f-PG is not measured: 50,000 replay transitions with up to 10,000 pretraining updates were not enough in 34 of the 36 finite runs. Across the snapshots only the number of updates varied; the replay size was fixed. A cost comparison at equal quality is therefore not available, and the amortized figures should not be read as an advantage of FB. Amortized over its three goals, FB uses fewer task-blind transitions per task (a third of the f-PG budget) but more wall time on this CPU: in per-task phase times (Table 14: pretraining with its replay generation divided by three plus the final-snapshot adaptation, against f-PG training, evaluation excluded), about 45 to 46 times the wall time of f-PG in the finite mazes and about 2.1 times in the continuous maze. And it buys a worse policy. In the continuous maze neither family reached the thresholds within its budget, so no samples-to-threshold figure exists for either.

Evaluation is the largest consumer of transitions. The core suite gave 1,800,000 transitions to the 27 f-PG runs and 600,000 to the 9 FB models, while its evaluations generated 19,260,000 transitions, and the ablations another 1,560,000. Evaluating every checkpoint with hundreds of fresh rollouts is a property of this study, not of the methods; in the finite mazes the oracle makes most of it unnecessary, and the rollouts serve as a cross-check.

Total compute. In whole-job wall times (rule (iv)), the 144 core jobs ran for 932.7 s in total: 618.7 s of FB pretraining, 160.2 s of f-PG training, 90.2 s of evaluation, 61.2 s of latent adaptation and 2.3 s of expert evaluation. The adaptation jobs compute the $\mathrm{ER}_{\mathrm{FB}}$ latent and the $\mathrm{DM}_{\mathrm{FB}}$ steps at all four FB snapshots for the learning curves, about 2.3 s per job; the per-task cost of Table 14 counts the final snapshot only. The 21 new ablation jobs took 174.4 s. The peak process memory was 823 MiB.

14.3 Where the comparison is not symmetric

- **Resets.** FB's replay starts its 50-transition segments from states drawn uniformly over the free space, a simulator capability that f-PG does not use: every f-PG rollout starts at $(1, 1)$. This favours FB's coverage of the state space.
- **Network sizes.** The FB networks are larger than the f-PG policies, and finite f-PG is tabular. The wall-time ratio reflects these choices and the CPU; the pilot measured the laptop CPU faster than its GPU for these networks, and a different device would change the ratio.
- **Tuning.** Both f-PG learning rates and the FB temperature were tuned on one pilot seed and goal; the FB network sizes, update counts and $\mathrm{DM}_{\mathrm{FB}}$ settings were not tuned (Section 10.4).
- **Number of tasks.** With three goals per environment the amortization factor is three. FB's per-task cost falls further with more tasks; f-PG's does not. This is the regime where FB imitation is designed to pay off, and the one this study could not reach at equal quality.

15 Discussion

15.1 What the theory predicted, and what the runs show

f-PG. The fixed-target f-PG objective has an exact gradient formula (Theorem 4.2). The implementation replaces the occupancies by plug-in estimates, removes the same-batch dependence by cross-fitting, and smooths both arguments, so that its sampled update is, conditionally on the fitted signal, an unbiased but noisy estimate of the gradient of a surrogate, not of the divergence (Theorem 4.4). No theorem here says that the iteration converges, let alone to the expert. In the finite mazes, where the signal is built from exact counts of simulated transitions, it nevertheless did so reliably: every run, goal and seed met the thresholds by the checkpoint at 40 to 60% of its budget (an upper bound at the 10,000-transition evaluation interval), with smooth curves and standard deviations across seeds of no more than about 0.014 nats. In the continuous maze the signal depends on a kernel density estimate and the policy on an MLP; the method progressed steadily but slowly and produced very stochastic policies that reach the goal without staying there. The divergence was still falling when the budget ended.

FB imitation. The FB theory is exact only for exact representations, and Theorem 6.3 shows that a rank-16 model cannot be exact in the finite mazes, whose kernels have rank 40 or 49. So the finite experiments test FB imitation outside the regime of its guarantees, and the oracle shows how far outside: over state–action pairs, the largest total-variation error of the model's successor measures is 36 to 80, for a true mass of 20. The finite results are consistent with this picture. $\mathrm{ER}_{\mathrm{FB}}$, which only reads a latent off the demonstrations, often reaches the goal but rarely stays. $\mathrm{DM}_{\mathrm{FB}}$ trusts the model further: it minimizes the model's estimate of the successor-measure distance, which it did successfully in all 27 adaptations, and in 16 of the 18 finite ones the policy moved *away* from the expert while doing so. We read this as the latent optimizer exploiting the model's error, but the runs show the correlation, not the mechanism. This is not a contradiction of FI24, whose experiments use much larger models and data; it shows that the matching step is only as good as the model it is computed with. In the continuous maze, a nearly singular backward Gram matrix appeared in the pilot and again in every final model, and neither FB method imitated the expert.

Stability. Numerically, every method was stable: no failure, no non-finite value, no restart. Statistically, the two families differ sharply. f-PG's outcome hardly depends on the seed. FB imitation depends on the seed and on the pretraining snapshot: the divergence of a run moved by up to 0.53 nats between consecutive snapshots, and a third of the finite curves reached a

success of 0.80 at some snapshot but lost it by the last. A user of FB imitation would need a criterion to choose the snapshot, and none is available without evaluation rollouts or an oracle.

15.2 Limitations

- **Three seeds** per configuration describe these runs; they do not support confidence intervals, significance tests or rankings beyond the observed counts.
- **Small, simple environments.** Two 7×7 mazes and one continuous maze, with three goals each. Whether the conclusions transfer to larger state spaces, other dynamics or high-dimensional observations is untested.
- **Narrow tuning.** The f-PG learning rates and the FB temperature were chosen on one seed and one goal; both learning rates are at the edge of their grid, the KL ablations reuse the JS rate, and the FB network sizes, update counts and $\mathrm{DM}_{\mathrm{FB}}$ settings were not tuned at all. A better-tuned FB model could change the comparison.
- **Fixed budgets.** The budgets are the ceilings of the specification. They were not chosen to be sufficient; continuous f-PG in particular was still improving.
- **Evaluation choices.** The operational thresholds are study choices; success counts a single visit to the goal, so we also report staying in the goal and the return. The continuous divergence is a biased plug-in estimate on histogram cells, with a measured floor of 0.0005 to 0.0013 nats.
- **State matching.** All objectives compare state distributions; none identifies the expert's actions, and the state–action divergences are diagnostics only.
- **Scope of the theory.** The proofs cover population objectives and exact models. No statement here concerns the convergence of neural-network training or the quality of a trained model, and f-IRL was analysed but not run.

15.3 What would test the explanations

The runs suggest experiments that this study did not perform. A finite FB model with $d \geq 49$, for which an exact representation exists, would show whether $\mathrm{DM}_{\mathrm{FB}}$ improves on $\mathrm{ER}_{\mathrm{FB}}$ once the model error is small, and whether the oracle's model error predicts the imitation error. Tracking the true $\mathrm{DM}_{\mathrm{FB}}$ objective along the latent path, which the oracle can compute in the finite mazes, would test the model-exploitation reading directly. In the continuous maze, radial-basis inputs to B as in FB21, a larger budget and a larger rank would test the candidate causes of the degenerate Gram matrix. Longer continuous f-PG runs and a learning-rate grid that extends past its current edge would show where f-PG stops. More seeds, and goals not used for tuning, would turn the descriptive counts into estimates.

16 Conclusion: answers to the research questions

Q1. Mathematical principles, proofs, and exact versus approximate

Occupancy matching. The discounted next-state occupancy determines the return of every state reward (Theorem 3.3), so matching it imitates the expert as far as state rewards can tell, and no further: it does not determine the actions (Theorem B.12). **f-PG**, adapted to a fixed expert distribution, has the exact gradient of Theorem 4.2: a policy gradient whose per-step signal is $g = f'(d/q)$ at the next state, with discount weights and baselines (Theorem 4.3). It is exact for exact occupancies. The algorithm replaces them by plug-in estimates, which bias the nonlinear signal; cross-fitting removes the additional same-batch dependence, so that the sampled update is, conditionally on the fitted signal, an unbiased but noisy estimate of the gradient of a surrogate with that signal held fixed, not of the divergence; and smoothing both arguments changes the objective (Theorem 4.4, Table 3).

f-IRL's reward gradient is a covariance with the weight $h_f = (f^\circ)'(1/u)$ (Theorem 5.1). It is exact under the hypothesis (5.1) that the learner's law over state sequences is the Gibbs law $\bar{p}(\tau)\exp(\sum_t r_\theta(s_t)/\alpha)/Z(\theta)$ with a base weight $\bar{p}$ that does not depend on θ. An exact maximum-entropy policy is not shown to realize this law under stochastic dynamics [SA U-7], and a practical inner loop only approximates the policy; the learned reward is not unique in general. **FB.** Successor measures satisfy a linear Bellman equation (Theorem 6.1); if a model $F^\top B\nu$ is exact for every latent and the policies are greedy, the single vector $z_r = \mathbb{E}_\nu[rB]$ optimizes any reward r (Theorem 6.2). A rank-16 model cannot be exact in the finite mazes (Theorem 6.3), softmax policies replace optimality by a temperature bound (Theorem 6.4), and only the conditional bound of Theorem 6.5 applies to trained models. If the transition kernel has a square-integrable density with respect to the replay measure, the pretraining loss is a Bellman residual up to a constant; its minibatch gradients are unbiased for the loss on the stored buffer, and there is no convergence guarantee (Theorem 6.6). **FB imitation.** $\mathrm{ER}_{\mathrm{FB}}$ is reward maximization of $\mathrm{d}\rho_e/\,\mathrm{d}\nu$, not matching (Theorem 7.2); the $\mathrm{DM}_{\mathrm{FB}}$ loss expansion holds for exact models without greedy policies (Theorem 7.4), and the algorithm minimizes its model-based, truncated estimate (Theorem 7.5).

Q2. Similarities, differences and use cases

Both families imitate from expert states without rewards or actions, through discounted next-state objects that are linked by $d_\pi^+ = (1-\gamma)\int \mu_0(\,\mathrm{d}x)M^{\pi,+}(x,\cdot)$, and neither solves a Bellman optimality equation for the imitation task. They are not the same objective: f-PG and f-IRL compare *marginals*, $\mathrm{DM}_{\mathrm{FB}}$ compares *conditional* successor measures in a replay-weighted norm, and $\mathrm{ER}_{\mathrm{FB}}$ compares nothing (Table 5). The main difference is where the approximation sits. f-PG samples the real dynamics and approximates the densities inside its signal; its sampled update is conditionally unbiased only for a surrogate gradient with the fitted signal held fixed. FB imitation does not interact after pretraining: $\mathrm{ER}_{\mathrm{FB}}$ reads a latent off the demonstrations, and $\mathrm{DM}_{\mathrm{FB}}$ takes a finite number of projected Adam steps on an empirical, truncated estimate of a model-based objective, computed with a model that cannot be exact in the finite mazes. f-PG suits single tasks with a cheap simulator; f-IRL suits the case where a reward is wanted, although the reward it learns is not unique in general; FB imitation suits many tasks in one environment with expensive interaction, provided the model is accurate on the relevant states (Section 8).

Q3. The role of the hyperparameters

In f-PG and f-IRL the generator f acts only through the shape of $v \mapsto f'(v)$ at the occupancy ratio: JS is bounded, $\mathrm{KL}(d\|q)$ is zero-forcing and needs smoothing, $\mathrm{KL}(q\|d)$ is mass-covering (Section 9.1). On one goal with three seeds, $\mathrm{KL}(d\|q)$ behaved like JS, and $\mathrm{KL}(q\|d)$ first met the thresholds at the 40,000- to 50,000-transition checkpoints instead of 20,000 to 30,000 and kept more stochastic policies, at a learning rate tuned for JS (Section 13). The successor measure is learned by temporal differences on its measure-valued Bellman equation, with target networks, an orthonormality penalty and a latent distribution (Section 9.4). Its rank decides whether exactness is possible at all. For an exact model, the temperature and the latent norm act together as one inverse temperature (Theorem 6.4(d)); the trained forward network also depends on the latent, so for it this equivalence is not guaranteed. The orthonormality penalty pushes the Gram matrix of B towards the identity, which encourages good conditioning and a fixed scale but does not guarantee them (Theorem 6.7(c)): in the finite four-rooms ablation the models trained with it were well conditioned, and those trained without it were poorly conditioned and gave nearly uniform policies, an observation on three seeds that did not change whether the thresholds were met; in the continuous maze, with the

penalty on, the Gram matrix was nearly singular ($\lambda_{\min}$ between 1.7×10^{-5} and 4.2×10^{-5}). The learned model was not stationary at the end of pretraining: imitation quality moved by up to 0.53 nats between snapshots.

Q4. The finite scenario with exact planning

With an exact oracle, three seeds and predeclared thresholds: all 18 f-PG runs met the thresholds, first at the 20,000- or 30,000-transition checkpoint (evaluated every 10,000 transitions), and kept them; $\mathrm{ER}_{\mathrm{FB}}$ met them at the end in 2 of 18 runs and $\mathrm{DM}_{\mathrm{FB}}$ in none (Section 11). $\mathrm{DM}_{\mathrm{FB}}$ lowered its model objective in every adaptation, yet moved away from the expert in 16 of 18, while the FB model's measured errors made every FB guarantee vacuous. All methods were numerically stable; FB imitation was statistically unstable across seeds and pretraining snapshots, f-PG was not. For one task, both families used the same 50,000 transitions; FB amortizes to 16,667 per task over three goals but did not reach f-PG's quality, so no cost comparison at equal quality exists (Section 14).

Q5. The continuous state space with a DQN-style approximation

With a forward network that outputs one vector per action, FB pretraining on 100,000 task-blind transitions produced a backward representation with a nearly singular Gram matrix ($\lambda_{\min}$ between 1.7×10^{-5} and 4.2×10^{-5}), and neither $\mathrm{ER}_{\mathrm{FB}}$ nor $\mathrm{DM}_{\mathrm{FB}}$ imitated the expert: both sometimes reach the goal, neither stays. f-PG with a KDE signal and an MLP policy ended closest (divergence 0.24 to 0.41 nats, success 0.92 to 0.99) but no method met the thresholds within its budget (Section 12).

In one sentence. In these small mazes, matching occupancies directly with policy gradients was reliable where the occupancies could be counted and made steady but incomplete progress in the continuous maze, while forward–backward imitation was limited by the accuracy of its learned successor measures, which neither its pretraining loss nor its matching objective could reveal.

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A About the appendices

Contents. Sections B to H contain the complete statements and proofs of every result used in the main text, in the same notation: occupancy measures and the finite oracle; f-divergences, their orientation and smoothing; the fixed-target f-PG gradient, its estimator and approximations; the f-IRL reward gradient (theory only); forward–backward representations; $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$; and the implemented environments, estimators, experts and replay law. They were written before the experiments, as the project's proof compendium, and are reproduced with only editorial changes (cross-references, and the pilot outcome where a remark anticipated it). The main-text statements name the appendix result they summarize. Section I maps every implemented update to its result, equation and code; Section J lists the discrepancies found in the sources and the design decisions that resolve them; Section K records how the results were produced and checked; Section L gives every seed of every run.

Claim tags. Each result carries an identifier and a type, for example [FPG-2 | D]: S for a source result reproved under its own assumptions, D for our derived adaptation, I for an identity or definition-level fact, and C for a source result cited for comparison only, from which nothing is concluded; combined types such as S+D mark a source part extended by our part. A tag says where a statement comes from, not that it holds for the trained models: every result is stated with its hypotheses, and where a hypothesis fails for the trained models (exactness of FB, the Gibbs hypothesis of f-IRL, exact occupancies), the text says so.

Standard results used without proof. The Ionescu–Tulcea and Kolmogorov extension theorems; Dynkin's π–λ theorem; monotone and dominated convergence; Tonelli and Fubini; Banach's and Brouwer's fixed-point theorems; the mean value theorem; the Radon–Nikodym theorem with the Hahn and Lebesgue decompositions; Cramer's rule; the strong law of large numbers; and the data-processing inequality for f-divergences [5].

What the appendices do not contain. No statement concerns the convergence of stochastic-gradient training, of Adam, or of target-network iterations, and none asserts that a trained network is exact or optimal. The numerical checks of the proof compendium (tiny random MDPs in `theory/checks/`) test the mathematics, not the experiments.

B Occupancy measures, timing and the finite oracle

Status. Everything in this section is either a definition-level identity (I) or a derived statement (D) about the conventions of Section 2. No result here is copied from a source. The sources only fix the conventions:

- FI24 Eq. (1), p. 3: next-state successor measure;
- FPG pp. 2–3: return $\sum_t \gamma^t r(s_{t+1}, g)$;
- FB21 Eq. (1), p. 2: state–action successor measure from $t = 0$.

The finite oracle results of Section B.7 are standard dynamic programming. They are proved here because the finite expert and every "gap to optimum" depend on them.

B.1 Setting and the trajectory law

Assumption B.1 (A-MDP)**.**

(i) The state space $\mathcal{S}$ is either a finite set or a Borel subset of $\mathbb{R}^2$ with its Borel σ-algebra. The action set $\mathcal{A}$ is finite (here $|\mathcal{A}| = 5$).

(ii) $P(\cdot \mid s, a)$ is a Markov kernel from $\mathcal{S} \times \mathcal{A}$ to $\mathcal{S}$.

(iii) μ_0 is a probability measure on $\mathcal{S}$.

(iv) $\gamma \in (0,1)$ and $H \in \mathbb{N}$ with $H \geq 1$. We write $Z_H := \sum_{t=0}^{H-1} \gamma^t = (1-\gamma^H)/(1-\gamma)$.
(v) A (stationary Markov) policy is a kernel $\pi(\cdot \mid s)$ from $\mathcal{S}$ to $\mathcal{A}$, measurable in s.
(vi) Rewards $r : \mathcal{S} \to \mathbb{R}$ are bounded and measurable, with $\|r\|_\infty := \sup_s |r(s)|$.

Let $\Omega = (\mathcal{S} \times \mathcal{A})^{\mathbb{N}}$ carry the coordinates $(s_0, a_0, s_1, a_1, \dots)$. Let $\mathcal{F}_t := \sigma(s_0, a_0, \dots, a_{t-1}, s_t)$ denote the history up to and including s_t, before a_t is drawn. By the Ionescu–Tulcea theorem there is a unique probability measure $\mathbb{P}^\pi_{\mu_0}$ on Ω such that

$$s_0 \sim \mu_0, \qquad a_t \mid \mathcal{F}_t \sim \pi(\cdot \mid s_t), \qquad s_{t+1} \mid (\mathcal{F}_t, a_t) \sim P(\cdot \mid s_t, a_t) \qquad (t \geq 0). \tag{B.1}$$

For $(s,a) \in \mathcal{S} \times \mathcal{A}$ we write $\mathbb{P}^\pi_{(s,a)}$ for the analogous law started from $s_0 = s$, $a_0 = a$, after which actions are drawn from π. Expectations are written $\mathbb{E}^\pi_{\mu_0}$ and $\mathbb{E}^\pi_{(s,a)}$.

Let $\mathcal{G}$ be the product σ-algebra on Ω. A *cylinder* is a set $E = \{(s_0, a_0, \dots, s_T, a_T) \in B\}$ with $T \geq 0$ and B measurable in $(\mathcal{S} \times \mathcal{A})^{T+1}$. The cylinders form a π-system that generates $\mathcal{G}$. The Ionescu–Tulcea theorem defines $\mathbb{P}^\pi_{(s,a)}(E)$ on cylinders by the iterated integral

$$\mathbb{P}^\pi_{(s,a)}(E) = \int P(\,\mathrm{d}s_1 \mid s,a) \sum_{a_1} \pi(a_1 \mid s_1) \cdots \\ \cdots \int P(\,\mathrm{d}s_T \mid s_{T-1}, a_{T-1}) \sum_{a_T} \pi(a_T \mid s_T)\, \mathbf{1}_B(s, a, s_1, a_1, \dots, s_T, a_T), \tag{B.2}$$

and $\mathbb{P}^\pi_{\mu_0}(E)$ by the same formula preceded by $\int \mu_0(\,\mathrm{d}s_0) \sum_{a_0} \pi(a_0 \mid s_0)$. It extends each of them uniquely to a probability measure on $(\Omega, \mathcal{G})$. We use two consequences of this construction.

(M1) *First-pair disintegration.* For every $E \in \mathcal{G}$, the map $(s,a) \mapsto \mathbb{P}^\pi_{(s,a)}(E)$ is measurable and $\mathbb{P}^\pi_{\mu_0}(E) = \int \mu_0(\,\mathrm{d}s) \sum_a \pi(a \mid s)\, \mathbb{P}^\pi_{(s,a)}(E)$.

(M2) *Markov shift.* For every $t \geq 1$ and measurable $X \subseteq \mathcal{S}$,

$$\mathbb{P}^\pi_{(s,a)}(s_{t+1} \in X) = \int P(\,\mathrm{d}s_1 \mid s,a) \sum_{a_1} \pi(a_1 \mid s_1)\, \mathbb{P}^\pi_{(s_1,a_1)}(s_t \in X).$$

Proof of (M1) and (M2). Measurability. Let $\mathcal{D}$ be the class of $E \in \mathcal{G}$ for which $(s,a) \mapsto \mathbb{P}^\pi_{(s,a)}(E)$ is measurable. For a cylinder, (B.2) is an iterated integral of a bounded measurable function against Markov kernels. Each integration step preserves measurability in the remaining variables, so every cylinder lies in $\mathcal{D}$. $\mathcal{D}$ contains Ω and is closed under proper differences ($E \subseteq F$ gives $\mathbb{P}(F \setminus E) = \mathbb{P}(F) - \mathbb{P}(E)$). It is closed under increasing unions, because the pointwise limit of measurable functions is measurable (continuity from below). So $\mathcal{D}$ is a Dynkin system containing the generating π-system, and Dynkin's π–λ theorem gives $\mathcal{D} = \mathcal{G}$.

(M1). Define $Q(E) := \int \mu_0(\,\mathrm{d}s) \sum_a \pi(a \mid s) \mathbb{P}^\pi_{(s,a)}(E)$ for $E \in \mathcal{G}$; the integrand is measurable by the first step. $Q(\Omega) = 1$, and Q is countably additive by monotone convergence, so Q is a probability measure. On a cylinder, $Q(E)$ is (B.2) preceded by $\int \mu_0(\,\mathrm{d}s) \sum_a \pi(a \mid s)$, which is the defining formula of $\mathbb{P}^\pi_{\mu_0}(E)$. Two probability measures that agree on a generating π-system agree on $\mathcal{G}$, so $Q = \mathbb{P}^\pi_{\mu_0}$.

(M2). Let $\theta(s_0, a_0, s_1, a_1, \dots) := (s_1, a_1, s_2, a_2, \dots)$; it is measurable since each coordinate of θ is a coordinate of Ω. Fix (s,a). Both $E \mapsto \mathbb{P}^\pi_{(s,a)}(\theta^{-1}E)$ and $E \mapsto R(E) := \int P(\,\mathrm{d}s_1 \mid s,a) \sum_{a_1} \pi(a_1 \mid s_1) \mathbb{P}^\pi_{(s_1,a_1)}(E)$ are probability measures on $\mathcal{G}$; for R this uses the measurability step and monotone convergence. For a cylinder $E = \{(s_0, \dots, a_T) \in B\}$ we have $\theta^{-1}E = \{(s_1, a_1, \dots, s_{T+1}, a_{T+1}) \in B\}$. By (B.2) at length $T+1$, $\mathbb{P}^\pi_{(s,a)}(\theta^{-1}E)$ is the integral over (s_1, a_1) of (B.2) started at (s_1, a_1), which is $R(E)$. This uses that P and π do not depend on t. Hence the two measures agree on cylinders and therefore on $\mathcal{G}$. Taking $E = \{s_t \in X\}$, for which $\theta^{-1}E = \{s_{t+1} \in X\}$, gives (M2). □

When $\mathcal{S}$ is finite, (B.2) with the μ_0 prefix and B a singleton is the explicit product formula $\mathbb{P}^\pi_{\mu_0}(s_{0:T} = x_{0:T}, a_{0:T} = b_{0:T}) = \mu_0(x_0)\big[\prod_{k=0}^{T-1} \pi(b_k \mid x_k) P(x_{k+1} \mid x_k, b_k)\big] \pi(b_T \mid x_T)$, and (M1)–(M2) can also be read off by summing over the first or the second pair.

Definition B.2 (Marginals, occupancies, successor measures)**.** Let $\mathrm{Law}_t := \mathbb{P}^\pi_{\mu_0}(s_t \in \cdot)$; the policy is suppressed from the notation when it is clear. For measurable $X \subseteq \mathcal{S}$ and $Y \subseteq \mathcal{S} \times \mathcal{A}$:

$$d^+_\pi(X) := (1-\gamma)\sum_{t\geq 0}\gamma^t \mathrm{Law}_{t+1}(X), \qquad d^+_{\pi,H}(X) := Z_H^{-1}\sum_{t=0}^{H-1}\gamma^t \mathrm{Law}_{t+1}(X),$$
$$d^{\mathrm{SA}}_\pi(Y) := (1-\gamma)\sum_{t\geq 0}\gamma^t \mathbb{P}^\pi_{\mu_0}((s_t,a_t)\in Y), \qquad d^{\mathrm{SA}}_{\pi,H}(Y) := Z_H^{-1}\sum_{t=0}^{H-1}\gamma^t \mathbb{P}^\pi_{\mu_0}((s_t,a_t)\in Y),$$
$$d^{\mathrm{pre}}_\pi(X) := (1-\gamma)\sum_{t\geq 0}\gamma^t \mathrm{Law}_t(X), \qquad d^{\mathrm{pre}}_{\pi,H}(X) := Z_H^{-1}\sum_{t=0}^{H-1}\gamma^t \mathrm{Law}_t(X),$$

$$M^{\pi,+}(s,a,X) := \sum_{t\geq 0}\gamma^t\, \mathbb{P}^\pi_{(s,a)}(s_{t+1}\in X), \qquad M^{\pi,+}(s,X) := \sum_a \pi(a\mid s)\, M^{\pi,+}(s,a,X).$$

Lemma B.3 (Marginal recursion)**.** *Under Theorem B.1,* $\mathrm{Law}_{t+1}(X) = \int \mathrm{Law}_t(\mathrm{d}s)\sum_a \pi(a \mid s)P(X \mid s,a)$ *for all* $t \geq 0$*. If* $\mathcal{S}$ *is finite, write* $P_\pi(s,s') := \sum_a \pi(a\mid s)P(s'\mid s,a)$ *and regard laws as row vectors. Then* $\mathrm{Law}_{t+1} = \mathrm{Law}_t P_\pi$*, and hence* $\mathrm{Law}_t = \mu_0^\top P_\pi^t$*.*

Proof. By (B.1) and the tower property,

$$\mathbb{P}(s_{t+1}\in X) = \mathbb{E}\big[\mathbb{P}(s_{t+1}\in X \mid \mathcal{F}_t)\big] = \mathbb{E}\Big[\textstyle\sum_a \pi(a\mid s_t)P(X\mid s_t,a)\Big].$$

The integrand is a bounded measurable function of s_t, so the right-hand side is its integral against Law_t. In the finite case this reads $\mathrm{Law}_{t+1}(s') = \sum_s \mathrm{Law}_t(s)P_\pi(s,s')$. Induction from $\mathrm{Law}_0 = \mu_0$ gives $\mathrm{Law}_t = \mu_0^\top P_\pi^t$. □

B.2 Normalization

Proposition B.4. [OCC-1 | I] *Under Theorem B.1:*

(a) for every (s,a)*,* $M^{\pi,+}(s,a,\cdot)$ *and* $M^{\pi,+}(s,\cdot)$ *are measures on* $\mathcal{S}$ *of total mass* $1/(1-\gamma)$*;*

(b) d^+_π*,* $d^+_{\pi,H}$*,* d^{pre}_π*,* $d^{\mathrm{pre}}_{\pi,H}$ *are probability measures on* $\mathcal{S}$*, and* d^{SA}_π*,* $d^{\mathrm{SA}}_{\pi,H}$ *are probability measures on* $\mathcal{S}\times\mathcal{A}$*;*

(c) $d^+_\pi = (1-\gamma)\int \mu_0(\mathrm{d}s)\sum_a \pi(a\mid s)M^{\pi,+}(s,a,\cdot) = (1-\gamma)\int \mu_0(\mathrm{d}s)M^{\pi,+}(s,\cdot)$*.*

Proof. (a) Fix (s,a) and measurable X. The terms $\gamma^t\mathbb{P}^\pi_{(s,a)}(s_{t+1}\in X)$ lie in $[0,\gamma^t]$, so the series converges. Its value lies in $[0,1/(1-\gamma)]$, and it vanishes for $X=\emptyset$. For countable additivity, let $X = \bigsqcup_i X_i$ be a disjoint union. Countable additivity of $\mathbb{P}^\pi_{(s,a)}$ gives $\mathbb{P}^\pi_{(s,a)}(s_{t+1}\in X) = \sum_i \mathbb{P}^\pi_{(s,a)}(s_{t+1}\in X_i)$. The double series $\sum_t\sum_i \gamma^t\mathbb{P}^\pi_{(s,a)}(s_{t+1}\in X_i)$ has nonnegative terms, so Tonelli's theorem for series allows interchanging the order of summation: $M^{\pi,+}(s,a,X) = \sum_i M^{\pi,+}(s,a,X_i)$. Taking $X=\mathcal{S}$ gives total mass $\sum_t\gamma^t = 1/(1-\gamma)$. $M^{\pi,+}(s,\cdot)$ is a convex combination of finitely many such measures, so the same holds for it.

(b) The argument of (a), applied to Law_{t+1}, Law_t and the state–action marginals, shows that each object is a measure. The total masses are $(1-\gamma)\sum_{t\geq 0}\gamma^t = 1$ and $Z_H^{-1}\sum_{t=0}^{H-1}\gamma^t = 1$.

(c) By (M1) with $E = \{s_{t+1}\in X\}$, $\mathrm{Law}_{t+1}(X) = \int\mu_0(\mathrm{d}s)\sum_a\pi(a\mid s)\mathbb{P}^\pi_{(s,a)}(s_{t+1}\in X)$. Multiply by $(1-\gamma)\gamma^t$ and sum over t. Since every term is nonnegative, Tonelli's theorem lets the sum pass inside $\int\mu_0(\mathrm{d}s)\sum_a$. □

B.3 Time shift between pre-action and next-state occupancies

Proposition B.5. [OCC-2 | I] *Under Theorem B.1, for all measurable* $X \subseteq \mathcal{S}$*:*

(a) $d^{\mathrm{SA}}_\pi(X\times\mathcal{A}) = d^{\mathrm{pre}}_\pi(X)$ *and* $d^{\mathrm{SA}}_{\pi,H}(X\times\mathcal{A}) = d^{\mathrm{pre}}_{\pi,H}(X)$*;*

(b) $d^{\mathrm{pre}}_\pi = (1-\gamma)\mu_0 + \gamma\, d^+_\pi$*;*

(c) $Z_H\, d^{\mathrm{pre}}_{\pi,H} = \mu_0 + \gamma Z_H\, d^+_{\pi,H} - \gamma^H \mathrm{Law}_H$*.*

Proof. (a) Use $\mathbb{P}((s_t,a_t)\in X\times\mathcal{A}) = \mathbb{P}(s_t\in X)$ termwise.

(b) Separate the $t=0$ term and shift the index $u = t-1$:

$$d^{\mathrm{pre}}_\pi = (1-\gamma)\mathrm{Law}_0 + (1-\gamma)\sum_{t\geq 1}\gamma^t\mathrm{Law}_t = (1-\gamma)\mu_0 + \gamma(1-\gamma)\sum_{u\geq 0}\gamma^u\mathrm{Law}_{u+1} = (1-\gamma)\mu_0 + \gamma d^+_\pi.$$

(c) Similarly, $Z_H d^{\mathrm{pre}}_{\pi,H} = \mu_0 + \gamma \sum_{u=0}^{H-2} \gamma^u \mathrm{Law}_{u+1}$. Moreover $\sum_{u=0}^{H-2} \gamma^u \mathrm{Law}_{u+1} = Z_H d^{+}_{\pi,H} - \gamma^{H-1}\mathrm{Law}_H$. Substituting gives the claim. □

Remark B.6. The pre-action occupancy contains the uncontrolled start through the atom $(1-\gamma)\mu_0$. The next-state occupancy does not. This is the timing difference between FB21 (from $t = 0$) and FI24/FPG (next state). Theorem B.19 gives the corresponding statement for successor measures.

B.4 Finite resolvent and Bellman identities

Lemma B.7 (Neumann series)**.** *Let $\mathcal{S}$ be finite and P_π row-stochastic. Then $I - \gamma P_\pi$ is invertible, and $(I - \gamma P_\pi)^{-1} = \sum_{t\geq 0} \gamma^t P_\pi^t$, the series converging absolutely in the induced norm $\|\cdot\|_\infty$ (maximum absolute row sum). Its entries are nonnegative, every row sums to $1/(1-\gamma)$, and it commutes with P_π.*

Proof. $\|P_\pi\|_\infty = 1$ and the norm is submultiplicative, so $\|\gamma^t P_\pi^t\|_\infty \leq \gamma^t$. The partial sums $S_n := \sum_{t\leq n} \gamma^t P_\pi^t$ are therefore Cauchy and converge to a limit S. Telescoping gives $(I-\gamma P_\pi)S_n = S_n(I - \gamma P_\pi) = I - \gamma^{n+1} P_\pi^{n+1} \to I$. Hence $(I - \gamma P_\pi)S = S(I - \gamma P_\pi) = I$. The entries of S are nonnegative because every term is nonnegative. For the row sums, $S\mathbf{1} = \sum_t \gamma^t P_\pi^t \mathbf{1} = \sum_t \gamma^t \mathbf{1}$. Each S_n commutes with P_π, so the limit does too. □

Proposition B.8. [OCC-3 | I] *Let $\mathcal{S}$ be finite, write $p_{sa} := P(\cdot \mid s, a)$ as a row vector, and let $R_\pi := (I - \gamma P_\pi)^{-1}$.*

(a) $M^{\pi,+}(s, a, \cdot) = p_{sa} R_\pi$ and $M^{\pi,+}(s, \cdot) = e_s^\top P_\pi R_\pi$. Writing $\bar{M}_\pi$ for the matrix with rows $M^{\pi,+}(s, \cdot)$, we have $\bar{M}_\pi = P_\pi R_\pi = R_\pi P_\pi$.

(b) (Bellman identity) $M^{\pi,+}(s, a, \cdot) = p_{sa} + \gamma \sum_{s_1} P(s_1 \mid s, a) \sum_{a_1} \pi(a_1 \mid s_1) M^{\pi,+}(s_1, a_1, \cdot)$. Moreover $\bar{M}_\pi$ is the unique matrix solution X of $X = P_\pi + \gamma P_\pi X$.

(c) $d^+_\pi = (1 - \gamma)\mu_0^\top P_\pi R_\pi$.

(d) $d^+_{\pi,H} = Z_H^{-1} \sum_{t=0}^{H-1} \gamma^t \mu_0^\top P_\pi^{t+1} = Z_H^{-1} \mu_0^\top P_\pi (I - \gamma^H P_\pi^H) R_\pi$.

Proof. (a) Under $\mathbb{P}^\pi_{(s,a)}$ the law of s_1 is p_{sa}. Theorem B.3, applied from time 1, gives that the law of s_{t+1} is $p_{sa} P_\pi^t$. Hence $M^{\pi,+}(s, a, \cdot) = \sum_t \gamma^t p_{sa} P_\pi^t = p_{sa} R_\pi$, where the series is passed through the linear map $x \mapsto p_{sa} x$ (continuous in finite dimension). Averaging over a uses $\sum_a \pi(a \mid s) p_{sa} = e_s^\top P_\pi$. The commutation is Theorem B.7.

(b) Theorem B.7 gives $R_\pi = I + \gamma P_\pi R_\pi$. Hence

$$p_{sa} R_\pi = p_{sa} + \gamma\, p_{sa} P_\pi R_\pi = p_{sa} + \gamma \sum_{s_1} P(s_1 \mid s, a)\, e_{s_1}^\top P_\pi R_\pi.$$

By (a), $e_{s_1}^\top P_\pi R_\pi = M^{\pi,+}(s_1, \cdot) = \sum_{a_1} \pi(a_1 \mid s_1) M^{\pi,+}(s_1, a_1, \cdot)$. The matrix form follows by averaging over a with weights $\pi(a \mid s)$. For uniqueness, if $X = P_\pi + \gamma P_\pi X$ then $(I - \gamma P_\pi)X = P_\pi$, so $X = R_\pi P_\pi = \bar{M}_\pi$.

(c) By Theorem B.3, $\mathrm{Law}_{t+1} = \mu_0^\top P_\pi^{t+1}$. Sum with weights $(1 - \gamma)\gamma^t$ and use Theorem B.7.

(d) The first equality is the definition combined with Theorem B.3. For the second, telescoping gives $\big(\sum_{t<H} \gamma^t P_\pi^t\big)(I - \gamma P_\pi) = I - \gamma^H P_\pi^H$. Multiply on the right by R_π, then on the left by $\mu_0^\top P_\pi$. □

Proposition B.9 (measure form of the Bellman identity)**.** [OCC-3 | I] *Under Theorem B.1, for every (s, a) and measurable X:*

$$M^{\pi,+}(s, a, X) = P(X \mid s, a) + \gamma \int P(\mathrm{d}s_1 \mid s, a) \sum_{a_1} \pi(a_1 \mid s_1)\, M^{\pi,+}(s_1, a_1, X).$$

Proof. The $t = 0$ term of $M^{\pi,+}(s, a, X)$ is $\mathbb{P}^\pi_{(s,a)}(s_1 \in X) = P(X \mid s, a)$. For $t \geq 1$, (M2) gives $\gamma^t \mathbb{P}^\pi_{(s,a)}(s_{t+1} \in X) = \gamma \int P(\mathrm{d}s_1 \mid s, a) \sum_{a_1} \pi(a_1 \mid s_1)\, \gamma^{t-1} \mathbb{P}^\pi_{(s_1,a_1)}(s_{(t-1)+1} \in X)$. Sum over $t \geq 1$ and interchange the sum with the integral and the finite sum by Tonelli's theorem, since all terms are nonnegative. The remaining sum over $u = t - 1 \geq 0$ is $M^{\pi,+}(s_1, a_1, X)$. □

Remark B.10 (validation). The finite oracle evaluates $d^+_{\pi,H}$ both by the explicit finite sum and by the resolvent form in Theorem B.8(d). The two are algebraically identical; numerically they agree up to floating-point round-off, as checked in `theory/checks/check_2A.py`.

B.5 Return identity and the non-identifiability of actions

Proposition B.11. [OCC-4 | D] *Under Theorem B.1, let r be bounded and measurable, and define $J_H(\pi) := \mathbb{E}^\pi_{\mu_0} \sum_{t=0}^{H-1} \gamma^t r(s_{t+1})$ and $J_\infty(\pi) := \mathbb{E}^\pi_{\mu_0} \sum_{t\geq 0} \gamma^t r(s_{t+1})$.*

(a) $J_H(\pi) = Z_H \int r \, \mathrm{d}d^+_{\pi,H}$.

(b) $J_\infty(\pi)$ is well defined and $J_\infty(\pi) = (1-\gamma)^{-1} \int r \, \mathrm{d}d^+_\pi$.

(c) $d^+_{\pi,H} = d^+_{\pi',H}$ if and only if $J_H(\pi) = J_H(\pi')$ for every bounded measurable r. The same holds with (d^+_π, J_∞) in place of $(d^+_{\pi,H}, J_H)$.

(d) Suppose that for every s, $\sum_a \pi(a \mid s) P(\cdot \mid s, a) = \sum_a \pi'(a \mid s) P(\cdot \mid s, a)$. Then $\mathrm{Law}^\pi_t = \mathrm{Law}^{\pi'}_t$ for all t, so all state occupancies (next-state and pre-action, finite and infinite horizon) coincide, even when $\pi \neq \pi'$.

Proof. First, an integration fact. Let ν_t be probability measures and $c_t \geq 0$ with $\sum_t c_t < \infty$. Then

$$\int r \, \mathrm{d}\Big(\sum_t c_t \nu_t \Big) = \sum_t c_t \int r \, \mathrm{d}\nu_t \tag{B.3}$$

for bounded measurable r. It holds for indicators by definition of the series measure, and for simple functions by linearity. For measurable $r \geq 0$ it follows by monotone convergence applied to an increasing sequence of simple functions, on both sides. For general bounded $r = r^+ - r^-$, both sides are finite and the identity follows by subtraction.

(a) By linearity of expectation over the finite sum, and since $\mathbb{E} r(s_{t+1}) = \int r \, \mathrm{dLaw}_{t+1}$, we get $J_H = \sum_{t<H} \gamma^t \int r \, \mathrm{dLaw}_{t+1} = Z_H \int r \, \mathrm{d}d^+_{\pi,H}$ by (B.3).

(b) $\big| \sum_{t\leq n} \gamma^t r(s_{t+1}) \big| \leq \|r\|_\infty / (1-\gamma)$ for all n. So dominated convergence gives $J_\infty = \sum_{t\geq 0} \gamma^t \int r \, \mathrm{dLaw}_{t+1}$, which equals $(1-\gamma)^{-1} \int r \, \mathrm{d}d^+_\pi$ by (B.3).

(c) "Only if" follows from (a). For "if", take $r = \mathbf{1}_X$: then $Z_H d^+_{\pi,H}(X) = Z_H d^+_{\pi',H}(X)$ for every X. The infinite-horizon case is the same with (b).

(d) In Theorem B.3, the policy enters only through the kernel $\sum_a \pi(a \mid s) P(\cdot \mid s, a)$. Induction on t from $\mathrm{Law}_0 = \mu_0$ gives the equality of marginals, and every occupancy in Theorem B.2 restricted to states is a function of the marginals. □

Example B.12 (stay versus blocked move)**.** In the finite maze with deterministic movement, take the corner cell $c = (0, 0)$. There, "move left" is blocked and therefore stays, so $P(\cdot \mid c, \mathrm{left}) = P(\cdot \mid c, \mathrm{stay}) = \delta_c$. Two policies that agree everywhere except that one plays "left" and the other "stay" at c satisfy the hypothesis of Theorem B.11(d). They have identical state occupancies, hence by (c) identical returns for every state reward. If c is visited with positive probability, their state–action occupancies differ: $d^{\mathrm{SA}}(c, \mathrm{left}) > 0$ for one and $= 0$ for the other. State-only matching therefore cannot identify the expert's action choices. This is why the report never claims that state-only matching identifies the expert's action choices. The same holds for any movement-noise model in which the blocked move and stay induce the same next-state law.

B.6 Finite versus infinite horizon

Proposition B.13. [OCC-5 | D] *Under Theorem B.1, define*

$$d^{+,\mathrm{tail}}_\pi := (1-\gamma) \sum_{u\geq 0} \gamma^u \mathrm{Law}_{H+1+u},$$

a probability measure by the argument of Theorem B.4. Then:

(a) $d^+_\pi = (1-\gamma^H) d^+_{\pi,H} + \gamma^H d^{+,\mathrm{tail}}_\pi$;

(b) $\mathrm{TV}(d^+_\pi, d^+_{\pi,H}) := \sup_X |d^+_\pi(X) - d^+_{\pi,H}(X)| \leq \gamma^H$;

(c) $J_\infty(\pi) - J_H(\pi) = \gamma^H(1-\gamma)^{-1}\int r\,\mathrm{d}d_\pi^{+,\mathrm{tail}}$*, hence* $|J_\infty - J_H| \le \gamma^H\|r\|_\infty/(1-\gamma)$*.*
For $\gamma = 0.95$ *and* $H = 100$*:* $\gamma^H \approx 5.92\times10^{-3}$*,* $Z_H \approx 19.88$ *and* $\gamma^H/(1-\gamma) \approx 0.1184$*.*

Proof. (a) Split the series at $t = H$. The head is $(1-\gamma)\sum_{t<H}\gamma^t\mathrm{Law}_{t+1} = (1-\gamma)Z_H d_{\pi,H}^+ = (1-\gamma^H)d_{\pi,H}^+$. The tail is $(1-\gamma)\sum_{t\ge H}\gamma^t\mathrm{Law}_{t+1} = \gamma^H(1-\gamma)\sum_{u\ge0}\gamma^u\mathrm{Law}_{H+u+1}$.

(b) By (a), $d_\pi^+ - d_{\pi,H}^+ = \gamma^H(d_\pi^{+,\mathrm{tail}} - d_{\pi,H}^+)$. Both measures are probabilities, so for every X, $|d_\pi^{+,\mathrm{tail}}(X) - d_{\pi,H}^+(X)| \le 1$.

(c) Following (b) of Theorem B.11,

$$J_\infty - J_H = \sum_{t\ge H}\gamma^t\int r\,\mathrm{dLaw}_{t+1} = \gamma^H\sum_u\gamma^u\int r\,\mathrm{dLaw}_{H+u+1}.$$

This equals $\gamma^H(1-\gamma)^{-1}\int r\,\mathrm{d}d_\pi^{+,\mathrm{tail}}$ by (B.3), and $|\int r\,\mathrm{d}d_\pi^{+,\mathrm{tail}}| \le \|r\|_\infty$. □

Remark B.14. In the f-PG objective, the learner occupancy and the expert target are both defined at the same horizon H (Section 2). The H-step objective therefore involves no truncation approximation; d_π^+ serves only as a reference.

B.7 The finite oracle: stationary and H-step optima

Let $\mathcal{S}$ and $\mathcal{A}$ be finite and r a state reward. For $V \in \mathbb{R}^{\mathcal{S}}$ define the next-state Bellman operators

$$(T_\pi V)(s) := \sum_a \pi(a \mid s)\sum_{s'} P(s' \mid s,a)\big[r(s') + \gamma V(s')\big],$$
$$(TV)(s) := \max_a \sum_{s'} P(s' \mid s,a)\big[r(s') + \gamma V(s')\big].$$

General policies. In this subsection a *policy* σ may be history-dependent and randomized. It is a family of laws $\sigma(\cdot \mid h_t)$ on $\mathcal{A}$, one for each history $h_t = (s_0, a_0, \dots, s_t) \in (\mathcal{S}\times\mathcal{A})^t \times \mathcal{S}$ and $t \ge 0$. A stationary Markov π is the case $\sigma(\cdot \mid h_t) = \pi(\cdot \mid s_t)$. For $s \in \mathcal{S}$, $T \ge 0$, $x_{0:T} \in \mathcal{S}^{T+1}$ and $b_{0:T} \in \mathcal{A}^{T+1}$, write $h_k := (x_0, b_0, \dots, x_k)$ and set

$$p_{s,T}^\sigma(x_{0:T}, b_{0:T}) := \delta_s(x_0)\,\sigma(b_0 \mid h_0)\prod_{k=1}^{T} P(x_k \mid x_{k-1}, b_{k-1})\,\sigma(b_k \mid h_k). \tag{B.4}$$

These are probability vectors, and the family is consistent: summing $p_{s,T+1}^\sigma$ over (x_{T+1}, b_{T+1}) gives $p_{s,T}^\sigma$, because $\sum_x P(x \mid x_T, b_T)\sum_b \sigma(b \mid h_T, b_T, x) = 1$. All coordinate spaces are finite, so the Kolmogorov extension theorem gives a unique probability $\mathbb{P}_s^\sigma$ on Ω whose law of $(s_{0:T}, a_{0:T})$ is $p_{s,T}^\sigma$ for every T. For $\sigma = \pi$ stationary Markov, (B.4) is the product formula of Section B.1 with $\mu_0 = \delta_s$, so $\mathbb{P}_s^\pi = \mathbb{P}_{\delta_s}^\pi$ by uniqueness. Summing (B.4) over b_T gives $\mathbb{P}_s^\sigma(h_T) = \delta_s(x_0)\prod_{k=1}^T \sigma(b_{k-1} \mid h_{k-1})P(x_k \mid x_{k-1}, b_{k-1})$. Dividing $\mathbb{P}_s^\sigma(h_n, a_n = a, s_{n+1} = s')$ by $\mathbb{P}_s^\sigma(h_n)$ therefore gives, for every h_n with $\mathbb{P}_s^\sigma(h_n) > 0$,

$$\mathbb{P}_s^\sigma(a_n = a,\ s_{n+1} = s' \mid h_n) = \sigma(a \mid h_n)\,P(s' \mid s_n, a). \tag{B.5}$$

Let $V^\sigma(s) := \mathbb{E}_s^\sigma\sum_{t\ge0}\gamma^t r(s_{t+1})$, an absolutely convergent series since $|r| \le \|r\|_\infty$. For $n \ge 0$ let $V_n^\sigma(s) := \mathbb{E}_s^\sigma\sum_{t=0}^{n-1}\gamma^t r(s_{t+1})$; the H-step value is V_H^σ.

Lemma B.15. [OCC-6 | I]

(i) T and T_π are γ-contractions in $\|\cdot\|_\infty$. They are monotone ($V \le W \Rightarrow TV \le TW$) and satisfy $T(V + c\mathbf{1}) = TV + \gamma c\mathbf{1}$.

(ii) For a stationary Markov π, $V^\pi = (I - \gamma P_\pi)^{-1}P_\pi r$ is the unique fixed point of T_π.

(iii) T has a unique fixed point $V^\star$, and $\|V^\star\|_\infty \le \|r\|_\infty/(1-\gamma)$. Every stationary deterministic $\pi^\star$ that is greedy with respect to $V^\star$, with ties broken in any fixed way, satisfies $V^{\pi^\star} = V^\star$.

(iv) For every policy σ and every s, $V^\sigma(s) \le V^\star(s)$. Hence $\pi^\star$ is optimal among all policies.

Proof. (i) Since $|\max_a x_a - \max_a y_a| \le \max_a |x_a - y_a|$, we get

$$|(TV)(s) - (TW)(s)| \le \gamma \max_a \sum_{s'} P(s' \mid s,a)|V(s') - W(s')| \le \gamma \|V - W\|_\infty.$$

The same argument applies to T_π, with an average in place of the max. Monotonicity and the constant shift are immediate from the definitions.

(ii) $T_\pi V = P_\pi r + \gamma P_\pi V$, an affine γ-contraction, so by Banach's theorem it has a unique fixed point. By Theorem B.11(b) with $\mu_0 = \delta_s$ and Theorem B.3, $V^\pi = \sum_t \gamma^t P_\pi^{t+1} r = P_\pi r + \gamma P_\pi \sum_t \gamma^t P_\pi^{t+1} r = T_\pi V^\pi$. By Theorem B.7, $\sum_t \gamma^t P_\pi^{t+1} r = P_\pi (I - \gamma P_\pi)^{-1} r = (I - \gamma P_\pi)^{-1} P_\pi r$.

(iii) Existence and uniqueness of $V^\star$ follow from Banach's theorem on the complete space $(\mathbb{R}^{\mathcal{S}}, \|\cdot\|_\infty)$. For the bound, let $\rho := \|r\|_\infty/(1-\gamma)$. If $\|V\|_\infty \le \rho$, then $|(TV)(s)| \le \|r\|_\infty + \gamma\rho = \rho$. So T maps the closed ball $\{\|V\|_\infty \le \rho\}$ into itself. The Banach iterates $T^n 0$ stay in this ball and converge to $V^\star$, and the ball is closed. Greediness means $T_{\pi^\star} V^\star = TV^\star = V^\star$, so $V^\star$ is the fixed point of $T_{\pi^\star}$, which is $V^{\pi^\star}$ by (ii).

(iv) Fix σ and s. We show by induction that for every $n \ge 0$, $\mathbb{E}_s^\sigma\big[\sum_{t<n} \gamma^t r(s_{t+1}) + \gamma^n V^\star(s_n)\big] \le V^\star(s)$. The case $n = 0$ is an equality. For the induction step, condition on h_n with $\mathbb{P}_s^\sigma(h_n) > 0$. By (B.5), $\mathbb{E}[r(s_{n+1}) + \gamma V^\star(s_{n+1}) \mid h_n] = \sum_a \sigma(a \mid h_n) \sum_{s'} P(s' \mid s_n, a)[r(s') + \gamma V^\star(s')] \le (TV^\star)(s_n) = V^\star(s_n)$, since an average does not exceed a maximum. Multiplying by γ^n and averaging over h_n gives $\mathbb{E}[\gamma^n r(s_{n+1}) + \gamma^{n+1} V^\star(s_{n+1})] \le \mathbb{E}[\gamma^n V^\star(s_n)]$. Adding this to the induction hypothesis for n gives the statement for $n+1$. As $n \to \infty$, $|\gamma^n \mathbb{E} V^\star(s_n)| \le \gamma^n \|r\|_\infty/(1-\gamma) \to 0$ by (iii). The partial sums are bounded by $\|r\|_\infty/(1-\gamma)$, so dominated convergence gives $\mathbb{E}\sum_{t<n} \gamma^t r(s_{t+1}) \to V^\sigma(s)$. □

Lemma B.16 (residual certificates)**.** [OCC-6 | I]

(i) If $\|TV - V\|_\infty \le \eta$, then $\|V - V^\star\|_\infty \le \eta/(1-\gamma)$.

(ii) If moreover π is greedy with respect to V, then $0 \le V^\star - V^\pi \le 2\gamma\eta/(1-\gamma)^2$ componentwise.

(iii) For a stationary π, let $\eta_\pi := \|TV^\pi - V^\pi\|_\infty$. Then $0 \le V^\star - V^\pi \le \eta_\pi/(1-\gamma)$. Here η_π is computable from the linear solve in Theorem B.15(ii), up to the floating-point error of that solve.

Proof. (i) $\|V - V^\star\| \le \|V - TV\| + \|TV - TV^\star\| \le \eta + \gamma\|V - V^\star\|$.

(ii) The lower bound is Theorem B.15(iv). For the upper bound, $T_\pi V = TV$ by greediness and $T_\pi V^\pi = V^\pi$, so $V^\star - V^\pi = (TV^\star - TV) + (T_\pi V - T_\pi V^\pi)$. The first term is at most $\gamma\|V^\star - V\|\mathbf{1}$. The second equals $\gamma P_\pi(V - V^\pi) = \gamma P_\pi(V - V^\star) + \gamma P_\pi(V^\star - V^\pi) \le \gamma\|V - V^\star\|\mathbf{1} + \gamma\delta\mathbf{1}$, where $\delta := \max_s(V^\star - V^\pi)(s) \ge 0$. Taking the maximum over s gives $\delta \le 2\gamma\|V - V^\star\| + \gamma\delta$; then apply (i).

(iii) Apply (i) with $V = V^\pi$, and Theorem B.15(iv) for the lower bound. □

Proposition B.17 (H-step optimum by backward induction)**.** [OCC-6 | D] *Let $W_H := 0$ and $W_k := TW_{k+1}$ for $k = H-1, \dots, 0$.*

(a) For every policy σ and every s, $V_H^\sigma(s) \le W_0(s)$.

(b) Equality holds for any non-stationary Markov policy $\sigma^\star = (\sigma_0^\star, \dots, \sigma_{H-1}^\star)$ with $\sigma_k^\star(s) \in \arg\max_a \sum_{s'} P(s' \mid s,a)[r(s') + \gamma W_{k+1}(s')]$. Thus $V_H^\star := W_0$ is the H-step optimum.

(c) Let $\pi^\star$ be the stationary optimal policy of Theorem B.15. For bounded r, $0 \le V_H^\star - V_H^{\pi^\star} \le 2\gamma^H\|r\|_\infty/(1-\gamma)$. If $r \ge 0$, then $0 \le V_H^\star - V_H^{\pi^\star} \le \gamma^H\|r\|_\infty/(1-\gamma)$. For $r_g \in [0,1]$, $\gamma = 0.95$ and $H = 100$, the bound $\gamma^H\|r_g\|_\infty/(1-\gamma)$ is at most $\gamma^H/(1-\gamma) \approx 0.1184$ (< 0.119).

Proof. (a)–(b) Fix a policy σ. Define functions U_k on all histories by backward recursion: $U_H(h_H) := 0$ and, for $k = H-1, \dots, 0$,

$$U_k(h_k) := \sum_a \sigma(a \mid h_k) \sum_{s'} P(s' \mid s_k, a)\big[r(s') + \gamma U_{k+1}(h_k, a, s')\big].$$

Step 1: $U_0(s) = V_H^\sigma(s)$. We claim that for every $k \le H$ and every h_k,

$$U_k(h_k) = \sum_{b_k, x_{k+1}, \dots, b_{H-1}, x_H} \prod_{j=k}^{H-1} \sigma(b_j \mid h_j) P(x_{j+1} \mid x_j, b_j) \sum_{t=k}^{H-1} \gamma^{t-k} r(x_{t+1}),$$

where h_j extends h_k by $(b_k, x_{k+1}, \dots, b_{j-1}, x_j)$. For $k = H$ both sides are 0. Assume the claim for $k+1$ and insert it into the recursion. The weights $\prod_{j=k+1}^{H-1} \sigma(b_j \mid h_j) P(x_{j+1} \mid x_j, b_j)$ sum to 1 over $(b_{k+1}, \dots, x_H)$, so $r(x_{k+1})$ may be written as the same weighted sum of the constant $r(x_{k+1})$. Combining the two terms gives the claim for k. For $k = 0$ and $h_0 = s$, the right-hand side is the expectation of $\sum_{t<H} \gamma^t r(s_{t+1})$ under $p_{s,H}^\sigma$ of (B.4), summed over b_H. Hence $U_0(s) = V_H^\sigma(s)$.

Step 2: comparison. We show $U_k(h_k) \le W_k(s_k)$ for all h_k, by backward induction on k. The case $k = H$ is trivial. If $U_{k+1}(h_{k+1}) \le W_{k+1}(s_{k+1})$ for all h_{k+1}, then $U_k(h_k) \le \sum_a \sigma(a \mid h_k) \sum_{s'} P(s' \mid s_k, a)[r(s') + \gamma W_{k+1}(s')] \le W_k(s_k)$, since an average does not exceed a maximum. For $\sigma^\star$, i.e. $\sigma(\cdot \mid h_k) = \delta_{\sigma_k^\star(s_k)}$ for $k < H$, both inequalities are equalities at every stage. Indeed, by induction $U_{k+1}(h_{k+1}) = W_{k+1}(s_{k+1})$ for all h_{k+1}, and the greedy action attains the maximum. Taking $k = 0$ and using Step 1 gives (a) and (b).

(c) The lower bound follows from (a) applied to $\sigma = \pi^\star$. For the upper bound, let $\tilde\sigma$ follow $\sigma^\star$ for $t < H$ and $\pi^\star$ afterwards. Split its infinite-horizon value at H. In (B.4) for $\tilde\sigma$, the factors with index $k < H$ are those of $\sigma^\star$, and those with $k \ge H$ are those of $\pi^\star$. So the law of h_H is the same under $\tilde\sigma$ and $\sigma^\star$, and the first H terms contribute $V_H^\star(s)$ by (b). Given h_H, the law of $(a_H, s_{H+1}, a_{H+1}, \dots, s_{H+n})$ is that of $(a_0, s_1, a_1, \dots, s_n)$ under $\mathbb{P}_{s_H}^{\pi^\star}$. Hence $\mathbb{E}_s^{\tilde\sigma} \sum_{t=H}^{H+n-1} \gamma^t r(s_{t+1}) = \gamma^H \mathbb{E}_s^{\sigma^\star}[V_n^{\pi^\star}(s_H)]$ for every n. Letting $n \to \infty$ by dominated convergence (all terms are bounded by $\|r\|_\infty/(1-\gamma)$) gives $V^{\tilde\sigma}(s) = V_H^\star(s) + \gamma^H \mathbb{E}_s^{\sigma^\star}[V^{\pi^\star}(s_H)]$. In the same way, $V^{\pi^\star}(s) = V_H^{\pi^\star}(s) + \gamma^H \mathbb{E}_s^{\pi^\star}[V^{\pi^\star}(s_H)]$. By Theorem B.15(iv), $V^{\tilde\sigma} \le V^\star = V^{\pi^\star}$. Since $|V^{\pi^\star}| \le \|r\|_\infty/(1-\gamma)$, this gives $V_H^\star(s) - \gamma^H \|r\|_\infty/(1-\gamma) \le V_H^{\pi^\star}(s) + \gamma^H \|r\|_\infty/(1-\gamma)$. If $r \ge 0$, then $V^{\pi^\star} \ge 0$; dropping the first γ^H term gives the sharper bound. □

Remark B.18 (use in the protocol). The finite expert is the stationary greedy policy $\pi^\star$, which is Markov as required by $\mathrm{DM}_{\mathrm{FB}}$ [SA A4]. The oracle reports:

- $V^\star$ with its Bellman residual (Theorem B.16);
- the exact $V^{\pi^\star}$, with the certificate $\eta_{\pi^\star}$;
- $V_H^\star$;
- the expert's H-step return $V_H^{\pi^\star}$.

H-step gaps of learners are measured against $V_H^\star$ (Section 10). Ties in the greedy step do not affect optimality (Theorem B.15(iii)); they affect only which optimal policy is used, so they are broken deterministically.

B.8 The 2021 and 2024 successor conventions

FB21 (Eq. (1), p. 2) uses $M_{21}^\pi(s, a, Y) := \sum_{t \ge 0} \gamma^t \mathbb{P}_{(s,a)}^\pi((s_t, a_t) \in Y)$ for $Y \subseteq \mathcal{S} \times \mathcal{A}$, together with Q-functions $\sum_t \gamma^t \mathbb{E}[r(s_t, a_t)]$.

Proposition B.19. [OCC-7 | D] *Under Theorem B.1, with a state reward r:*

(a) $M_{21}^\pi(s, a, X \times \mathcal{A}) = \delta_s(X) + \gamma M^{\pi,+}(s, a, X)$*; both sides have total mass* $1/(1-\gamma)$.

(b) Let

$$Q_{21}^\pi(s, a) := \mathbb{E}_{(s,a)}^\pi \sum_{t \ge 0} \gamma^t r(s_t),$$
$$Q^{\pi,+}(s, a) := \int r \, \mathrm{d}M^{\pi,+}(s, a, \cdot) = \mathbb{E}_{(s,a)}^\pi \sum_{t \ge 0} \gamma^t r(s_{t+1}).$$

Then $Q_{21}^\pi = r(s) + \gamma Q^{\pi,+}$.

(c) $\arg\max_a Q_{21}^\pi(s, a) = \arg\max_a Q^{\pi,+}(s, a)$ *for every* π *and* s. *Also* $\sup_\pi Q_{21}^\pi = r(s) + \gamma \sup_\pi Q^{\pi,+}$*, so the two conventions have the same optimal policies.*

(d) $d_\pi^{\mathrm{SA}} = (1-\gamma)\int \mu_0(\,\mathrm{d}s)\sum_a \pi(a \mid s) M_{21}^\pi(s,a,\cdot)$.

(e) The Q-function printed in FI24 (p. 3), $\sum_t \gamma^t \mathbb{E}[r(s_t) \mid s_0, a_0, \pi]$*, equals* Q_{21}^π *by (b). It differs from* $\int r\,\mathrm{d}M^{\pi,+}$ *(FI24 Eq. 1) by* $r(s_0)$ *and the factor* γ*. FI24 Prop. 1's identity* $Q_r^\star = F^\top z_r$ *is consistent with* $M^{\pi,+}$ *only if the reward is read at* s_{t+1} [SA R1]*. We adopt that reading; Theorem F.6 completes it.*

Proof. (a) The $t=0$ term is $\mathbb{P}_{(s,a)}(s_0 \in X) = \delta_s(X)$. For $t \geq 1$, $\gamma^t \mathbb{P}(s_t \in X) = \gamma \cdot \gamma^{t-1}\mathbb{P}(s_{(t-1)+1} \in X)$; summing over $t \geq 1$ gives $\gamma M^{\pi,+}(s,a,X)$. The masses are $1 + \gamma/(1-\gamma) = 1/(1-\gamma)$.

(b) Under $\mathbb{P}^\pi_{(s,a)}$, dominated convergence (as in Theorem B.11(b)) gives

$$Q_{21}^\pi(s,a) = \textstyle\sum_{t\geq 0} \gamma^t \int r\,\mathrm{d}\mathbb{P}^\pi_{(s,a)}(s_t \in \cdot).$$

By (B.3) this is $\int r(x)\, M_{21}^\pi(s,a,\,\mathrm{d}x \times \mathcal{A})$, and in the same way $Q^{\pi,+}(s,a) = \int r\,\mathrm{d}M^{\pi,+}(s,a,\cdot)$. Integrating r against the identity of measures in (a) gives $Q_{21}^\pi = r(s) + \gamma Q^{\pi,+}$.

(c) $r(s)$ does not depend on a and $\gamma > 0$, so the two functions of a differ by a positive affine map. The identity of (b) holds for every π pointwise, so taking suprema preserves it.

(d) Use (M1) as in Theorem B.4(c).

(e) The printed expression is, term by term, the definition of Q_{21}^π in (b). Its relation to $\int r\,\mathrm{d}M^{\pi,+} = Q^{\pi,+}$ is then (b), and the two differ in general. For example, if s is left at $t=1$ and never revisited and $r = \mathbf{1}_{\{s\}}$, then $Q_{21}^\pi(s,a) = 1$ and $Q^{\pi,+}(s,a) = 0$. Any identity of the form $F(s,a)^\top z_r = \int r\,\mathrm{d}M^{\pi,+}(s,a,\cdot)$ therefore equals $Q^{\pi,+}$, the reward read at s_{t+1}, and not the printed Q. Theorem F.6(a) shows that an exact FB representation gives $F(s,a,z)^\top z_r = \int r\,\mathrm{d}M^{\pi_z,+}(s,a,\cdot)$, the reward read at s_{t+1}; with the printed Q the identity becomes $Q_{21,r}^\star = r(s) + \gamma F(s,a,z_r)^\top z_r$ (Theorem F.6(c)). □

Remark B.20 (continuous spaces, densities and the reference measure). All statements above are identities between measures; none needs a density. Density statements need domination. If $P(\cdot \mid s,a) \ll \lambda$ for every (s,a) and some σ-finite λ, then $\mathrm{Law}_{t+1}(X) = \int \mathrm{Law}_t(\,\mathrm{d}s)\sum_a \pi(a \mid s)P(X \mid s,a) = 0$ whenever $\lambda(X) = 0$. Hence every Law_{t+1}, and every countable nonnegative combination such as d_π^+, $d_{\pi,H}^+$ and $M^{\pi,+}(s,a,\cdot)$, is λ-absolutely continuous. The pre-action quantities additionally contain μ_0, and M_{21} contains the atom δ_s (FB21 B.6, p. 24). In the continuous maze, boundary clipping or collision rules can put positive mass on Lebesgue-null sets. Whether this matters depends on the reference measure used (Lebesgue, the replay law ν, or histogram cells); the implemented kernel is examined in Section H.2, which shows that it has no atom and no clipping. A histogram estimate never proves absolute continuity. Theorem F.25 states the consequences for FB representations, and Theorem F.26 those of the continuous-maze kernel design.

C f-divergences: convention, orientation and smoothing

Status and sources. We fix the convention $D_f(p\|q) = \sum q\, f(p/q)$. This is the convention used in FPG Table 1 and in its proofs (FPG pp. 3, 16 and 19), and in FIRL Eq. (18) (p. 12). FPG Eq. (3) (p. 3) prints the opposite orientation [SA D1]. The JS generator printed in FPG Table 1 and FIRL Table 2 (p. 4) is twice the half-weight generator used here [SA D3]. All statements in this section concern a *finite* set $\mathcal{X}$: the free cells of the finite maze, or the histogram cells of the continuous maze. Continuous plug-in density estimates are treated as approximations in Section D.4.

C.1 Definition and nonnegativity

Let $f : (0,\infty) \to \mathbb{R}$ be convex with $f(1) = 0$.

Lemma C.1 (boundary values)**.** *The limits* $f(0) := \lim_{u\downarrow 0} f(u) \in (-\infty, +\infty]$ *and* $f'(\infty) := \lim_{u\to\infty} f(u)/u \in (-\infty, +\infty]$ *exist. The extension of* f *to* $[0, \infty)$ *by* $f(0)$ *is convex. If* f *is differentiable,* $\lim_{u\to\infty} f'(u)$ *exists and equals* $f'(\infty)$*. For* $0 \le a < b$*, every secant slope satisfies* $\frac{f(b)-f(a)}{b-a} \le f'(\infty)$*.*

Proof. Convexity makes the secant slope $\sigma(u) := \frac{f(1)-f(u)}{1-u}$ nondecreasing in $u \in (0, 1)$. So $\sigma(u)$ has a limit $L \in [-\infty, +\infty)$ as $u \downarrow 0$, and $f(u) = f(1) - (1-u)\sigma(u) \to -L \in (-\infty, +\infty]$.

Similarly, $\tau(u) := \frac{f(u)-f(1)}{u-1}$ is nondecreasing in $u > 1$, with a limit $L' \in (-\infty, +\infty]$. Since $f(u)/u = \frac{f(1)}{u} + \frac{u-1}{u}\tau(u)$, we get $f(u)/u \to L'$.

Convexity of the extension follows by letting the endpoint tend to 0 in the convexity inequality.

If f is differentiable, f' is nondecreasing, so $\lim f' =: L'' \in (-\infty, \infty]$ exists. Convex functions are absolutely continuous on compact subintervals, so $f(u) = f(1) + \int_1^u f'(v)\,\mathrm{d}v$. Hence $f(u)/u$ is, up to an $O(1/u)$ term, a Cesàro mean of the monotone function f', and it converges to L''; so $L'' = f'(\infty)$.

For the secant bound: by convexity, for every $w > b$, $\frac{f(b)-f(a)}{b-a} \le \frac{f(w)-f(b)}{w-b} = \frac{w}{w-b}\big(\frac{f(w)}{w} - \frac{f(b)}{w}\big)$, and the right-hand side tends to $f'(\infty)$ as $w \to \infty$. □

Definition C.2. For probability vectors p, q on $\mathcal{X}$, let $\mathcal{X}_+ := \{q > 0\}$ and $m_0 := p(\{q = 0\})$, and define

$$D_f(p\|q) := \sum_{x\in\mathcal{X}_+} q(x)\, f\Big(\frac{p(x)}{q(x)}\Big) + f'(\infty)\, m_0.$$

Conventions: $f(0)$ from Theorem C.1 is used where $p(x) = 0 < q(x)$; terms of the form $\infty \cdot 0$ are 0; $D_f \in (-\infty, +\infty]$. In our objectives the learner is always the first argument (Section 2).

Proposition C.3. [DIV-2 | I]

(a) $D_f(p\|q) \ge 0$*.*

(b) Suppose f *is* strictly convex at 1*: there is* $c \in \partial f(1)$ *with* $f(u) > c(u-1)$ *for every* $u \in (0, \infty) \setminus \{1\}$*. Then* $D_f(p\|q) = 0$ *if and only if* $p = q$*.*

Proof. (a) If $D_f = +\infty$ there is nothing to prove. Otherwise, the weights $q(x)$, $x \in \mathcal{X}_+$, sum to 1. Jensen's inequality for the convex extension of f to $[0, \infty)$ (Theorem C.1) gives $\sum_{\mathcal{X}_+} q\, f(p/q) \ge f\big(\sum_{\mathcal{X}_+} p\big) = f(1 - m_0)$. If $m_0 = 0$ we are done, since $f(1) = 0$. If $m_0 > 0$, the secant bound of Theorem C.1 gives $f(1) - f(1 - m_0) \le m_0 f'(\infty)$. Hence $D_f \ge f(1 - m_0) + m_0 f'(\infty) \ge f(1) = 0$.

(b) Let $g(u) := f(u) - c(u-1)$. Then g is convex, $g(1) = 0$, and $g > 0$ on $(0, \infty) \setminus \{1\}$. Also $g(0) > 0$, because $g(\frac{1}{2}) \le \frac{1}{2}g(0) + \frac{1}{2}g(1)$. Moreover $g'(\infty) = f'(\infty) - c > 0$: the secant slope of g over $[1, 2]$ equals $g(2) > 0$ and bounds $g'(\infty)$ from below. Substituting $f = g + c(u-1)$ gives

$$D_f(p\|q) = \sum_{\mathcal{X}_+} q\, g(p/q) + c(p(\mathcal{X}_+) - 1) + f'(\infty)m_0 = \sum_{\mathcal{X}_+} q\, g(p/q) + m_0(f'(\infty) - c).$$

Both terms are nonnegative. The sum is 0 if and only if $p = q$ on $\mathcal{X}_+$; the second term is 0 if and only if $m_0 = 0$. Together these mean $p = q$. □

Proposition C.4 (the three generators used)**.** [DIV-2 | I] *Let* $\bar{m} := \frac{1}{2}(p + q)$*. With the usual conventions* $0 \log 0 = 0$ *and* $a \log(a/0) = +\infty$ *for* $a > 0$*:*

name (learner p *first)*	$f(u)$	$f'(u)$	$f(0)$	$f'(\infty)$
$\mathrm{KL}(p\|q)$	$u \log u$	$1 + \log u$	0	$+\infty$
$\mathrm{KL}(q\|p)$	$-\log u$	$-1/u$	$+\infty$	0
$\mathrm{JS}(p, q)$	$f_{\mathrm{JS}}(u) = \frac{1}{2}\big[u \log u - (1+u) \log \frac{1+u}{2}\big]$	$\frac{1}{2} \log \frac{2u}{1+u}$	$\frac{1}{2} \log 2$	$\frac{1}{2} \log 2$

(a) With these generators, $D_f(p\|q)$ *equals, respectively:* $\mathrm{KL}(p\|q) = \sum_{p>0} p \log \frac{p}{q}$*;* $\mathrm{KL}(q\|p) = \sum_{q>0} q \log \frac{q}{p}$*; and* $\mathrm{JS}(p, q) = \frac{1}{2}\mathrm{KL}(p\|\bar{m}) + \frac{1}{2}\mathrm{KL}(q\|\bar{m}) = \mathcal{H}(\bar{m}) - \frac{1}{2}\mathcal{H}(p) - \frac{1}{2}\mathcal{H}(q)$*, where* $\mathcal{H}(p) = -\sum p \log p$*.*

(b) All three generators are strictly convex ($f'' > 0$), so Theorem C.3(b) applies.
(c) $0 \le \mathrm{JS}(p,q) \le \log 2$*, and* JS *is jointly continuous on (simplex)×(simplex).* $\mathrm{KL}(p\|q) < \infty$ *if and only if* $\mathrm{supp}p \subseteq \mathrm{supp}q$*;* $\mathrm{KL}(q\|p) < \infty$ *if and only if* $\mathrm{supp}q \subseteq \mathrm{supp}p$*.*
(d) The generator printed in FPG Table 1 and FIRL Table 2 is $2f_{\mathrm{JS}}$ *and yields* 2JS*. FIRL's* $h_{\mathrm{JS}} = -\log(1+u)$ *equals* $h_{2f_{\mathrm{JS}}} - \log 2$*.*

Proof. (a) For $u\log u$: on $\{q > 0\}$, $q \cdot \frac{p}{q}\log\frac{p}{q} = p\log\frac{p}{q}$. Points with $p = 0 < q$ contribute $qf(0) = 0$. Points with $q = 0 < p$ contribute $f'(\infty)p = +\infty$, matching the KL convention.

For $-\log u$: on $\{q > 0\}$, $q(-\log\frac{p}{q}) = q\log\frac{q}{p}$. If $p = 0 < q$, the term is $qf(0) = +\infty$. The $q = 0$ points contribute $f'(\infty)m_0 = 0$.

For f_{JS}, a point with $p, q > 0$ contributes $qf_{\mathrm{JS}}(p/q) = \frac{1}{2}\big[p\log\frac{p}{q} - (p+q)\log\frac{p+q}{2q}\big] = \frac{1}{2}\big[p\log\frac{2p}{p+q} + q\log\frac{2q}{p+q}\big]$. This is the pointwise JS term. A point with $p = 0 < q$ contributes $qf_{\mathrm{JS}}(0) = \frac{q}{2}\log 2$, and the JS term is $\frac{1}{2}q\log\frac{2q}{q}$, the same. A point with $q = 0 < p$ contributes $f'_{\mathrm{JS}}(\infty)p = \frac{p}{2}\log 2$, and the JS term is $\frac{1}{2}p\log\frac{2p}{p}$, the same.

The entropy form follows from $\frac{1}{2}\sum p\log\frac{p}{\bar m} + \frac{1}{2}\sum q\log\frac{q}{\bar m} = -\frac{1}{2}\mathcal{H}(p) - \frac{1}{2}\mathcal{H}(q) - \sum \bar m\log\bar m$, where the last step uses $\frac{1}{2}(p+q) = \bar m$. The derivatives and limits are direct computations; $f'_{\mathrm{JS}}(\infty) = \lim\frac{1}{2}\log\frac{2u}{1+u} = \frac{1}{2}\log 2$.

(b) The second derivatives are $f'' = 1/u$, $1/u^2$ and $\frac{1}{2}(\frac{1}{u} - \frac{1}{1+u}) = \frac{1}{2u(1+u)}$, all positive. For $f \in C^2$ with $f'' > 0$, $f(u) - f(1) - f'(1)(u-1) = \int_1^u (u-v)f''(v)\,\mathrm{d}v > 0$ for $u \ne 1$.

(c) Nonnegativity is Theorem C.3. For the upper bound, $\mathrm{KL}(p\|\bar m) = \sum_{p>0} p\log\frac{2p}{p+q} \le \sum p\log 2 = \log 2$, since $p \le p+q$; similarly for $\mathrm{KL}(q\|\bar m)$. Continuity follows from the entropy form, because $x \mapsto x\log x$ is continuous on $[0,1]$. The KL statements are read off the formulas in (a).

(d) This is a direct computation: $2f_{\mathrm{JS}}(u) - u\cdot 2f'_{\mathrm{JS}}(u) = -\log(1+u) + \log 2$. □

C.2 Orientation: the perspective map

Proposition C.5. [DIV-1 | D] *Let* $f^\circ(u) := u\,f(1/u)$ *for* $u > 0$*.*
(a) f° *is convex with* $f^\circ(1) = 0$*; moreover* $f^\circ(0) = f'(\infty)$*,* $(f^\circ)'(\infty) = f(0)$ *and* $(f^\circ)^\circ = f$*.*
(b) $D_f(p\|q) = D_{f^\circ}(q\|p)$ *for all probability vectors* p, q *on* $\mathcal{X}$*, including when they have zeros.*
(c) $(u\log u)^\circ = -\log u$ *and* $f^\circ_{\mathrm{JS}} = f_{\mathrm{JS}}$*. Hence* $\mathrm{KL}(p\|q) = D_{-\log}(q\|p)$*, and JS is symmetric.*
(d) If f *is differentiable,* $(f^\circ)'(v) = f(1/v) - \frac{1}{v}f'(1/v) = h_f(1/v)$*, where* $h_f(u) := f(u) - uf'(u)$ *is FIRL's weight (Thm 4.1, p. 3).*

Proof. (a) Let $u = \lambda u_1 + (1-\lambda)u_2$ with $u_1, u_2 > 0$ and $\lambda \in [0,1]$. Then $\frac{1}{u} = \frac{\lambda u_1}{u}\cdot\frac{1}{u_1} + \frac{(1-\lambda)u_2}{u}\cdot\frac{1}{u_2}$, a convex combination since the two weights sum to 1. Convexity of f gives $f(1/u) \le \frac{\lambda u_1}{u}f(1/u_1) + \frac{(1-\lambda)u_2}{u}f(1/u_2)$; multiplying by u gives $f^\circ(u) \le \lambda f^\circ(u_1) + (1-\lambda)f^\circ(u_2)$. Next, $f^\circ(1) = f(1) = 0$. With $v = 1/u$, $f^\circ(0) = \lim_{u\downarrow 0} uf(1/u) = \lim_{v\to\infty} f(v)/v = f'(\infty)$. Also $(f^\circ)'(\infty) = \lim_{u\to\infty} f^\circ(u)/u = \lim f(1/u) = f(0)$. Finally $(f^\circ)^\circ(u) = u\cdot\frac{1}{u}f(u) = f(u)$.

(b) Partition $\mathcal{X}$ into $A = \{p > 0, q > 0\}$, $B = \{p = 0 < q\}$, $C = \{q = 0 < p\}$ and $\{p = q = 0\}$; the last set contributes nothing to either side. Then

$$D_f(p\|q) = \textstyle\sum_A q\,f(p/q) + f(0)\,q(B) + f'(\infty)\,p(C),$$
$$D_{f^\circ}(q\|p) = \textstyle\sum_A p\,f^\circ(q/p) + f^\circ(0)\,p(C) + (f^\circ)'(\infty)\,q(B).$$

On A, $p\,f^\circ(q/p) = p\cdot\frac{q}{p}f(p/q) = qf(p/q)$. The boundary terms match by (a), including when they are $+\infty$.

(c) $(u\log u)^\circ = u\cdot\frac{1}{u}\log\frac{1}{u} = -\log u$. For JS, $f^\circ_{\mathrm{JS}}(u) = \frac{u}{2}\big[\frac{1}{u}\log\frac{1}{u} - (1+\frac{1}{u})\log\frac{1+1/u}{2}\big] = \frac{1}{2}\big[-\log u - (1+u)\log\frac{1+u}{2u}\big] = \frac{1}{2}\big[u\log u - (1+u)\log\frac{1+u}{2}\big]$.

(d) Differentiate $vf(1/v)$ by the product and chain rules. □

Corollary C.6 (orientation dictionary)**.** *FIRL minimizes $D_f(\rho_E\|\rho_\theta)$, with the expert first (FIRL Eq. 2, p. 3). By Theorem C.5(b) this equals $D_{f^\circ}(\rho_\theta\|\rho_E)$ in our learner-first convention. FIRL's "FKL" ($f = u\log u$) is therefore our $\mathrm{KL}(q\|d)$ with generator $-\log u$; FIRL's "RKL" ($f = -\log u$) is our $\mathrm{KL}(d\|q)$ with generator $u\log u$. FPG's "fkl" is $\mathrm{KL}(d\|q)$ and "rkl" is $\mathrm{KL}(q\|d)$ (FPG Table 1). The two papers' names are thus opposite* [SA D2]*. Our text names both directions by their arguments.*

C.3 Mixture smoothing

Let υ be a fixed *strictly positive* reference law on $\mathcal{X}$, and $\upsilon_{\min} := \min_x \upsilon(x) > 0$. In the finite maze υ is uniform on the free cells, so $\upsilon_{\min} = 1/n$ with $n := |\mathcal{X}|$. In the continuous maze it is uniform on free space projected to the histogram cells, i.e. proportional to the free area of each cell (Section 2). This requires every cell of $\mathcal{X}$ to have positive free area; the retained cells of the implemented histogram have this property, and the implemented kernel has no clipping (Theorems H.4 and H.6). Here $\upsilon(b)$ is a probability *mass*. A KDE objective works with densities instead, and must be mixed with the reference density $\upsilon_\lambda = \mathbf{1}_{\mathcal{S}}/A_{\text{free}}$ (Section 2); this section covers only the finite case. For $\varepsilon \in (0,1]$ write $p_\varepsilon := (1-\varepsilon)p + \varepsilon\upsilon$. The letter u is reserved for the argument of f.

Proposition C.7. [DIV-3 | D] *Let f be convex, C^1 on $(0,\infty)$, with $f(1) = 0$; for example any of the three generators above.*

(a) p_ε and q_ε are probability vectors with all entries at least $\varepsilon\upsilon_{\min}$. Every ratio $p_\varepsilon(x)/q_\varepsilon(x)$ lies in $[\varepsilon\upsilon_{\min},\, 1/(\varepsilon\upsilon_{\min})]$, so $D_f(p_\varepsilon\|q_\varepsilon)$ is finite.

(b) (both arguments smoothed) Let $\phi \mapsto d_\phi$ be C^1 into the simplex and let q be fixed. Then $J_\varepsilon(\phi) := D_f((d_\phi)_\varepsilon\|q_\varepsilon)$ is C^1, and

$$\nabla_\phi J_\varepsilon = (1-\varepsilon)\sum_x f'\Big(\frac{(d_\phi)_\varepsilon(x)}{q_\varepsilon(x)}\Big)\nabla_\phi d_\phi(x).$$

(c) (target only smoothed) If $d_\phi(x) > 0$ for all x, then $\tilde{J}_\varepsilon(\phi) := D_f(d_\phi\|q_\varepsilon)$ is finite and C^1, with $\nabla\tilde{J}_\varepsilon = \sum_x f'(d_\phi(x)/q_\varepsilon(x))\nabla d_\phi(x)$. Without that positivity, $\mathrm{KL}(q_\varepsilon\|d_\phi) = +\infty$ as soon as d_ϕ has a zero, whereas $\mathrm{KL}(d_\phi\|q_\varepsilon)$ and JS remain finite.

(d) (comparison with the unsmoothed objective) If $p, q > 0$ entrywise, then $D_f(p_\varepsilon\|q_\varepsilon) \le (1-\varepsilon)D_f(p\|q)$, and $D_f(p_\varepsilon\|q_\varepsilon) \to D_f(p\|q)$ as $\varepsilon \downarrow 0$. For JS both statements hold for all p, q, with or without zeros.

Proof. (a) Each entry is $(1-\varepsilon)p(x) + \varepsilon\upsilon(x) \ge \varepsilon\upsilon_{\min}$, and the entries sum to 1. Entries are at most 1, so each ratio lies between $\varepsilon\upsilon_{\min}/1$ and $1/(\varepsilon\upsilon_{\min})$. f is continuous on this compact subinterval of $(0,\infty)$, so every term is bounded.

(b) $\partial_\phi(d_\phi)_\varepsilon(x) = (1-\varepsilon)\partial_\phi d_\phi(x)$. Each term $q_\varepsilon(x)f\big((d_\phi)_\varepsilon(x)/q_\varepsilon(x)\big)$ is a composition of C^1 maps, because its argument stays in $(0,\infty)$. The chain rule gives $q_\varepsilon f'(\cdot)\cdot\frac{(1-\varepsilon)\nabla d_\phi}{q_\varepsilon}$. Summing over the finitely many x gives the formula.

(c) Since $q_\varepsilon > 0$, every term is $q_\varepsilon f(d_\phi/q_\varepsilon)$, with argument in $(0,\infty)$ when $d_\phi > 0$; the chain rule gives the formula. For the finiteness statements, use Theorem C.4(c) with $\mathrm{supp} q_\varepsilon = \mathcal{X}$.

(d) The perspective $g(a,b) := b\,f(a/b)$ is jointly convex on $(0,\infty)^2$. To see this, let $(a,b) = \lambda(a_1,b_1) + (1-\lambda)(a_2,b_2)$. Then $\frac{a}{b} = \frac{\lambda b_1}{b}\frac{a_1}{b_1} + \frac{(1-\lambda)b_2}{b}\frac{a_2}{b_2}$, a convex combination; apply convexity of f and multiply by b. Hence $D_f(p_\varepsilon\|q_\varepsilon) = \sum_x g((1-\varepsilon)(p(x),q(x)) + \varepsilon(\upsilon(x),\upsilon(x))) \le (1-\varepsilon)D_f(p\|q) + \varepsilon D_f(\upsilon\|\upsilon)$, and $D_f(\upsilon\|\upsilon) = 0$. For convergence, every term $q_\varepsilon f(p_\varepsilon/q_\varepsilon)$ is continuous in ε at 0 when $p, q > 0$.

For JS with zeros, apply the positive case to $p^{(k)} := p_{1/k}$ and $q^{(k)} := q_{1/k}$, for which $\mathrm{JS}(p_\varepsilon^{(k)}, q_\varepsilon^{(k)}) \le (1-\varepsilon)\mathrm{JS}(p^{(k)}, q^{(k)})$. As $k \to \infty$, $p^{(k)} \to p$ and $p_\varepsilon^{(k)} \to p_\varepsilon$, and likewise for q; JS is continuous (Theorem C.4(c)), so the inequality passes to the limit. The convergence as $\varepsilon \downarrow 0$ is again continuity. □

Remark C.8 (what smoothing changes). Smoothing changes the objective. The minimizers of J_ε need not be those of the unsmoothed divergence, so no claim is made that they coincide. Smoothing is used to make $\mathrm{KL}(d\|q)$ finite when the expert target has zeros (FPG-8 below), and to keep f' finite. Reported evaluation divergences are the unsmoothed exact finite JS, or the common held-out estimator in the continuous case. The value of ε is reported with every result.

D f-policy gradient with a fixed expert target

Status and relation to the source. FPG minimizes $J(\theta) = D_f(p_\theta\|p_g)$ for a Dirac goal target $p_g = \delta_g$ (FPG Eq. 4, p. 4; target p. 3). Its gradient is stated in Theorem 4.1 (p. 4; proof in App. B.1, pp. 18–19), and its discounted variants appear in App. B.2–B.3 (pp. 20–21). This section replaces p_g by a *fixed* expert target q, and replaces the paper's visitation by the finite-horizon next-state occupancy $d^+_{\pi,H}$. All results below are derived independently. Their labels are:

- D (our adaptation) for FPG-1, FPG-2, FPG-5, FPG-6 and FPG-8;
- I (standard identities) for the baseline and Monte Carlo statements FPG-3 and FPG-4;
- S+D for FPG-7: parts (a)–(b) are FPG Lemma 5.1 (p. 6, proof p. 17) reproved in our normalization, and parts (c)–(d) are ours;
- C for FPG-C1, a comparison with the source's Dirac-target theorems from which no guarantee is drawn.

Section D.8 records exactly where the printed source formulas agree and where they do not; the full findings are [SA F1–F10].

D.1 Setting

Assumption D.1 (A-FPG)**.** Theorem B.1 holds, together with the following.

(i) *Cells.* $\{B_1, \dots, B_n\}$ is a finite measurable partition of $\mathcal{S}$. If $\mathcal{S}$ is finite the cells are the singletons; in the continuous maze they are histogram cells. $b(s)$ denotes the index of the cell containing s, and $\nu(b) := \nu(B_b)$ for a measure ν.

(ii) *Policies.* $\{\pi_\phi : \phi \in \Phi\}$ with $\Phi \subseteq \mathbb{R}^m$ open satisfies, for all (s,a): $\pi_\phi(a \mid s) > 0$; $\phi \mapsto \pi_\phi(a \mid s)$ is C^1 with $\nabla_\phi \pi_\phi(a \mid s)$ jointly measurable; and the score is locally uniformly bounded. The last condition means that every $\phi_0 \in \Phi$ has a neighbourhood N and a constant C with $\sup_{\phi\in N,\, s\in\mathcal{S},\, a\in\mathcal{A}} \|\nabla_\phi \log \pi_\phi(a \mid s)\| \le C$.

(iii) *Target.* q is a fixed probability vector on the cells; it does not depend on ϕ.

(iv) *Generator.* f is convex and C^1 on $(0,\infty)$, with $f(1) = 0$.

(v) *Positivity.* $q(b) > 0$ and $d_\phi(b) > 0$ for every cell b and every $\phi \in \Phi$, where $d_\phi(b) := d^+_{\pi_\phi,H}(B_b) = Z_H^{-1} \sum_{t=0}^{H-1} \gamma^t \mathbb{P}_\phi(s_{t+1} \in B_b)$.

Item (v) holds, for example, when q is a smoothed target $q_{E,\varepsilon}$ (Theorem C.7(a)) and every cell satisfies $\mathbb{P}^{\pi'}_{\mu_0}(s_{t+1} \in B_b) > 0$ for some stationary policy π' and some $t < H$. By Theorem D.12(a), every π_ϕ then reaches every cell, so $d_\phi > 0$. The case of a target with zeros is treated in Theorem D.22.

Lemma D.2 (tabular softmax)**.** *For $\pi_\phi(a \mid s) = e^{\phi_{s,a}} / \sum_b e^{\phi_{s,b}}$ with $\phi \in \mathbb{R}^{\mathcal{S}\times\mathcal{A}}$, $\partial \log \pi_\phi(a \mid s)/\partial \phi_{s',b} = \mathbf{1}[s' = s](\mathbf{1}[b = a] - \pi_\phi(b \mid s))$. Consequently $\|\nabla \log \pi_\phi(a \mid s)\|_2 \le \sqrt{2}$, and Theorem D.1(ii) holds on $\Phi = \mathbb{R}^{\mathcal{S}\times\mathcal{A}}$.*

Proof. Differentiate $\log \pi_\phi(a \mid s) = \phi_{s,a} - \log \sum_b e^{\phi_{s,b}}$. The squared norm is $(1 - \pi(a \mid s))^2 + \sum_{b\neq a} \pi(b \mid s)^2 \le 1 + 1$. The map $\phi \mapsto \pi_\phi$ is C^∞. □

Remark D.3 (continuous policy class). A categorical MLP policy $\pi_\phi(a \mid s) = \mathrm{softmax}_a \ell_\phi(s,a)$ satisfies Theorem D.1(ii) under the following conditions: the activation is C^1 (for example tanh or ELU); $\mathcal{S}$ is bounded; $(\phi, s) \mapsto \nabla_\phi \ell_\phi(s,a)$ is continuous, so that it is bounded on $\bar{N} \times \bar{\mathcal{S}}$ for a

compact neighbourhood $\bar{N}$. The score $\nabla\ell(s,a) - \sum_b \pi(b \mid s)\nabla\ell(s,b)$ is then bounded. Only C^1 activations are covered by this remark; for other activations the results below are not claimed. The implemented MLP policy uses tanh hidden layers, which are C^∞ [SA U-12].

Notation for the rest of the section:

- $\chi_k := \nabla_\phi \log \pi_\phi(a_k \mid s_k)$ is the score at step k;
- $J(\phi) := D_f(d_\phi \| q) = \sum_b q(b) f(d_\phi(b)/q(b))$;
- $g_\phi(b) := f'(d_\phi(b)/q(b))$ is the *f-PG signal*, extended to states by $g_\phi(s) := g_\phi(b(s))$;
- $G_k := \sum_{t=k}^{H-1} \gamma^{t-k} g_\phi(s_{t+1})$ is the discounted signal-to-go.

The signal g_ϕ is evaluated at the current ϕ and is not differentiated.

D.2 Two lemmas: likelihood ratio and causality

Lemma D.4 (likelihood ratio)**.** *Under Theorem D.1(ii), let h be a bounded measurable function of $(s_0, a_0, \dots, a_{H-1}, s_H)$ that does not depend on ϕ. Then $\phi \mapsto \mathbb{E}_\phi h$ is C^1 and $\nabla \mathbb{E}_\phi h = \mathbb{E}_\phi\big[h \sum_{k=0}^{H-1} \chi_k\big]$.*

Proof. By the iterated-integral form of (B.1),

$$\mathbb{E}_\phi h = \sum_{a_0,\dots,a_{H-1}} \int \mu_0(\,\mathrm{d}s_0) \prod_{k=0}^{H-1} P(\,\mathrm{d}s_{k+1} \mid s_k, a_k)\; \Pi_\phi(a_{0:H-1}, s_{0:H-1})\, h,$$

where $\Pi_\phi := \prod_{k=0}^{H-1} \pi_\phi(a_k \mid s_k)$. The measure $\mu_0 \otimes \prod_k P$ does not depend on ϕ. For fixed $(a_{0:H-1}, s_{0:H})$ the integrand is C^1 in ϕ, with $\nabla\Pi_\phi = \Pi_\phi \sum_k \nabla \log \pi_\phi(a_k \mid s_k)$ by the product rule (all factors are positive). Fix ϕ_0 and let N, C be as in Theorem D.1(ii). Shrinking N, we may take it to be an open ball around ϕ_0, so it contains the segment between any two of its points. On N, $0 < \Pi_\phi \le 1$, so $\|\nabla_\phi(\Pi_\phi h)\| \le HC\|h\|_\infty$ uniformly in $\phi \in N$ and in the trajectory. By the mean value theorem along segments in N, the difference quotients of the integrand are bounded by the same constant. Dominated convergence, with respect to the finite measure $\mu_0 \otimes \prod_k P$ times counting measure on actions, therefore allows differentiation under the integral. Continuity of $\phi \mapsto \nabla \mathbb{E}_\phi h$ follows from dominated convergence again, since $\nabla\pi_\phi$ is continuous in ϕ. When $\mathcal{S}$ is finite the integral is a finite sum and the argument is elementary. □

Lemma D.5 (zero-mean score; causality)**.** *Under Theorem D.1(ii):*

(a) $\sum_a \nabla_\phi \pi_\phi(a \mid s) = 0$ for every s, and $\mathbb{E}_\phi[\chi_k \mid \mathcal{F}_k] = 0$.

(b) If Y is bounded and $\mathcal{F}_k$-measurable, then $\mathbb{E}_\phi[Y \chi_k] = 0$. In particular, $\mathbb{E}_\phi[\mathbf{1}[s_{t+1} \in B]\, \chi_k] = 0$ for $k \ge t+1$, and $\mathbb{E}_\phi[\beta_k \chi_k] = 0$ for every bounded $\mathcal{F}_k$-measurable baseline β_k.

Proof. (a) Since $\sum_a \pi_\phi(a \mid s) = 1$ for all ϕ, differentiating the finite sum gives the first identity. Given $\mathcal{F}_k$, a_k has law $\pi_\phi(\cdot \mid s_k)$ by (B.1). Hence $\mathbb{E}[\chi_k \mid \mathcal{F}_k] = \sum_a \pi_\phi(a \mid s_k) \nabla \log \pi_\phi(a \mid s_k) = \sum_a \nabla \pi_\phi(a \mid s_k) = 0$.

(b) By the tower property, $\mathbb{E}[Y\chi_k] = \mathbb{E}\big[Y\, \mathbb{E}[\chi_k \mid \mathcal{F}_k]\big] = 0$; integrability holds since $|Y\chi_k| \le \|Y\|_\infty C$. For $k \ge t+1$, s_{t+1} is $\mathcal{F}_k$-measurable. □

D.3 The exact gradient

Theorem D.6. [FPG-1 | D] *Under Theorem D.1, $d_\phi(b)$ and J are C^1 on Φ, and*

$$\nabla d_\phi(b) = Z_H^{-1} \sum_{t=0}^{H-1} \gamma^t\, \mathbb{E}_\phi\Big[\mathbf{1}[s_{t+1} \in B_b] \sum_{k=0}^{t} \chi_k\Big], \tag{D.1}$$

$$\nabla J(\phi) = \sum_b g_\phi(b)\, \nabla d_\phi(b), \tag{D.2}$$

$$\nabla J(\phi) = Z_H^{-1} \sum_{t=0}^{H-1} \gamma^t\, \mathbb{E}_\phi\Big[g_\phi(s_{t+1}) \sum_{k=0}^{t} \chi_k\Big] = Z_H^{-1}\, \mathbb{E}_\phi\Big[\Big(\sum_{k=0}^{H-1} \chi_k\Big)\Big(\sum_{t=0}^{H-1} \gamma^t g_\phi(s_{t+1})\Big)\Big]. \tag{D.3}$$

Proof. Step 1 (∇d_ϕ). Apply Theorem D.4 with $h = \mathbf{1}[s_{t+1} \in B_b]$, which is bounded and ϕ-free: $\nabla \mathbb{P}_\phi(s_{t+1} \in B_b) = \mathbb{E}_\phi[\mathbf{1}[s_{t+1} \in B_b] \sum_{k=0}^{H-1} \chi_k]$. By Theorem D.5(b), the terms $k \geq t+1$ vanish. Multiplying by γ^t / Z_H and summing over the finitely many t gives (D.1). $d_\phi(b)$ is C^1 as a finite combination of C^1 functions.

Step 2 (chain rule). By Theorem D.1(v), $d_\phi(b)/q(b) \in (0, \infty)$ for every ϕ. Each term $q(b) f(d_\phi(b)/q(b))$ is therefore a composition of C^1 maps, with gradient

$$q(b)\, f'(d_\phi(b)/q(b))\, \nabla d_\phi(b)/q(b) = g_\phi(b)\, \nabla d_\phi(b).$$

Sum over b.

Step 3 (trajectory form). Substitute (D.1) into (D.2) and use linearity of expectation. Since $\sum_b g_\phi(b) \mathbf{1}[s_{t+1} \in B_b] = g_\phi(s_{t+1})$, this gives the first expression in (D.3). For the second, add the terms $k \geq t+1$. Each vanishes by Theorem D.5(b), because $g_\phi(s_{t+1})$ is bounded (finitely many finite values) and $\mathcal{F}_k$-measurable. □

Theorem D.7 (causal form and discount weights)**.** [FPG-2 | D] *Under Theorem D.1,*

$$\nabla J(\phi) = Z_H^{-1}\, \mathbb{E}_\phi\Big[\sum_{k=0}^{H-1} \gamma^k\, \chi_k\, G_k \Big], \qquad G_k = \sum_{t=k}^{H-1} \gamma^{t-k} g_\phi(s_{t+1}). \tag{D.4}$$

Equivalently, in policy-gradient form,

$$\nabla J(\phi) = Z_H^{-1} \sum_{k=0}^{H-1} \gamma^k\, \mathbb{E}_\phi[\chi_k\, Q_k^g(s_k, a_k)]. \tag{D.5}$$

Here $Q_{H-1}^g(s,a) = \int P(\mathrm{d}s' \mid s,a) g_\phi(s')$ and, for $k < H-1$, $Q_k^g(s,a) = \int P(\mathrm{d}s' \mid s,a)\big[g_\phi(s') + \gamma \sum_{a'} \pi_\phi(a' \mid s') Q_{k+1}^g(s', a')\big]$.

Proof. Start from the first expression in (D.3). The index set $\{(t,k) : 0 \leq k \leq t \leq H-1\}$ can be enumerated either way, so $\sum_{t=0}^{H-1} \sum_{k=0}^{t} = \sum_{k=0}^{H-1} \sum_{t=k}^{H-1}$. Writing $\gamma^t = \gamma^k \gamma^{t-k}$ gives (D.4).

For (D.5), condition on $(\mathcal{F}_k, a_k)$. By the Markov structure (B.1), the conditional law of $(s_{k+1}, a_{k+1}, \dots)$ depends only on (s_k, a_k). Since $G_k = g_\phi(s_{k+1}) + \gamma G_{k+1}$ for $k < H-1$, and $G_{H-1} = g_\phi(s_H)$, backward induction gives $\mathbb{E}[G_k \mid \mathcal{F}_k, a_k] = Q_k^g(s_k, a_k)$ with the stated recursion. Then use $\mathbb{E}[\chi_k G_k] = \mathbb{E}[\chi_k\, \mathbb{E}[G_k \mid \mathcal{F}_k, a_k]]$, which holds because χ_k is $(\mathcal{F}_k, a_k)$-measurable. □

Signs and weights (summary).

- The update is *descent*: $\phi \leftarrow \phi - \eta\, \widehat{\nabla J}$. In REINFORCE terms the per-step "reward" is $-g_\phi(s_{t+1})$.
- Because f' is nondecreasing (by convexity), g_ϕ is larger on cells where the learner over-visits relative to the target. As a first-order heuristic, not a theorem, descent lowers the probability of actions whose signal-to-go is above the baseline.
- The score at step k carries the weight γ^k, and the signal at s_{t+1} carries γ^{t-k} inside G_k. Equivalently, the signal carries the absolute weight γ^t and the score carries none.
- Dropping γ^k, which is common practice for discounted policy gradients, does *not* give ∇J in general; `check_2A.py` exhibits a counterexample.

Proposition D.8 (baselines)**.** [FPG-3 | I] *Under Theorem D.1:*

(a) For bounded $\mathcal{F}_k$-measurable β_k, such as $\beta_k(s_k)$, time-dependent state values, or any function of the history up to s_k, $\nabla J = Z_H^{-1} \mathbb{E}_\phi\big[\sum_k \gamma^k \chi_k (G_k - \beta_k)\big]$.

(b) Replacing g_ϕ by $g_\phi + c$ for a constant c leaves ∇J unchanged.

(c) A "baseline" that depends on a_k generally changes the gradient.

Proof. (a) Apply Theorem D.5(b) to $Y = \gamma^k \beta_k$.

(b) In (D.2), $\sum_b c\nabla d_\phi(b) = c\nabla \sum_b d_\phi(b) = c\nabla 1 = 0$.

(c) For $\beta_k = \mathbf{1}[a_k = a^\star]$, $\mathbb{E}[\chi_k \beta_k \mid \mathcal{F}_k] = \nabla \pi_\phi(a^\star \mid s_k)$. For tabular softmax this is nonzero whenever $\pi_\phi(a^\star \mid s_k) \in (0,1)$, by Theorem D.2. □

Proposition D.9 (Monte Carlo estimator and descent)**.** [FPG-4 | I] *Assume Theorem D.1 and let β_k be as in Theorem D.8.*

(a) With the exact *signal g_ϕ, define $\hat{\nabla}_N := \frac{1}{N}\sum_{i=1}^N Z_H^{-1} \sum_k \gamma^k \chi_k^{(i)} (G_k^{(i)} - \beta_k^{(i)})$ for N i.i.d. trajectories from $\mathbb{P}_\phi$. Then $\mathbb{E}\hat{\nabla}_N = \nabla J(\phi)$, $\mathrm{Cov}(\hat{\nabla}_N) = \mathrm{Cov}(\hat{\nabla}_1)/N$, and $\hat{\nabla}_N \to \nabla J(\phi)$ almost surely.*

(b) If $\nabla J(\phi) \neq 0$, then $J(\phi - \eta \nabla J(\phi)) < J(\phi)$ for all sufficiently small $\eta > 0$. If ∇J is L-Lipschitz on the segment $[\phi, \phi - \eta \nabla J(\phi)]$, then $J(\phi - \eta \nabla J) \leq J(\phi) - \eta(1 - L\eta/2)\|\nabla J\|^2$.

No statement is made about global convergence. No convexity of J in ϕ is assumed or claimed, and the proposition says nothing about the stationary points that gradient descent may reach.

Proof. (a) By Theorem D.8(a), the single-trajectory estimator has mean ∇J. It is bounded: $\|\chi_k\| \leq C$, $|G_k| \leq \|g_\phi\|_\infty/(1-\gamma)$, and β_k is bounded. Hence it has finite covariance. Independence gives the covariance identity, and the strong law of large numbers gives the almost sure limit.

(b) Differentiability gives $J(\phi - \eta\nabla J) = J(\phi) - \eta\|\nabla J\|^2 + o(\eta)$. Under the Lipschitz condition, put $v = -\eta \nabla J(\phi)$. Then $J(\phi + v) - J(\phi) = \int_0^1 \nabla J(\phi + sv) \cdot v \, \mathrm{d}s \leq \nabla J(\phi) \cdot v + \int_0^1 Ls\|v\|^2 \, \mathrm{d}s$. □

D.4 Exact objective versus implemented approximations

Proposition D.10. [FPG-5 | D] *Assume Theorem D.1.*

(a) Plug-in signal from independent data. *Let $\hat{g}$ be a random function on cells, independent of the trajectories entering the score sums. This is the case under sample splitting or cross-fitting. Then $\mathbb{E}[\hat{\nabla}_N \mid \hat{g}] = \sum_b \hat{g}(b) \nabla d_\phi(b)$, and*

$$\left\| \mathbb{E}[\hat{\nabla}_N \mid \hat{g}] - \nabla J(\phi) \right\| \leq \max_b |\hat{g}(b) - g_\phi(b)| \cdot C\kappa_H, \qquad \kappa_H := Z_H^{-1} \sum_{t=0}^{H-1} (t+1)\gamma^t.$$

(b) Bias. *Even when $\hat{d}(b)$ is unbiased for $d_\phi(b)$, in general*

$$\mathbb{E}\hat{g}(b) = \mathbb{E} f'(\hat{d}(b)/\hat{q}(b)) \neq f'(d_\phi(b)/q(b)),$$

because f' is nonlinear: the plug-in signal *is biased. Let* $\mathrm{bias}(b) := \mathbb{E}\hat{g}(b) - g_\phi(b)$. *With $\hat{g}$ independent of the score trajectories and integrable, (a) gives $\mathbb{E}\hat{\nabla}_N - \nabla J(\phi) = \sum_b \mathrm{bias}(b) \nabla d_\phi(b)$. This vanishes when* $\mathrm{bias}(\cdot)$ *is constant across cells, because $\sum_b \nabla d_\phi(b) = 0$; otherwise no unbiasedness of the* gradient estimator *is claimed, and (a) bounds the conditional error. If $\hat{g}$ is computed from the same trajectories as the score, the conditional factorization of (a) fails.* `check_2A.py` *computes an exact example with $N = 1$ whose bias is nonzero.*

(c) Smoothing. *Assume the learner occupancy and the target are both smoothed as in Theorem C.7(b). Then $J_\varepsilon(\phi) = \sum_b q_\varepsilon(b) f((d_\phi)_\varepsilon(b)/q_\varepsilon(b))$ satisfies Theorems D.6 and D.7 and Theorems D.8 and D.9 verbatim, with the signal*

$$g_\phi^{(\varepsilon)}(b) := (1 - \varepsilon)\, f'((d_\phi)_\varepsilon(b)/q_\varepsilon(b)).$$

No positivity assumption on d_ϕ is needed for this, only $q_\varepsilon > 0$. The minimized objective is then J_ε, not J (Theorem C.8).

(d) Optional PPO surrogate. *Fix ϕ', compute $c_k := \gamma^k(G_k - \beta_k)$ under ϕ', and let $\rho_k(\phi) := \pi_\phi(a_k \mid s_k)/\pi_{\phi'}(a_k \mid s_k)$. Set*

$$L(\phi) := Z_H^{-1} \mathbb{E}_{\phi'} \sum_k \rho_k(\phi) c_k,$$
$$L^{\mathrm{clip}}(\phi) := Z_H^{-1} \mathbb{E}_{\phi'} \sum_k \max\left(\rho_k c_k,\ \mathrm{clip}(\rho_k, 1-\epsilon_c, 1+\epsilon_c) c_k\right).$$

Here β_k is a bounded baseline as in Theorem D.8(a), and the clip parameter $\epsilon_c \in (0,1)$ is unrelated to the smoothing weight ε. Then L and L^{clip} coincide on a neighbourhood of ϕ', $\nabla L(\phi') = \nabla L^{\text{clip}}(\phi') = \nabla J(\phi')$, and $L^{\text{clip}} \geq L$ everywhere. Minimizing L^{clip} is the same as maximizing the standard PPO objective with advantage $-c_k$, since $-\max(x,y) = \min(-x,-y)$. Away from ϕ' both surrogates ignore the change of state distribution, and their gradients are not ∇J in general.

Proof. (a) Conditionally on $\hat{g}$, the computation of Theorem D.6 Step 3 and Theorem D.7 applies with the fixed bounded function $\hat{g}$ in place of g_ϕ. Steps 1 and 3 never used the definition of g_ϕ, only that it is a bounded function on cells. This gives $\mathbb{E}[\hat{\nabla}_N \mid \hat{g}] = \sum_b \hat{g}(b)\nabla d_\phi(b)$. Subtract (D.2) and bound $\sum_b \|\nabla d_\phi(b)\|$ using (D.1): $\|\nabla d_\phi(b)\| \leq Z_H^{-1} \sum_t \gamma^t (t+1) C\, \mathbb{P}_\phi(s_{t+1} \in B_b)$; summing over b gives $C\kappa_H$.

(b) *Signal bias.* Take $f'(u) = 1 + \log u$, which is strictly concave, and let $\hat{d}(b)$ be non-degenerate and unbiased with $\hat{d}(b) > 0$. Jensen's inequality is then strict: $\mathbb{E}[1 + \log(\hat{d}/q)] < 1 + \log(d/q)$.

Gradient bias with an independent signal. Taking expectations in the identity of (a), which is allowed since $\hat{g}$ is integrable and the score sums are bounded, gives $\mathbb{E}\hat{\nabla}_N = \sum_b \mathbb{E}[\hat{g}(b)]\nabla d_\phi(b)$. Subtracting (D.2) gives $\sum_b \text{bias}(b)\nabla d_\phi(b)$. If $\text{bias}(b) = c$ for all b, this equals $c\nabla \sum_b d_\phi(b) = c\nabla 1 = 0$.

Same batch: an analytic counterexample. Let $\mu_0 = \delta_x$ and $H = 1$, so $Z_H = 1$. Two actions $a^{(0)}, a^{(1)}$ lead deterministically from x to the states 0 and 1, which are the cells. Let $\pi_\phi(a^{(1)} \mid x) = p := e^\phi/(1+e^\phi)$ (logits $(0, \phi)$), let the target be $q = (\frac{1}{2}, \frac{1}{2})$, and take $\text{KL}(d\|q)$, i.e. $f(u) = u \log u$. Then $d_\phi = (1-p, p)$ and $J(\phi) = (1-p)\log(2(1-p)) + p\log(2p)$, so $\partial J/\partial p = \log \frac{p}{1-p}$. Since $\partial p/\partial \phi = p(1-p)$, the true gradient is $\nabla J = p(1-p)\log\frac{p}{1-p}$, which equals $\frac{3\log 3}{16} > 0$ at $p = \frac{3}{4}$. Now let $N = 1$ and estimate the occupancy from the same single trajectory, so $\hat{d} = \delta_{s_1}$. The signal is evaluated only at the sampled state, where it equals $f'(\hat{d}(s_1)/q(s_1)) = f'(2) = 1 + \log 2$, whichever state was sampled. With zero baseline the estimator is therefore $\chi_0(1 + \log 2)$, whose mean is 0 by Theorem D.5(a). Hence the same-batch estimator is biased. `check_2A.py` verifies both numbers, and gives a second exact example with nonzero bias on a random three-state MDP for JS and $\text{KL}(d\|q)$.

(c) J_ε has the form $\sum_b \Psi_b(d_\phi(b))$ with $\Psi_b(x) = q_\varepsilon(b) f(((1-\varepsilon)x + \varepsilon \upsilon(b))/q_\varepsilon(b))$, where υ is the positive reference law of Theorem C.7. Each Ψ_b is C^1 on a neighbourhood of $[0,1]$, because for $x \in [0,1]$ its argument stays in $[\varepsilon\upsilon_{\min}, 1/(\varepsilon\upsilon_{\min})]$, and $\Psi_b'(d_\phi(b)) = g_\phi^{(\varepsilon)}(b)$. Steps 1 and 3 of the proof of Theorem D.6 use only (D.1), which does not need positivity, and the boundedness of the signal.

(d) The c_k are computed once under ϕ' and do not depend on ϕ; they are bounded, $|c_k| \leq \|g_{\phi'}\|_\infty/(1-\gamma) + \sup|\beta_k| =: K$. Let C and a neighbourhood of ϕ' be as in Theorem D.1(ii) for $\phi_0 = \phi'$; shrink it to an open ball $U := B(\phi', r_1)$, and enlarge C if necessary so that $C > 0$. For $\phi \in U$, the mean value theorem along the segment from ϕ' to ϕ gives $|\log \rho_k(\phi)| \leq C\|\phi - \phi'\| < Cr_1$ for every trajectory and every k. Hence $\rho_k \leq e^{Cr_1}$ and $\|\nabla \rho_k(\phi)\| = \rho_k(\phi)\|\nabla \log \pi_\phi(a_k \mid s_k)\| \leq Ce^{Cr_1}$ on U. The integrand $\sum_k \rho_k(\phi)c_k$ is therefore C^1 on U with gradient bounded by $HCe^{Cr_1}K$, uniformly in the trajectory. As in Theorem D.4, dominated convergence allows differentiation under $\mathbb{E}_{\phi'}$: $\nabla L(\phi) = Z_H^{-1}\mathbb{E}_{\phi'} \sum_k \rho_k(\phi)\nabla \log \pi_\phi(a_k \mid s_k)c_k$. At $\phi = \phi'$, $\rho_k = 1$, so $\nabla L(\phi') = Z_H^{-1}\mathbb{E}_{\phi'} \sum_k \chi_k \gamma^k (G_k - \beta_k) = \nabla J(\phi')$ by Theorem D.8(a), where G_k uses the signal $g_{\phi'}$.

For the clipped surrogate, let $r_0 := \min\{r_1,\ \log(1+\epsilon_c)/C,\ -\log(1-\epsilon_c)/C\} > 0$. If $\|\phi - \phi'\| < r_0$, then $1 - \epsilon_c < e^{-C\|\phi-\phi'\|} \leq \rho_k(\phi) \leq e^{C\|\phi-\phi'\|} < 1 + \epsilon_c$ for every trajectory and every k. On this ball $\text{clip}(\rho_k, 1-\epsilon_c, 1+\epsilon_c) = \rho_k$, so both arguments of the max equal $\rho_k c_k$ and $L^{\text{clip}} = L$. Hence the two surrogates have the same gradient at ϕ'. $L^{\text{clip}} \geq L$ is immediate from the max. The identity $-\max(x,y) = \min(-x,-y)$ gives the PPO equivalence. □

Remark D.11 (continuous states and kernel density estimates). Theorem D.6 is exact for any finite cell partition, in particular for the histogram used for diagnostics. The continuous implementation instead evaluates $\hat{g}(s) = f'(\hat{d}(s)/\hat{q}(s))$ with weighted Gaussian KDEs at the sampled points s_{t+1}.

This is the plug-in version of the formal density gradient $\int f'(\delta_\phi/\delta_q)\nabla\delta_\phi \,\mathrm{d}\lambda$, where δ_ϕ and δ_q are Lebesgue densities. Making that formula exact would require further assumptions that we do not verify: existence of the densities, positivity of δ_q, and domination allowing differentiation under $\int \mathrm{d}\lambda$. The continuous f-PG update is therefore labelled a plug-in approximation (implementation approximation, not a theorem). The cell objective is a rigorous finite-partition analogue. These results do not establish that the KDE plug-in update is unbiased for the gradient of that cell objective. If the KDE objective is smoothed, the mixture uses the reference *density* υ_λ, not the cell masses υ_b (Section 2). No finite-horizon approximation is involved, since learner and target use the same H (Theorem B.13). The optimizer (Adam), minibatching and any normalization of the signal further depart from exact gradient descent on J. Only Theorem D.9(b) is claimed for exact gradient steps.

D.5 State–action extension (finite diagnostic)

The following lemma holds under Theorem B.1 alone; finiteness of $\mathcal{S}$ is used only in part (b). Part (a) is also used for the continuous maze (after Theorem D.1).

Lemma D.12 (support domination)**.** *Under Theorem B.1, let π satisfy $\pi(a \mid s) > 0$ for all (s, a), and let π' be any stationary policy, both started from μ_0.*

(a) $\mathrm{Law}_t^{\pi'} \ll \mathrm{Law}_t^{\pi}$ *for every $t \geq 0$. Consequently $d^+_{\pi',H} \ll d^+_{\pi,H}$, $d^{\mathrm{pre}}_{\pi',H} \ll d^{\mathrm{pre}}_{\pi,H}$ and $d^{\mathrm{SA}}_{\pi',H} \ll d^{\mathrm{SA}}_{\pi,H}$. In particular, a cell charged by $d^+_{\pi',H}$ is charged by $d^+_{\pi,H}$, and two positive policies charge the same cells.*

(b) If $\mathcal{S}$ is finite, then $\mathrm{supp} d^{\mathrm{SA}}_{\pi,H} = R_H \times \mathcal{A}$ *with $R_H := \bigcup_{t<H} \mathrm{supp}\mathrm{Law}_t^{\pi}$, and R_H is the same for every positive policy.*

Proof. (a) Induction on t. For $t = 0$ both marginals are μ_0. Assume $\mathrm{Law}_t^{\pi'} \ll \mathrm{Law}_t^{\pi}$ and let X satisfy $\mathrm{Law}_{t+1}^{\pi}(X) = 0$. By Theorem B.3, $0 = \int \mathrm{Law}_t^{\pi}(\mathrm{d}s) \sum_a \pi(a \mid s) P(X \mid s, a)$. The integrand is nonnegative, so it vanishes outside a Law_t^{π}-null set N_X. Since every $\pi(a \mid s) > 0$, each $P(X \mid s, a)$ vanishes for $s \notin N_X$. By the induction hypothesis $\mathrm{Law}_t^{\pi'}(N_X) = 0$, so $\mathrm{Law}_{t+1}^{\pi'}(X) = \int_{\mathcal{S}\setminus N_X} \mathrm{Law}_t^{\pi'}(\mathrm{d}s) \sum_a \pi'(a \mid s) P(X \mid s, a) = 0$.

For the occupancies, $d^+_{\pi,H}(X) = 0$ forces $\mathrm{Law}_{t+1}^{\pi}(X) = 0$ for every $t < H$, since all weights γ^t/Z_H are positive; hence $\mathrm{Law}_{t+1}^{\pi'}(X) = 0$ and $d^+_{\pi',H}(X) = 0$. The pre-action case is identical with Law_t. For the state–action case, let $Y \subseteq \mathcal{S} \times \mathcal{A}$ with $d^{\mathrm{SA}}_{\pi,H}(Y) = 0$ and write $Y_a := \{s : (s, a) \in Y\}$. Then $\int_{Y_a} \pi(a \mid s) \mathrm{Law}_t^{\pi}(\mathrm{d}s) = 0$ for every $t < H$ and a, and positivity of $\pi(a \mid s)$ gives $\mathrm{Law}_t^{\pi}(Y_a) = 0$. Hence $\mathrm{Law}_t^{\pi'}(Y_a) = 0$, and $d^{\mathrm{SA}}_{\pi',H}(Y) = Z_H^{-1} \sum_{t<H} \gamma^t \sum_a \int_{Y_a} \pi'(a \mid s) \mathrm{Law}_t^{\pi'}(\mathrm{d}s) = 0$. The statement about cells is the case $X = B_b$; applying it in both directions gives the last claim.

(b) $d^{\mathrm{SA}}_{\pi,H}(s, a) = Z_H^{-1} \sum_{t<H} \gamma^t \mathrm{Law}_t^{\pi}(s)\pi(a \mid s)$ is positive if and only if $\mathrm{Law}_t^{\pi}(s) > 0$ for some $t < H$. By (a) applied in both directions, $\mathrm{supp}\mathrm{Law}_t^{\pi}$ is the same for all positive policies. □

For the rest of this subsection let $\mathcal{S}$ and $\mathcal{A}$ be finite, write $d^{\mathrm{SA}}_\phi := d^{\mathrm{SA}}_{\pi_\phi,H}$, let q^{SA} be a fixed probability vector on $\mathcal{S} \times \mathcal{A}$, and set $J^{\mathrm{SA}}(\phi) := D_f(d^{\mathrm{SA}}_\phi \| q^{\mathrm{SA}})$.

Proposition D.13. [FPG-6 | D] *Assume Theorem D.1(ii),(iv). Let $\mathcal{X} \subseteq \mathcal{S} \times \mathcal{A}$ be the set of pairs on which the divergence is computed. Assume $\mathcal{X} \supseteq \mathrm{supp} d^{\mathrm{SA}}_\phi$ for every ϕ and $q^{\mathrm{SA}}(\mathcal{X}) = 1$, so that both are probability vectors on $\mathcal{X}$, and that $q^{\mathrm{SA}} > 0$ and $d^{\mathrm{SA}}_\phi > 0$ on $\mathcal{X}$. For instance, $\mathcal{X} = R_H \times \mathcal{A}$ with a smoothed target satisfies this by Theorem D.12(b). Let $g^{\mathrm{SA}}_\phi := f'(d^{\mathrm{SA}}_\phi/q^{\mathrm{SA}})$. Then*

$$\nabla J^{\mathrm{SA}}(\phi) = Z_H^{-1}\, \mathbb{E}_\phi\Big[\sum_{k=0}^{H-1} \gamma^k \chi_k G^{\mathrm{SA}}_k\Big], \qquad G^{\mathrm{SA}}_k := \sum_{t=k}^{H-1} \gamma^{t-k} g^{\mathrm{SA}}_\phi(s_t, a_t).$$

The current-step term $g^{\mathrm{SA}}_\phi(s_k, a_k)$ is not *a baseline: in general $\mathbb{E}[\chi_k g^{\mathrm{SA}}_\phi(s_k, a_k)] \neq 0$.*

Proof. Theorem D.4 with $h = \mathbf{1}[s_t = s, a_t = a]$ gives $\nabla \mathbb{P}(s_t = s, a_t = a) = \mathbb{E}[\mathbf{1}[s_t = s, a_t = a] \sum_{k<H} \chi_k]$. For $k \geq t+1$, the indicator is $\mathcal{F}_k$-measurable, because $\mathcal{F}_k$ contains a_t when $k > t$; those terms vanish by Theorem D.5(b). The term $k = t$ is *not* removed, because $\mathbf{1}[a_t = a]$ is not $\mathcal{F}_t$-measurable. Hence $\nabla d_\phi^{\mathrm{SA}}(s,a) = Z_H^{-1} \sum_t \gamma^t \mathbb{E}[\mathbf{1}[s_t = s, a_t = a] \sum_{k \leq t} \chi_k]$. The chain rule and the exchange of summation order proceed exactly as in Theorems D.6 and D.7.

For the last claim, $\mathbb{E}[\chi_k g^{\mathrm{SA}}(s_k, a_k) \mid \mathcal{F}_k] = \sum_a \nabla \pi_\phi(a \mid s_k)\, g^{\mathrm{SA}}(s_k, a)$. For tabular softmax, its (s_k, a)-component equals $\pi(a \mid s_k)(g^{\mathrm{SA}}(s_k, a) - \sum_b \pi(b \mid s_k) g^{\mathrm{SA}}(s_k, b))$ by Theorem D.2. This is nonzero unless $g^{\mathrm{SA}}(s_k, \cdot)$ is constant. □

Corollary D.14 (finiteness with a deterministic expert target)**.** *Let $\pi_\phi > 0$ and let $q_E^{\mathrm{SA}} = d_{\pi_E,H}^{\mathrm{SA}}$ for an expert π_E, unsmoothed.*

(a) $\mathrm{JS}(d_\phi^{\mathrm{SA}}, q_E^{\mathrm{SA}})$ *is finite.*

(b) $\mathrm{KL}(q_E^{\mathrm{SA}} \| d_\phi^{\mathrm{SA}})$ *is finite, by Theorem D.12, because the learner covers the expert's support.*

(c) $\mathrm{KL}(d_\phi^{\mathrm{SA}} \| q_E^{\mathrm{SA}}) = +\infty$ *as soon as the learner puts mass outside the expert's support. This is always the case when π_E is deterministic and $|\mathcal{A}| \geq 2$, because every non-expert action at every expert-visited state has positive learner mass.*

(d) With both *arguments smoothed by a strictly positive reference law υ on $\mathcal{X} = R_H \times \mathcal{A}$, both KL directions are finite (Theorem C.7(a)). With only the target smoothed,* $\mathrm{KL}(d_\phi^{\mathrm{SA}} \| q_{E,\varepsilon}^{\mathrm{SA}})$ *is finite, and so is* $\mathrm{KL}(q_{E,\varepsilon}^{\mathrm{SA}} \| d_\phi^{\mathrm{SA}})$*, because $d_\phi^{\mathrm{SA}} > 0$ on $R_H \times \mathcal{A}$ by Theorem D.12(b).*

For the state *occupancies d_ϕ and $q_E = d_{\pi_E,H}^+$, let $R'_H := \bigcup_{1 \leq t \leq H} \mathrm{supp}\mathrm{Law}_t^{\pi_\phi}$. By Theorem D.12(a), R'_H does not depend on ϕ, $d_\phi > 0$ exactly on R'_H, and $\mathrm{supp} q_E \subseteq R'_H$. Then (a) and (b) hold verbatim, and (d) holds with $\mathcal{X}$ replaced by R'_H. If a smoothed target charges a cell outside R'_H, then $\mathrm{KL}(q_{E,\varepsilon} \| d_\phi) = +\infty$ there; so smoothing over all cells needs every cell to lie in R'_H, as in the example after Theorem D.1. The analogue of (c) is* not *automatic: a deterministic expert may visit every state of R'_H. It holds exactly when $R'_H \not\subseteq \mathrm{supp} q_E$; this case is Theorem D.18(c).*

Proof. (a) is Theorem C.4(c). (b) and (c) combine Theorem C.4(c) with Theorem D.12. For the last sentence of (c), let $\pi_E(\cdot \mid s) = \delta_{a_E(s)}$ and let s satisfy $\mathrm{Law}_t^{\pi_E}(s) > 0$ for some $t < H$; such an s exists since $\mathrm{Law}_0 = \mu_0$. Then $q_E^{\mathrm{SA}}(s,a) = 0$ for $a \neq a_E(s)$. By Theorem D.12(a), $\mathrm{Law}_t^{\pi_\phi}(s) > 0$, so $d_\phi^{\mathrm{SA}}(s,a) \geq Z_H^{-1} \gamma^t \mathrm{Law}_t^{\pi_\phi}(s) \pi_\phi(a \mid s) > 0$. (d) is Theorem C.7(a) for both arguments smoothed; for the target alone, use Theorem C.4(c) with the stated supports. The state statements follow in the same way from Theorem D.12(a). □

Remark D.15. The state–action quantities are *diagnostics* only. Learners never receive expert actions; they are stored only for this evaluation (Section 10). The finite diagnostic is computed exactly by the oracle from P, which is legitimate because it is evaluation only.

D.6 State-MaxEnt identity

Proposition D.16 (FPG Lemma 5.1, p. 6; proof p. 17)**.** [FPG-7 | S+D] *Let d, q be probability vectors on a finite set with* $\mathrm{supp} d \subseteq \mathrm{supp} q$*. Then:*

(a) $\mathrm{KL}(d \| q) = -\mathcal{H}(d) - \mathbb{E}_d[\log q]$*. Minimizing* $\mathrm{KL}(d_{\pi,H}^+ \| q)$ *over policies is therefore equivalent to maximizing $\mathcal{H}(d_{\pi,H}^+) + \mathbb{E}_{d_{\pi,H}^+}[\log q]$. The* state-occupancy entropy *enters the maximized objective with a plus sign.*

(b) If $q(s) = e^{w(s)}/Z_w$ for a potential w, then $\mathrm{KL}(d \| q) = -\mathcal{H}(d) - \mathbb{E}_d w + \log Z_w$*. By Theorem B.11(a), $\mathbb{E}_{d_{\pi,H}^+} w = J_H^w(\pi)/Z_H$, so minimizing the KL means maximizing $J_H^w(\pi)/Z_H + \mathcal{H}(d_{\pi,H}^+)$. This is FPG Eq. (7), p. 7, with our normalization.*

(c) The state entropy is not the action entropy. In general neither maximizes the other (Theorem D.17).

(d) In the gradient of Theorem D.6 for $f(u) = u \log u$, the signal is $g_\phi = 1 + \log(d_\phi/q)$. The constant 1 does not contribute (Theorem D.8(b)), so the effective per-state signal is $\log(d_\phi/q)$. It depends on ϕ through d_ϕ, so it is not a fixed reward.

Proof. (a) $\mathrm{KL}(d\|q) = \sum_{d>0} d \log d - \sum_{d>0} d \log q = -\mathcal{H}(d) - \mathbb{E}_d \log q$, where every $\log q$ term is finite because $\mathrm{supp} d \subseteq \mathrm{supp} q$.

(b) Substitute $\log q = w - \log Z_w$.

(c) and (d) are proved by the example below and by Theorem D.8(b). □

Example D.17 (action entropy versus state entropy)**.** Let $\mathcal{S} = \{x, y_1, y_2\}$, $\mu_0 = \delta_x$ and $H = 1$. Actions a_1, a_2 lead deterministically to y_1, and a_3 leads to y_2. The uniform policy maximizes the action entropy ($\log 3$) and gives $d^+_{\pi,1} = (\frac{2}{3}, \frac{1}{3})$ on (y_1, y_2), with state entropy $\log 3 - \frac{2}{3}\log 2 \approx 0.637$. The policy $(\frac{1}{4}, \frac{1}{4}, \frac{1}{2})$ has lower action entropy ($\approx 1.040 < \log 3$) but gives $d^+ = (\frac{1}{2}, \frac{1}{2})$ with the maximal state entropy $\log 2 \approx 0.693$.

D.7 The fixed-target optimum and its attainability

Proposition D.18. [FPG-8 | D] *Let $\mathcal{S}$ and $\mathcal{A}$ be finite, let π_ϕ be tabular softmax, and let f be one of the three generators.*

(a) $J(\phi) \geq 0$, and $J(\phi) = 0$ if and only if $d_\phi = q$.

(b) If $q = q_E := d^+_{\pi_E,H}$ and $J(\phi) = 0$, then $J^r_H(\pi_\phi) = J^r_H(\pi_E)$ for every bounded state reward r.

(c) Suppose $q = q_E$ is unsmoothed and some state of $R'_H := \bigcup_{1\leq t\leq H} \mathrm{suppLaw}^{\pi_\phi}_t$ lies outside $\mathrm{supp} q_E$. Then $\mathrm{KL}(d_\phi\|q_E) = +\infty$ for every *ϕ. This is the case when the expert goes straight to a goal while the learner can reach other states.*

(d) For JS *and* $\mathrm{KL}(q_E\|d_\phi)$, $\inf_\phi J(\phi) = 0$.

(e) The infimum need not be attained (Theorem D.19).

No claim is made that gradient descent reaches the infimum, nor even a global minimizer of J over the policy class.

Proof. (a) is Theorem C.3 together with Theorem C.4(b).

(b) is Theorem B.11(c).

(c) By Theorem D.12(a), every ϕ gives $d_\phi > 0$ on R'_H, and R'_H is independent of ϕ. Some such state has $q_E = 0$, so Theorem C.4(c) applies.

(d) Choose logits $\phi^{(n)}_{s,a} := \log \pi_E(a \mid s)$ if $\pi_E(a \mid s) > 0$ and $\phi^{(n)}_{s,a} := -n$ otherwise. Let k_s be the number of actions with $\pi_E(a \mid s) = 0$. Then $\pi_{\phi^{(n)}}(a \mid s) = \pi_E(a \mid s)/(1 + k_s e^{-n})$ on the support and $e^{-n}/(1 + k_s e^{-n})$ off it, so $\pi_{\phi^{(n)}}(\cdot \mid s) \to \pi_E(\cdot \mid s)$ for every s. This covers deterministic experts, partially supported stochastic experts, and (with $k_s = 0$) fully supported ones, which are then represented exactly. d_ϕ is a polynomial in the policy entries (Theorem B.3), so $d_{\phi^{(n)}} \to q_E$. JS is continuous (Theorem C.4(c)), so $\mathrm{JS}(d_{\phi^{(n)}}, q_E) \to 0$. Also $\mathrm{KL}(q_E\|d_{\phi^{(n)}}) = \sum_{q_E>0} q_E \log(q_E/d_{\phi^{(n)}}) \to 0$, because $d_{\phi^{(n)}} \to q_E > 0$ on $\mathrm{supp} q_E$. Since $J \geq 0$, the infimum is 0.

(e) See the example below. □

Example D.19 (non-attainment)**.** Let $\mathcal{S} = \{0, 1\}$ and $\mu_0 = \delta_0$. Action a_{go} moves $0 \to 1$, action a_{stay} keeps $0 \to 0$, and state 1 is absorbing under both actions. The expert always plays a_{go}, so $q_E = d^+_{\pi_E,H} = \delta_1$. For any softmax policy, $p := \pi_\phi(a_{\mathrm{stay}} \mid 0) > 0$, so $\mathbb{P}(s_1 = 0) = p$ and $d_\phi(0) \geq p/Z_H > 0$. Hence $d_\phi \neq q_E$ and $J(\phi) > 0$ for every strictly convex f, while $J \to 0$ along $p \to 0$ for JS and $\mathrm{KL}(q_E\|d)$ (Theorem D.18(d)). Since $d_\phi(0) > 0 = q_E(0)$, $\mathrm{KL}(d_\phi\|q_E) = +\infty$ for all ϕ, which illustrates (c).

D.8 Correspondence with the printed source formulas

Remark D.20 (re-indexing FPG App. B.3). FPG's discounted visitation (B.3, p. 20) is $\eta_\tau(s) = \sum_{t=1}^{T} \gamma^t \mathbf{1}[s_t = s]$. With $T = H$, $\sum_{t=1}^{H} \gamma^t \mathbf{1}[s_t = s] = \gamma \sum_{t=0}^{H-1} \gamma^t \mathbf{1}[s_{t+1} = s]$, so its normalization by γZ_H is exactly our d_ϕ. Against this reference:

- (D.3) is the structure of FPG Theorem 4.1 (Eq. 5, p. 4; Eq. 8, p. 18), corrected as follows. Scores are indexed $k = 0, \ldots, H-1$; the source sums $t = 1, \ldots, T$ (F3). The normalization is Z_H^{-1}; Eq. 5 omits $1/T$ (F2). The signal carries the weights γ^t.
- FPG Eq. (10) (p. 20), on-policy, with $s_{t'}$ in place of the printed s_t inside the sum and normalized by $(\gamma Z_H)^{-1}$, equals (D.4). This identification also needs the score range $k = 0, \ldots, H-1$ (F3). The printed range $t = 1, \ldots, T$ omits the $k = 0$ term, which is not zero-mean in general, and includes a_T, which lies outside the trajectory. The extra term $t' = k$ is $\gamma^k g_\phi(s_k)\chi_k$, a zero-mean baseline by Theorem D.5(b).
- FPG Eq. (12) (p. 21) also discounts the score, as $\sum_t \gamma^t \chi_t$, and Eq. (13) uses $\gamma^k \gamma^{t'}$. In general neither equals (D.3)–(D.4); the numerical check exhibits a nonzero difference.
- FPG's practical Eqs. (6) and (11) are clipped surrogates written for maximization; Theorem D.10(d) gives the minimization form.

Remark D.21 (the Dirac-target theorems). [FPG-C1 | C] FPG Theorem 5.1 (p. 6; proof pp. 16–17) and Theorem 4.2 (p. 5; proof p. 20) use $D_f(p\|\delta_g) = f(p(g)) + f'(\infty)(1 - p(g))$, which depends on p only through $p(g)$. The objective is then a monotone function of the goal visitation. For a non-degenerate expert target, $D_f(d\|q)$ depends on all of d, so this argument has no analogue. Moreover, Lemma A.1 identifies the optimal policy for the *discounted* return H_g with the policy maximizing $p_\pi(g)$. This needs p_π to carry the same time weights as the return, whereas FPG §2 defines p_θ undiscounted. We use Theorem D.18 instead and make no optimality claim.

Remark D.22 (target with zeros). If $q(b) = 0$ for some cells and $f'(\infty) < \infty$ (JS: $\frac{1}{2}\log 2$; $\mathrm{KL}(q\|d)$: 0), then the contribution of those cells is $f'(\infty) d_\phi(b)$, which is linear in d_ϕ. Theorems D.6 and D.7 then hold with $g_\phi(b) := f'(\infty)$ on $\{q = 0\}$, provided $d_\phi(b) > 0$ on $\{q > 0\}$. For $\mathrm{KL}(d\|q)$ the objective is $+\infty$ whenever d_ϕ charges $\{q = 0\}$; by Theorem D.12(a) this then holds for every positive policy (Theorem D.18(c)), so smoothing is required.

E f-IRL: the analytic reward gradient (theory only)

Status and sources. f-IRL is not implemented in this study. We reprove its central result, the covariance form of the reward gradient (FIRL Theorem 4.1, p. 3; proof App. A.1–A.2, pp. 11–13), under its exact MaxEnt hypothesis, and we relate it to f-PG. FIRL's setting differs from ours in four respects:

- its horizon T is finite and undiscounted;
- its marginal counts the states $s_1, \ldots, s_T$, excluding s_0 (A.1, p. 11);
- its objective puts the expert first, $D_f(\rho_E\|\rho_\theta)$ (Eq. 2, p. 3);
- its parameters θ parameterize the *reward*.

E.1 Setting and the exact MaxEnt hypothesis

Assumption E.1 (A-FIRL)**.**

(i) $\mathcal{S}$ is finite, $T \in \mathbb{N}$ with $T \geq 1$, and $\mathcal{T} := \mathcal{S}^{T+1}$ is the set of state sequences $\tau = (s_0, \ldots, s_T)$.

(ii) *(Gibbs hypothesis.)* There is a weight $\bar{p} : \mathcal{T} \to [0, \infty)$ that does not depend on θ such that the agent's law over state sequences is

$$\rho_\theta(\tau) = \frac{\bar{p}(\tau) \exp\left(R_\theta(\tau)/\alpha\right)}{Z(\theta)}, \qquad R_\theta(\tau) := \sum_{t=1}^{T} r_\theta(s_t), \qquad Z(\theta) := \sum_{\tau} \bar{p}(\tau) e^{R_\theta(\tau)/\alpha},$$

with temperature $\alpha > 0$.

(iii) For each s, $\theta \mapsto r_\theta(s)$ is C^1 on an open set $\Theta \subseteq \mathbb{R}^m$.

(iv) *(Visitability.)* For every $s \in \mathcal{S}$ there is τ with $\bar{p}(\tau) > 0$ and $\eta_\tau(s) > 0$, where $\eta_\tau(s) := \sum_{t=1}^{T} \mathbf{1}[s_t = s]$.

(v) f is convex and C^1 on $(0, \infty)$ with $f(1) = 0$; ρ_E is a probability vector on $\mathcal{S}$. If ρ_E has zeros, also assume that $f(0)$ is finite and $\lim_{u \downarrow 0} u f'(u) = 0$, and set $h_f(0) := f(0)$. These conditions hold for $u \log u$ and f_{JS}. The proof uses only the finiteness of $f(0)$. The limit condition is used only to make $h_f(0) := f(0)$ the continuous extension, since then $h_f(u) = f(u) - u f'(u) \to f(0)$ as $u \downarrow 0$; it plays no other role.

Remark E.2 (what the hypothesis does and does not say). FIRL writes the base weight as $p(\tau) = \rho_0(s_0) \prod_t p(s_{t+1} \mid s_t, a_t)$ (p. 3 and Eq. 7, p. 11), although τ contains no actions; the action weighting is left implicit. Theorem E.1(ii) states the property the proof actually uses: the agent's state-trajectory law is an exponential tilt of a fixed base weight by the cumulative state reward. We do not claim that any policy produced by MaxEnt RL realizes this law. In particular we make no claim about soft-Q or SAC policies under stochastic dynamics, nor about approximately trained ones. The theorem below is a statement about the Gibbs family. When the agent's law is only approximately Gibbs, the formula is only an approximation of the gradient [SA U-7]. The objective of FIRL's preliminaries, $\sum_t \mathbb{E}[r(s_t, a_t) + \alpha \mathcal{H}(\cdot \mid s_t)]$ with *action* entropy (p. 3), is a different object. The paper links it to the Gibbs form by citing Ziebart [28]; that link is not used here.

Define the agent marginal $\rho_\theta(s) := \frac{1}{T} \sum_\tau \rho_\theta(\tau) \eta_\tau(s)$ and the loss $L_f(\theta) := D_f(\rho_E \| \rho_\theta) = \sum_s \rho_\theta(s) f\big(\rho_E(s)/\rho_\theta(s)\big)$, where the second form uses $\rho_\theta > 0$, proved next. Let $u_\theta(s) := \rho_E(s)/\rho_\theta(s)$ and $h_f(u) := f(u) - u f'(u)$.

Lemma E.3. *Under Theorem E.1:*

(a) $\rho_\theta(\cdot)$ *is a probability vector with* $\rho_\theta(s) > 0$ *for every* s *and every* θ*;*

(b) $\nabla_\theta \rho_\theta(\tau) = \frac{1}{\alpha} \rho_\theta(\tau) \big(\nabla R_\theta(\tau) - \mathbb{E}_{\rho_\theta} \nabla R_\theta\big)$*;*

(c) $\nabla_\theta \rho_\theta(s) = \frac{1}{\alpha T} \mathrm{Cov}_{\tau \sim \rho_\theta}\big(\eta_\tau(s), \nabla R_\theta(\tau)\big)$*, where* $\nabla R_\theta(\tau) = \sum_{t=1}^{T} \nabla r_\theta(s_t)$*.*

Proof. (a) $\sum_s \eta_\tau(s) = T$ for every τ, so $\sum_s \rho_\theta(s) = \frac{1}{T} \sum_\tau \rho_\theta(\tau) T = 1$. Positivity: by (iv) some τ with $\eta_\tau(s) > 0$ has $\bar{p}(\tau) > 0$, hence $\rho_\theta(\tau) > 0$, because the exponential factor is positive and $Z(\theta) \in (0, \infty)$ is a finite sum.

(b) $\log \rho_\theta(\tau) = \log \bar{p}(\tau) + R_\theta(\tau)/\alpha - \log Z(\theta)$ wherever $\bar{p}(\tau) > 0$, and $\rho_\theta(\tau) = 0$ for all θ otherwise. Moreover $\nabla \log Z = Z^{-1} \sum_\tau \bar{p}\, e^{R/\alpha} \nabla R/\alpha = \mathbb{E}_{\rho_\theta}[\nabla R]/\alpha$. All sums are finite, so differentiation is termwise.

(c) Differentiate the finite sum defining $\rho_\theta(s)$ and insert (b): $\nabla \rho_\theta(s) = \frac{1}{\alpha T} \mathbb{E}_{\rho_\theta}\big[\eta_\tau(s)(\nabla R - \mathbb{E} \nabla R)\big]$, which is the stated covariance. □

Theorem E.4 (FIRL Theorem 4.1, p. 3)**.** [FIRL-1 | S] *Under Theorem E.1,* L_f *is* C^1 *on* Θ *and*

$$\nabla_\theta L_f(\theta) = \frac{1}{\alpha T} \mathrm{Cov}_{\tau \sim \rho_\theta}\Big(\sum_{t=1}^{T} h_f(u_\theta(s_t)),\ \sum_{t=1}^{T} \nabla_\theta r_\theta(s_t) \Big).$$

Proof. Step 1 (derivative through the marginal). Fix s with $\rho_E(s) > 0$. The term $\rho_\theta(s) f(u_\theta(s))$ is C^1 in θ, because $\rho_\theta(s) > 0$ is C^1 (Theorem E.3) and f is C^1 on $(0, \infty)$. Its gradient is

$$\nabla \rho_\theta(s)\, f(u) + \rho_\theta(s) f'(u) \cdot \Big(- \frac{\rho_E(s)}{\rho_\theta(s)^2} \Big) \nabla \rho_\theta(s) = \big(f(u) - u f'(u)\big) \nabla \rho_\theta(s) = h_f(u_\theta(s))\, \nabla \rho_\theta(s).$$

This is FIRL Eq. (20), p. 12. If $\rho_E(s) = 0$, the term is $\rho_\theta(s) f(0)$, with gradient $f(0) \nabla \rho_\theta(s) = h_f(0) \nabla \rho_\theta(s)$ under the convention of Theorem E.1(v). Summing over the finitely many s gives $\nabla L_f = \sum_s h_f(u_\theta(s)) \nabla \rho_\theta(s)$.

Step 2 (Gibbs covariance). By Theorem E.3(c) and bilinearity of the covariance over the finite sum,

$$\nabla L_f = \frac{1}{\alpha T}\sum_s h_f(u_\theta(s))\mathrm{Cov}\big(\eta_\tau(s), \nabla R_\theta\big) = \frac{1}{\alpha T}\mathrm{Cov}\Big(\sum_s h_f(u_\theta(s))\eta_\tau(s),\ \nabla R_\theta\Big).$$

Step 3 (counting identity). $\sum_s h_f(u_\theta(s))\eta_\tau(s) = \sum_{t=1}^{T} h_f(u_\theta(s_t))$, by the definition of η_τ. □

Remark E.5 (sign and update direction). FIRL performs descent, $\theta \leftarrow \theta - \lambda\hat{\nabla}L_f$ (Algorithm 1, p. 4). Take tabular rewards $r_\theta(s) = \theta_s$. Then $\nabla R_\theta(\tau) = \eta_\tau$, and the s'-component of the gradient is $\frac{1}{\alpha T}\mathrm{Cov}(\sum_t h_f(u(s_t)), \eta_\tau(s'))$. For a tabular reward, if a state's visit count has negative covariance with the trajectory signal $\sum_t h_f(u(s_t))$, gradient descent increases that state's reward. Co-occurrence alone does not guarantee this sign. Since h_f is nonincreasing (Theorem E.6), trajectories along which the expert/agent ratio is large have a small $\sum_t h_f$. FIRL's reading on p. 4, that states visited by such trajectories gain reward, is therefore a heuristic about the typical sign of this covariance, not a consequence of the formula. Adding a constant to h_f does not change a covariance; this is why FIRL's $h_{\mathrm{JS}} = -\log(1+u)$ may drop $\log 2$ [SA D3].

E.2 Properties of h_f and the orientation dictionary

Proposition E.6. [FIRL-2 | S+D] *Let f be C^2 and convex on $(0,\infty)$. Part (a) and the left half of the table are FIRL's (p. 4; Table 2, p. 4); part (b) and the learner-first half of the table are ours.*

(a) $h_f'(u) = -uf''(u) \le 0$, so h_f is nonincreasing (FIRL p. 4).

(b) $h_f(u) = (f^\circ)'(1/u)$, where f° is the perspective of Theorem C.5.

(c) In both orientations:

FIRL (expert first), $u = \rho_E/\rho_\theta$			*ours (learner first), $v = \rho_\theta/\rho_E$*		
name	f	$h_f(u)$	*divergence*	f°	$(f^\circ)'(v)$
FKL	$u\log u$	$-u$	$\mathrm{KL}(q\|d)$	$-\log v$	$-1/v$
RKL	$-\log u$	$1-\log u$	$\mathrm{KL}(d\|q)$	$v\log v$	$1+\log v$
JS	f_{JS}	$\frac{1}{2}\log\frac{2}{1+u}$	JS	f_{JS}	$\frac{1}{2}\log\frac{2v}{1+v}$

In each row, $h_f(u) = (f^\circ)'(1/u)$.

Proof. (a) $h_f' = f' - f' - uf''$.

(b) This is Theorem C.5(d).

(c) Direct computation from $h_f = f - uf'$:

- $u\log u - u(1+\log u) = -u$;
- $-\log u - u(-1/u) = 1 - \log u$;
- $f_{\mathrm{JS}}(u) - u\cdot\frac{1}{2}\log\frac{2u}{1+u} = \frac{1}{2}\big[u\log u - (1+u)\log(1+u) + (1+u)\log 2 - u\log 2 - u\log u + u\log(1+u)\big] = \frac{1}{2}\log\frac{2}{1+u}$.

The right-hand column is Theorem C.5(c) together with the derivatives of Theorem C.4. Evaluating at $v = 1/u$ reproduces each h_f; for example $1+\log(1/u) = 1-\log u$, and $\frac{1}{2}\log\frac{2/u}{1+1/u} = \frac{1}{2}\log\frac{2}{u+1}$. □

E.3 Comparison with f-PG

Proposition E.7 (common structure)**.** [FIRL-3 | D] *Let $\vartheta \mapsto m_\vartheta$ be any C^1 parameterization of a learner marginal on a finite set, with $m_\vartheta > 0$, and let $q > 0$ be a fixed expert marginal. For convex C^1 f,*

$$\nabla_\vartheta D_f(q\|m_\vartheta) = \nabla_\vartheta D_{f^\circ}(m_\vartheta\|q) = \sum_x h_f\Big(\frac{q(x)}{m_\vartheta(x)}\Big)\nabla m_\vartheta(x) = \sum_x (f^\circ)'\Big(\frac{m_\vartheta(x)}{q(x)}\Big)\nabla m_\vartheta(x).$$

Hence the two methods weight each state by the same *function of the ratio, namely the derivative of the learner-first generator. They differ in three respects:*

(i) how ∇m is computed*: f-IRL uses the Gibbs covariance over reward parameters (Theorem E.3(c)); f-PG uses the likelihood ratio over policy parameters (Theorem D.6);*

(ii) which marginal is matched*: f-IRL matches the undiscounted uniform marginal over $s_1, \dots, s_T$; our f-PG matches the discounted next-state occupancy $d^+_{\pi,H}$;*

(iii) what is learned*: f-IRL learns a reward and needs a MaxEnt RL inner loop after every reward update (FIRL Algorithm 1, p. 4); f-PG updates the policy directly.*

Proof. The first equality is Theorem C.5(b). For the second, apply Theorem E.4 Step 1 to $D_f(q\|m) = \sum m\, f(q/m)$. The third is Theorem E.6(b), or Theorem D.6 Step 2 applied to $\sum q\, f^\circ(m/q)$. The three differences are read off Theorems D.6 and E.4 and the cited algorithm. □

Remark E.8 (exact versus approximate in f-IRL, and reward recovery). The covariance formula is exact only under Theorem E.1. The implemented f-IRL departs from it in several ways:

- the agent is an approximately trained SAC policy, so its trajectory law is not exactly Gibbs;
- ρ_θ is estimated by KDE (FIRL App. C.1, p. 18);
- the covariance is estimated from samples;
- the practical Eq. (6), p. 5, mixes in expert trajectories, which biases the gradient (the paper says so, p. 5).

FIRL's claim that a state-only reward is recovered up to a constant (§4.3, p. 4; App. A.4, p. 13) rests on AIRL Theorem 5.1, which is external to the supplied PDFs and not reproved here. The report presents it as the source's claim, with that dependence stated. Since f-IRL is not implemented, none of these approximations affects our experiments.

F Forward–backward representations

Status and sources. Sources are FB21 (Touati & Ollivier, arXiv v3) and FI24 (Pirotta et al., ICLR 2024). Every theorem number, equation number and page below was checked in the supplied PDFs on 5 October 2026; the mathematics of FB21 pp. 14, 24, 29 and FI24 pp. 17–18 was also read on the rendered pages.

- FB21: Definition 1 and Theorem 2 (p. 3); Eq. (4) (p. 4); Algorithm 1 (p. 14); Definition 3 and Theorem 4 (pp. 15–16); Remark 5 (p. 16); Proposition 6 (p. 17); Remark 7 (p. 18); Theorem 8 (p. 19); §B.6 (p. 24); proofs of Theorems 2 and 4 (pp. 25–26), of Proposition 6 (pp. 26–27), Propositions 17–18 (pp. 27–28) and the proof of Theorem 8 (pp. 29–30, Eqs. 97–109); Eqs. (164)–(166) (pp. 36–37).
- FI24: Eqs. (1)–(3) (p. 3); Proposition 1 (p. 4); the measure Bellman equation (p. 6).

Type labels:

- S: FB-1 (FB21 Eq. 4 and FI24's averaged Bellman equation), FB-2 (FB21 Theorem 2), FB-7 (FI24 Proposition 1, second part) and FB-9 (orthonormality), each reproved here under its own assumptions;
- S+D: FB-3 (FB21 Proposition 6, and our state-only analogue with a rank obstruction) and FB-6 (FB21 Theorem 8 in our convention with a direct proof, and our softmax and scaled versions);
- D: FB-4, FB-5, FB-8 and FB-11, which are ours;
- I+D: FB-10 (Gram matrix; its part (d) is S+D).

Throughout, the *next-state convention* of Section B is used: $M^{\pi,+}$ (FI24 Eq. 1) and rewards read at s_{t+1}. The 2021 convention of FB21 Eq. (1) is written M^π_{21} and is translated by Theorem B.19. Nothing in this section is a statement about a trained network unless it says so; Theorem F.27 lists what changes for the implemented model. In proofs, θ denotes the shift map of the proof of (M2), not the f-IRL reward parameter.

F.1 Setting

Assumption F.1 (A-FB)**.**

(i) Theorem B.1 holds; $\mathcal{A}$ is finite.

(ii) ν is a nonzero finite measure on $\mathcal{S}$, the *reference measure*. In the implementation ν is the population law of the replay next states (Section 2), a probability measure; the finite buffer replaces it by its empirical law $\hat{\nu}$ (Theorem F.27(2)).

(iii) $B : \mathcal{S} \to \mathbb{R}^d$ is measurable and bounded: $\beta_B := \sup_x \|B(x)\|_2 < \infty$.

(iv) For every $(a, z) \in \mathcal{A} \times \mathbb{R}^d$, $s \mapsto F(s, a, z) \in \mathbb{R}^d$ is measurable and $\sup_{s,a} \|F(s, a, z)\|_2 < \infty$.

(v) $\mathcal{Z} \subseteq \mathbb{R}^d$ is the *latent domain*, and $z \mapsto \pi_z$ assigns a stationary Markov policy to every $z \in \mathcal{Z}$.

Items (iii)–(iv) hold for continuous networks on a bounded state space (and, for (iv), at each fixed z). They only make every integral below finite; FB21 instead assumes that the integrals exist (p. 3, below Eq. 3).

Definition F.2 (FB objects)**.** Under Theorem F.1:

(a) $G_B := \int BB^\top \, \mathrm{d}\nu \in \mathbb{R}^{d\times d}$ (the *Gram matrix*; FB21 and FI24 call it $\operatorname{Cov} B$, Theorem F.20); and $z_r := \int rB \, \mathrm{d}\nu \in \mathbb{R}^d$ for every bounded measurable $r : \mathcal{S} \to \mathbb{R}$.

(b) The *model kernel* is $\hat{M}_z(s, a, X) := \int_X F(s, a, z)^\top B(x) \, \nu(\mathrm{d}x)$, a finite signed measure for every (s, a, z), with density $\hat{m}_z(s, a, x) := F(s, a, z)^\top B(x)$ with respect to ν.

(c) The *greedy family* is $\pi^{\mathrm{gr}}_z(s) :=$ the first maximizer of $a \mapsto F(s, a, z)^\top z$ in a fixed order on $\mathcal{A}$. The *softmax family* with temperature $\tau > 0$ is

$$\pi_{z,\tau}(a \mid s) := \frac{\exp(F(s, a, z)^\top z/\tau)}{\sum_b \exp(F(s, b, z)^\top z/\tau)}. \tag{F.1}$$

Both are measurable in s. For the greedy family, $\{\pi^{\mathrm{gr}}_z(s) = a\}$ is a finite intersection of sets of the form $\{F(s, a, z)^\top z > F(s, b, z)^\top z\}$ and $\{F(s, a, z)^\top z \geq F(s, b, z)^\top z\}$.

(d) $(F, B, (\pi_z)_{z\in\mathcal{Z}})$ is an *exact FB representation with respect to ν at z* if $M^{\pi_z,+}(s, a, \cdot) = \hat{M}_z(s, a, \cdot)$ for every $(s, a) \in \mathcal{S} \times \mathcal{A}$; it is exact *on $\mathcal{Z}$* if this holds for every $z \in \mathcal{Z}$. We call it *greedy* if $\pi_z = \pi^{\mathrm{gr}}_z$ and *softmax* if $\pi_z = \pi_{z,\tau}$.

FI24 Eq. (2) (p. 3) is (d) with "$\approx$" in place of equality and an approximately greedy policy. FB21 Definition 1 (p. 3) is the analogue in the 2021 convention: a measure ρ on $\mathcal{S} \times \mathcal{A}$, $B(s, a)$, M^π_{21} from $t = 0$ and greedy policies. FB21 states argmax policies with "ties may be broken any way" (p. 2); a fixed tie-breaking rule is one such choice, and every statement below holds for any fixed measurable rule.

General policies. A (history-dependent, randomized) policy σ is a sequence of kernels $\sigma_t(\mathrm{d}a \mid h_t)$ from $\mathcal{H}_t := (\mathcal{S} \times \mathcal{A})^t \times \mathcal{S}$ to $\mathcal{A}$. By the Ionescu–Tulcea theorem there is a unique probability $\mathbb{P}^\sigma_s$ on Ω with $s_0 = s$, $a_t \mid h_t \sim \sigma_t(\cdot \mid h_t)$ and $s_{t+1} \mid (h_t, a_t) \sim P(\cdot \mid s_t, a_t)$. A stationary Markov policy is the case $\sigma_t(\cdot \mid h_t) = \pi(\cdot \mid s_t)$, and then $\mathbb{P}^\sigma_s = \mathbb{P}^\pi_{\delta_s}$. For finite $\mathcal{S}$ this is the construction of Section B.7.

The shift identity. The proof of (M2) showed that the image of $\mathbb{P}^\pi_{(s,a)}$ under the shift θ is $R_{(s,a)} := \int P(\mathrm{d}s_1 \mid s, a) \sum_{a_1} \pi(a_1 \mid s_1)\mathbb{P}^\pi_{(s_1,a_1)}$. Integrating a bounded measurable $\Psi : \Omega \to \mathbb{R}$ against both measures gives

$$\mathbb{E}^\pi_{(s,a)}[\Psi \circ \theta] = \int P(\mathrm{d}s_1 \mid s, a) \sum_{a_1} \pi(a_1 \mid s_1) \, \mathbb{E}^\pi_{(s_1,a_1)}[\Psi] = \int P(\mathrm{d}s_1 \mid s, a) \, \mathbb{E}^\pi_{\delta_{s_1}}[\Psi]. \tag{F.2}$$

The first equality holds for indicators by the identity of measures, for simple Ψ by linearity, and for bounded Ψ by uniform approximation with simple functions. The second is (M1) with

$\mu_0 = \delta_{s_1}$, extended from events to bounded functions in the same way. The same argument shows that $(s,a) \mapsto \mathbb{E}^{\pi}_{(s,a)}[\Psi]$ is measurable for every bounded measurable Ψ, and that (M1) holds with a bounded measurable Ψ in place of an indicator.

F.2 Two lemmas on value functions

Lemma F.3 (next-state Q-functions)**.** *Under Theorem B.1, let r be bounded and measurable, let π be stationary Markov, and set $c_r(s,a) := \int P(\mathrm{d}x \mid s,a)\, r(x)$.*

(a) $Q_r^{+,\pi}(s,a) := \mathbb{E}^{\pi}_{(s,a)} \sum_{t\geq 0} \gamma^t r(s_{t+1}) = \int r \,\mathrm{d}M^{\pi,+}(s,a,\cdot)$. It is measurable in (s,a) and $|Q_r^{+,\pi}| \leq \|r\|_\infty/(1-\gamma)$.

(b) With $V_r^{\pi}(s) := \sum_a \pi(a \mid s) Q_r^{+,\pi}(s,a)$, we have $Q_r^{+,\pi}(s,a) = c_r(s,a) + \gamma \int P(\mathrm{d}s_1 \mid s,a) V_r^{\pi}(s_1)$.

(c) For every policy σ and every s, $\mathbb{E}^{\sigma}_s \sum_{t\geq 0} \gamma^t r(s_{t+1}) = \mathbb{E}^{\sigma}_s \sum_{t\geq 0} \gamma^t c_r(s_t,a_t)$.

Proof. (a) The partial sums are bounded by $\|r\|_\infty/(1-\gamma)$, so dominated convergence gives $Q_r^{+,\pi}(s,a) = \sum_t \gamma^t \int r \,\mathrm{d}\mathbb{P}^{\pi}_{(s,a)}(s_{t+1} \in \cdot)$, and (B.3) identifies this with $\int r \,\mathrm{d}M^{\pi,+}(s,a,\cdot)$. Each $(s,a) \mapsto \mathbb{E}^{\pi}_{(s,a)} r(s_{t+1})$ is measurable (end of Section F.1), so $Q_r^{+,\pi}$ is a pointwise limit of measurable functions.

(b) Write $\sum_{t\geq 0} \gamma^t r(s_{t+1}) = r(s_1) + \gamma\, \Psi \circ \theta$ with $\Psi := \sum_{u\geq 0} \gamma^u r(s_{u+1})$, which is bounded and measurable on Ω. Under $\mathbb{P}^{\pi}_{(s,a)}$ the law of s_1 is $P(\cdot \mid s,a)$, so $\mathbb{E}^{\pi}_{(s,a)} r(s_1) = c_r(s,a)$. By (F.2), $\mathbb{E}^{\pi}_{(s,a)}[\Psi \circ \theta] = \int P(\mathrm{d}s_1 \mid s,a) \sum_{a_1} \pi(a_1 \mid s_1) Q_r^{+,\pi}(s_1,a_1) = \int P(\mathrm{d}s_1 \mid s,a) V_r^{\pi}(s_1)$.

(c) By construction of $\mathbb{P}^{\sigma}_s$, $\mathbb{E}^{\sigma}_s[r(s_{t+1}) \mid h_t, a_t] = c_r(s_t,a_t)$. Take expectations, multiply by γ^t and sum; the infinite sums are exchanged with the expectation by dominated convergence, as in (a). □

Lemma F.4 (Bellman optimality on a general state space)**.** *Let $c : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ be bounded and measurable. For a policy σ let $V_c^{\sigma}(s) := \mathbb{E}^{\sigma}_s \sum_{t\geq 0} \gamma^t c(s_t,a_t)$; for stationary π let $Q_c^{\pi}(s,a) := c(s,a) + \gamma \int P(\mathrm{d}s' \mid s,a) V_c^{\pi}(s')$. On the Banach space $\mathcal{B}_b(\mathcal{S})$ of bounded measurable functions with the norm $\|\cdot\|_\infty$, define*

$$(T_c V)(s) := \max_a \Big[c(s,a) + \gamma \int P(\mathrm{d}s' \mid s,a) V(s') \Big],$$
$$(T_{\pi,c} V)(s) := \sum_a \pi(a \mid s) \Big[c(s,a) + \gamma \int P(\mathrm{d}s' \mid s,a) V(s') \Big].$$

(i) T_c and $T_{\pi,c}$ map $\mathcal{B}_b(\mathcal{S})$ into itself and are monotone γ-contractions.

(ii) V_c^{π} is the unique fixed point of $T_{\pi,c}$; equivalently, $V_c^{\pi}(s) = \sum_a \pi(a \mid s) Q_c^{\pi}(s,a)$.

(iii) T_c has a unique fixed point $V_c^{\star}$, and $V_c^{\sigma} \leq V_c^{\star}$ for every policy σ. With $Q_c^{\star} := c + \gamma \int P(\mathrm{d}s' \mid \cdot,\cdot) V_c^{\star}(s')$ we have $V_c^{\star}(s) = \max_a Q_c^{\star}(s,a)$.

(iv) For every $V \in \mathcal{B}_b(\mathcal{S})$, $\|V - V_c^{\star}\|_\infty \leq \|T_c V - V\|_\infty/(1-\gamma)$.

(v) A stationary π is optimal, i.e. $V_c^{\pi} = V_c^{\star}$, if and only if $\sum_a \pi(a \mid s) Q_c^{\pi}(s,a) = \max_a Q_c^{\pi}(s,a)$ for every s, that is, if and only if every $\pi(\cdot \mid s)$ is supported on $\arg\max_a Q_c^{\pi}(s,\cdot)$. In that case $Q_c^{\pi} = Q_c^{\star}$. Every stationary π whose every $\pi(\cdot \mid s)$ is supported on $\arg\max_a Q_c^{\star}(s,\cdot)$ is optimal.

Proof. (i) For $V \in \mathcal{B}_b(\mathcal{S})$, $s \mapsto \int P(\mathrm{d}s' \mid s,a) V(s')$ is measurable (kernel property, extended from indicators to bounded functions as in (F.2)) and bounded by $\|V\|_\infty$. Finite maxima and averages of measurable functions are measurable. Since $|\max_a x_a - \max_a y_a| \leq \max_a |x_a - y_a|$ and $|\int P(\mathrm{d}s' \mid s,a)(V-W)(s')| \leq \|V-W\|_\infty$, both operators are γ-Lipschitz; monotonicity is immediate.

(ii) V_c^{π} is bounded by $\|c\|_\infty/(1-\gamma)$ and measurable as in Theorem F.3(a). By (M1) with $\mu_0 = \delta_s$, $V_c^{\pi}(s) = \sum_a \pi(a \mid s) \mathbb{E}^{\pi}_{(s,a)} \sum_t \gamma^t c(s_t,a_t)$. Under $\mathbb{P}^{\pi}_{(s,a)}$ the sum equals $c(s,a) + \gamma\, \Psi \circ \theta$ with

$\Psi := \sum_{u\ge 0}\gamma^u c(s_u,a_u)$, and (F.2) gives $\mathbb{E}^\pi_{(s,a)}[\Psi\circ\theta] = \int P(\,\mathrm{d}s_1 \mid s,a)V^\pi_c(s_1)$. Hence $V^\pi_c = T_{\pi,c}V^\pi_c$, which is the stated identity; uniqueness follows from (i) and Banach's fixed-point theorem.

(iii) Existence and uniqueness of $V^\star_c$ follow from Banach's theorem. Fix σ and s, and let $\mathcal{F}_n$ be the σ-algebra generated by h_n. By construction of $\mathbb{P}^\sigma_s$,

$$\begin{aligned}\mathbb{E}\big[c(s_n,a_n)+\gamma V^\star_c(s_{n+1})\,\big|\,\mathcal{F}_n\big] &= \int \sigma_n(\,\mathrm{d}a \mid h_n)\Big[c(s_n,a)+\gamma\int P(\,\mathrm{d}s' \mid s_n,a)V^\star_c(s')\Big]\\ &\le (T_cV^\star_c)(s_n) = V^\star_c(s_n),\end{aligned}$$

since an average does not exceed a maximum. Multiply by γ^n, take expectations and sum over $n<N$; the terms telescope to $\mathbb{E}^\sigma_s\big[\sum_{t<N}\gamma^t c(s_t,a_t)+\gamma^N V^\star_c(s_N)\big] \le V^\star_c(s)$. Let $N\to\infty$: $|\gamma^N\mathbb{E}V^\star_c(s_N)| \le \gamma^N\|V^\star_c\|_\infty \to 0$, and dominated convergence handles the partial sums. The identity $V^\star_c = \max_a Q^\star_c$ is $T_cV^\star_c = V^\star_c$ rewritten.

(iv) $\|V-V^\star_c\| \le \|V-T_cV\| + \|T_cV-T_cV^\star_c\| \le \|T_cV-V\| + \gamma\|V-V^\star_c\|$.

(v) By definition $(T_cV^\pi_c)(s) = \max_a Q^\pi_c(s,a)$, and by (ii) $V^\pi_c(s) = \sum_a \pi(a\mid s)Q^\pi_c(s,a)$. If these agree for every s, then V^π_c is a fixed point of T_c, so $V^\pi_c = V^\star_c$ by (iii). Conversely, $V^\pi_c = V^\star_c$ gives $T_cV^\pi_c = T_cV^\star_c = V^\star_c = V^\pi_c$. An average of $Q^\pi_c(s,\cdot)$ equals its maximum exactly when no mass sits on non-maximizing actions. If π is optimal, $Q^\pi_c = c+\gamma PV^\pi_c = c+\gamma PV^\star_c = Q^\star_c$. For the last claim, support on $\arg\max Q^\star_c(s,\cdot)$ gives $T_{\pi,c}V^\star_c = T_cV^\star_c = V^\star_c$, so $V^\star_c$ is the fixed point V^π_c of $T_{\pi,c}$ by (ii). □

Both timing conventions are instances. For a next-state reward r, Theorem F.3(b)–(c) give $Q^{+,\pi}_r = Q^\pi_{c_r}$ and $\mathbb{E}^\sigma_s\sum_t\gamma^t r(s_{t+1}) = V^\sigma_{c_r}(s)$. For the 2021 convention with a reward $r(s,a)$, FB21's $Q^\pi_r(s_0,a_0) = \sum_t\gamma^t\mathbb{E}[r(s_t,a_t)\mid s_0,a_0,\pi]$ (p. 2) equals Q^π_c with $c=r$, by the computation of (ii). We write $V^\star_r := V^\star_{c_r}$ and $Q^\star_r := Q^\star_{c_r}$ for next-state rewards. FI24 defines $Q^\star_r := \sup_\pi Q^\pi_r$ (p. 3). With the reward read at s_{t+1} [SA R1] the two agree, since $Q^\pi_{c_r} = c_r+\gamma PV^\pi_{c_r} \le Q^\star_r$ by (iii), with equality for the optimal policies of (v); with the printed $r(s_t)$ the supremum is $r(s)+\gamma Q^\star_r$ (Theorem F.6(c)). For finite $\mathcal{S}$, Theorem F.4 reduces to Theorem B.15 and Theorem B.16(i),(iii).

F.3 The successor Bellman identity

Proposition F.5. [FB-1 | S] *Under Theorem B.1, for every stationary Markov policy π:*

(a) (next-state convention) For all (s,a,X),

$$M^{\pi,+}(s,a,X) = P(X\mid s,a) + \gamma\int P(\,\mathrm{d}s_1\mid s,a)\sum_{a_1}\pi(a_1\mid s_1)M^{\pi,+}(s_1,a_1,X).$$

This per-pair identity is Theorem B.9, the next-state analogue of FB21 Eq. (4). Averaging over $a\sim\pi(\cdot\mid s)$ gives the measure Bellman equation of FI24 (p. 6): $M^{\pi,+}(s,\cdot) = P_\pi(s,\cdot)+\gamma\int P_\pi(s,\,\mathrm{d}s_1)M^{\pi,+}(s_1,\cdot)$.

(b) (uniqueness) Let $\mathcal{K}(s,a,\cdot)$ be finite signed measures with $(s,a)\mapsto\mathcal{K}(s,a,X)$ measurable for every X and $\sup_{s,a,X}|\mathcal{K}(s,a,X)|<\infty$. If $\mathcal{K}$ satisfies the identity in (a), then $\mathcal{K}=M^{\pi,+}$.

(c) (2021 convention; FB21 Eq. 4, p. 4) $M^\pi_{21}(s,a,Y) = \delta_{(s,a)}(Y)+\gamma\int P(\,\mathrm{d}s_1\mid s,a)\sum_{a_1}\pi(a_1\mid s_1)M^\pi_{21}(s_1,a_1,Y)$ for measurable $Y\subseteq\mathcal{S}\times\mathcal{A}$, and M^π_{21} is the unique bounded solution in the sense of (b).

(d) (time shift) $M^\pi_{21}(s,a,X\times\mathcal{A}) = \delta_s(X)+\gamma M^{\pi,+}(s,a,X)$ (Theorem B.19(a)). The $t=0$ term of M^π_{21} is the atom $\delta_{(s,a)}$; the $t=0$ term of $M^{\pi,+}$ is $P(\cdot\mid s,a)$.

Proof. (a) is Theorem B.9; multiply by $\pi(a\mid s)$ and sum over a for the averaged form.

(b) Let $D := \mathcal{K}-M^{\pi,+}$. It inherits the measurability, and it is bounded because $0\le M^{\pi,+}(s,a,X)\le 1/(1-\gamma)$. Subtracting the two identities gives $D(s,a,X) = \gamma\int P(\,\mathrm{d}s_1\mid s,a)\sum_{a_1}\pi(a_1\mid s_1)D(s_1,a_1,X)$. With $\delta := \sup_{s,a,X}|D(s,a,X)|<\infty$, the integrand is bounded by δ, so $|D(s,a,X)|\le\gamma\delta$ for all (s,a,X). Hence $\delta\le\gamma\delta$ and $\delta=0$.

(c) The $t = 0$ term of $M_{21}^{\pi}(s, a, Y)$ is $\mathbb{P}_{(s,a)}^{\pi}((s_0, a_0) \in Y) = \delta_{(s,a)}(Y)$. For $t \geq 1$, apply (F.2) to $\Psi = \mathbf{1}[(s_{t-1}, a_{t-1}) \in Y]$, for which $\Psi \circ \theta = \mathbf{1}[(s_t, a_t) \in Y]$, and sum over $t \geq 1$ by Tonelli's theorem as in Theorem B.9. FB21 Eq. (4) writes a deterministic $\pi(s_1)$; for a stochastic π the next action is averaged. Uniqueness is proved as in (b).

(d) is Theorem B.19(a). □

Theorem F.5(a) is the regression target of FB pretraining (Theorem F.22); (d) explains why the "diagonal" term of the 2021 training loss (FB21 Eq. 164) differs from that of our next-state loss (Theorem F.23).

F.4 Reward-to-task optimality

Theorem F.6 (FB21 Theorem 2, p. 3; FI24 Proposition 1, p. 4, first part)**.** [FB-2 | S] *Assume Theorem F.1. Let $r : \mathcal{S} \to \mathbb{R}$ be bounded and measurable, and $z_r = \int rB \, \mathrm{d}\nu$.*

(a) (policy evaluation; no greediness) If the representation is exact at $z \in \mathcal{Z}$, for any policy family, then

$$Q_r^{+,\pi_z}(s, a) = F(s, a, z)^\top z_r \qquad \text{for all } (s, a).$$

(b) (optimality) If $z_r \in \mathcal{Z}$ and the greedy representation $(F, B, (\pi_z^{\mathrm{gr}}))$ is exact at z_r, then $\pi_{z_r}^{\mathrm{gr}}$ is optimal for the next-state reward r among all policies, and $Q_r^\star(s, a) = F(s, a, z_r)^\top z_r$.

(c) (2021 reading) Under the hypothesis of (b), $\pi_{z_r}^{\mathrm{gr}}$ is also optimal in the 2021 convention, among all policies, and $Q_{21,r}^\star(s, a) = r(s) + \gamma F(s, a, z_r)^\top z_r$.

(d) (the source statement) FB21 Theorem 2 as printed (a measure ρ on $\mathcal{S} \times \mathcal{A}$, $B(s, a)$, M_{21}^{π}, a bounded reward $r(s, a)$, $z_R = \int rB \, \mathrm{d}\rho$, greedy policies, exactness for all z) holds, with its standing assumptions made explicit: r, B, $F(\cdot, \cdot, z)$ and the tie-breaking rule are measurable, and z_R exists (FB21: "assuming the integral exists"). Its proof is the proof of (a)–(b) with M_{21}^{π} in place of $M^{\pi,+}$ and $c = r$.

Proof. (a) By Theorem F.3(a) and exactness at z,

$$Q_r^{+,\pi_z}(s, a) = \int r \, \mathrm{d}M^{\pi_z,+}(s, a, \cdot) = \int r(x) \, F(s, a, z)^\top B(x) \, \nu(\mathrm{d}x) = F(s, a, z)^\top \int rB \, \mathrm{d}\nu.$$

The last step is linearity of the integral applied to the d functions rB_i, which are ν-integrable because $|rB_i| \leq \|r\|_\infty \beta_B$ and ν is finite.

(b) Let $\pi := \pi_{z_r}^{\mathrm{gr}}$. By (a) at $z = z_r$, $F(s, \cdot, z_r)^\top z_r = Q_r^{+,\pi}(s, \cdot)$ for every s. By definition, $\pi(s)$ maximizes the left-hand side, so $\pi(\cdot \mid s) = \delta_{\pi(s)}$ is supported on $\arg\max_a Q_r^{+,\pi}(s, a) = \arg\max_a Q_{c_r}^{\pi}(s, a)$ (Theorem F.3(b)). Theorem F.4(v) with $c = c_r$ gives $V_{c_r}^{\pi} = V_r^\star$ and $Q_r^{+,\pi} = Q_r^\star$. By Theorem F.4(iii) and Theorem F.3(c), no policy has a larger next-state return from any state. Combining with (a) gives $Q_r^\star = F^\top z_r$.

(c) For every policy σ and state s, the pathwise identity $\sum_t \gamma^t r(s_t) = r(s_0) + \gamma \sum_t \gamma^t r(s_{t+1})$ gives $\mathbb{E}_s^\sigma \sum_t \gamma^t r(s_t) = r(s) + \gamma \, \mathbb{E}_s^\sigma \sum_t \gamma^t r(s_{t+1})$. So the 2021 value of every policy is $r(s)$ plus γ times its next-state value, and the two conventions have the same optimal policies among all policies. For stationary π, Theorem B.19(b) gives $Q_{21}^{\pi} = r(s) + \gamma Q_r^{+,\pi}$; at the optimal policy of (b) this is $Q_{21,r}^\star = r(s) + \gamma Q_r^\star = r(s) + \gamma F^\top z_r$.

(d) FB21 Proposition 16 (p. 25) gives $Q_{21}^{\pi}(s, a) = \int r \, \mathrm{d}M_{21}^{\pi}(s, a, \cdot)$, which is proved exactly as Theorem F.3(a). Under exactness, $F(s, a, z)^\top B$ is a ρ-density of the nonnegative measure $M_{21}^{\pi_z}(s, a, \cdot)$ of mass $1/(1 - \gamma)$; hence it is nonnegative ρ-a.e., $rF^\top B \in L^1(\rho)$, and, since each $rB_i \in L^1(\rho)$ by the existence of z_R, $\int rF(s, a, z)^\top B \, \mathrm{d}\rho = F(s, a, z)^\top z_R$. So $Q_{21}^{\pi_z} = F^\top z_R$ for every z, as in (a). For the greedy π_{z_R}, $\pi_{z_R}(s) \in \arg\max_a Q_{21}^{\pi_{z_R}}(s, a)$, and $Q_{21}^{\pi} = Q_c^{\pi}$ with $c = r$; Theorem F.4(v) concludes. □

Remark F.7 (what the proof uses).

- Exactness is needed only at the single latent z_r. FB21 assumes it for every z so that the conclusion holds for every reward.

- Greediness enters only in (b). For the softmax family, (a) still holds and Theorem F.12 replaces (b).
- FB21 Remark 5 (p. 16) notes that the FB equation requires the support of M^{π_z} to be included in that of ρ. Exactness in fact forces the stronger condition $M^{\pi_z,+}(s,a,\cdot) \ll \nu$, because $\hat{M}_z(s,a,\cdot) \ll \nu$ always (Theorem F.25(a), ours).
- (a)–(b) complete Theorem B.19(e): FI24 Proposition 1's identity $Q_r^\star = F^\top z_r$, which is (b), holds when the reward is read at s_{t+1}, as in FI24 Eq. (1). With the Q printed on FI24 p. 3, which reads $r(s_t)$, the identity becomes $Q^\star_{21,r} = r(s) + \gamma F^\top z_r$ by (c).

F.5 Exact finite representations and the rank obstruction

Proposition F.8. [FB-3 | S+D] *Let $\mathcal{S}$ and $\mathcal{A}$ be finite.*

(a) (S; FB21 Proposition 6, p. 17; proof pp. 26–27) Let ρ be a measure on $\mathcal{S}\times\mathcal{A}$ with $\rho(s,a) > 0$ everywhere and $d \geq \#\mathcal{S}\cdot\#\mathcal{A}$. Then there are $F : \mathcal{S}\times\mathcal{A}\times\mathbb{R}^d \to \mathbb{R}^d$ and $B : \mathcal{S}\times\mathcal{A} \to \mathbb{R}^d$ such that, for every z, $F(s,a,z)^\top B(s',a') = M_{21}^{\pi_z}(s,a,\{(s',a')\})/\rho(s',a')$ for all pairs, where π_z is the greedy policy of $F(\cdot,\cdot,z)^\top z$.

(b) (D; next-state, state-only analogue) Let $\nu(s) > 0$ for every s and $d \geq \#\mathcal{S}$. Then there are $F : \mathcal{S}\times\mathcal{A}\times\mathbb{R}^d \to \mathbb{R}^d$ and $B : \mathcal{S} \to \mathbb{R}^d$, with $B(s)$ the s-th unit vector padded with zeros, such that $(F, B, (\pi_z^{\mathrm{gr}})_{z\in\mathbb{R}^d})$ is an exact greedy representation on $\mathbb{R}^d$. This holds for any fixed tie-breaking rule.

(c) (D; rank obstruction) Let $\tilde{P} \in \mathbb{R}^{(\mathcal{S}\times\mathcal{A})\times\mathcal{S}}$ have rows $p_{sa} = P(\cdot \mid s,a)$ and $k := \operatorname{rank}\tilde{P}$. For every stationary π, the matrix $\mathbf{M}^\pi := [M^{\pi,+}(s,a,\{s'\})]$ has rank k, and an exact representation at z with $B : \mathcal{S} \to \mathbb{R}^d$ requires $d \geq k$. If some action a_0 has a nonsingular matrix $[P(s' \mid s,a_0)]_{s,s'}$, then $k = \#\mathcal{S}$. This is the case for a stay action of the form $P(\cdot \mid s,a_0) = (1-\epsilon)\delta_s + \epsilon u_s$, with probability vectors u_s and $0 \leq \epsilon < \frac{1}{2}$.

Proof. (b) Let $d = \#\mathcal{S}$. For larger d, pad B with zero coordinates and set the extra coordinates of F to zero; then $F^\top B$ is unchanged and $F^\top z$ depends on z only through its first $\#\mathcal{S}$ coordinates, to which the argument below applies. With $B(s) = e_s$, the coordinates of z_r are $(z_r)_s = \nu(s)r(s)$. For $z \in \mathbb{R}^d$ define the reward $r_z(s) := z_s/\nu(s)$, so that $z_{r_z} = z$. Let $\pi^{(z)}$ be the greedy policy of $Q^\star_{r_z}$ under the fixed tie-breaking rule; it is optimal by the last claim of Theorem F.4(v). Define $F(s,a,z)_{s'} := M^{\pi^{(z)},+}(s,a,\{s'\})/\nu(s')$. Then:

- $F(s,a,z)^\top B(s')\nu(s') = M^{\pi^{(z)},+}(s,a,\{s'\})$, i.e. $\hat{M}_z = M^{\pi^{(z)},+}$;
- $F(s,a,z)^\top z = \sum_{s'} M^{\pi^{(z)},+}(s,a,\{s'\})\, r_z(s') = Q^{+,\pi^{(z)}}_{r_z}(s,a) = Q^\star_{r_z}(s,a)$, by Theorem F.3(a)–(b) and optimality (Theorem F.4(v)).

Hence the first maximizer of $F(s,\cdot,z)^\top z$ is the first maximizer of $Q^\star_{r_z}(s,\cdot)$, that is $\pi_z^{\mathrm{gr}} = \pi^{(z)}$, and $M^{\pi_z^{\mathrm{gr}},+} = \hat{M}_z$.

(a) The same construction in the 2021 convention: $B(s,a) := e_{(s,a)}$, so $(z_R)_{(s,a)} = r(s,a)\rho(s,a)$ and $r_z(s,a) := z_{(s,a)}/\rho(s,a)$ inverts $r \mapsto z_R$. Let $\pi^{(z)}$ be the greedy policy of $Q^\star_{21,r_z} = Q^\star_c$ with $c = r_z$ (Theorem F.4), and $F(s,a,z)_{(s',a')} := M_{21}^{\pi^{(z)}}(s,a,\{(s',a')\})/\rho(s',a')$. As in (b), $F^\top B(s',a') = M_{21}^{\pi^{(z)}}(s,a,\{(s',a')\})/\rho(s',a')$ and $F(s,a,z)^\top z = Q^{\pi^{(z)}}_{21,r_z}(s,a) = Q^\star_{21,r_z}(s,a)$, whose first maximizer is $\pi^{(z)}(s)$. Under FB21's $\in$-reading of $\arg\max$ (p. 2), the source proof, which takes "an optimal policy", is complete. Our greedy family uses a fixed tie-breaking rule. Then the optimal policy must be chosen as the greedy policy of $Q^\star$ under that same rule; otherwise the greedy policy of $F^\top z$ could be a different optimal policy with a different successor measure.

(c) By Theorem B.8(a), $\mathbf{M}^\pi = \tilde{P}R_\pi$ with R_π invertible (Theorem B.7), so $\operatorname{rank}\mathbf{M}^\pi = k$. Exactness at z means $\mathbf{M}^{\pi_z}(s,a,s') = F(s,a,z)^\top B(s')\nu(s')$, i.e. $\mathbf{M}^{\pi_z} = \mathbf{F}_z\mathbf{B}_\nu^\top$, where $\mathbf{F}_z \in \mathbb{R}^{(\mathcal{S}\times\mathcal{A})\times d}$ has rows $F(s,a,z)^\top$ and $\mathbf{B}_\nu \in \mathbb{R}^{\mathcal{S}\times d}$ has rows $\nu(s')B(s')^\top$. A product through $\mathbb{R}^d$ has rank at most d. If $[P(s' \mid s,a_0)]$ is nonsingular, its $\#\mathcal{S}$ rows are linearly independent rows of $\tilde{P}$, so $k = \#\mathcal{S}$. For the stay action, let $W := (1-\epsilon)I + \epsilon U$ with U row-stochastic. In row s, $W_{ss} \geq 1-\epsilon$ and $\sum_{s'\neq s}|W_{ss'}| = \epsilon(1-U_{ss}) \leq \epsilon < 1-\epsilon$, so W is strictly diagonally

dominant. Such a matrix is nonsingular: if $Wx = 0$ with $x \neq 0$, pick s with $|x_s| = \max|x|$; then $|W_{ss}||x_s| = |\sum_{s' \neq s} W_{ss'} x_{s'}| \leq \sum_{s' \neq s} |W_{ss'}|\,|x_s| < |W_{ss}||x_s|$, a contradiction. The threshold is sharp: for $\epsilon = \frac{1}{2}$ and U a transposition, W is singular. □

Remark F.9 (consequence for the finite mazes). The finite mazes have $\#\mathcal{S} = 40$ (four rooms) or 49 (open layout), and $d = 16$. If the configured kernel has a stay action of the form in (c) (deterministic moves, $\epsilon = 0$; or a noise model that replaces the chosen action by a random one with probability $\epsilon < \frac{1}{2}$), then $k = \#\mathcal{S} > 16$ and no exact rank-16 representation exists, for any policy family. Every result that assumes exactness (FB-2, FB-5(c)–(d), FB-7, FB-8(c), FB-10(c), FBIL-1(b), FBIL-2(a),(b),(d), FBIL-5(d)) then has a false hypothesis for the implemented model; only the conditional bounds of FB-6 apply. The implemented finite kernel has a stay action of this form with $\epsilon = p_{\text{slip}} = 0.1$, and its rank is $\#\mathcal{S}$ (Theorem H.2(b)).

Proposition F.10. [FB-4 | D] *Let $\mathcal{S}$ and $\mathcal{A}$ be finite and $\tau > 0$.*

(a) For every reward $r : \mathcal{S} \to \mathbb{R}$ there is a stationary policy π such that, for every s, $\pi(\cdot \mid s) = \operatorname{softmax}(Q_r^{+,\pi}(s,\cdot)/\tau)$.

(b) If $\nu(s) > 0$ for every s and $d \geq \#\mathcal{S}$, there are F and B (with B one-hot as in Theorem F.8(b)) such that $(F, B, (\pi_{z,\tau})_{z \in \mathbb{R}^d})$ is an exact softmax representation on $\mathbb{R}^d$.

(c) Uniqueness of the fixed point in (a) is not claimed, so the construction in (b) depends on the selection of fixed points.

Proof. (a) $\Pi := \prod_{s \in \mathcal{S}} \Delta(\mathcal{A}) \subset \mathbb{R}^{\mathcal{S} \times \mathcal{A}}$ is nonempty, compact and convex. For $\pi \in \Pi$, P_π depends linearly on π, and $I - \gamma P_\pi$ is invertible (Theorem B.7). By Cramer's rule the entries of $(I - \gamma P_\pi)^{-1}$ are ratios of polynomials in the entries of π with nonvanishing denominator, so they are continuous on Π. Hence $\pi \mapsto Q_r^{+,\pi}(s,a) = p_{sa}(I - \gamma P_\pi)^{-1} r$ (Theorem B.8(a), Theorem F.3(a)) is continuous, and so is $\Phi(\pi)(\cdot \mid s) := \operatorname{softmax}(Q_r^{+,\pi}(s,\cdot)/\tau)$. Φ maps Π into Π, so Brouwer's fixed-point theorem gives $\pi = \Phi(\pi)$.

(b) Proceed as in Theorem F.8(b), with $\pi^{(z)}$ a fixed point of (a) for the reward r_z (one fixed point chosen for each z). With $F(s,a,z)_{s'} := M^{\pi^{(z)},+}(s,a,\{s'\})/\nu(s')$ we get $\hat{M}_z = M^{\pi^{(z)},+}$ and $F(s,a,z)^\top z = Q_{r_z}^{+,\pi^{(z)}}(s,a)$. Hence $\pi_{z,\tau}(\cdot \mid s) = \operatorname{softmax}(Q_{r_z}^{+,\pi^{(z)}}(s,\cdot)/\tau) = \pi^{(z)}(\cdot \mid s)$, and $M^{\pi_{z,\tau},+} = \hat{M}_z$ for every z.

(c) The construction uses an arbitrary selection among fixed points; whether the fixed point is unique is not settled here. □

Theorem F.8(b) and Theorem F.10(b) show that the exactness hypotheses of FB-2 (greedy) and FB-5 (softmax) can be met in the finite maze when $d \geq \#\mathcal{S}$. Theorem F.8(c) shows that they cannot be met at $d = 16$ when $\operatorname{rank} \tilde{P} > 16$, for instance under the stay-action condition of (c).

F.6 What replaces Theorem 2 for the softmax family

Lemma F.11 (log-sum-exp). *For $x \in \mathbb{R}^{\mathcal{A}}$, $\tau > 0$ and $p := \operatorname{softmax}(x/\tau)$: $\sum_a p_a x_a = \tau \log \sum_b e^{x_b/\tau} - \tau \mathcal{H}(p)$ and $\max_a x_a - \tau \log|\mathcal{A}| \leq \sum_a p_a x_a \leq \max_a x_a$.*

Proof. $\log p_a = x_a/\tau - \log \sum_b e^{x_b/\tau}$; multiply by τp_a and sum over a to get the identity. Moreover $\tau \log \sum_b e^{x_b/\tau} \geq \tau \log e^{\max_b x_b/\tau} = \max_b x_b$, and $\mathcal{H}(p) = \sum_a p_a \log(1/p_a) \leq \log \sum_a p_a (1/p_a) \leq \log|\mathcal{A}|$ by Jensen's inequality. The upper bound holds because an average does not exceed a maximum. □

Proposition F.12. [FB-5 | D] *Let c be bounded and measurable, and $\tau > 0$.*

(a) If a stationary π satisfies $\pi(\cdot \mid s) = \operatorname{softmax}(Q_c^\pi(s,\cdot)/\tau)$ for every s, then

$$0 \leq V_c^\star - V_c^\pi \leq \frac{\tau \log|\mathcal{A}|}{1-\gamma}, \qquad 0 \leq Q_c^\star - Q_c^\pi \leq \frac{\gamma\,\tau \log|\mathcal{A}|}{1-\gamma}.$$

(b) Such a π is optimal if and only if $Q_c^\pi(s,\cdot)$ is constant on $\mathcal{A}$ for every s. More precisely, $V_c^\pi(s) < V_c^\star(s)$ at every s where $Q_c^\pi(s,\cdot)$ is not constant.

(c) (exact softmax FB) Assume Theorem F.1, a bounded reward r, and an exact softmax representation at $z_r \in \mathcal{Z}$. Then $\pi_{z_r,\tau}$ satisfies (a) with $c = c_r$: $V_r^\star - V_r^{\pi_{z_r,\tau}} \le \tau \log|\mathcal{A}|/(1-\gamma)$, in the units of r.

(d) (scale of z, under exactness) If instead the softmax representation is exact at $\zeta z_r \in \mathcal{Z}$ for some $\zeta > 0$, then $\pi := \pi_{\zeta z_r,\tau}$ satisfies $\pi(\cdot \mid s) = \mathrm{softmax}\big(Q_r^{+,\pi}(s,\cdot)/(\tau/\zeta)\big)$, so $V_r^\star - V_r^\pi \le (\tau/\zeta)\log|\mathcal{A}|/(1-\gamma)$. On the sphere $\mathcal{Z}_R := \{z : \|z\|_2 = R\}$ with $z_r \ne 0$, the representative $Rz_r/\|z_r\|_2$ therefore has effective temperature *$\tau\|z_r\|_2/R$ relative to r.*

(e) (orthonormal B) If $G_B = I$, then z_r is the coefficient vector of the $L^2(\nu)$-orthogonal projection $\Pi_B r$ of r onto $\mathrm{span}\{B_1,\dots,B_d\}$, and $\|z_r\|_2 = \|\Pi_B r\|_{L^2(\nu)} \le \|r\|_{L^2(\nu)}$.

Proof. (a) By Theorem F.4(ii), $V_c^\pi(s) = \sum_a \pi(a \mid s) Q_c^\pi(s,a)$. Theorem F.11 with $x = Q_c^\pi(s,\cdot)$ gives $V_c^\pi(s) \ge \max_a Q_c^\pi(s,a) - \tau\log|\mathcal{A}| = (T_c V_c^\pi)(s) - \tau\log|\mathcal{A}|$; also $V_c^\pi \le T_c V_c^\pi$. Hence $\|T_c V_c^\pi - V_c^\pi\|_\infty \le \tau\log|\mathcal{A}|$, and Theorem F.4(iv) gives $\|V_c^\pi - V_c^\star\|_\infty \le \tau\log|\mathcal{A}|/(1-\gamma)$; Theorem F.4(iii) gives the sign. Finally $Q_c^\star - Q_c^\pi = \gamma \int P(\,\mathrm{d}s' \mid \cdot,\cdot)(V_c^\star - V_c^\pi)(s')$.

(b) If every $Q_c^\pi(s,\cdot)$ is constant, every $\pi(\cdot \mid s)$ is supported on $\arg\max_a Q_c^\pi(s,\cdot) = \mathcal{A}$, and Theorem F.4(v) applies. Conversely, a softmax policy has full support, so $\sum_a \pi(a \mid s) Q_c^\pi(s,a) < \max_a Q_c^\pi(s,a)$, i.e. $(T_c V_c^\pi)(s) > V_c^\pi(s)$, at every s where $Q_c^\pi(s,\cdot)$ is not constant. Since $T_c V_c^\pi \ge V_c^\pi$ and T_c is monotone, $T_c^n V_c^\pi$ is nondecreasing in n and converges to $V_c^\star$; hence $V_c^\star(s) \ge (T_c V_c^\pi)(s) > V_c^\pi(s)$ at such s.

(c) By Theorem F.6(a) at $z = z_r$, $F(s,\cdot,z_r)^\top z_r = Q_r^{+,\pi}(s,\cdot)$ for $\pi = \pi_{z_r,\tau}$, so $\pi(\cdot \mid s) = \mathrm{softmax}(Q_r^{+,\pi}(s,\cdot)/\tau)$, and $Q_r^{+,\pi} = Q_{c_r}^\pi$ (Theorem F.3(b)). Apply (a).

(d) By Theorem F.6(a) at $z = \zeta z_r$ with the reward r, $F(s,\cdot,\zeta z_r)^\top z_r = Q_r^{+,\pi}(s,\cdot)$, so $F(s,\cdot,\zeta z_r)^\top(\zeta z_r)/\tau = Q_r^{+,\pi}(s,\cdot)/(\tau/\zeta)$. Apply (a) with temperature τ/ζ. For the sphere, take $\zeta = R/\|z_r\|_2$.

(e) For $w \in \mathbb{R}^d$, $\|r - w^\top B\|_{L^2(\nu)}^2 = \|r\|_{L^2(\nu)}^2 - 2w^\top z_r + w^\top G_B w$. With $G_B = I$ this strictly convex quadratic is minimized at $w = z_r$, so $\Pi_B r = z_r^\top B$ and $\|\Pi_B r\|_{L^2(\nu)}^2 = z_r^\top G_B z_r = \|z_r\|_2^2$. The orthogonal projection does not increase the norm. □

Remark F.13. Part (b) is why FB21 Theorem 2 does not apply to the softmax family unchanged: even under exact FB, $\pi_{z_r,\tau}$ is strictly suboptimal wherever its action values differ. Parts (a), (c) bound the loss, in the units of r. As $\tau \to 0$ the bound vanishes, which matches the greedy case; but $\pi_{z,\tau}$ with $\tau > 0$ is never the greedy family. Parts (d)–(e) give the precise sense in which τ and the scale of z are coupled *under exactness* [SA T2]: only τ/ζ matters, and on a sphere the reward enters through $\|z_r\|_2$, which equals $\|\Pi_B r\|_{L^2(\nu)}$ when $G_B = I$. For a trained network, $F(s,a,\zeta z)^\top \zeta z \ne \zeta F(s,a,z)^\top z$ in general (unless the latent input of F is scale-normalized, as in FI24 D.3), so for the implemented model τ and R are coupled only empirically.

Corollary F.14 (finite-horizon evaluation)**.** [FB-5 | D] *Let $r \ge 0$ be bounded, π stationary and $s \in \mathcal{S}$. Let $V_H^\sigma(s) := \mathbb{E}_s^\sigma \sum_{t<H} \gamma^t r(s_{t+1})$ and $V_H^\star(s) := \sup_\sigma V_H^\sigma(s)$, the supremum over all policies. Then $0 \le V_H^\star(s) - V_H^\pi(s) \le V_r^\star(s) - V_r^\pi(s) + \gamma^H\|r\|_\infty/(1-\gamma)$. For finite $\mathcal{S}$, $V_H^\star$ is the backward-induction value of Theorem B.17.*

Proof. The lower bound holds because π is one of the policies in the supremum. For every σ, $V_H^\sigma(s) \le \mathbb{E}_s^\sigma \sum_{t\ge 0} \gamma^t r(s_{t+1}) \le V_r^\star(s)$, by $r \ge 0$, Theorem F.3(c) and Theorem F.4(iii); hence $V_H^\star \le V_r^\star$. Also $V_r^\pi(s) - V_H^\pi(s) = \mathbb{E}_s^\pi \sum_{t\ge H} \gamma^t r(s_{t+1}) \le \gamma^H\|r\|_\infty/(1-\gamma)$. Combine the two inequalities. The finite case is Theorem B.17(a)–(b). □

F.7 Approximate representations

Lemma F.15 (the dual $L^2(\nu)$ norm; FI24 Eq. 30, p. 17)**.** *Let ν be a finite measure and μ a finite signed measure on $\mathcal{S}$. Then*

$$\|\mu\|_{L^2(\nu)^\star} := \sup\Big\{ \int r \,\mathrm{d}\mu : \ r \text{ bounded measurable}, \ \|r\|_{L^2(\nu)} \le 1 \Big\}$$

equals $\|\,\mathrm{d}\mu/\,\mathrm{d}\nu\|_{L^2(\nu)}$ if $\mu \ll \nu$ with $\mathrm{d}\mu/\,\mathrm{d}\nu \in L^2(\nu)$, and $+\infty$ otherwise. In particular $|\int r\,\mathrm{d}\mu| \le \|r\|_{L^2(\nu)}\|\mu\|_{L^2(\nu)^\star}$ for every bounded measurable r whenever $\|\mu\|_{L^2(\nu)^\star} < \infty$.

Proof. Case $\mu = \dot\mu\,\nu$ with $\dot\mu \in L^2(\nu)$. Cauchy–Schwarz gives $\int r\dot\mu\,\mathrm{d}\nu \le \|\dot\mu\|_{L^2(\nu)}$. If $\dot\mu \neq 0$ in $L^2(\nu)$, the bounded functions $r_n := \dot\mu\mathbf{1}[|\dot\mu| \le n]/\|\dot\mu\|_{L^2(\nu)}$ satisfy $\|r_n\|_{L^2(\nu)} \le 1$ and $\int r_n\,\mathrm{d}\mu = \int \dot\mu^2\mathbf{1}[|\dot\mu| \le n]\,\mathrm{d}\nu/\|\dot\mu\|_{L^2(\nu)} \to \|\dot\mu\|_{L^2(\nu)}$ by monotone convergence. *Case $\mu = \dot\mu\,\nu$ with $\dot\mu \notin L^2(\nu)$.* Here $\dot\mu \in L^1(\nu)$ because μ is finite. With $\dot\mu_n := \dot\mu\mathbf{1}[|\dot\mu| \le n]$, the functions $r_n := \dot\mu_n/\|\dot\mu_n\|_{L^2(\nu)}$ (for n large, so that $\dot\mu_n \neq 0$) give $\int r_n\,\mathrm{d}\mu = \|\dot\mu_n\|_{L^2(\nu)} \to \infty$. *Case μ not absolutely continuous.* There is a ν-null set N with $|\mu|(N) > 0$. With a Hahn decomposition $\mathcal{S} = S^+ \cup S^-$ of μ, one of $\mu(N \cap S^+) > 0$ or $\mu(N \cap S^-) < 0$ holds; call that set N', a ν-null set with $\mu(N') \neq 0$. For $r_c := c\,\mathrm{sign}(\mu(N'))\mathbf{1}_{N'}$ we have $\|r_c\|_{L^2(\nu)} = 0$ and $\int r_c\,\mathrm{d}\mu = c|\mu(N')| \to \infty$ as $c \to \infty$. The inequality holds in the first case by Cauchy–Schwarz; the other two cases are excluded by its hypothesis. (Without that hypothesis it can fail: if μ charges a ν-null set N', then $r = \mathbf{1}_{N'}$ has $\|r\|_{L^2(\nu)} = 0$ but $\int r\,\mathrm{d}\mu \neq 0$.) □

Lemma F.16 (approximately greedy and soft-greedy policies)**.** *Let c be bounded and measurable, π stationary, and $\hat Q : \mathcal{S} \times \mathcal{A} \to \mathbb{R}$ bounded and measurable with $\|\hat Q - Q_c^\pi\|_\infty \le \delta$. Let $\kappa \ge 0$ satisfy $\sum_a \pi(a \mid s)\hat Q(s,a) \ge \max_a \hat Q(s,a) - \kappa$ for every s. Then*

$$0 \le V_c^\star - V_c^\pi \le \frac{\kappa + 2\delta}{1-\gamma}, \qquad 0 \le Q_c^\star - Q_c^\pi \le \frac{\gamma(\kappa + 2\delta)}{1-\gamma},$$

and $\|\hat Q - Q_c^\star\|_\infty \le \delta + \gamma(\kappa + 2\delta)/(1-\gamma)$. $\kappa = 0$ for the greedy policy of $\hat Q$, and $\kappa = \tau \log|\mathcal{A}|$ for $\mathrm{softmax}(\hat Q/\tau)$ *(Theorem F.11).*

Proof. By Theorem F.4(ii) and the two hypotheses,

$$\begin{aligned} V_c^\pi(s) = \sum_a \pi(a \mid s) Q_c^\pi(s,a) &\ge \sum_a \pi(a \mid s)\hat Q(s,a) - \delta \\ &\ge \max_a \hat Q(s,a) - \kappa - \delta \ge \max_a Q_c^\pi(s,a) - \kappa - 2\delta. \end{aligned}$$

The right-hand side is $(T_c V_c^\pi)(s) - \kappa - 2\delta$, and $V_c^\pi \le T_c V_c^\pi$. So $\|T_c V_c^\pi - V_c^\pi\|_\infty \le \kappa + 2\delta$, and Theorem F.4(iv),(iii) give the first bound. The second follows from $Q_c^\star - Q_c^\pi = \gamma \int P(\,\mathrm{d}s' \mid \cdot,\cdot)(V_c^\star - V_c^\pi)(s')$, and the third from the triangle inequality. For the greedy policy of $\hat Q$, $\sum_a \pi(a \mid s)\hat Q(s,a) = \max_a \hat Q(s,a)$. □

Theorem F.17. [FB-6 | S+D] *Assume Theorem F.1, without any exactness, and let (π_z) be any policy family. Define the* error kernel $\mathcal{E}_z(s,a,\cdot) := M^{\pi_z,+}(s,a,\cdot) - \hat M_z(s,a,\cdot)$ *(FB21 Eq. 98). Let $\|\cdot\|_A$ (on functions) and $\|\cdot\|_B$ (on finite signed measures) satisfy $|\int f\,\mathrm{d}\mu| \le \|f\|_A\|\mu\|_B$ whenever both factors are finite. Examples:*

(N1) $\|f\|_\infty$ and the total variation $\|\mu\|_{\mathrm{TV}} := |\mu|(\mathcal{S})$;

(N2) $\|f\|_{L^2(\nu)}$ and $\|\mu\|_{L^2(\nu)^\star}$ (Theorem F.15);

(N3) for a metric on $\mathcal{S}$ compatible with its topology, $\max(\|f\|_\infty, \|f\|_{\mathrm{Lip}})$ and the Kantorovich–Rubinstein norm of FB21 Eq. (23) (p. 19).

Let r be bounded and measurable with $\|r\|_A < \infty$.

(a) (S; FB21 Eqs. 100–103, p. 29) For every $z \in \mathcal{Z}$ and every (s,a) with $\|\mathcal{E}_z(s,a,\cdot)\|_B < \infty$,

$$|Q_r^{+,\pi_z}(s,a) - F(s,a,z)^\top z_r| \le \|r\|_A\,\|\mathcal{E}_z(s,a,\cdot)\|_B.$$

Greediness is not used.

Let $z_r \in \mathcal{Z}$, assume $\sup_{s,a} \|\mathcal{E}_{z_r}(s,a,\cdot)\|_B < \infty$, and set $\delta := \|r\|_A \sup_{s,a} \|\mathcal{E}_{z_r}(s,a,\cdot)\|_B$.

- *(b) (S+D; FB21 Theorem 8.3, Eqs. 27–28, p. 19, restated in our convention with a direct proof) For the greedy family, with $\pi := \pi^{\mathrm{gr}}_{z_r}$: $0 \le V^\star_r - V^\pi_r \le 2\delta/(1-\gamma)$, $0 \le Q^\star_r - Q^{+,\pi}_r \le 2\gamma\delta/(1-\gamma)$, and $|F(s,a,z_r)^\top z_r - Q^\star_r(s,a)| \le (1+\gamma)\delta/(1-\gamma)$. The same proof in the 2021 convention ($c = r$, M^π_{21}, ρ on $\mathcal{S}\times\mathcal{A}$, error level δ_{21}) gives $2\delta_{21}/(1-\gamma)$, $2\gamma\delta_{21}/(1-\gamma)$ and $(1+\gamma)\delta_{21}/(1-\gamma)$, which imply FB21's bounds $3\delta_{21}/(1-\gamma)$ (Eq. 27, proved as Eq. 108) and $2\delta_{21}/(1-\gamma)$ (Eq. 28, proved as Eq. 109).*
- *(c) (D) For the softmax family, with $\pi := \pi_{z_r,\tau}$: $0 \le V^\star_r - V^\pi_r \le (\tau \log|\mathcal{A}| + 2\delta)/(1-\gamma)$ and $|F(s,a,z_r)^\top z_r - Q^\star_r(s,a)| \le \delta + \gamma(\tau\log|\mathcal{A}| + 2\delta)/(1-\gamma)$.*
- *(d) (D) If $r \ge 0$, Theorem F.14 turns (b) and (c) into bounds on the H-step gap $V^\star_H - V^\pi_H$ by adding $\gamma^H \|r\|_\infty/(1-\gamma)$.*
- *(e) (D; scaled latent) Suppose instead that $\zeta z_r \in \mathcal{Z}$ for some $\zeta > 0$ (for instance the representative of z_r on a sphere), $\sup_{s,a} \|\mathcal{E}_{\zeta z_r}(s,a,\cdot)\|_B < \infty$, and $\eta := \|r\|_A \sup_{s,a} \|\mathcal{E}_{\zeta z_r}(s,a,\cdot)\|_B$. Then, for $\pi := \pi_{\zeta z_r,\tau}$,*

$$0 \le V^\star_r - V^\pi_r \le \frac{(\tau/\zeta)\log|\mathcal{A}| + 2\eta}{1-\gamma},$$
$$|F(s,a,\zeta z_r)^\top z_r - Q^\star_r(s,a)| \le \eta + \frac{\gamma((\tau/\zeta)\log|\mathcal{A}| + 2\eta)}{1-\gamma},$$

and the same bounds without the τ terms hold for $\pi := \pi^{\mathrm{gr}}_{\zeta z_r}$.

Proof. (a) By Theorem F.3(a), $Q^{+,\pi_z}_r(s,a) = \int r \,\mathrm{d}M^{\pi_z,+}(s,a,\cdot)$; as in the proof of Theorem F.6(a), $\int r \,\mathrm{d}\hat{M}_z(s,a,\cdot) = F(s,a,z)^\top z_r$. The difference is $\int r \,\mathrm{d}\mathcal{E}_z(s,a,\cdot)$, which the norm inequality bounds since both factors are finite. The examples satisfy the inequality: (N1) because $|\int f \,\mathrm{d}\mu| \le \int |f| \,\mathrm{d}|\mu|$; (N2) by Theorem F.15; (N3) because $f/\max(\|f\|_\infty, \|f\|_{\mathrm{Lip}})$ is 1-Lipschitz and bounded by 1, which is the class in the definition of the Kantorovich–Rubinstein norm.

(b)–(c) Apply Theorem F.16 with $c = c_r$, $\hat{Q} := F(\cdot,\cdot,z_r)^\top z_r$ (bounded and measurable by Theorem F.1(iv)), and δ from (a), using $Q^{+,\pi}_r = Q^\pi_{c_r}$ (Theorem F.3(b)). For the greedy family $\kappa = 0$; for the softmax family $\kappa = \tau\log|\mathcal{A}|$. In the 2021 convention the same lemma applies with $c = r$ and $\hat{Q} = F^\top z_R$, and FB21's constants follow from $2 \le 3$, $2\gamma \le 3$ and $1+\gamma \le 2$.

(d) is Theorem F.14.

(e) By (a) at $z = \zeta z_r$, $\|Q^{+,\pi}_r - F(\cdot,\cdot,\zeta z_r)^\top z_r\|_\infty \le \eta$, where π is either policy at the latent ζz_r. Since $\pi_{\zeta z_r,\tau}(\cdot \mid s) = \mathrm{softmax}(F(s,\cdot,\zeta z_r)^\top z_r/(\tau/\zeta))$, Theorem F.16 with $c = c_r$, $\hat{Q} := F(\cdot,\cdot,\zeta z_r)^\top z_r$, $\delta := \eta$ and $\kappa := (\tau/\zeta)\log|\mathcal{A}|$ gives both bounds. The first maximizers of $a \mapsto F(s,a,\zeta z_r)^\top(\zeta z_r)$ and of $\hat{Q}(s,\cdot)$ coincide because $\zeta > 0$, so $\pi^{\mathrm{gr}}_{\zeta z_r}$ is the greedy policy of $\hat{Q}$ and $\kappa = 0$ applies. □

Remark F.18 (what FB-6 does and does not give).

- FB-6 is conditional on the model error (δ, or η in (e)) of a given (F,B) at the latent actually used. Nothing in Theorem F.22 bounds that error by the training loss. That loss is not a norm of $\mathcal{E}_z$, since it omits a parameter-independent and possibly infinite constant; it is minimized only approximately, on the replay distribution, and not uniformly over (s,a).
- In the finite maze, the error of a trained network can be computed exactly from P by the oracle. This is an evaluation-only diagnostic and never enters a learner.
- In the continuous maze, if $M^{\pi_z,+}(s,a,\cdot)$ has a part singular to ν (Theorem F.25), then the TV error of the model kernel is at least the singular mass, so the (N1) error level is at least $\|r\|_\infty$ times that mass, and the (N2) error level is $+\infty$. These are lower bounds on errors of measures, not on the error $|\int r \,\mathrm{d}\mathcal{E}_z(s,a,\cdot)|$ for a particular reward, which can still be small (for instance when r vanishes near the singular part). The (N3) error level can also be small: FB21 §B.6 calls this weak approximation, and it remains useful, in particular for the smooth evaluation reward r_g.

F.8 Successor features of B

Proposition F.19 (FI24 Proposition 1, p. 4, second part)**.** [FB-7 | S] *Assume Theorem F.1 and exactness at z, for any family. Then for all (s,a),*

$$G_B F(s,a,z) = \int B \,\mathrm{d}M^{\pi_z,+}(s,a,\cdot) = \mathbb{E}^{\pi_z}_{(s,a)} \sum_{t\geq 0} \gamma^t B(s_{t+1}), \qquad G_B F(s,z) = \int B \,\mathrm{d}M^{\pi_z,+}(s,\cdot),$$

where $F(s,z) := \sum_a \pi_z(a \mid s) F(s,a,z)$ (FI24 Eq. 3). If G_B is invertible, $F(s,a,z) = \mathbb{E}^{\pi_z}_{(s,a)} \sum_t \gamma^t \varphi(s_{t+1})$ with $\varphi := G_B^{-1} B$ (FI24 Eq. 5). Greediness is not used (FI24 states the result under its exact model (2), whose policies are greedy).

Proof. For each coordinate i, $\int B_i \,\mathrm{d}\hat{M}_z(s,a,\cdot) = \int B_i B^\top F(s,a,z) \,\mathrm{d}\nu = (G_B F(s,a,z))_i$. Exactness replaces $\hat{M}_z$ by $M^{\pi_z,+}$, and Theorem F.3(a) with the bounded reward B_i gives the expectation. Average over $a \sim \pi_z(\cdot \mid s)$ for the second identity, and multiply by G_B^{-1} for the last. □

F.9 The Gram matrix

Proposition F.20. [FB-10 | I+D] *Assume Theorem F.1.*

(a) (I) G_B is symmetric positive semidefinite, with $w^\top G_B w = \|w^\top B\|^2_{L^2(\nu)}$. It is invertible if and only if $B_1, \dots, B_d$ are linearly independent in $L^2(\nu)$. If $\mathcal{S}$ is finite and $d > \#\{s : \nu(s) > 0\}$, it is singular.

(b) (I) G_B is uncentered*. With $m_B := \nu(\mathcal{S})^{-1} \int B \,\mathrm{d}\nu$, the covariance is $\mathrm{Cov}_\nu(B) := \nu(\mathcal{S})^{-1} \int (B - m_B)(B - m_B)^\top \,\mathrm{d}\nu = \nu(\mathcal{S})^{-1} G_B - m_B m_B^\top$.*

(c) (D) Under exactness at z, for any family, $F(s,a,z)^\top \int B \,\mathrm{d}\nu = 1/(1-\gamma)$ for every (s,a). Hence $\int B \,\mathrm{d}\nu \neq 0$ and $\mathrm{Cov}_\nu(B) \neq \nu(\mathcal{S})^{-1} G_B$. The identities derived from exactness that involve a second-moment matrix of B (FB-7, FBIL-2) involve G_B, never the covariance, and B must not be constrained to have ν-mean zero if exactness is the target.

(d) (S+D; FB21 Remark 7, p. 18, for greedy policies; the softmax family and the statement on exactness are ours) For invertible $C \in \mathbb{R}^{d\times d}$, let $B' := CB$ and $F'(s,a,z) := C^{-\top} F(s,a,C^{-1}z)$. Then $\hat{M}'_z = \hat{M}_{C^{-1}z}$ and $F'(s,a,z)^\top z = F(s,a,C^{-1}z)^\top (C^{-1}z)$, so $\pi'^{\mathrm{gr}}_z = \pi^{\mathrm{gr}}_{C^{-1}z}$ and $\pi'_{z,\tau} = \pi_{C^{-1}z,\tau}$ with the same τ. Moreover $z'_r = Cz_r$, $F'(s,a,z'_r)^\top z'_r = F(s,a,z_r)^\top z_r$, and $G_{B'} = C G_B C^\top$. The primed representation is exact on $C\mathcal{Z}$ if and only if the original one is exact on $\mathcal{Z}$. If G_B is invertible, $C := G_B^{-1/2}$ gives $G_{B'} = I$; within this family of transformations, two representations with Gram matrix I differ by an orthogonal C. (Exact softmax representations with the same B need not be related by any C if softmax fixed points are not unique, Theorem F.10(c).)

(e) (I) Let B be fixed. If ν is a probability and $x_1, \dots, x_N$ are i.i.d. $\sim \nu$ and independent of B, then $\hat{G}_B := N^{-1} \sum_j B(x_j) B(x_j)^\top$ is unbiased and converges to G_B almost surely. For the trained B_ω evaluated on the buffer it was trained on, $\hat{G}_B$ is the exact Gram matrix of the empirical measure $\hat{\nu}$ of the buffer next states; no unbiasedness for the population G_B is claimed, because the buffer is a single dependent sample and B_ω depends on it. The implemented updates never invert G_B (Theorems F.22 and F.24, and FBIL-1 to FBIL-5). G_B^{-1} appears only in interpretations (FB-7, FBIL-1(d), FBIL-2(b),(c)), where invertibility is assumed. A diagnostic that needs an inverse must use an explicitly stated ridge $(G_B + \lambda I)^{-1}$, $\lambda > 0$, which only approximates G_B^{-1}.

Proof. (a) $w^\top G_B w = \int (w^\top B)^2 \,\mathrm{d}\nu \geq 0$. This vanishes for some $w \neq 0$ exactly when $w^\top B = 0$ ν-a.e., i.e. when the coordinates are linearly dependent in $L^2(\nu)$. In the finite case $G_B = \sum_{\nu(s)>0} \nu(s) B(s) B(s)^\top$ has rank at most $\#\{\nu > 0\}$.

(b) Expand $(B - m_B)(B - m_B)^\top$ and integrate.

(c) $M^{\pi_z,+}(s,a,\mathcal{S}) = 1/(1-\gamma)$ by Theorem B.4(a), and $\hat{M}_z(s,a,\mathcal{S}) = F(s,a,z)^\top \int B \,\mathrm{d}\nu$; exactness equates the two. If $\int B \,\mathrm{d}\nu = 0$, the right-hand side would vanish. Then $m_B \neq 0$, so $m_B m_B^\top \neq 0$.

(d) $\hat{M}'_z(s,a,\mathrm{d}x) = F(s,a,C^{-1}z)^\top C^{-1} CB(x)\nu(\mathrm{d}x) = \hat{M}_{C^{-1}z}(s,a,\mathrm{d}x)$, and $F'(s,a,z)^\top z = F(s,a,C^{-1}z)^\top C^{-1}z$. Both policy families are functions of these scores, so $\pi'_z = \pi_{C^{-1}z}$. Next, $z'_r = \int rCB \,\mathrm{d}\nu = Cz_r$, hence $C^{-1}z'_r = z_r$; and $G_{B'} = \int CBB^\top C^\top \,\mathrm{d}\nu$. Since $M^{\pi'_z,+} = M^{\pi_{C^{-1}z},+}$ and $\hat{M}'_z = \hat{M}_{C^{-1}z}$, the primed representation is exact at z if and only if the original is exact at $C^{-1}z$. With the symmetric positive definite square root $G_B^{1/2}$, $C = G_B^{-1/2}$ gives $CG_BC^\top = I$. If $G_{B'} = I$ and $G_{\tilde{C}B'} = \tilde{C}\tilde{C}^\top = I$, then $\tilde{C}$ is orthogonal.

(e) The entries of $BB^\top$ are bounded by β_B^2; linearity of expectation and the strong law of large numbers apply entrywise. □

F.10 The pretraining loss in the next-state convention

Assumption F.21 (A-DATA, A-NET)**.**

(i) Replay transitions (s,a,s_1) have a law D of the form $\mathsf{D}(\mathrm{d}s, \mathrm{d}a, \mathrm{d}s_1) = \mathsf{D}_{\mathrm{SA}}(\mathrm{d}s, \mathrm{d}a)\, P(\mathrm{d}s_1 \mid s,a)$. This holds when D is the law of a transition drawn from the population replay process (the mixture over collection times and segments) and neither the storage nor the sampling of a transition depends on its next state s_1 beyond (s,a). Uniform sampling from a buffer of real environment steps, collected by a task-blind behaviour policy with any reset scheme, satisfies this; discarding transitions according to where they land would not. Reset times are not terminal: the environment is continuing (Section 2), so no transition is masked as an episode end.

(ii) ν is the law of s_1 under D (the replay next states): $\nu = \int \mathsf{D}_{\mathrm{SA}}(\mathrm{d}s, \mathrm{d}a) P(\cdot \mid s,a)$.

(iii) Latents $z \sim \mathcal{D}_z$, a law on $\mathcal{Z}$, are independent of the transitions; target states $x \sim \nu$ are independent of everything else.

(iv) The online parameters (ξ,ω) range over open sets. For every (s,a,z,x) in the supports, $(\xi,\omega) \mapsto (F_\xi(s,a,z), B_\omega(x))$ is C^1, and the values and parameter gradients are bounded uniformly in (s,a,z,x), locally uniformly in (ξ,ω). The target parameters (ξ^-,ω^-) are fixed during a gradient step, and $F^- := F_{\xi^-}$, $B^- := B_{\omega^-}$ are bounded.

Multilayer perceptrons with C^1 activations (for example tanh, ELU or SiLU) on a bounded state domain and a compact latent domain satisfy (iv). ReLU networks, used by FB21 (D.2, p. 35) and FI24 (D.3, p. 28), are only piecewise C^1: for them, parts (b)–(c) below and Theorem F.24 are not proved [SA U-14]; the implemented networks use LayerNorm, tanh and SiLU, which are C^∞. The target policy and the target expected features are those of FB21 Algorithm 1, line 18:

$$\pi_z^-(a \mid s) := \mathrm{softmax}_a(F^-(s,a,z)^\top z/\tau), \qquad \bar{F}^-(s,z) := \sum_a \pi_z^-(a \mid s) F^-(s,a,z). \tag{F.3}$$

The per-sample, population and minibatch losses are

$$\begin{aligned}\ell(\xi,\omega;s,a,s_1,z) &:= \tfrac{1}{2}\int \left(F_\xi(s,a,z)^\top B_\omega(x) - \gamma \bar{F}^-(s_1,z)^\top B^-(x)\right)^2 \nu(\mathrm{d}x) \\ &\quad - F_\xi(s,a,z)^\top B_\omega(s_1), \qquad (\text{F.4})\\ L(\xi,\omega) &:= \mathbb{E}_{z\sim\mathcal{D}_z}\mathbb{E}_{(s,a,s_1)\sim\mathsf{D}}\, \ell(\xi,\omega;s,a,s_1,z), \qquad (\text{F.5})\\ \hat{L} &:= \frac{1}{2bb'}\sum_{i=1}^{b}\sum_{j=1}^{b'} \left(F_\xi(s_i,a_i,z_i)^\top B_\omega(x_j) - \gamma \bar{F}^-(s_{i+1},z_i)^\top B^-(x_j)\right)^2 \\ &\quad - \frac{1}{b}\sum_{i=1}^{b} F_\xi(s_i,a_i,z_i)^\top B_\omega(s_{i+1}), \qquad (\text{F.6})\end{aligned}$$

where $(s_i,a_i,s_{i+1}) \sim \mathsf{D}$, $z_i \sim \mathcal{D}_z$ and $x_j \sim \nu$ are i.i.d. and mutually independent. For fixed z, the *target measure* is the Bellman operator of Theorem F.5(a) applied to the target model with the

target policy:

$$(\mathcal{T}_z\hat{M}^-)(s,a,\mathrm{d}x) := P(\mathrm{d}x \mid s,a) + \gamma \int P(\mathrm{d}s_1 \mid s,a)\, \bar{F}^-(s_1,z)^\top B^-(x)\, \nu(\mathrm{d}x).$$

Proposition F.22. [FB-8 | D] *Assume Theorem F.1 for (F_ξ, B_ω) and Theorem F.21.*

(a) (Bellman residual) Fix (s,a,z) and write $\hat{m}(x) := F_\xi(s,a,z)^\top B_\omega(x)$ and $\bar{n}(x) := \gamma \int P(\mathrm{d}s_1 \mid s,a)\bar{F}^-(s_1,z)^\top B^-(x)$. Then

$$\int P(\mathrm{d}s_1 \mid s,a)\, \ell(\xi,\omega;s,a,s_1,z) = \tfrac{1}{2}\int \hat{m}^2\,\mathrm{d}\nu - \int \hat{m}\,\mathrm{d}P(\cdot \mid s,a) - \int \hat{m}\,\bar{n}\,\mathrm{d}\nu + C_1,$$

where C_1 does not depend on (ξ,ω). If $P(\cdot \mid s,a) \ll \nu$ with $p_\nu := \mathrm{d}P(\cdot \mid s,a)/\mathrm{d}\nu \in L^2(\nu)$, this equals $\frac{1}{2}\|\hat{M}_z(s,a,\cdot) - (\mathcal{T}_z\hat{M}^-)(s,a,\cdot)\|^2_{L^2(\nu)^\star} - C_z(s,a)$, with $C_z(s,a)$ finite and independent of (ξ,ω). Otherwise the residual norm is $+\infty$ (Theorem F.15), and L is not a residual minus a finite constant: it is defined by (F.4)*. If moreover $P(N \mid s,a) > 0$ for a ν-null set N, then over model classes that can concentrate mass near N the left-hand side is unbounded below (the term $-\int \hat{m}\,\mathrm{d}P$ rewards $\hat{m}$ large on N at no $L^2(\nu)$ cost).*

(b) (unbiased minibatch; closed form) $\mathbb{E}\hat{L} = L$ and $\mathbb{E}\nabla_{\xi,\omega}\hat{L} = \nabla_{\xi,\omega}L$. The double sum in $\hat{L}$ equals $\frac{1}{2b}\sum_i [F_i^\top \hat{G} F_i - 2\gamma F_i^\top \hat{G}_\times \bar{F}_i^- + \gamma^2 \bar{F}_i^{-\top}\hat{G}^-\bar{F}_i^-]$, where $F_i := F_\xi(s_i,a_i,z_i)$, $\bar{F}_i^- := \bar{F}^-(s_{i+1},z_i)$, $\hat{G} := \frac{1}{b'}\sum_j B_\omega(x_j)B_\omega(x_j)^\top$, $\hat{G}_\times := \frac{1}{b'}\sum_j B_\omega(x_j)B^-(x_j)^\top$ and $\hat{G}^- := \frac{1}{b'}\sum_j B^-(x_j)B^-(x_j)^\top$. If the target states are not independent of the transitions (for instance $x_i := s_{i+1}$), the terms with $i = j$ are biased. These identities use only the independence of the draws, not Theorem F.21(i): they hold for whatever law D the transitions are drawn from, with ν the law of its next states. With minibatches drawn from a stored buffer, D and ν are the empirical laws $\hat{\mathsf{D}}$ and $\hat{\nu}$, so (b) gives unbiasedness for the empirical (buffer) loss and its gradient, not for the population loss (Theorem F.27(2)).

(c) (exactness implies stationarity) Suppose $(\xi^-,\omega^-) = (\xi,\omega)$ and the softmax representation $(F_\xi, B_\omega, (\pi_{z,\tau}))$ is exact on $\mathcal{S}\times\mathcal{A}$ for every z in the support of $\mathcal{D}_z$. Then $\nabla_{\xi,\omega}L = 0$, the gradient being taken with the target held fixed.

Proof. (a) Expand the square in (F.4) and integrate over $s_1 \sim P(\cdot \mid s,a)$; all terms are bounded, so Fubini's theorem exchanges $\int P(\mathrm{d}s_1 \mid s,a)$ and $\int \nu(\mathrm{d}x)$. This gives the displayed form with $C_1 := \frac{\gamma^2}{2}\int P(\mathrm{d}s_1 \mid s,a)\int (\bar{F}^-(s_1,z)^\top B^-(x))^2 \nu(\mathrm{d}x)$. If $P(\cdot \mid s,a) = p_\nu \nu$ with $p_\nu \in L^2(\nu)$, then $\int \hat{m}\,\mathrm{d}P(\cdot \mid s,a) = \int \hat{m} p_\nu\,\mathrm{d}\nu$, and $(\mathcal{T}_z\hat{M}^-)(s,a,\cdot)$ has density $p_\nu + \bar{n} \in L^2(\nu)$. Expanding,

$$\tfrac{1}{2}\|\hat{m} - p_\nu - \bar{n}\|^2_{L^2(\nu)} = \tfrac{1}{2}\int \hat{m}^2\,\mathrm{d}\nu - \int \hat{m}(p_\nu + \bar{n})\,\mathrm{d}\nu + \tfrac{1}{2}\|p_\nu + \bar{n}\|^2_{L^2(\nu)},$$

so the claim holds with $C_z(s,a) := \frac{1}{2}\|p_\nu + \bar{n}\|^2_{L^2(\nu)} - C_1$, which depends only on the kernel and the target parameters. By Theorem F.15, the first expression is $\frac{1}{2}\|\hat{M}_z(s,a,\cdot) - (\mathcal{T}_z\hat{M}^-)(s,a,\cdot)\|^2_{L^2(\nu)^\star}$. If $P(\cdot \mid s,a)$ has no square-integrable ν-density, the difference of the two measures does not have one either, since $\hat{M}_z$ and the γ-part have bounded densities; the norm is then $+\infty$ by Theorem F.15. For the last claim, a model with $\hat{m} \geq 0$, $\hat{m} = K$ on an open neighbourhood O of N with $\nu(O) \leq K^{-2}$, and $\hat{m} = 0$ far from N has $\int \hat{m}^2\,\mathrm{d}\nu \leq 1$ and $-\int \hat{m}\,\mathrm{d}P \leq -KP(N \mid s,a)$; letting $K \to \infty$ makes the expression tend to $-\infty$ when the class contains such $\hat{m}$.

(b) In the double sum, every pair (i,j) consists of independent samples from $\mathsf{D}\otimes\mathcal{D}_z$ and from ν, so its expectation is $\mathbb{E}_{z,\mathsf{D}}\frac{1}{2}\int(\cdots)^2\,\mathrm{d}\nu$; the single sum has expectation $\mathbb{E}_{z,\mathsf{D}}F_\xi(s,a,z)^\top B_\omega(s_1)$. Hence $\mathbb{E}\hat{L} = L$. Under Theorem F.21(iv) the integrands have parameter gradients bounded uniformly in the data, locally uniformly in the parameters. The mean value theorem and dominated convergence then allow differentiation under the expectation, as in Theorem D.4. The closed form follows from $(F^\top B_j)^2 = F^\top B_j B_j^\top F$ and $(F^\top B_j)(\bar{F}^{-\top}B_j^-) = F^\top B_j B_j^{-\top}\bar{F}^-$. If $x_i = s_{i+1}$, the $i = j$ term pairs a target state with its own transition, so the independence used above fails.

(c) Fix (s, a, z) and a direction v in parameter space, and write $\partial_v \hat{m}$ for the directional derivative of $\hat{m}$, with the target held fixed. Differentiating under the integrals (justified as in (b)),

$$\partial_v \int P(\mathrm{d}s_1 \mid s, a)\, \ell = \int \partial_v \hat{m}\ \mathrm{d}\Gamma, \quad \text{where}$$
$$\Gamma(\mathrm{d}x) := \hat{M}_z(s, a, \mathrm{d}x) - \gamma \int P(\mathrm{d}s_1 \mid s, a)\, \bar{F}^-(s_1, z)^\top B^-(x)\, \nu(\mathrm{d}x) - P(\mathrm{d}x \mid s, a).$$

With $(\xi^-, \omega^-) = (\xi, \omega)$ we have $\pi_z^- = \pi_{z,\tau}$, and $\bar{F}^-(s_1, z)^\top B(x)\nu(\mathrm{d}x) = \sum_{a_1} \pi_{z,\tau}(a_1 \mid s_1)\hat{M}_z(s_1, a_1, \mathrm{d}x)$. Exactness on all of $\mathcal{S} \times \mathcal{A}$ replaces every $\hat{M}_z$ by $M^{\pi_{z,\tau},+}$, so $\Gamma = 0$ by Theorem F.5(a). Integrating over $\mathsf{D}_{\mathrm{SA}} \otimes \mathcal{D}_z$ and differentiating under the expectation gives $\partial_v L = 0$ for every v. □

Remark F.23 (what the loss does not give; the diagonal term).

- *No converse, no convergence.* (c) is one-directional: stationary points of the semi-gradient field need not be exact. At $d = 16$ in the finite maze exactness is impossible under the condition of Theorem F.9. The semi-gradient need not be the gradient of any fixed objective, because the target moves (TD learning; FB21 p. 2: "learning seeks a fixed point but the loss itself is not observable"). No convergence of Adam with Polyak targets is claimed, and the loss value, which omits C_z, cannot certify the error of FB-6.
- *The implemented ω-update.* (c) concerns L only. The ω-update descends $L + \lambda_{\mathrm{orth}} L_{\mathrm{reg}}$ (Theorem F.24); an exact representation is stationary for it only if moreover $\nabla_\omega \mathcal{R} = 0$, for instance if $G_B = I$.
- *Only the replay support is constrained.* L weights (s, a, z) by $\mathsf{D}_{\mathrm{SA}} \otimes \mathcal{D}_z$. Pairs or latents outside the support are extrapolation. Exactness at all next states is needed in (c), because the target is evaluated at s_1.
- *Diagonal term and time shift.* The diagonal term of FB21 Eq. (164) (p. 36) is

  $$-\frac{1}{b} \sum_i F(s_i, a_i, z_i)^\top B(s_i, a_i),$$

  the integral of $\hat{m}$ against the $t = 0$ atom $\delta_{(s,a)}$ of M_{21}^π (Theorem F.5(c)–(d)). In the next-state convention the $t = 0$ term is $P(\cdot \mid s, a)$, whose sample is the observed s_{i+1}. Hence our diagonal term is

  $$-\frac{1}{b} \sum_i F(s_i, a_i, z_i)^\top B(s_{i+1}).$$

  FB21 §B.6 (p. 24) describes a "$t = 1$" variant whose measure is $\sum_{t \geq 1} \gamma^t \mathbb{P}(s_t \in \cdot) = \gamma M^{\pi,+}$ for state-only targets, which is why its diagonal term carries a factor γ; ours does not. We read the pair (s', a') in that bullet as the next pair (s_1, a_1) of the transition, not as the independent target sample of FB21 Eq. (5).
- *FI24.* FI24 Theorem 5 (p. 19) is a population analogue in a different setting: it concerns z-gradients of a Bellman-gap loss with expert transitions, action-averaged $F(s, z)$ and the population $\mathrm{Cov}\, B$. Its quadratic (Eq. 39) equals twice ours after replacing $\mathrm{Cov}\, B$ by $\hat{G}$ and setting $\hat{G}_\times = \hat{G}^- = \hat{G}$.

F.11 The orthonormality penalty

Proposition F.24 (FB21 Eqs. 165–166, pp. 36–37)**.** [FB-9 | S] *Let ν be a probability and let B_ω satisfy Theorem F.21(iv). Let $\mathcal{R}(\omega) := \frac{1}{4}\|G_{B_\omega} - I\|_F^2$.*

(a) With $\partial_\omega B_\omega(x) \in \mathbb{R}^{d \times p}$ the parameter Jacobian,

$$\nabla_\omega \mathcal{R} = \mathbb{E}_{(x,x') \sim \nu \otimes \nu}\Big[\partial_\omega B_\omega(x)^\top \big(B_\omega(x') B_\omega(x')^\top B_\omega(x) - B_\omega(x)\big)\Big]. \tag{F.7}$$

(b) Let $x_1, \dots, x_b$ and $x'_1, \dots, x'_{b'}$ be independent i.i.d. samples of ν, and let $\lfloor\cdot\rfloor$ denote a stop-gradient (a value treated as a constant when differentiating). Then

$$L_{\mathrm{reg}}(\omega) := \frac{1}{bb'} \sum_{i,j} B_\omega(x_i)^\top \lfloor B_\omega(x'_j)\rfloor \, \lfloor B_\omega(x_i)^\top B_\omega(x'_j)\rfloor - \frac{1}{b} \sum_i B_\omega(x_i)^\top \lfloor B_\omega(x_i)\rfloor \tag{F.8}$$

satisfies $\mathbb{E}\nabla_\omega L_{\mathrm{reg}} = \nabla_\omega \mathcal{R}$. With batches drawn from a stored buffer, ν is the empirical law $\hat{\nu}$, and $\mathcal{R}$ is the penalty for the Gram matrix of $\hat{\nu}$.

(c) $\mathcal{R} = 0$ if and only if $G_B = I$. The penalty is soft: the ω-update descends $L + \lambda_{\mathrm{orth}} L_{\mathrm{reg}}$, $G_B \neq I$ in general, and no claim is made that its fixed points satisfy $G_B = I$ or minimize L. Only on $\{\mathcal{R} = 0\}$ does the freedom of Theorem F.20(d) reduce to orthogonal transformations.

Proof. (a) $\|G - I\|_F^2 = \mathrm{tr}(G^2) - 2\,\mathrm{tr}\,G + d$. Since $G_{ij} = \mathbb{E}B_i(x)B_j(x)$, $\mathrm{tr}(G^2) = \sum_{ij} G_{ij}^2 = \mathbb{E}_{x,x'}[(B(x)^\top B(x'))^2]$, and $\mathrm{tr}\,G = \mathbb{E}\|B(x)\|^2$. Differentiating under the expectation (dominated convergence, as in Theorem F.22(b)), $\nabla \mathrm{tr}(G^2) = 2\mathbb{E}[(B(x)^\top B(x'))(\partial B(x)^\top B(x') + \partial B(x')^\top B(x))] = 4\mathbb{E}[(B(x)^\top B(x'))\,\partial B(x)^\top B(x')]$, where the second step exchanges the i.i.d. x and x'; and $\nabla \mathrm{tr}\,G = 2\mathbb{E}[\partial B(x)^\top B(x)]$. Divide by 4 and use $(B(x)^\top B(x'))B(x') = B(x')B(x')^\top B(x)$.

(b) $\nabla L_{\mathrm{reg}} = \frac{1}{bb'} \sum_{i,j} \partial B(x_i)^\top B(x'_j)(B(x_i)^\top B(x'_j)) - \frac{1}{b}\sum_i \partial B(x_i)^\top B(x_i)$. Every pair (x_i, x'_j) is independent, so the expectation is (a).

(c) $\|G - I\|_F = 0$ if and only if $G = I$. If $G_B = I$ and $G_{CB} = CC^\top = I$, then C is orthogonal. □

In FB21 Algorithm 1 (line 21), the first batch is the transition batch (s_i, a_i) and the second the target batch, both distributed as ρ there. In our convention both batches must be ν-distributed: we take $x_i := s_{i+1}$, the next states of the transition batch (ν-distributed by Theorem F.21(ii)), and x'_j the independent target batch. Using the current states s_i instead would whiten B under a different law and break the single-reference requirement [SA A3].

F.12 Domination and singular measures

Proposition F.25. [FB-11 | D] *Under Theorem B.1, let ν be a finite measure on $\mathcal{S}$.*

(a) $\hat{M}_z(s, a, \cdot) \ll \nu$ for all (s, a, z) and every (F, B) satisfying Theorem F.1. Hence if $M^{\pi_z,+}(s, a, \cdot)$ is not absolutely continuous with respect to ν for some (s, a), no exact representation at z exists. This happens for every π and z as soon as $P(X \mid s, a) > 0$ for some X with $\nu(X) = 0$, because $M^{\pi,+}(s, a, \cdot) \geq P(\cdot \mid s, a)$.

(b) (sufficient condition) Let λ be a σ-finite measure on $\mathcal{S}$. Suppose $P(\mathrm{d}x \mid s, a) = p(x \mid s, a)\lambda(\mathrm{d}x)$ with p jointly measurable and $p \leq p_{\max}$, and $\nu(\mathrm{d}x) = n(x)\lambda(\mathrm{d}x)$ with $n \geq n_{\min} > 0$ on $\mathcal{S}$ (so $\lambda(\mathcal{S}) \leq \nu(\mathcal{S})/n_{\min} < \infty$). Then for every stationary π and every (s, a), $M^{\pi,+}(s, a, \cdot) \ll \nu$, with density $m^\pi(s, a, \cdot) \leq p_{\max}/(n_{\min}(1 - \gamma))$; in particular $m^\pi(s, a, \cdot) \in L^2(\nu)$.

(c) (singular mass) Let $M^{\pi_z,+}(s, a, \cdot) = M_{\mathrm{ac}} + M_{\mathrm{sing}}$ be the Lebesgue decomposition with respect to ν. For every (F, B) satisfying Theorem F.1, $\|M^{\pi_z,+}(s, a, \cdot) - \hat{M}_z(s, a, \cdot)\|_{\mathrm{TV}} \geq M_{\mathrm{sing}}(\mathcal{S})$, and the $L^2(\nu)^\star$ norm of the difference is $+\infty$ if $M_{\mathrm{sing}} \neq 0$.

(d) (finite case) If $\mathcal{S}$ is finite and $\nu(s) > 0$ for every s, every measure on $\mathcal{S}$ is absolutely continuous with respect to ν, densities are ratios $M(s, a, \{x\})/\nu(x)$, and all $L^2(\nu)$ quantities are finite sums.

Proof. (a) If $\nu(X) = 0$, then $\hat{M}_z(s, a, X) = \int_X F^\top B \,\mathrm{d}\nu = 0$, while exactness would give $M^{\pi_z,+}(s, a, X) = \hat{M}_z(s, a, X)$. The $t = 0$ term of $M^{\pi,+}(s, a, \cdot)$ is $P(\cdot \mid s, a)$.

(b) Fix (s, a) and let μ_t be the law of (s_t, a_t) under $\mathbb{P}^\pi_{(s,a)}$. By the trajectory construction, $\mathbb{P}^\pi_{(s,a)}(s_{t+1} \in X) = \int \mu_t(\mathrm{d}\bar{s}, \mathrm{d}\bar{a}) P(X \mid \bar{s}, \bar{a}) = \int_X \left[\int \mu_t(\mathrm{d}\bar{s}, \mathrm{d}\bar{a})\, p(x \mid \bar{s}, \bar{a})\right]\lambda(\mathrm{d}x)$ by Tonelli's theorem, and the inner integral is at most $p_{\max}$. By monotone convergence, $M^{\pi,+}(s, a, \cdot) =$

$\sum_t \gamma^t \mathbb{P}^\pi_{(s,a)}(s_{t+1} \in \cdot)$ has a λ-density $g \le p_{\max}/(1-\gamma)$. Since $n \ge n_{\min}$, $\lambda(X) \le \nu(X)/n_{\min}$, so $\lambda \ll \nu$ on $\mathcal{S}$ with $\mathrm{d}\lambda/\,\mathrm{d}\nu = 1/n$. Hence $M^{\pi,+}(s,a,\cdot) \ll \nu$ with density $g/n \le p_{\max}/(n_{\min}(1-\gamma))$. A bounded function is in $L^2(\nu)$ for finite ν.

(c) Let N be a ν-null set with $M_{\text{sing}}(\mathcal{S} \setminus N) = 0$. Then $M_{\text{ac}}(N) = \hat{M}_z(s,a,N) = 0$, so the difference μ satisfies $\mu(N) = M_{\text{sing}}(\mathcal{S})$. Hence $|\mu|(\mathcal{S}) \ge |\mu(N)| = M_{\text{sing}}(\mathcal{S})$. If $M_{\text{sing}} \ne 0$, then μ charges the ν-null set N, so it is not absolutely continuous and Theorem F.15 gives $+\infty$.

(d) A ν-null set is empty when every point has positive mass. □

Remark F.26 (the continuous maze and the 2021 atom). These consequences depend on the design of the kernel; the implemented kernel is analysed in Section H.2.

- The Gaussian part of a move has a Lebesgue density bounded by $1/(2\pi\sigma^2_{\text{noise}})$, but on the bounded free space a kernel needs a rule for moves that leave it or cross a wall, and that rule decides (b). A rule that redraws only the noise while the deterministic part of the move is admissible, and replaces a blocked deterministic move by "stay" before adding noise, keeps a bounded density, because the acceptance probability of the noise near a straight wall or corner is bounded below (about 1/4). Redrawing a full noisy move into an adjacent wall would have acceptance probability of order $\Phi(-(1/7)/0.01)$, which gives no useful bound.
- "Undoing" a move that would cross a wall (FB21 D.1, p. 35) gives $P(\cdot \mid s,a)$ an atom at s. Domination fails from that atom only if $\nu(\{s\}) = 0$. This holds when the replay law of the states from which such moves start has no atom at s (for instance with diffuse resets and Gaussian noise), but not at a state that carries an atom of the replay law, such as a fixed reset state from which a move is undone. Where $\nu(\{s\}) = 0$, (a) and (c) apply at (s,a): no exact representation exists, the $L^2(\nu)^\star$ error of the model kernel is infinite, and its TV error is at least the singular mass, hence at least the probability of the undone move. These are errors of measures; weak (Kantorovich–Rubinstein) approximation, (N3) of FB-6, can still be small and useful (FB21 §B.6).
- Boundary clipping (FB21 D.1: "the agent's position is clipped") puts positive probability on the boundary lines, a Lebesgue-null set. With ν the law of replay next states, clipping during replay collection gives these lines positive ν-mass as well, so clipping alone need not trigger (a) and (c); it does break the bounded-density condition of (b), whose conclusion then has to be checked directly.
- In the 2021 convention, $M^\pi_{21}(s,a,\cdot \times \mathcal{A}) \ge \delta_s$ (Theorem F.5(d)), which is singular to every diffuse reference measure (FB21 §B.6, p. 24). The next-state convention removes this atom but not the atoms created by the kernel. FB21 §B.6 gives sufficient conditions for an ordinary density in the $t = 1$ variant: non-singular kernels, a reference measure with positive density and non-singular policies.
- Empirical measures, such as demonstration states or a finite replay buffer, are atomic and hence singular to every diffuse ν (FBIL-1).

F.13 The implemented model: approximations and design decisions

Remark F.27 (approximations in FB pretraining). Each item names what changes and which result stops applying. [I] marks an implementation approximation.

(1) [I] Finite rank and neural parametrization. $d = 16$, two hidden layers of width 64, float32. Exactness is impossible in the finite maze under the condition of Theorem F.9 and is not claimed in the continuous maze. The results that assume exactness (listed in Theorem F.9) do not apply to the trained model; FB-6 applies with an unknown error, in the scaled form (e) when the latent is projected onto a sphere.

(2) [I] Finite replay buffer. D and ν are replaced by the empirical laws of the buffer. The loss then defines densities with respect to the empirical $\hat{\nu}$, so $\hat{G}_B$, z_r and the orthonormality penalty must use the same buffer next states [SA A3]. In the continuous maze $\hat{\nu}$ is atomic,

and a smooth network relates to the population ν only through generalization; no statement is made about it. The unbiasedness statements of Theorem F.22(b) and Theorem F.24(b) then refer to the empirical buffer loss and the Gram matrix of $\hat{\nu}$, and $\hat{G}_B$ is exact for $\hat{\nu}$ (Theorem F.20(e)). None of them is unbiased for its population counterpart in general: the buffer is a single dependent sample, and the networks are trained on it.

(3) [I] Minibatches, Adam and moving targets. The minibatch gradient $\nabla\hat{L}$ is unbiased for the gradient of the empirical buffer loss (Theorem F.22(b)), not for the population ∇L; an Adam step is not an unbiased step of anything, the targets move, and no convergence is claimed. Theorem F.22(c) shows only that an exact representation with $\xi^- = \xi$ is stationary for L (no converse), and Theorem F.23 adds the condition $\nabla_\omega \mathcal{R} = 0$ for the implemented ω-update.

(4) [I] Latent sampling. The loss constrains the model only on the support of $\mathcal{D}_z$. A latent outside it, such as an unprojected $\mathrm{ER}_{\mathrm{FB}}$ vector, is extrapolation.

(5) [I] Soft orthonormality. $G_B \neq I$ in general; $\mathrm{DM}_{\mathrm{FB}}$ uses the measured $\hat{G}_B$, not I.

(6) [I] Activation. With ReLU networks the gradient statements of Theorem F.22(b)–(c) and Theorem F.24 are not proved [SA U-14].

(7) No finite-horizon truncation enters pretraining: transitions bootstrap through segment ends (Theorem F.21(i)). Truncation enters only in $\mathrm{DM}_{\mathrm{FB}}$ (FBIL-3) and in H-step evaluation (Theorem F.14).

(8) Temperature. $\tau > 0$: the policy family is not greedy, and FB-5 replaces FB-2(b).

Remark F.28 (design decisions: network copy, latent domain, temperature, targets).

D1. [SA U-3] After pretraining, the *online* copy (ξ, ω) defines F, B, $\pi_{z,\tau}$, $\hat{G}_B$, z_{ER}, the $\mathrm{DM}_{\mathrm{FB}}$ objective and evaluation: one copy for all of them [SA A2–A3]. Target copies appear only inside the pretraining target, with the target policy π_z^- computed from F^- (FB21 Algorithm 1, lines 18–19). FB21 also acts with the online network (line 8). Theorem F.22(c) assumes $\xi^- = \xi$; it does not show that the two copies coincide after training.

D2. [SA U-2] Under exactness at ζz_r, Theorem F.12(d)–(e) show that only τ/ζ matters, and that on a sphere $\mathcal{Z}_R$ the effective temperature for a reward r is $\tau\|z_r\|_2/R$, which equals $\tau\|\Pi_B r\|_{L^2(\nu)}/R$ when $G_B = I$. For the trained network this coupling is only empirical. Recommendation: $\mathcal{Z} = \mathcal{Z}_R$ with $R = \sqrt{d}$ (the normalization of FI24 D.3, p. 28), $\mathcal{D}_z$ uniform on $\mathcal{Z}_R$, and ER/DM outputs projected onto $\mathcal{Z}_R$ (FBIL-5). The pilot fixed $\tau = 1$ together with $R = \sqrt{d} = 4$ (Section 10); τ is reported with the measured $\|z_{ER}\|_2$. FB21 instead samples $z = \sqrt{d}\, u\, x/\|x\|$ with Cauchy u and feeds $z/\sqrt{1 + \|z\|^2/d}$ to F (D.3, p. 36). Whatever is chosen must be identical in pretraining, $\mathrm{ER}_{\mathrm{FB}}$, $\mathrm{DM}_{\mathrm{FB}}$ and evaluation.

D3. (targets) Polyak averaging $\xi^- \leftarrow \alpha_{\mathrm{tgt}}\xi^- + (1 - \alpha_{\mathrm{tgt}})\xi$, and likewise for ω^-, after a stated number of updates. FB21 uses $\alpha = 0.95$ after every 40 updates (D.3 and Table 1, pp. 36–37); FI24 uses momentum 0.99 (D.4, p. 29). The implementation uses momentum 0.99 after every update (Section 10).

G FB imitation: $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$

Status and sources. FI24: Eq. (8) and footnote 3 (pp. 4–5) and the reward reading $r = \rho_e/\rho$ (p. 5) for $\mathrm{ER}_{\mathrm{FB}}$; Eq. (12) (p. 5) and Eqs. (30)–(32) with Theorem 3 (p. 17; proof p. 18, Eqs. 33–37) for $\mathrm{DM}_{\mathrm{FB}}$; the warm start of other methods (p. 7; E.2, p. 31); latent normalization (D.3, p. 28). Type labels:

- S+D: FBIL-1 (FI24 Eq. 8 and its reward reading; our additions are the domination condition, the span reward $\tilde{r}_e$ and the limits) and FBIL-2 (FI24 Theorem 3 reproved under our hypotheses, with our successor-feature reading, the extension to approximate models, and the occupancy bound from the start state);
- D: FBIL-3 (estimator, truncation and Gram estimate), FBIL-4 (latent gradient) and FBIL-5 (latent constraint), which are ours.

The section uses the objects of Section F. The pretrained (F, B) is *frozen*: only the latent z is optimized, and no simulator data is collected during latent adaptation. FI24's ρ is our ν; a bar $(\bar{\rho})$ always denotes a start law for $\mathrm{DM}_{\mathrm{FB}}$.

G.1 Setting

Assumption G.1 (A-FBIL)**.**

(i) Theorem F.1 holds for the frozen online copy $(F, B) = (F_\xi, B_\omega)$ (Theorem F.28, decision D1), with the softmax family $\pi_{z,\tau}$, the action-averaged features $F(x, z) := \sum_a \pi_{z,\tau}(a \mid x) F(x, a, z)$, and the averaged model $\hat{M}_z(x, \cdot) := \sum_a \pi_{z,\tau}(a \mid x)\hat{M}_z(x, a, \cdot) = F(x, z)^\top B\, \nu$.

(ii) [SA A4] The expert π_E is a stationary Markov policy.

(iii) The demonstrations are N_E i.i.d. trajectories of π_E from μ_0, of which the states $s_0^i, \dots, s_H^i$ are observed $(i = 1, \dots, N_E)$. Actions are stored for diagnostics only and are not used by the learners.

Definition G.2 (imitation objects)**.**

(a) (FI24 Eq. 8 and footnote 3) $\rho_e := \frac{1}{N_E} \sum_i \frac{1}{H} \sum_{t=1}^H \delta_{s_t^i}$, the empirical expert state law (equal trajectory weight, uniform over $s_1, \dots, s_H$), and its mean $\bar{\rho}_e := \mathbb{E}\rho_e = \frac{1}{H} \sum_{t=1}^H \mathrm{Law}_t^{\pi_E}$. The $\mathrm{ER}_{\mathrm{FB}}$ latent is

$$z_{ER} := \int B \, \mathrm{d}\rho_e = \frac{1}{N_E} \sum_{i=1}^{N_E} \frac{1}{H} \sum_{t=1}^{H} B(s_t^i). \tag{G.1}$$

(b) The expert successor features are $\bar{\psi}_E(x) := \int B \, \mathrm{d}M^{\pi_E,+}(x, \cdot) = \mathbb{E}_{\delta_x}^{\pi_E} \sum_{k \geq 0} \gamma^k B(s_{k+1})$, with $M^{\pi_E,+}(x, \cdot) = \sum_a \pi_E(a \mid x) M^{\pi_E,+}(x, a, \cdot)$. Along demonstration i the targets are

$$\psi_t^{(K),i} := \sum_{k=0}^{K-1} \gamma^k B(s_{t+k+1}^i) \quad (t + K \leq H), \qquad \psi_t^{(\infty),i} := \sum_{k \geq 0} \gamma^k B(s_{t+k+1}^i), \tag{G.2}$$

the latter computed on the infinite (unobserved) continuation. (K is a truncation length, not FI24's number of demonstrations.)

(c) For a probability $\bar{\rho}$ on $\mathcal{S}$ (the *start law*),

$$\mathcal{L}_{DM}(z; \bar{\rho}) := \int \bar{\rho}(\mathrm{d}x) \, \|M^{\pi_{z,\tau},+}(x, \cdot) - M^{\pi_E,+}(x, \cdot)\|_{L^2(\nu)^\star}^2 \qquad \text{(FI24 Eq. 31)}, \tag{G.3}$$

$$\ell_{DM}(z; \bar{\rho}) := \int \bar{\rho}(\mathrm{d}x) \, [F(x, z)^\top G_B F(x, z) - 2F(x, z)^\top \bar{\psi}_E(x)] \qquad \text{(FI24 Eq. 32)}. \tag{G.4}$$

$\mathcal{L}_{DM}$ is used only where its integrand is measurable, for instance under (A1) and (A6) of Theorem G.6, where it is an integral of jointly measurable functions. More generally $\mathcal{L}(\pi; \bar{\rho})$ denotes the same expression for an arbitrary stationary policy π in place of $\pi_{z,\tau}$.

(d) For a finite nonempty deterministic set of start times $\mathcal{T} \subseteq \mathbb{N}$, let $\bar{\rho}_{\mathcal{T}} := |\mathcal{T}|^{-1} \sum_{t \in \mathcal{T}} \mathrm{Law}_t^{\pi_E}$. For $\mathcal{T} \subseteq \{0, \dots, H - K\}$ and a symmetric matrix G, define the estimator

$$\hat{\ell}^{(K)}(z; G) := \frac{1}{N_E |\mathcal{T}|} \sum_{i=1}^{N_E} \sum_{t \in \mathcal{T}} \left[F(s_t^i, z)^\top G F(s_t^i, z) - 2F(s_t^i, z)^\top \psi_t^{(K),i} \right]. \tag{G.5}$$

The implemented objective is $\hat{\ell}^{(K)}(z; \hat{G}_B)$, with $\hat{G}_B$ the frozen Gram estimate of Theorem F.20(e). $\hat{\ell}^{(\infty)}(z; G)$ is the same expression with $\psi_t^{(\infty),i}$, for any finite nonempty deterministic $\mathcal{T} \subseteq \mathbb{N}$.

Lemma G.3 (Markov property at a fixed time)**.** *Under Theorem B.1, let π be stationary Markov, $t \geq 0$, $g : \mathcal{S} \to \mathbb{R}$ and $\Psi : \Omega \to \mathbb{R}$ bounded and measurable, and θ^t the t-fold shift of the proof of (M2). Then $\mathbb{E}_{\mu_0}^{\pi}[g(s_t)\, \Psi \circ \theta^t] = \mathbb{E}_{\mu_0}^{\pi}[g(s_t)\, \bar{\Psi}(s_t)]$ with $\bar{\Psi}(x) := \mathbb{E}_{\delta_x}^{\pi}[\Psi]$.*

Proof. $\bar{\Psi}$ is measurable, and (M1) holds for bounded measurable functions, by the remarks after (F.2). $t = 0$. By (M1), $\mathbb{E}_{\mu_0}[g(s_0)\Psi] = \int \mu_0(\,\mathrm{d}x) \sum_a \pi(a \mid x)\mathbb{E}_{(x,a)}[g(s_0)\Psi] = \int \mu_0(\,\mathrm{d}x)\, g(x) \sum_a \pi(a \mid x)\mathbb{E}_{(x,a)}[\Psi] = \int \mu_0(\,\mathrm{d}x)\, g(x)\bar{\Psi}(x)$, where the last step is (M1) with $\mu_0 = \delta_x$. *One shift.* For any bounded measurable Ξ on Ω, (M1) and (F.2) give $\mathbb{E}_{\mu_0}[\Xi \circ \theta] = \int \mu_0(\,\mathrm{d}x) \sum_a \pi(a \mid x) \int P(\,\mathrm{d}x_1 \mid x, a)\mathbb{E}_{\delta_{x_1}}[\Xi] = \mathbb{E}_{\mu_0}\big[\mathbb{E}_{\delta_{s_1}}[\Xi]\big]$, since s_1 has law $\int \mu_0(\,\mathrm{d}x) \sum_a \pi(a \mid x)P(\cdot \mid x, a)$ (Theorem B.3) and $x_1 \mapsto \mathbb{E}_{\delta_{x_1}}[\Xi]$ is bounded and measurable. *Induction.* Assume the claim for t and every initial law. Apply the one-shift identity to $\Xi := g(s_t)\,\Psi \circ \theta^t$, then the induction hypothesis under δ_{s_1}, then the one-shift identity backwards to $\Xi' := g(s_t)\bar{\Psi}(s_t)$:

$$\mathbb{E}_{\mu_0}[g(s_{t+1})\Psi \circ \theta^{t+1}] = \mathbb{E}_{\mu_0}[\mathbb{E}_{\delta_{s_1}}\Xi] = \mathbb{E}_{\mu_0}[\mathbb{E}_{\delta_{s_1}}\Xi'] = \mathbb{E}_{\mu_0}[\Xi' \circ \theta] = \mathbb{E}_{\mu_0}[g(s_{t+1})\bar{\Psi}(s_{t+1})]. \qquad \square$$

G.2 $\mathbf{ER}_{\mathrm{FB}}$

Proposition G.4. [FBIL-1 | S+D] *Assume Theorem G.1.*

(a) (S; FI24 Eq. 8 and p. 5) If $\rho_e \ll \nu$, then $z_{ER} = z_{r_e}$ with $r_e := \mathrm{d}\rho_e/\,\mathrm{d}\nu$, which is bounded: ρ_e has finitely many atoms x, each with $\nu(\{x\}) > 0$, and $r_e = \rho_e(\{x\})/\nu(\{x\})$ on them and 0 elsewhere (ν-a.e.).

(b) (S+D) Hence, under exactness at z_{ER}:

- *for the greedy family, $\pi^{\mathrm{gr}}_{z_{ER}}$ maximizes $\mathbb{E}\sum_t \gamma^t r_e(s_{t+1})$ from every state (Theorem F.6(b));*
- *for the softmax family, $V^\star_{r_e} - V^{\pi_{z_{ER},\tau}}_{r_e} \le \tau \log|\mathcal{A}|/(1-\gamma)$ in the units of r_e (Theorem F.12(c)), whose maximum is $\|r_e\|_\infty = \max_x \rho_e(\{x\})/\nu(\{x\})$;*
- *after projection onto $\mathcal{Z}_R$, and with exactness there, the effective temperature relative to r_e is $\tau\|z_{ER}\|_2/R$ (Theorem F.12(d)).*

(c) (D) This is a reward-maximization statement, not distribution matching: optimality for r_e does not imply that the next-state occupancy of the policy is close to the expert's occupancy $q_E = d^+_{\pi_E,H}$ (Theorem G.5).

(d) (D; no domination needed) If G_B is invertible, then $z_{ER} = z_{\tilde{r}_e}$ for the bounded reward $\tilde{r}_e := B^\top G_B^{-1} z_{ER} \in \mathrm{span}\{B_1, \dots, B_d\}$, which is as regular as B and is, up to ν-null modifications, the reward of minimal $L^2(\nu)$ norm among all bounded r with $z_r = z_{ER}$. If moreover $\rho_e \ll \nu$, then $\tilde{r}_e = \Pi_B r_e$. When ν is diffuse (as the population replay law of the continuous maze is, Theorem H.10), the atomic ρ_e is singular to ν and $\mathrm{d}\rho_e/\,\mathrm{d}\nu$ does not exist. With the empirical buffer law $\hat{\nu}$, if the kernel has a Lebesgue density (as the implemented kernel has, Theorem H.4) and the demonstrations are independent of the buffer, $\rho_e \ll \hat{\nu}$ fails almost surely. Under exactness, ER_{FB} then corresponds to $\tilde{r}_e$, not to a density ratio.

(e) (D) ρ_e (uniform over $s_1, \dots, s_H$) and the f-PG target $\hat{q}_E$ (weights γ^t/Z_H on s_{t+1}, $t = 0, \dots, H-1$) differ in general, and so do their means $\bar{\rho}_e$ and q_E. They are never interchanged.

(f) (D; sampling) Conditionally on the frozen B, which does not depend on the demonstrations, z_{ER} is the average of N_E i.i.d. trajectory means, each bounded by β_B; it is unbiased for $\int B\,\mathrm{d}\bar{\rho}_e$, with covariance of order $1/N_E$. No expert-recovery claim is made.

Proof. (a) A finite atomic measure is absolutely continuous with respect to ν exactly when ν charges each of its atoms; its density is then the stated ratio on the atoms and 0 elsewhere, which is bounded. By the definition of the Radon–Nikodym derivative, $\int r_e B\,\mathrm{d}\nu = \int B\,\mathrm{d}\rho_e$. (b) Apply Theorem F.6(b) and Theorem F.12(c),(d) to the bounded reward r_e. (c) See Theorem G.5. (d) $\int \tilde{r}_e B\,\mathrm{d}\nu = (\int BB^\top\,\mathrm{d}\nu)G_B^{-1} z_{ER} = z_{ER}$, and $|\tilde{r}_e(x)| \le \beta_B \|G_B^{-1} z_{ER}\|_2$. The set $\{r : z_r = z_{ER}\}$ is an affine subspace of $L^2(\nu)$ whose direction is the orthogonal complement of $\mathrm{span}(B)$; its element of minimal norm is its unique element in $\mathrm{span}(B)$, which is $\tilde{r}_e$. If $\rho_e \ll \nu$, the minimizers w of $\|r_e - w^\top B\|^2_{L^2(\nu)} = \|r_e\|^2 - 2w^\top z_{ER} + w^\top G_B w$ solve $G_B w = z_{ER}$, so $\Pi_B r_e = B^\top G_B^{-1} z_{ER} = \tilde{r}_e$. A finite atomic measure charges a finite set, which is ν-null when ν has no atoms, so $\rho_e \perp \nu$. When

the kernel has a Lebesgue density and the demonstrations are independent of the buffer, each s_t^i with $t \geq 1$ has a Lebesgue density given the buffer, so it almost surely differs from the finitely many buffer states. (e) Example: an expert that reaches the goal at step n and stays there (deterministic dynamics) has $\bar{\rho}_e(\text{goal}) = (H-n+1)/H$ and $q_E(\text{goal}) = Z_H^{-1} \sum_{t=n-1}^{H-1} \gamma^t = (\gamma^{n-1} - \gamma^H)/(1-\gamma^H)$. For $n = 4$, $H = 100$, $\gamma = 0.95$ these are 0.97 and 0.8565. They coincide, for instance, when the expert stays at an absorbing start state. (f) Linearity of expectation over the i.i.d. demonstrations, and $\|H^{-1} \sum_t B(s_t^i)\|_2 \leq \beta_B$. □

Example G.5 (ER_{FB} is not distribution matching)**.** Let $\mathcal{S} = \{x, y\}$ and $\mathcal{A} = \{\text{stay}, \text{switch}\}$ with deterministic effects, $\mu_0 = \delta_x$, and an expert that always switches. For even H, $\rho_e = \bar{\rho}_e = (\frac{1}{2}, \frac{1}{2})$, while the expert's next-state occupancy is $q_E = (\frac{\gamma}{1+\gamma}, \frac{1}{1+\gamma})$ on (x, y). With $\nu = (\frac{2}{3}, \frac{1}{3})$, the ER reward is $r_e = (\frac{3}{4}, \frac{3}{2})$. The unique optimal policy for r_e switches at x and stays at y: every other policy has some $s_{t+1} = x$, which earns $\frac{3}{4} < \frac{3}{2}$. Its next-state occupancy is δ_y, at total variation $\gamma/(1+\gamma)$ from q_E, and the expert itself is strictly suboptimal for r_e. With $\nu = (\frac{1}{2}, \frac{1}{2})$, r_e is constant, every policy is optimal, and the greedy tie-breaking rule alone decides the behaviour.

G.3 DM_{FB}: the expansion

Theorem G.6. [FBIL-2 | S+D] *Assume Theorem G.1 and let $\bar{\rho}$ be a start law. Consider the hypotheses (A1) exactness of the softmax representation at z (on all of $\mathcal{S} \times \mathcal{A}$); (A3) G_B computed under the same ν; (A4) as in Theorem G.1(ii); (A6) for $\bar{\rho}$-a.e. x, $M^{\pi_E,+}(x, \cdot) \ll \nu$ with a density $m_E(x, y)$ that is jointly measurable in (x, y) and square integrable in y, and $C_E := \int \bar{\rho}(\mathrm{d}x) \|m_E(x, \cdot)\|^2_{L^2(\nu)} < \infty$.*

(a) (S+D; FI24 Theorem 3, p. 17, with its proof on p. 18, under our hypotheses) Under (A1), (A3), (A4) and (A6), $\mathcal{L}_{DM}(z; \bar{\rho}) = \ell_{DM}(z; \bar{\rho}) + C_E$, where C_E does not depend on z. FI24 assumes the exact model for all z with greedy policies and does not state (A6); the proof uses only $M^{\pi_{z,\tau},+}(x, \cdot) = F(x, z)^\top B\, \nu$ [SA A7].

(b) (D; successor-feature reading) Assume (A1), (A4) and G_B invertible. Then $\bar{\psi}^z(x) := \int B\, \mathrm{d}M^{\pi_{z,\tau},+}(x, \cdot)$ equals $G_B F(x, z)$, and, with $\|v\|^2_{G_B^{-1}} := v^\top G_B^{-1} v$,

$$\ell_{DM}(z; \bar{\rho}) = \int \bar{\rho}(\mathrm{d}x)\, \|\bar{\psi}^z(x) - \bar{\psi}_E(x)\|^2_{G_B^{-1}} - \int \bar{\rho}(\mathrm{d}x)\, \|\bar{\psi}_E(x)\|^2_{G_B^{-1}},$$

where the subtracted term is finite and does not depend on z. Under (A1), with $\Lambda_z := \beta_B \sup_{s,a} \|F(s, a, z)\|_2$ (finite by Theorem F.1(iv)), $P(\cdot \mid s, a) \leq \Lambda_z \nu$ for every (s, a), and every successor measure of every stationary policy, the expert's included, satisfies $M^{\pi,+}(x, \cdot) \leq \Lambda_z \nu/(1-\gamma)$ for every x. Its ν-density is bounded by $\Lambda_z/(1-\gamma)$. So the square integrability in (A6) and $C_E \leq \Lambda_z^2 \nu(\mathcal{S})/(1-\gamma)^2 < \infty$ hold automatically, and (A6) reduces to the joint measurability of a version of m_E. Hence (b) is not an extension of (a) to non-square-integrable expert densities, which cannot occur under (A1); it is an equivalent successor-feature reading of the same objective. If (A6) holds, the subtracted term equals $\int \bar{\rho}\, \|\Pi_B m_E\|^2_{L^2(\nu)}$, and

$$\mathcal{L}_{DM}(z; \bar{\rho}) = \int \bar{\rho}(\mathrm{d}x) \Big[\|F(x, z)^\top B - \Pi_B m_E(x, \cdot)\|^2_{L^2(\nu)} + \|m_E(x, \cdot) - \Pi_B m_E(x, \cdot)\|^2_{L^2(\nu)} \Big] :$$

DM_{FB} can only match the projection of the expert's successor density onto $\mathrm{span}(B)$.

(c) (D; without A1) Without exactness, and for invertible G_B, $\ell_{DM}(z; \bar{\rho}) = \int \bar{\rho}\, \|G_B F(x, z) - \bar{\psi}_E(x)\|^2_{G_B^{-1}} - \text{const}$ is, by definition, the population objective that the implemented $\hat{\ell}^{(K)}(z; \hat{G}_B)$ estimates (bias: Theorem G.8(d)): a mismatch between the model's predicted features $G_B F(x, z) = \int B\, \mathrm{d}\hat{M}_z(x, \cdot)$ and the expert's. With $\mathcal{E}_z(x, \cdot) := \sum_a \pi_{z,\tau}(a \mid x) \mathcal{E}_z(x, a, \cdot)$ (Theorem F.17), for every x

$$\Big| \|\bar{\psi}^z(x) - \bar{\psi}_E(x)\|_{G_B^{-1}} - \|G_B F(x, z) - \bar{\psi}_E(x)\|_{G_B^{-1}} \Big| \leq \lambda_{\min}(G_B)^{-1/2}\, \beta_B\, \|\mathcal{E}_z(x, \cdot)\|_{\text{TV}}$$

(here $\|\mu\|_{\mathrm{TV}} = |\mu|(\mathcal{S})$ as in (N1), twice the distance TV *used in (e)). The quantity $\|\bar{\psi}^z(x) - \bar{\psi}_E(x)\|_{G_B^{-1}}$ is the dual seminorm $\sup\{\int w^\top B \,\mathrm{d}(M^{\pi_{z,\tau},+}(x,\cdot) - M^{\pi_E,+}(x,\cdot)) : \|w^\top B\|_{L^2(\nu)} \le 1\}$ over $\mathrm{span}(B)$. It is at most the $L^2(\nu)^\star$ norm, and it can be strictly smaller. The bound is pointwise in x and between norms: it compares the model's feature mismatch with the feature-projected distance of the true successor measures at each x. It is not a bound between objectives; squaring and integrating over $\bar{\rho}$ needs further factors, for example $|a^2 - b^2| \le |a-b|\,(|a-b|+2b)$ for $a, b \ge 0$. So (c) relates ℓ_{DM} to a feature-projected version of the true distance, not to $\mathcal{L}_{DM}$, and nothing ensures that the model error is small.*

(d) (D; value-gap reading) Under (A1), (A3), (A4) and (A6), for every bounded r and every x, $|V_r^{\pi_{z,\tau}}(x) - V_r^{\pi_E}(x)| \le \|r\|_{L^2(\nu)} \|M^{\pi_{z,\tau},+}(x,\cdot) - M^{\pi_E,+}(x,\cdot)\|_{L^2(\nu)^\star}$, where the right-hand side is finite by (b). Hence $\int \bar{\rho}(\,\mathrm{d}x)(V_r^{\pi_{z,\tau}}(x) - V_r^{\pi_E}(x))^2 \le \|r\|^2_{L^2(\nu)}(\ell_{DM}(z;\bar{\rho}) + C_E)$. This controls value gaps only at states charged by $\bar{\rho}$ (Theorem G.7).

(e) (D; the occupancy from the start state) Let π be any stationary policy and suppose $\bar{\rho} \ge c\,\mu_0$ for some $c > 0$. Assume that, for $\bar{\rho}$-a.e. x, $M^{\pi,+}(x,\cdot) - M^{\pi_E,+}(x,\cdot)$ has a ν-density $\dot{\mu}(x,\cdot)$ that is jointly measurable in (x,y) (automatic in the finite case when ν has full support). Then

$$\mathrm{TV}(d_\pi^+, d_{\pi_E}^+) \le \frac{1-\gamma}{2c}\,\nu(\mathcal{S})^{1/2}\,\mathcal{L}(\pi;\bar{\rho})^{1/2},$$
$$\mathrm{TV}(d_{\pi,H}^+, q_E) \le 2\gamma^H + \frac{1-\gamma}{2c}\,\nu(\mathcal{S})^{1/2}\,\mathcal{L}(\pi;\bar{\rho})^{1/2}.$$

In particular $\mathcal{L}_{DM}(z;\bar{\rho}) = 0$ implies $d_{\pi_{z,\tau}}^+ = d_{\pi_E}^+$. If $0 \in \mathcal{T}$, then $\bar{\rho}_{\mathcal{T}} \ge \mu_0/|\mathcal{T}|$, so $c = 1/|\mathcal{T}|$. This concerns the true objective $\mathcal{L}$; nothing here says that minimizing the model's $\hat{\ell}$ makes $\mathcal{L}$ small.

Proof. (a) Under (A1), $M^{\pi_{z,\tau},+}(x,\cdot) = \sum_a \pi_{z,\tau}(a \mid x)\hat{M}_z(x,a,\cdot)$ has the bounded density $F(x,z)^\top B$ with respect to ν. With (A6) the difference of the two successor measures has the density $\dot{\mu}_x := F(x,z)^\top B - m_E(x,\cdot) \in L^2(\nu)$, whose dual norm is $\|\dot{\mu}_x\|_{L^2(\nu)}$ (Theorem F.15). Expanding,

$$\|\dot{\mu}_x\|^2_{L^2(\nu)} = F(x,z)^\top G_B F(x,z) - 2F(x,z)^\top \int B m_E(x,\cdot)\,\mathrm{d}\nu + \|m_E(x,\cdot)\|^2_{L^2(\nu)},$$

and $\int B m_E(x,\cdot)\,\mathrm{d}\nu = \int B\,\mathrm{d}M^{\pi_E,+}(x,\cdot) = \bar{\psi}_E(x)$. The integrand is jointly measurable, so Tonelli's theorem allows integration over $\bar{\rho}$; the first two terms are bounded in x (since $\|\bar{\psi}_E\|_2 \le \beta_B/(1-\gamma)$) and the third is integrable by (A6). Only $M^{\pi_{z,\tau},+}(x,\cdot) = F(x,z)^\top B\,\nu$ was used, not greediness. FI24's proof states after its Eq. (33) that the constant "does not depend on s". The constant is $\|M^{\pi_e}(s)\|^2_{L^2(\rho)^\star}$, which depends on the start state s and not on z; after averaging over s_t it is a z-independent constant, so the conclusion is unaffected.

(b) The identity $\bar{\psi}^z = G_B F(x,z)$ is Theorem F.19, averaged over actions. Then $(G_B F - \bar{\psi}_E)^\top G_B^{-1}(G_B F - \bar{\psi}_E) = F^\top G_B F - 2F^\top \bar{\psi}_E + \bar{\psi}_E^\top G_B^{-1}\bar{\psi}_E$; integrate over $\bar{\rho}$. For the density bound: under (A1), $P(\cdot \mid s,a) \le M^{\pi_{z,\tau},+}(s,a,\cdot) = F(s,a,z)^\top B\,\nu$ (the $t=0$ term of the successor measure), and the density satisfies $F(s,a,z)^\top B(y) \le \|F(s,a,z)\|_2 \|B(y)\|_2 \le \Lambda_z$, so $P(\cdot \mid s,a) \le \Lambda_z \nu$ for every (s,a). For any policy and any start, $\mathbb{P}(s_{t+1} \in X) = \mathbb{E}[P(X \mid s_t, a_t)] \le \Lambda_z \nu(X)$; summing γ^t over $t \ge 0$ gives $M^{\pi,+}(x,\cdot) \le \Lambda_z \nu/(1-\gamma)$ for every x. A measure bounded by $c\,\nu$ has a ν-density with values in $[0,c]$, which lies in $L^2(\nu)$ with squared norm at most $c^2\nu(\mathcal{S})$ because ν is finite. This gives the bound on C_E. Under (A6) the $L^2(\nu)$-projection of $m_E(x,\cdot)$ onto $\mathrm{span}(B)$ is $w^\top B$ with $G_B w = \int B m_E\,\mathrm{d}\nu = \bar{\psi}_E(x)$, so $\|\Pi_B m_E\|^2 = w^\top G_B w = \bar{\psi}_E^\top G_B^{-1}\bar{\psi}_E$. Since $F(x,z)^\top B \in \mathrm{span}(B)$, Pythagoras' theorem splits $\|F^\top B - m_E\|^2$ as stated.

(c) The reverse triangle inequality for the norm $\|\cdot\|_{G_B^{-1}}$ bounds the left-hand side by $\|\bar{\psi}^z(x) - G_B F(x,z)\|_{G_B^{-1}}$. As in the proof of Theorem F.19, but without exactness, $G_B F(x,z) = \int B\,\mathrm{d}\hat{M}_z(x,\cdot)$; hence $\bar{\psi}^z(x) - G_B F(x,z) = \int B\,\mathrm{d}\mathcal{E}_z(x,\cdot)$. For a unit vector e, $|e^\top \int B\,\mathrm{d}\mathcal{E}| \le$

$\beta_B\|\mathcal{E}\|_{\mathrm{TV}}$, so $\|\int B\,\mathrm{d}\mathcal{E}\|_2 \le \beta_B\|\mathcal{E}\|_{\mathrm{TV}}$; and $\|v\|_{G_B^{-1}} \le \lambda_{\min}(G_B)^{-1/2}\|v\|_2$. For the seminorm: with $v := \bar\psi^z(x) - \bar\psi_E(x) = \int B\,\mathrm{d}(M^{\pi_z,\tau,+} - M^{\pi_E,+})(x,\cdot)$, Cauchy–Schwarz in the inner product $\langle w, w'\rangle_{G_B}$ gives $\sup\{w^\top v : w^\top G_B w \le 1\} = (v^\top G_B^{-1} v)^{1/2}$, attained at $w \propto G_B^{-1}v$, and $\|w^\top B\|^2_{L^2(\nu)} = w^\top G_B w$. The functions $w^\top B$ are bounded, so they belong to the class defining $\|\cdot\|_{L^2(\nu)^\star}$.

(d) By Theorem F.3(a) averaged over actions, $V_r^\pi(x) = \int r\,\mathrm{d}M^{\pi,+}(x,\cdot)$. The difference of values is therefore $\int r\,\mathrm{d}\mu$ for the difference μ of the two measures, which has a bounded ν-density for every x by (b); Theorem F.15 applies. Square, integrate over $\bar\rho$ and use (a).

(e) If $\mathcal{L}(\pi;\bar\rho) = \infty$ there is nothing to prove. Otherwise, by Theorem F.15, for $\bar\rho$-a.e. x the difference μ_x has the density $\dot\mu(x,\cdot) \in L^2(\nu)$, and $f(x) := \|\dot\mu(x,\cdot)\|_{L^2(\nu)}$ is measurable by Tonelli's theorem. Since $\mu_0 \le c^{-1}\bar\rho$, $\bar\rho$-null sets are μ_0-null, so $\mu_x = \dot\mu(x,\cdot)\,\nu$ and $f(x) < \infty$ for μ_0-a.e. x; Tonelli's theorem also makes $x \mapsto |\mu_x|(\mathcal{S}) = \int |\dot\mu(x,\cdot)|\,\mathrm{d}\nu$ measurable. Cauchy–Schwarz gives $|\mu_x|(\mathcal{S}) = \int |\dot\mu(x,\cdot)|\,\mathrm{d}\nu \le \nu(\mathcal{S})^{1/2} f(x)$. By Theorem B.4(c), $d_\pi^+ - d_{\pi_E}^+ = (1-\gamma)\int \mu_0(\,\mathrm{d}x)\,\mu_x$, so its total variation norm is at most $(1-\gamma)\int \mu_0(\,\mathrm{d}x)|\mu_x|(\mathcal{S}) \le (1-\gamma)\nu(\mathcal{S})^{1/2}\int \mu_0 f$. Since $\mu_0 \le c^{-1}\bar\rho$ and $f \ge 0$, $\int \mu_0 f \le c^{-1}\int \bar\rho f \le c^{-1}(\int \bar\rho f^2)^{1/2} = c^{-1}\mathcal{L}(\pi;\bar\rho)^{1/2}$ by Jensen's inequality. Halve the total variation norm to get TV. The second bound adds $\mathrm{TV}(d^+_{\pi,H}, d^+_\pi) \le \gamma^H$ and $\mathrm{TV}(d^+_{\pi_E}, q_E) \le \gamma^H$ (Theorem B.13(b)). Finally $\bar\rho_{\mathcal{T}} \ge |\mathcal{T}|^{-1}\mathrm{Law}_0 = |\mathcal{T}|^{-1}\mu_0$ when $0 \in \mathcal{T}$. □

Example G.7 (the loss is blind outside the start law)**.** Let $\mathcal{S} = \{x_0, x, y\}$, $\mu_0 = \delta_{x_0}$ and $\mathcal{A} = \{a, b\}$. From x_0, a leads to x and b to y; x and y are absorbing. The expert plays a at x_0, so its demonstrations are $x_0, x, x, \ldots$, and a start law built from $s_1, \ldots, s_H$ is δ_x, which excludes x_0. Every policy has $M^{\pi,+}(x,\cdot) = \delta_x/(1-\gamma)$, so $\mathcal{L}(\pi;\delta_x) = 0$ for every policy, including the policy that plays b at x_0. That policy has $d^+_{\pi,H} = \delta_y$ while $q_E = \delta_x$: the state-occupancy JS is $\log 2$, the maximum, and its return for $r = \mathbf{1}_{\{x\}}$ is 0 instead of Z_H. Here x_0 cannot be reached from the later expert states, so its action is invisible to the loss. In the maze, the start state can be revisited through return paths, and its action then enters the loss only through those discounted paths. Including $t = 0$ puts it in the loss directly, and (e) then controls the occupancy from μ_0.

G.4 The estimator, tail truncation and the Gram estimate

Proposition G.8. [FBIL-3 | D] *Assume Theorem G.1, and let z be fixed (not chosen from the demonstrations).*

(a) (unbiasedness with the infinite continuation) For any finite nonempty deterministic $\mathcal{T} \subseteq \mathbb{N}$, $\mathbb{E}\,\hat\ell^{(\infty)}(z; G_B) = \ell_{DM}(z;\bar\rho_{\mathcal{T}})$. This makes FI24 Theorem 3's "unbiased" precise: the start law is the population law $\bar\rho_{\mathcal{T}}$, and the targets use the infinite continuation under π_E.

For the remaining parts let $\mathcal{T} \subseteq \{0, \ldots, H-K\}$ be nonempty, with $K \ge 1$, so that $\psi^{(K)}$ is observed, and let $\hat G_B$ be symmetric and independent of the demonstrations.

(b) (truncation) For every demonstration and $t \in \mathcal{T}$, $\|\psi_t^{(\infty),i} - \psi_t^{(K),i}\|_2 \le \gamma^K\beta_B/(1-\gamma)$. With $F_{\max}(z) := \max_{i,t}\|F(s_t^i, z)\|_2$, deterministically, for every symmetric G, $|\hat\ell^{(K)}(z;G) - \hat\ell^{(\infty)}(z;G)| \le 2\gamma^K\beta_B F_{\max}(z)/(1-\gamma)$. If $z \mapsto F(x,a,z)$ is C^1 and $\mathcal{J}_{\max}(z) := \max_{i,t}\|\partial_z F(s_t^i, z)\|_{\mathrm{op}}$, the z-gradients differ by at most $2\gamma^K\beta_B\mathcal{J}_{\max}(z)/(1-\gamma)$.

(c) (Gram estimate) $|\hat\ell^{(K)}(z;\hat G_B) - \hat\ell^{(K)}(z;G_B)| \le \|\hat G_B - G_B\|_{\mathrm{op}} F_{\max}(z)^2$, and the z-gradients differ by at most $2\|\hat G_B - G_B\|_{\mathrm{op}}\mathcal{J}_{\max}(z)F_{\max}(z)$.

(d) (bias of the implemented objective) With $\bar F(z) := \sup_x\|F(x,z)\|_2$, conditionally on $\hat G_B$,

$$\left|\mathbb{E}\,\hat\ell^{(K)}(z;\hat G_B) - \ell_{DM}(z;\bar\rho_{\mathcal{T}})\right| \le \frac{2\gamma^K\beta_B\bar F(z)}{1-\gamma} + \|\hat G_B - G_B\|_{\mathrm{op}}\bar F(z)^2.$$

FI24's unbiasedness does not transfer unchanged to the truncated estimator with an estimated Gram matrix.

(e) (dependence on future states) $\psi_t^{(K),i} = \sum_{k<K}\gamma^k b_{t+k+1}$ with the slot values $b_j := B(s_j^i)$, so $\partial\psi_t^{(K),i}/\partial b_{t+k+1} = \gamma^k I$ for $0 \le k < K$: changing the expert state in a future slot changes

the target of every earlier start time within distance K. (When a state repeats, as at a goal, its contributions add over the slots it occupies.)

Proof. (a) Fix i, $t \in \mathcal{T}$ and a coordinate j. With $\Psi_j := \sum_{k\geq 0} \gamma^k B_j(s_{k+1})$, which is bounded and measurable on Ω, we have $\psi_{t,j}^{(\infty),i} = \Psi_j \circ \theta^t$ along demonstration i. Theorem G.3 with $g := F_j(\cdot, z)$ (bounded and measurable) and $\pi = \pi_E$ gives $\mathbb{E}[F_j(s_t, z)\, \psi_{t,j}^{(\infty)}] = \mathbb{E}[F_j(s_t, z)\, \bar{\Psi}_j(s_t)]$. By Theorem F.3(a) with the reward B_j, averaged over the first action, $\bar{\Psi}_j(x) = \bar{\psi}_{E,j}(x)$. The quadratic term depends on s_t only, and s_t^i has law $\mathrm{Law}_t^{\pi_E}$. Averaging over $t \in \mathcal{T}$ and over the identically distributed demonstrations gives $\ell_{DM}(z; \bar{\rho}_{\mathcal{T}})$.

(b) $\| \sum_{k\geq K} \gamma^k B(s_{t+k+1}) \|_2 \leq \beta_B \sum_{k\geq K} \gamma^k$. The two estimators differ by -2 times the average of $F(s_t^i, z)^\top (\psi_t^{(K),i} - \psi_t^{(\infty),i})$, and $\nabla_z [F(x,z)^\top v] = \partial_z F(x,z)^\top v$ for a fixed vector v.

(c) $|F^\top (\hat{G}_B - G_B) F| \leq \|\hat{G}_B - G_B\|_{\mathrm{op}} \|F\|_2^2$ and $\nabla_z [F^\top (\hat{G}_B - G_B) F] = 2\, \partial_z F^\top (\hat{G}_B - G_B) F$.

(d) Write $\hat{\ell}^{(K)}(z; \hat{G}_B) - \ell_{DM} = [\hat{\ell}^{(K)}(z; \hat{G}_B) - \hat{\ell}^{(K)}(z; G_B)] + [\hat{\ell}^{(K)}(z; G_B) - \hat{\ell}^{(\infty)}(z; G_B)] + [\hat{\ell}^{(\infty)}(z; G_B) - \ell_{DM}]$. Take expectations conditionally on $\hat{G}_B$ (independent of the demonstrations): the last bracket has mean 0 by (a), and the first two are bounded by (c) and (b) with $F_{\max} \leq \bar{F}$.

(e) $\psi_t^{(K),i}$ is linear in the slot values $b_{t+1}, \ldots, b_{t+K}$ with coefficients $\gamma^0, \ldots, \gamma^{K-1}$. □

Remark G.9 (decisions D4–D5: tail treatment and start times; [SA U-4]).

- **D4 (decided).** The implemented start-time set is $\mathcal{T}_K := \{0, 1, \ldots, H - K\}$, with $K = K_{DM} = 60$ by default (a configuration value, frozen before final runs and not tuned on final test outcomes). For $\gamma = 0.95$, the truncation term of (d) has the factor $\gamma^K/(1-\gamma) \approx 0.92$, against $1/(1-\gamma) = 20$ for the full sum: the worst-case bound on the missing tail is the fraction $\gamma^K \approx 4.6\%$ of the worst-case bound on the full sum. This is not a bound on the relative bias of the objective. With $H = 100$ there are 41 start times per demonstration. Rejected alternatives: all start times with per-start truncation (at $t = H - 1$ only one of about twenty effective terms remains); and a model bootstrap of the tail, which would use the learner's own model as a surrogate for the expert's continuation and make the target depend on z.
- **D5 (decided, configurable).** $t = 0$ is included. This deviates from the ρ_e of FI24 Eq. (8), which starts at s_1. FI24 is not explicit about the DM start law: p. 4 calls ρ_e "the empirical distribution of states visited", footnote 3 samples "a state in that trajectory", and Eq. (12) writes $s_0 \sim \rho_e$; FI24 also does not say how the infinite sum of Eq. (32) is evaluated near trajectory ends. With $t = 0$ included, $\bar{\rho}_{\mathcal{T}_K} \geq \mu_0/41$, so Theorem G.6(e) controls the occupancy from μ_0 by the true DM objective; Theorem G.7 shows what can be lost otherwise. $\mathrm{ER}_{\mathrm{FB}}$ keeps FI24's ρ_e ($t = 1, \ldots, H$).
- The DM start law is therefore $\bar{\rho}_{\mathcal{T}_K}$, a stated change of the weighting of FI24 Eq. (31). The objective that is estimated is $\ell_{DM}(z; \bar{\rho}_{\mathcal{T}_K})$ up to the bias of Theorem G.8(d), for a fixed z.

G.5 The latent gradient with frozen networks

Proposition G.10. [FBIL-4 | D] *Assume that $z \mapsto F(x, a, z)$ is C^1 for every (x, a) (true for networks with C^1 activations; with ReLU the formulas hold only where $F(x, a, \cdot)$ is differentiable,* [SA U-14]*), and freeze (ξ, ω), $\hat{G}_B$ and the targets. For $x \in \mathcal{S}$ write $\hat{Q}_a := F(x, a, z)^\top z$, $\mathcal{J}_a := \partial_z F(x, a, z) \in \mathbb{R}^{d \times d}$ and $\pi_a := \pi_{z,\tau}(a \mid x)$.*

(a) $\nabla_z \hat{Q}_a = F(x, a, z) + \mathcal{J}_a^\top z$.

(b) $\nabla_z \pi_a = \frac{\pi_a}{\tau} (\nabla_z \hat{Q}_a - \sum_b \pi_b \nabla_z \hat{Q}_b)$.

(c) $\mathcal{J}(x, z) := \partial_z F(x, z) = \sum_a [\pi_a \mathcal{J}_a + F(x, a, z)(\nabla_z \pi_a)^\top]$.

(d) For $\hat{\ell}(z) = \frac{1}{n} \sum_{(i,t)} [F(s_t^i, z)^\top \hat{G}_B F(s_t^i, z) - 2F(s_t^i, z)^\top \psi_t^i]$,

$$\nabla_z \hat{\ell}(z) = \frac{2}{n} \sum_{(i,t)} \mathcal{J}(s_t^i, z)^\top (\hat{G}_B F(s_t^i, z) - \psi_t^i). \tag{G.6}$$

(e) *Dropping the second term of (c), i.e. treating the policy probabilities as constants ("detaching" them), gives a vector that differs from $\nabla_z \hat{\ell}$ in general;* `check_2B.py` *exhibits a nonzero difference.*

(f) *If $\nabla_z \hat{\ell}(z) \neq 0$, a small enough* exact *gradient step decreases $\hat{\ell}$, as in Theorem D.9(b). No convexity in z is claimed or used. No descent, convergence or global-optimality property is claimed for the implemented optimizer (Adam, with projection, at most 300 updates): its direction is not the gradient.*

Proof. (a) $\mathrm{d}(F^\top z) = (\mathrm{d}F)^\top z + F^\top \mathrm{d}z = z^\top \mathcal{J}_a \,\mathrm{d}z + F^\top \mathrm{d}z$.

(b) $\partial \pi_a / \partial \hat{Q}_b = \pi_a(\mathbf{1}[a=b] - \pi_b)/\tau$ for the softmax of $\hat{Q}/\tau$; apply the chain rule.

(c) Product rule applied to $\sum_a \pi_a F(x,a,z)$.

(d) $\nabla_z[F^\top \hat{G}_B F] = 2\mathcal{J}^\top \hat{G}_B F$ because $\hat{G}_B$ is symmetric, and $\nabla_z[F^\top \psi] = \mathcal{J}^\top \psi$ for a fixed ψ.

(e) The omitted term is $\frac{2}{n}\sum \big(\sum_a F(x,a,z)(\nabla_z \pi_a)^\top\big)^\top (\hat{G}_B F - \psi)$, which is nonzero in general.

(f) $\hat{\ell}(z - \eta\nabla\hat{\ell}) = \hat{\ell}(z) - \eta\|\nabla\hat{\ell}\|^2 + o(\eta)$ by differentiability. For a direction $u \neq \nabla\hat{\ell}$, the first-order change is $-\eta\nabla\hat{\ell}^\top u$, whose sign is not controlled in general. □

G.6 Constraining the latent

Proposition G.11. [FBIL-5 | D] *Let ℓ be C^1 on a neighbourhood of the sphere $\mathcal{Z}_R = \{\|z\|_2 = R\}$, $R > 0$, and let $g_R(z) := (I - zz^\top/R^2)\nabla\ell(z)$ be the Riemannian gradient.*

(a) If $z^\star$ minimizes ℓ on $\mathcal{Z}_R$, then $g_R(z^\star) = 0$, i.e. $\nabla\ell(z^\star)$ is parallel to $z^\star$.

(b) For $z \in \mathcal{Z}_R$, the projected step

$$z^+ := R\,\frac{z - \eta\nabla\ell(z)}{\|z - \eta\nabla\ell(z)\|_2} \tag{G.7}$$

satisfies $z^+ = z - \eta g_R(z) + O(\eta^2)$ and $\ell(z^+) = \ell(z) - \eta\|g_R(z)\|^2 + o(\eta)$; it decreases ℓ for small η when $g_R(z) \neq 0$. This concerns the exact projected gradient step only.

(c) With the reparametrization $z = Ru/\|u\|_2$, $\nabla_u \ell = (R/\|u\|_2)(I - \hat{u}\hat{u}^\top)\nabla\ell(z)$ with $\hat{u} = u/\|u\|_2 = z/R$; its zeros are those of g_R. The map is singular at $u = 0$.

(d) The constrained problem is a different problem. Minimizers on the sphere need not be stationary points of the unconstrained ℓ: for $\ell(z) = w^\top z$ with $w \neq 0$, the minimizer on $\mathcal{Z}_R$ is $-Rw/\|w\|_2$, where $\nabla\ell = w \neq 0$. Under exactness on $\mathcal{Z}_R$ and invertible G_B, every $z \in \mathcal{Z}_R$ satisfies $\pi_{z,\tau} = \mathrm{softmax}(Q_{r_z}^{+,\pi_{z,\tau}}/\tau)$ for the reward $r_z := B^\top G_B^{-1} z$ that it encodes (Theorem F.6(a), $z_{r_z} = z$): relative to that reward the temperature is pinned at τ, and DM_{FB} cannot sharpen the policy for a fixed reward direction by rescaling z. Even under exactness, the temperature relative to the normalized reward $r_z/\|r_z\|_{L^2(\nu)}$ is $\tau(z^\top G_B^{-1} z)^{-1/2}$, since $\|r_z\|^2_{L^2(\nu)} = z^\top G_B^{-1} z$; it is constant on $\mathcal{Z}_R$ only if $G_B \propto I$. Without exactness nothing is claimed.

(e) (warm start; our choice) $z_0 := Rz_{ER}/\|z_{ER}\|_2$ if $z_{ER} \neq 0$. FI24 states the ER warm start explicitly for BC_{FB}, $BBELL_{\mathrm{FB}}$ and FM_{FB} (p. 7); E.2 (p. 31) describes it for the FB methods that use gradient descent over z, a class that includes DM_{FB}; FI24 never names DM_{FB} in this context, so how DM_{FB} is initialized is not stated explicitly. If $z_{ER} = 0$, a stated fallback must be used (for example a z drawn from $\mathcal{D}_z$ with a recorded seed).

Proof. (a) For $v \perp z^\star$, the curve $c(h) := R(z^\star + hv)/\|z^\star + hv\|_2$ lies on $\mathcal{Z}_R$, with $c(0) = z^\star$ and $c'(0) = v$, because $z^{\star\top} v = 0$ and $\|z^\star\|_2 = R$. Minimality gives $(\ell \circ c)'(0) = \nabla\ell(z^\star)^\top v = 0$ for every $v \perp z^\star$, so $\nabla\ell(z^\star) \in \mathrm{span}(z^\star)$.

(b) Let $u := z - \eta\nabla\ell$. Since $\|z\|_2 = R$, $\|u\|^2 = R^2 - 2\eta z^\top\nabla\ell + \eta^2\|\nabla\ell\|^2$, so $R/\|u\| = 1 + \eta z^\top\nabla\ell/R^2 + O(\eta^2)$ and $z^+ = (z - \eta\nabla\ell)(1 + \eta z^\top\nabla\ell/R^2) + O(\eta^2) = z - \eta g_R + O(\eta^2)$. Differentiability gives $\ell(z^+) = \ell(z) - \eta\nabla\ell^\top g_R + o(\eta)$, and $\nabla\ell^\top g_R = \|g_R\|^2$ because g_R is the orthogonal projection of $\nabla\ell$ onto $z^\perp$.

(c) The Jacobian of $u \mapsto Ru/\|u\|_2$ is the symmetric matrix $(R/\|u\|_2)(I - \hat{u}\hat{u}^\top)$; apply the chain rule.

(d) For the linear example, $\min_{\|z\|=R} w^\top z = -R\|w\|_2$ by Cauchy–Schwarz, attained at $-Rw/\|w\|_2$. Under exactness, $z_{r_z} = G_B G_B^{-1} z = z$, and Theorem F.6(a) gives $F(x,\cdot,z)^\top z = Q_{r_z}^{+,\pi_{z,\tau}}(x,\cdot)$. Moreover $\|r_z\|_{L^2(\nu)}^2 = z^\top G_B^{-1}(\int BB^\top \mathrm{d}\nu) G_B^{-1} z = z^\top G_B^{-1} z$, and Q-functions are linear in the reward, so $\pi_{z,\tau} = \mathrm{softmax}(Q_{\bar{r}_z}^{+,\pi_{z,\tau}}/(\tau\|r_z\|_{L^2(\nu)}^{-1}))$ with $\bar{r}_z := r_z/\|r_z\|_{L^2(\nu)}$. (e) is a choice. □

G.7 Approximation ledger and the relation to f-PG

The approximations in the FB pipeline, with the result each affects. [I] implementation approximation; [C] stated change of objective.

Quantity	Exact object	Implemented	Consequence
FB model	$M^{\pi_z,+}$ on $\mathcal{Z}$ (FB-2, FB-5, FB-7)	rank-16 network, float32 [I]	exactness impossible in the finite maze under the condition of Theorem F.9; FB-6 with unknown error covered; ReLU would not be: FB-8(b)–(c) and FB-9 not proved, FBIL-4 only where F is differentiable [SA U-14]
Activation	C^1 networks (FB-8, FB-9, FBIL-4, FBIL-5)	LayerNorm, tanh, SiLU (C^∞) [I]	
Reference ν	law of replay next states	empirical buffer law $\hat{\nu}$ [I]	one buffer for loss, $\hat{G}_B$, z_r (Theorem F.27)
Pretraining	L (FB-8)	minibatches, Adam, Polyak targets [I]	minibatch gradient unbiased for the gradient of the empirical buffer loss, with C^1 activations and independent target states (FB-8(b)); not for the population ∇L; no convergence
Policy	greedy (FB21)	softmax, $\tau > 0$ [C]	FB-5 replaces FB-2(b)
Gram matrix	G_B	$\hat{G}_B$ from buffer next states [I]	exact for $\hat{\nu}$; unbiased for G_B only for a fixed B and an independent i.i.d. sample (FB-10(e)); FBIL-3(c)–(d)
ER latent	$\int B \, \mathrm{d}\bar{\rho}_e$	z_{ER} from $N_E = 4$ demonstrations [I]	FBIL-1(f); reward reading FBIL-1(a),(d)
Latent domain	$z \in \mathbb{R}^d$	projection onto $\mathcal{Z}_R$ [C]	FB-5(d), FB-6(e), FBIL-5
DM start law	$\bar{\rho}_e$ (FI24, $t \geq 1$)	$\bar{\rho}_{\mathcal{T}_K}$, $t = 0, \ldots, H-K$ [C], empirical over 4 demonstrations [I]	FBIL-3(a), FBIL-2(e); D4–D5
DM target	$\bar{\psi}_E(s_t)$, infinite continuation	$\psi^{(K)}$, $K = 60$ [I]	for fixed z, conditionally on $\hat{G}_B$: bias $\leq 2\gamma^K \beta_B \bar{F}/(1-\gamma) + \|\hat{G}_B - G_B\| \bar{F}^2$ (FBIL-3(d))
Expert measures	A6	automatic under A1 up to measurability (FBIL-2(b)); it holds for the population replay law (Theorem H.10) but fails for the atomic $\hat{\nu}$ of the continuous maze (Theorem H.5), and then A1 fails too	if $M^{\pi_{z,\tau},+}(x,\cdot) - M^{\pi_E,+}(x,\cdot)$ has no $L^2(\nu)$ density on a $\bar{\rho}$-non-null set, $\mathcal{L}_{DM} = +\infty$ (Theorem F.15); ℓ_{DM} keeps its model-feature meaning (FBIL-2(c))
Data reuse	fixed z	z_0 and $\hat{\ell}$ from the same demonstrations [I]	FBIL-3(a) holds for a fixed z only; an exact global minimum is optimistically biased, $\mathbb{E}\min_z \hat{\ell} \leq \min_z \mathbb{E}\hat{\ell}$; nothing is claimed for the output of ≤ 300 Adam steps
DM optimization	minimizer over $\mathcal{Z}$	≤ 300 projected Adam steps from the projected z_{ER}, float32 [I]	no descent or convergence claim (FBIL-4(f), FBIL-5(b) cover exact gradient steps only)
Evaluation	V^π from every state, infinite horizon; exact occupancies	H-step returns from μ_0; Monte Carlo rollouts and a density estimator in the continuous maze [I]	Theorem F.14, OCC-5; no guarantee for plug-in evaluation metrics (cf. FPG-5(b))

If minibatches over the pairs (i,t) are used in the latent step, they add sampling noise to the gradient of $\hat{\ell}$ without changing its expectation over the minibatch draw.

Remark G.12 (relation to the f-PG objective; see also Section 8). f-PG minimizes a divergence between *marginal* next-state occupancies from μ_0 over H steps (Section D). Under (A1) and (A6), the population objective of $\mathrm{DM}_{\mathrm{FB}}$ is, up to the constant C_E, a replay-weighted $L^2(\nu)$ distance between *conditional* successor measures at expert states (Theorem G.6(a)); without exactness it is the feature mismatch of Theorem G.6(c), through a finite-rank model. Neither objective reduces to the other. With a start law that excludes the start state, Theorem G.7 shows that $\mathcal{L}=0$ does not imply $d^+_{\pi,H}=q_E$. With the implemented start law $\bar{\rho}_{\mathcal{T}_K} \geq \mu_0/41$, Theorem G.6(e) shows that a small *true* DM objective does control the occupancy from μ_0. Nothing transfers this to the model's estimate $\hat{\ell}$: even under exactness, $\hat{\ell}$ determines $\mathcal{L}_{DM}$ only up to the unknown constant C_E and the estimation, truncation and Gram errors of Theorem G.8, and $\min_z \mathcal{L}_{DM}$ need not be small (for instance for a softmax policy with $\tau>0$ against a deterministic expert). Theorem G.5 shows that $\mathrm{ER}_{\mathrm{FB}}$ is not distribution matching at all. Conversely, $d^+_{\pi,H}=q_E$ does not imply equal conditional successor measures at every expert state, because the marginal averages over the trajectory (FI24 p. 5 makes the same point about averaging over the start law).

H The implemented environments, kernels and estimators

Status. This appendix fixes the environment items [SA U-5], [SA U-10] and [SA U-13] for the implemented code, and proves the properties of that code on which the other appendices rely. Nothing here is a source result. The environments are study choices (Section 10, `configs/environment.yaml`). They differ deliberately from FB21 D.1 (p. 35), whose discrete maze is deterministic and whose continuous maze clips positions at the boundary and undoes wall-crossing moves. Our continuous kernel (H.2) has neither clipping nor an atom, and the implemented sampler raises an error rather than return an atom (Theorem H.4(e)). Type labels: ENV-1 I+D, ENV-2 D, ENV-3 D, ENV-4 I, ENV-5 I+D. Section H.6 proves the domination property of the task-blind replay law used by FB pretraining (ENV-6, D), which Theorem H.4(f) left open. Implementation: `il_memoir/env_tabular.py`, `env_continuous.py`, `density.py`, `expert.py`, `tasks.py`; numerical checks in `tests/`, `scripts/check_environments.py` and `audit/prompt3A/`. Section H.7 proves the cross-fitting statement FPG-9 used by the implemented f-PG learner. Throughout, $L := 1/7$, Φ is the standard normal distribution function and φ_σ the density of $\mathcal{N}(0,\sigma^2 I_2)$.

H.1 Finite mazes

Definition H.1 (finite mazes)**.** The lattice is $G := \{0,\dots,6\}^2$, with cell (x,y) in column x and row y. The blocked sets are $\mathcal{W}_{\mathrm{open}} := \emptyset$ and

$$\mathcal{W}_{4\mathrm{R}} := \{(3,y) : y \notin \{1,5\}\} \cup \{(x,3) : x \notin \{1,5\}\},$$

which contains $(3,3)$ and has 9 cells, so $\mathcal{S} := G \setminus \mathcal{W}$ has 49 or 40 states. The actions, in the fixed order used for tie-breaking, are stay, right, up, left and down, with displacements $\delta_a \in \{(0,0),(1,0),(0,1),(-1,0),(0,-1)\}$. Let $\mathrm{nx}(s,a) := s+\delta_a$ if $s+\delta_a \in \mathcal{S}$ and $\mathrm{nx}(s,a) := s$ otherwise. With slip probability $p_{\mathrm{slip}} := 0.1$ [SA U-10],

$$P(\cdot \mid s,a) := (1-p_{\mathrm{slip}})\,\delta_{\mathrm{nx}(s,a)} + \frac{p_{\mathrm{slip}}}{5}\sum_{a'\in\mathcal{A}} \delta_{\mathrm{nx}(s,a')}. \tag{H.1}$$

Equivalently, the executed action is drawn uniformly from all five actions with probability p_{slip} and is the chosen one otherwise. The start law is $\mu_0 := \delta_{(1,1)}$ and the goals are $(5,1)$, $(5,5)$ and $(1,5)$. With the cell centres $\bar{s} := (s+\frac{1}{2})L \in (0,1)^2$ and the width $w := L/2$, the evaluation reward of goal g is $r_g(s) := \exp(-|\bar{s}-\bar{g}|^2/(2w^2))$. It is read at the next state, as everywhere in this compendium.

Proposition H.2. [ENV-1 | I+D]

(a) (H.1) *defines a Markov kernel on* $\mathcal{S}$.

(b) For both layouts and every $p_{\text{slip}} \in [0, \frac{1}{2})$, *in particular* $p_{\text{slip}} = 0.1$, $\operatorname{rank} \tilde{P} = \#\mathcal{S}$ *(49 or 40). Hence no exact FB representation of rank* $d = 16$ *exists, for any policy family (Theorem F.9).*

(c) $0 < r_g \le 1$, *with* $r_g(s) = 1$ *if and only if* $s = g$. *Every non-goal cell has* $r_g(s) \le e^{-2}$, *since distinct cell centres are at least* $L = 2w$ *apart.*

Proof. (a) Each row is a convex combination of Dirac masses on $\mathcal{S}$.

(b) $\mathrm{nx}(s, \text{stay}) = s$, so the stay rows form the matrix $(1 - p_{\text{slip}})I + p_{\text{slip}}U$ with $U := \frac{1}{5}\sum_{a'} N_{a'}$, where $N_{a'}$ is the 0–1 matrix of $s \mapsto \mathrm{nx}(s, a')$. U is row-stochastic. Theorem F.8(c) shows that such a matrix is nonsingular when $p_{\text{slip}} < \frac{1}{2}$. Its $\#\mathcal{S}$ rows are rows of $\tilde{P}$, so $\operatorname{rank} \tilde{P} \ge \#\mathcal{S}$; $\tilde{P}$ has $\#\mathcal{S}$ columns, so equality holds. Both 49 and 40 exceed 16.

(c) The exponent is 0 only at $\bar{s} = \bar{g}$. For $s \ne g$, $|\bar{s} - \bar{g}| \ge L$, so the exponent is at most $-L^2/(2w^2) = -2$. □

`oracle.transition_rank` computes $\operatorname{rank} \tilde{P}$ numerically: 40 and 49, with smallest singular value about 2.01 at $p_{\text{slip}} = 0.1$ and $\sqrt{5}$ at $p_{\text{slip}} = 0$. `tests/test_oracle.py` checks the rank and a smallest singular value above 0.1 for both slip values.

H.2 The continuous kernel

Definition H.3 (continuous maze; [SA U-5])**.** For a lattice cell c let $\bar{C}_c := [c_x L, (c_x + 1)L] \times [c_y L, (c_y + 1)L]$ (closed) and C_c° its interior. The free space is the open set

$$\mathcal{S} := (0,1)^2 \setminus \textstyle\bigcup_{c \in \mathcal{W}_{4\mathrm{R}}} \bar{C}_c, \qquad \lambda(\mathcal{S}) = 1 - 9L^2 = \tfrac{40}{49},$$

where λ is Lebesgue measure. For $p, q \in \mathbb{R}^2$, $[p, q]$ is the closed segment, $\mathcal{O} := \{(y, x) \in \mathbb{R}^2 \times \mathbb{R}^2 : [y, x] \subseteq \mathcal{S}\}$, and $V(y) := \{x : (y, x) \in \mathcal{O}\}$ is the set of points visible from y. Fix the step $\ell := L$ and the noise level $\sigma = 0.01$. For $s \in \mathcal{S}$ and an action a, let $m := s + \ell\delta_a$ and define the *base point* $b(s, a) := m$ if $[s, m] \subseteq \mathcal{S}$, and $b(s, a) := s$ otherwise (a blocked move becomes "stay"). With $\alpha(y) := \mathcal{N}(y, \sigma^2 I)(V(y))$, the kernel is

$$P(X \mid s, a) := \frac{\mathcal{N}(b, \sigma^2 I)(X \cap V(b))}{\alpha(b)}, \qquad b = b(s, a). \tag{H.2}$$

The start is the centre $s_0 := (1.5L, 1.5L)$ of cell $(1, 1)$. Reward, goal region and success use the positions themselves: $r_g(x) = \exp(-|x - \bar{g}|^2/(2w^2))$; the goal region is $\{|x - \bar{g}| \le \rho\}$ with $\rho := L/2$; success means that some s_t with $1 \le t \le H$ lies in the goal region.

The implementation (`ContinuousMaze.step`) draws $\xi \sim \mathcal{N}(0, \sigma^2 I)$ until $b + \xi \in V(b)$, at most $K_{\text{cap}} + 1$ times with $K_{\text{cap}} = 1000$, and raises an error if every draw fails. Visibility is decided by a slab test of the closed segment against every closed wall square, together with a strict end-point test for the open unit square, which suffices because the square is convex. The test is exact in real arithmetic. In floating point it can misclassify a segment within rounding distance of a wall boundary, an event of negligible probability that the proofs below do not cover.

Proposition H.4. [ENV-2 | D] *Let* $s \in \mathcal{S}$, $a \in \mathcal{A}$, $b := b(s, a)$, *and* $\alpha_\star := (\Phi(L/(2\sigma)) - \frac{1}{2})^2$.

(a) $b \in \mathcal{S}$, *the sets* $\mathcal{O}$ *and* $V(b)$ *are open, and* $\alpha(b) \ge \alpha_\star$. *For* $\sigma = 0.01$, $\alpha_\star = \frac{1}{4} - 4.6 \times 10^{-13}$.

(b) Uncapped rejection sampling returns exactly the law (H.2), *after a geometric number of draws with mean* $1/\alpha(b) \le 1/\alpha_\star$, *and its output is independent of the number of draws.*

(c) $P(\cdot \mid s, a)$ *is concentrated on* $\mathcal{S}$ *and has the Lebesgue density*

$$p(x \mid s, a) = \frac{\varphi_\sigma(x - b)\,\mathbf{1}_{V(b)}(x)}{\alpha(b)} \le p_{\max} := \frac{1}{2\pi\sigma^2\alpha_\star} \approx 6366.$$

In particular it has no atom and charges no Lebesgue-null set.

(d) (No tunnelling.) For every outcome s', the path $s \to b \to s'$ lies in $\mathcal{S}$: $[s,b] \subseteq \mathcal{S}$ and $[b,s'] \subseteq \mathcal{S}$. This holds for any step length ℓ, not only $\ell \le L$.

(e) (Implemented sampler.) The capped sampler fails, and raises, with probability $p_b := (1-\alpha(b))^{K_{\mathrm{cap}}+1} \le (1-\alpha_\star)^{K_{\mathrm{cap}}+1} < 10^{-125}$. Given that it does not fail, its output has exactly the law (H.2)*; it never returns an atom.*

(f) (Consequence for FB-11.) If a reference measure ν satisfies $\nu \ge n_{\min}\lambda$ on $\mathcal{S}$ for some $n_{\min} > 0$, then for every stationary π and every (s,a), $M^{\pi,+}(s,a,\cdot) \ll \nu$ with density at most $p_{\max}/(n_{\min}(1-\gamma))$ (Theorem F.25(b)). The replay law of FB pretraining has such a density (Theorem H.10); an atomic ν never has one (Theorem H.5).

(g) (Measurability.) $s \mapsto b(s,a)$ is Borel on $\mathcal{S}$, $(s,x) \mapsto p(x \mid s,a)$ is Borel on $\mathcal{S} \times \mathbb{R}^2$, and $s \mapsto P(X \mid s,a)$ is Borel for every Borel X. Hence (H.2) *is a Markov kernel, as Theorem B.1(ii) and Theorem F.25(b) require.*

Proof. (a) If $[s,m] \subseteq \mathcal{S}$ then $b = m \in \mathcal{S}$; otherwise $b = s \in \mathcal{S}$.

For openness of $\mathcal{O}$, let $(y,x) \in \mathcal{O}$. The compact set $[y,x]$ lies in the open set $\mathcal{S}$, so it has a positive distance r_0 from the closed set $\mathbb{R}^2 \setminus \mathcal{S}$. If $|y'-y| < r_0$ and $|x'-x| < r_0$, then for $t \in [0,1]$, $|(1-t)y' + tx' - ((1-t)y + tx)| < r_0$, so $[y',x'] \subseteq \mathcal{S}$. Hence $\mathcal{O}$ is open, and so is each section $V(b)$.

For the bound, the closed lattice squares cover $[0,1]^2$, so $b \in \bar{C}_c$ for some c. This c is free, because b lies in no closed wall square. In each coordinate k, at least one of the two sides of $\bar{C}_c$ orthogonal to e_k is at distance at least $L/2$ from b_k; let $\varsigma_k \in \{-1,+1\}$ point towards it. Then the open rectangle $R := \{b + (\varsigma_1 u_1, \varsigma_2 u_2) : u \in (0, L/2)^2\}$ lies in C_c°. C_c° is contained in $(0,1)^2$ and meets no closed wall square, since distinct closed lattice squares meet only on their boundaries; so $C_c^\circ \subseteq \mathcal{S}$. For $x = b + (\varsigma_1 u_1, \varsigma_2 u_2) \in R$ and $t \in (0,1]$, $b + t(x-b) = b + (\varsigma_1 t u_1, \varsigma_2 t u_2) \in R$; together with $b \in \mathcal{S}$ this gives $[b,x] \subseteq \mathcal{S}$. Hence $R \subseteq V(b)$, and since the coordinates of $\xi \sim \mathcal{N}(0, \sigma^2 I)$ are independent, $\alpha(b) \ge \mathbb{P}(\varsigma_1 \xi_1 \in (0, \frac{L}{2}))\, \mathbb{P}(\varsigma_2 \xi_2 \in (0, \frac{L}{2})) = \alpha_\star$. The numerical value uses $L/(2\sigma) = 7.14$.

(b) Let $\xi^{(1)}, \xi^{(2)}, \dots$ be i.i.d. $\mathcal{N}(0, \sigma^2 I)$ and J the first index with $b + \xi^{(J)} \in V(b)$; $V(b)$ is open, hence Borel. For $k \ge 1$ and Borel X, independence gives $\mathbb{P}(J = k,\ b + \xi^{(k)} \in X) = (1-\alpha)^{k-1}\, \mathcal{N}(b, \sigma^2 I)(X \cap V(b))$ with $\alpha = \alpha(b) > 0$. This factorizes as $\mathbb{P}(J = k)\, P(X \mid s,a)$, so the output $b + \xi^{(J)}$ is independent of J. Summing over k gives its law (H.2). J is geometric with mean $1/\alpha$.

(c) The density follows from (b). For $x \in V(b)$, $x \in [b,x] \subseteq \mathcal{S}$, so $V(b) \subseteq \mathcal{S}$. The bound uses $\varphi_\sigma \le 1/(2\pi\sigma^2)$ and (a).

(d) By construction, $b = m$ only if $[s,m] \subseteq \mathcal{S}$; if $b = s$, then $[s,b] = \{s\} \subseteq \mathcal{S}$. The output satisfies $s' \in V(b)$, i.e. $[b,s'] \subseteq \mathcal{S}$. The argument never uses the step length.

(e) The capped sampler fails exactly on $\{J > K_{\mathrm{cap}} + 1\}$, of probability $(1-\alpha(b))^{K_{\mathrm{cap}}+1} \le (1-\alpha_\star)^{K_{\mathrm{cap}}+1}$ by (a). On its complement it returns $b + \xi^{(J)}$, whose law given $\{J \le K_{\mathrm{cap}} + 1\}$ is (H.2) by the independence in (b). Numerically, $1001 \log_{10}(1-\alpha_\star) \approx -125.06$.

(f) Apply Theorem F.25(b) with λ restricted to $\mathcal{S}$, the bound of (c), and the measurability in (g).

(g) $A_a := \{s \in \mathcal{S} : (s, s + \ell\delta_a) \in \mathcal{O}\}$ is open, being the preimage of the open set $\mathcal{O}$ under a continuous map. So $b(s,a) = s + \ell\delta_a \mathbf{1}_{A_a}(s)$ is Borel. $\alpha(y) = \int \varphi_\sigma(x-y)\mathbf{1}_{\mathcal{O}}(y,x)\,\mathrm{d}x$ is Borel in y by Tonelli's theorem, because the integrand is jointly Borel; and $\alpha \ge \alpha_\star > 0$ on $\mathcal{S}$ by (a). Hence $p(x \mid s,a) = \varphi_\sigma(x - b(s,a))\mathbf{1}_{\mathcal{O}}(b(s,a),x)/\alpha(b(s,a))$ is a composition of Borel maps, and $P(X \mid s,a) = \int_X p(x \mid s,a)\,\mathrm{d}x$ is Borel in s, again by Tonelli. □

Remark H.5 (atoms and domination for this kernel). The kernel (H.2) has no atom, no clipping and no other part singular to λ. Hence Theorem F.25(a) cannot be triggered through P for a reference measure with $\lambda|_{\mathcal{S}} \ll \nu$, for instance $\nu \ge n_{\min}\lambda$ on $\mathcal{S}$ (Theorem H.4(f)). It is triggered at every (s,a) by any atomic ν, in particular by the empirical buffer law $\hat{\nu}$: $P(\mathcal{S} \setminus \mathrm{supp}\hat{\nu} \mid s,a) = 1$ while $\hat{\nu}(\mathcal{S} \setminus \mathrm{supp}\hat{\nu}) = 0$ (Theorem F.27(2)). Atoms appear elsewhere:

- the atom of $\mu_0 = \delta_{s_0}$ appears in $d^{\mathrm{pre}} = (1-\gamma)\mu_0 + \gamma d^+$ (Theorem B.5(b)), and the conditioning state appears as δ_s in $M_{21}(s, a, \cdot \times \mathcal{A})$ (Theorem B.19(a)); neither appears in d^+ or $M^{\pi,+}$;
- $\mathrm{DM}_{\mathrm{FB}}$'s start law $\bar{\rho}_{\mathcal{T}_K}$ is atomic, but it is a start law, not a successor measure;
- empirical measures (demonstrations, a replay buffer and its $\hat{\nu}$) are atomic (FBIL-1).

The implemented sampler never returns an atom. The tests and the recorded checks in `audit/prompt3A/env_check_*` completed, so the redraw cap was never reached in them. The bound of Theorem H.4(a) is essentially attained: near a 90° corner of $\mathcal{S}$ (the inside corner of a room, or a corner of the unit square) the visible set is, at the noise scale, a quadrant, so $\alpha(b) \approx \frac{1}{4}$. At the reflex corner of a doorway, $\alpha(b) \approx \frac{5}{8}$. `tests/test_env_continuous.py` measures acceptance rates $\frac{1}{4}$, $\frac{1}{2}$ and $\frac{5}{8}$ at a room corner, a straight wall face and a doorway corner; an end-point-only check would give $\frac{3}{4}$ at the doorway corner.

H.3 Positivity of histogram cells

Let $\mathcal{B}_{ij} := [\frac{i}{20}, \frac{i+1}{20}) \times [\frac{j}{20}, \frac{j+1}{20})$, and let the *retained* cells be those with $\lambda(\mathcal{B}_{ij} \cap \mathcal{S}) > 0$. For the four-rooms free space of Theorem H.3 there are 356 of them; the 44 dropped bins lie inside walls. (For the open layout, $\mathcal{S} = (0,1)^2$ and all 400 bins are retained.)

Proposition H.6. [ENV-3 | D]

(a) The sets $\mathcal{B}_{ij} \cap \mathcal{S}$ of the retained cells partition $\mathcal{S}$: a dropped bin contains no point of $\mathcal{S}$.

(b) Let π be a stationary policy with $\pi(\mathrm{stay} \mid x) > 0$ for every $x \in \mathcal{S}$, for example any softmax policy. Let D be the largest breadth-first-search distance from the start cell $(1,1)$ in the free-cell graph of the four-rooms lattice; $D = 10$. Then every retained cell $\mathcal{B}$ satisfies $\mathbb{P}^\pi(s_{t+1} \in \mathcal{B}) > 0$ for some $t \le D$. Hence $d^+_{\pi,H}(\mathcal{B}) > 0$ for every $H \ge D+1$, in particular $H = 100$.

(c) Consequently Theorem D.1(v) holds on the retained cells for the learner occupancy of every policy as in (b). An empirical expert target can have empty cells, so the target side needs the smoothing of Theorem C.7. With both arguments smoothed, positivity holds trivially (Theorem D.10(c)).

Proof. (a) Suppose $x \in \mathcal{B}_{ij} \cap \mathcal{S}$. Since $\mathcal{S}$ is open, it contains an open disc around x. The half-open square $\mathcal{B}_{ij}$ contains x, so the disc meets $\mathcal{B}_{ij}$ in a set containing a small square with corner x, which has positive area. Hence $\lambda(\mathcal{B}_{ij} \cap \mathcal{S}) > 0$ and the bin is retained. The bins partition $[0,1)^2 \supseteq \mathcal{S}$.

(b) *One step between adjacent free cells.* Let c, c' be 4-adjacent free cells and $x \in C^\circ_c$. The open rectangle $R_{cc'} := \mathrm{int}(\bar{C}_c \cup \bar{C}_{c'})$ is convex and lies in $(0,1)^2$. Its points off the shared edge lie in $C^\circ_c \cup C^\circ_{c'} \subseteq \mathcal{S}$. The interior points of the shared edge lie in no closed lattice square other than $\bar{C}_c$ and $\bar{C}_{c'}$, hence in no wall square. So $R_{cc'} \subseteq \mathcal{S}$, and for $y \in C^\circ_{c'}$, $[x, y] \subseteq R_{cc'}$, i.e. $C^\circ_{c'} \subseteq V(x)$. With $a = \mathrm{stay}$ we have $b = x$, and by Theorem H.4(c),

$$P(C^\circ_{c'} \mid x, \mathrm{stay}) = \mathcal{N}(x, \sigma^2 I)(C^\circ_{c'})/\alpha(x) > 0.$$

In the same way, $P(U \mid x, \mathrm{stay}) > 0$ for every nonempty open $U \subseteq C^\circ_c$, because C°_c is convex and contains x, so $C^\circ_c \subseteq V(x)$.

Chain. Let $\mathcal{B}$ be retained. $\mathcal{B} \cap \mathcal{S}$ has positive area and the cell boundaries are null, so some free cell c has $\lambda(\mathcal{B} \cap C^\circ_c) > 0$. Then $U := \mathrm{int}(\mathcal{B}) \cap C^\circ_c$ is open and nonempty. Let $c_0 = (1,1), c_1, \ldots, c_k = c$ be a shortest path in the free-cell graph, so $k \le D$. Let $E_i := \{a_j = \mathrm{stay},\ s_{j+1} \in C^\circ_{c_{j+1}} \text{ for } j < i\}$, with E_0 the sure event; note $s_0 \in C^\circ_{c_0}$. By the Markov property, $\mathbb{P}(E_{i+1}) = \mathbb{E}[\mathbf{1}_{E_i}\, g_i(s_i)]$ with $g_i(x) := \pi(\mathrm{stay} \mid x)\, P(C^\circ_{c_{i+1}} \mid x, \mathrm{stay})$. This g_i is Borel, by Theorem H.4(g) and the measurability of π, and strictly positive on $C^\circ_{c_i}$. On E_i, $s_i \in C^\circ_{c_i}$. So $\mathbb{P}(E_i) > 0$ implies $\mathbb{P}(E_{i+1}) > 0$, and by induction $\mathbb{P}(E_k) > 0$. One more stay step gives $\mathbb{P}(E_k,\ a_k = \mathrm{stay},\ s_{k+1} \in U) = \mathbb{E}[\mathbf{1}_{E_k} \pi(\mathrm{stay} \mid s_k) P(U \mid s_k, \mathrm{stay})] > 0$, since $s_k \in C^\circ_c$ on E_k. As

$U \subseteq \mathcal{B}$, $\mathbb{P}(s_{k+1} \in \mathcal{B}) > 0$ with $t = k \le D$, and $d^+_{\pi,H}(\mathcal{B}) \ge Z_H^{-1}\gamma^k \mathbb{P}(s_{k+1} \in \mathcal{B}) > 0$. $D = 10$ is the breadth-first-search distance from $(1,1)$ to $(6,6)$. It is at least the Manhattan distance 10, and the path $(1,1) \to (5,1) \to (5,5) \to (6,5) \to (6,6)$ through the doorways $(3,1)$ and $(5,3)$ has length 10. That no free cell is farther is computed by `GridLayout.bfs_distances` and reported by `scripts/check_environments.py`.

(c) is (b) together with Theorem C.7(a). □

The empirical coverage diagnostic of `scripts/check_environments.py` (a uniform random policy, 2000 trajectories) is a sanity check, not a proof: a finite sample can miss a cell of positive probability.

H.4 Estimators

Proposition H.7. [ENV-4 | I] *Let N trajectories of length H be given, and put $w_{i,t} := \gamma^t/(NZ_H)$ on the next state s^i_{t+1}, $0 \le t < H$.*

(a) $\hat{d} := \sum_{i,t} w_{i,t}\delta_{s^i_{t+1}}$ is a probability measure, each trajectory has total weight $1/N$, and for i.i.d. trajectories from μ_0 under π, $\mathbb{E}\hat{d}(X) = d^+_{\pi,H}(X)$ for every measurable X. The same weights define $\hat{q}_E$ from the demonstrations.

(b) (Cut-and-normalize KDE.) For points $x_i \in \mathcal{S}$ with weights $w_i \ge 0$, $\sum_i w_i = 1$, and bandwidth $h > 0$, let $m_i := \int_{\mathcal{S}} \varphi_h(y - x_i)\,\mathrm{d}y$ and $\hat{p}(x) := \sum_i w_i \varphi_h(x - x_i)\mathbf{1}_{\mathcal{S}}(x)/m_i$. Then $m_i \ge (\Phi(L/(2h)) - \frac{1}{2})^2$, which is 0.179 for $h = 0.05$, and $\int_{\mathcal{S}} \hat{p} = 1$. Moreover

$$m_i = G_{[0,1]^2}(x_i) - \textstyle\sum_{c\in\mathcal{W}} G_{\bar{C}_c}(x_i), \qquad G_{[a_1,b_1]\times[a_2,b_2]}(\mu) := \prod_{k=1,2}\Big[\Phi\big(\tfrac{b_k-\mu_k}{h}\big) - \Phi\big(\tfrac{a_k-\mu_k}{h}\big)\Big],$$

with $G_\emptyset := 0$. With the closure $Q \subseteq [0,1]^2$ of a histogram cell in place of $[0,1]^2$ and $Q \cap \bar{C}_c$ in place of $\bar{C}_c$, the same formula, weighted by w_i/m_i and summed, gives the exact $\hat{p}$-mass of the cell, up to the evaluation of Φ. The smoothed density $(1-\varepsilon)\hat{p} + \varepsilon\mathbf{1}_{\mathcal{S}}/\lambda(\mathcal{S})$ also integrates to 1.

(c) (Histogram.) The masses $\hat{d}(\mathcal{B} \cap \mathcal{S})$ of the retained cells sum to 1. The reference law $\upsilon_{\mathcal{B}} := \lambda(\mathcal{B} \cap \mathcal{S})/\lambda(\mathcal{S})$ is strictly positive, and the smoothed masses are at least $\varepsilon \min \upsilon > 0$.

No unbiasedness is claimed for $\hat{p}(x)$, for the plug-in signal built from it (Theorem D.11), or for the histogram density as an estimate of a Lebesgue density.

Proof. (a) $\sum_{i,t} w_{i,t} = \sum_i \frac{1}{N}\sum_t \gamma^t/Z_H = 1$, and the trajectory sums are $1/N$. By linearity, $\mathbb{E}\hat{d}(X) = \frac{1}{N}\sum_i \sum_t \frac{\gamma^t}{Z_H}\mathbb{P}^\pi_{\mu_0}(s_{t+1} \in X) = d^+_{\pi,H}(X)$ (Theorem B.2).

(b) The lower bound is the argument of Theorem H.4(a) with h in place of σ: the rectangle $R \subseteq \mathcal{S}$ built there at x_i has Gaussian mass $(\Phi(L/(2h)) - \frac{1}{2})^2$. By Tonelli, $\int_{\mathcal{S}} \hat{p} = \sum_i w_i \int_{\mathcal{S}} \varphi_h(y - x_i)\,\mathrm{d}y/m_i = \sum_i w_i = 1$. For the closed form, $\mathcal{S} = (0,1)^2 \setminus \bigcup_c \bar{C}_c$. The wall squares lie in $[0,1]^2$, the boundaries are null, and distinct wall squares meet in null sets. So λ-a.e. $\mathbf{1}_{\mathcal{S}} = \mathbf{1}_{[0,1]^2} - \sum_c \mathbf{1}_{\bar{C}_c}$. Integrating $\varphi_h(\cdot - x_i)$ and using independence of the coordinates gives the product of one-dimensional differences of Φ. The same identity holds with Q in place of $[0,1]^2$ and $Q \cap \bar{C}_c$ (a box, or empty) in place of $\bar{C}_c$. The smoothed density integrates to $(1-\varepsilon) + \varepsilon = 1$.

(c) The retained cells partition $\mathcal{S}$ (Theorem H.6(a)), so their masses sum to $\hat{d}(\mathcal{S}) = 1$. The reference masses are positive by definition of retained and sum to $\lambda(\mathcal{S})/\lambda(\mathcal{S}) = 1$. □

Remark H.8 (what the KDE gets wrong near walls). Truncation is Euclidean: a kernel ignores visibility. (i) For a centre x_i on one side of a wall of thickness at least $L \approx 2.9h$, at most $\Phi(-L/h)/m_i$ of the truncated kernel's mass lies beyond the far face of that wall: about 0.4% if $m_i \approx \frac{1}{2}$, at most 1.2%. (ii) This does not bound the mass on free points hidden behind a wall corner. `audit/prompt3A/kde_visibility.py` estimates, by Monte Carlo at $h = 0.05$, the hidden share of the truncated kernel. Over a 70×70 grid of centres in $\mathcal{S}$ it has median 0, mean 1.1% and 95th percentile 7.3%. It is 8% at the middle of a wall face next to a doorway and 16%

at a doorway corner. It reaches about 31.5% for centres within about $0.05L$ of a wall corner at a doorway entrance. Both effects are approximations made by the estimator; the pilot kept $h = 0.05$ (Section 10). They are not properties of the kernel P.

H.5 Experts

Proposition H.9. [ENV-5 | I+D]

(a) (Finite expert.) For every stationary deterministic π, $0 \le V^\star - V^\pi \le \eta_\pi/(1-\gamma)$ with $\eta_\pi := \|TV^\pi - V^\pi\|_\infty$, which `expert.finite_expert` *reports. If $Q^\pi(s,\pi(s)) \ge \max_a Q^\pi(s,a) - \kappa$ for every s, where $Q^\pi = Q^{+,\pi}_{r_g}$, then $\eta_\pi \le \kappa$.* `oracle.value_iteration` *refines the greedy policy of its final iterate as follows. An action is switched only where it is more than κ below $\max_a Q^\pi(s,a)$, and then to the first action, in the fixed order, within κ of that maximum. This refinement terminates, and its result satisfies the κ-greedy condition;* `finite_expert` *raises an error otherwise. The implementation uses $\kappa =$* `tie_tol` $= 10^{-10}$*; the claims hold up to the floating-point error of the linear solves.*

(b) (Route controller.) For $x \in \mathcal{S} \subseteq (0,1)^2$, the cell $c := \lfloor 7x \rfloor$ (componentwise) lies in G, and it is free: $x \in \bar{C}_c$, and $\mathcal{S}$ meets no closed wall square. The continuous expert $\pi_E(x) := \Gamma(\lfloor 7x \rfloor)$, with a deterministic table Γ on free cells, is therefore well defined. It is a stationary Markov policy, constant on half-open lattice cells, using no time index or history, as A4 requires [SA A4]*. Being deterministic, it does not belong to the learners' positive families π_ϕ and $\pi_{z,\tau}$ ($\tau > 0$). It is a constructed reference controller; no optimality is claimed.*

(c) (Planned moves are not blocked.) If $x \in C^\circ_c$ and $\Gamma(c) = a$ moves to a free neighbour $c' = c + \delta_a$, then $[x, x + L\delta_a] \subseteq R_{cc'} \subseteq \mathcal{S}$, so $b(x,a) = x + L\delta_a$. Since $s_0 \in C^\circ_{(1,1)}$ and s_t has a Lebesgue density for $t \ge 1$, almost surely no planned move of π_E is ever blocked.

Proof. (a) The first bound is Theorem B.16(iii). Under the κ-greedy condition, $(TV^\pi)(s) - V^\pi(s) = \max_a Q^\pi(s,a) - Q^\pi(s,\pi(s)) \in [0,\kappa]$, so $\eta_\pi \le \kappa$.

Termination. Let π' differ from π exactly at the switched states. At a switched state s, $Q^\pi(s,\pi'(s)) \ge \max_a Q^\pi(s,a) - \kappa > Q^\pi(s,\pi(s)) = V^\pi(s)$; elsewhere $Q^\pi(s,\pi'(s)) = V^\pi(s)$. So $T_{\pi'}V^\pi \ge V^\pi$, with strict inequality at the switched states. By monotonicity of $T_{\pi'}$ (Theorem B.15(i)), $T^n_{\pi'}V^\pi \ge T_{\pi'}V^\pi$ for every $n \ge 1$. Letting $n \to \infty$ gives $V^{\pi'} \ge T_{\pi'}V^\pi \ge V^\pi$ (Theorem B.15(ii)), strictly at the switched states. The value vector therefore increases strictly in the componentwise order at each round, so no deterministic policy is visited twice. There are finitely many, so the refinement stops, and it stops exactly when no state can be switched, i.e. at a κ-greedy policy.

(b) $x \mapsto \lfloor 7x \rfloor$ is Borel and Γ takes finitely many values, so π_E is a measurable deterministic Markov kernel. It uses no time index and no history. For $x \in (0,1)^2$, $\lfloor 7x \rfloor \in \{0,\dots,6\}^2$ and $x \in \bar{C}_{\lfloor 7x \rfloor}$.

(c) $x + L\delta_a \in C^\circ_{c'}$ because $C^\circ_{c'} = C^\circ_c + L\delta_a$. The segment joins two points of the convex set $R_{cc'} \subseteq \mathcal{S}$ (proof of Theorem H.6(b)). For $t \ge 1$, s_t has a Lebesgue density: Theorem H.4(c),(g) and Theorem B.20. Lattice-cell boundaries are Lebesgue-null, so s_t lies in an open cell almost surely. □

The checks in `expert.validate_route_controller` are empirical. They test the conclusions above on real noisy rollouts: no blocked planned move; both legs of every transition in $\mathcal{S}$, by dense sampling independent of the slab test; and actions recomputed from the stored positions equal to the stored actions. They also measure the success rate and how well the goal is held. Negative controls in `tests/test_expert.py` show that each criterion can fail. These are measurements, not proofs.

H.6 The task-blind replay law

FB pretraining (`fb.collect_replay`) collects N_{seg} segments of T_{seg} transitions each (the subscript keeps T free for the f-IRL horizon of Section E). Each segment starts from a reset $s_0 \sim \upsilon$, independent of everything else, where υ is uniform on the free cells (finite maze) or the normalized Lebesgue measure $\lambda(\cdot \cap \mathcal{S})/\lambda(\mathcal{S})$ (continuous maze, sampled by rejection from the unit square). The actions are i.i.d. uniform on $\mathcal{A}$, independent of the states, and $s_{t+1} \sim P(\cdot \mid s_t, a_t)$ with the kernel of Theorem H.1 or Theorem H.3. Minibatches pick stored transitions uniformly at random, so a transition has the law of (s_t, a_t, s_{t+1}) with t uniform on $\{0, \dots, T_{\text{seg}} - 1\}$, independent of the segment; neither storage nor sampling looks at s_{t+1}, which is Theorem F.21(i). The replay reference measure of Theorem F.21(ii) is therefore

$$\nu := \frac{1}{T_{\text{seg}}} \sum_{t=0}^{T_{\text{seg}}-1} \text{Law}(s_{t+1}).$$

Proposition H.10. [ENV-6 | D]

(a) (Finite mazes.) The uniform-action kernel $P_u(s, s') := |\mathcal{A}|^{-1} \sum_a P(s' \mid s, a)$ is a symmetric matrix. Hence $\text{Law}(s_t) = \upsilon$ for every t, and $\nu = \upsilon$ is uniform: $\nu(s) = 1/\#\mathcal{S} > 0$ for every state, for every segment length T_{seg} and slip probability.

(b) (Continuous maze.) With $\alpha_\star = (\Phi(L/(2\sigma)) - \frac{1}{2})^2$ as in Theorem H.4,

$$\nu \;\geq\; n_{\min}\lambda \quad \text{on } \mathcal{S}, \qquad n_{\min} := \frac{\alpha_\star}{|\mathcal{A}|\, T_{\text{seg}}\, \lambda(\mathcal{S})}.$$

Hence the hypothesis of Theorem H.4(f) holds: for every stationary π and every (s, a), $M^{\pi,+}(s, a, \cdot) \ll \nu$ with a ν-density bounded by $p_{\max}/(n_{\min}(1-\gamma))$. For $\sigma = 0.01$, $|\mathcal{A}| = 5$, $T_{\text{seg}} = 50$ and $\lambda(\mathcal{S}) = 40/49$, $n_{\min} \approx 1.2 \times 10^{-3}$.

(c) (Empirical buffer.) These statements concern the population law ν. The buffer's empirical law $\hat{\nu}$ is atomic. In the continuous maze it charges a finite set, so (b) does not transfer to it (Theorem H.5). In the finite mazes, $\hat{\nu}(s) > 0$ for every s is a property of each realized buffer, not a consequence of (a): `tests/test_fb_algebra.py` checks it for one 50 000-transition four-rooms buffer, and every buffer used in a run must be checked with `fb.replay_coverage`.

Proof. (a) By (H.1), $\sum_a P(\cdot \mid s, a) = \sum_a \left[(1 - p_{\text{slip}})\delta_{\text{nx}(s,a)} + \frac{p_{\text{slip}}}{|\mathcal{A}|} \sum_{a'} \delta_{\text{nx}(s,a')}\right] = \sum_a \delta_{\text{nx}(s,a)}$, so $P_u(s, s') = \#\{a : \text{nx}(s, a) = s'\}/|\mathcal{A}|$. Let $s \neq s'$. The displacements δ_a are distinct, so at most one action moves s to s', and it does so exactly when $s' - s = \delta_a$ and s' is free. The displacement set is symmetric ($\delta \mapsto -\delta$ exchanges right and left, up and down). Hence $\text{nx}(s, a) = s'$ for some a if and only if $\text{nx}(s', a') = s$ for the opposite action a', both cells being free. So $P_u(s, s') = P_u(s', s)$. A symmetric row-stochastic matrix is also column-stochastic, so $(\upsilon P_u)(s') = \#\mathcal{S}^{-1} \sum_s P_u(s', s) = \#\mathcal{S}^{-1}$: the uniform law is invariant. Given the states, the actions are uniform and independent, so $\text{Law}(s_{t+1}) = \text{Law}(s_t)P_u$, and by induction $\text{Law}(s_t) = \upsilon$ for all t. Averaging over t gives $\nu = \upsilon$.

(b) Let $X \subseteq \mathcal{S}$ be Borel. Since $\nu \geq T_{\text{seg}}^{-1}\text{Law}(s_1)$, it suffices to bound $\text{Law}(s_1)$. Keeping only the stay action, for which $b(s, \text{stay}) = s$, and using $\alpha(s) \leq 1$ in (H.2),

$$\text{Law}(s_1)(X) = \int \upsilon(\mathrm{d}s) \frac{1}{|\mathcal{A}|} \sum_a P(X \mid s, a) \;\geq\; \frac{1}{|\mathcal{A}|\lambda(\mathcal{S})} \int_{\mathcal{S}} \mathrm{d}s \int_X \varphi_\sigma(x - s)\, \mathbf{1}_{\mathcal{O}}(s, x)\, \mathrm{d}x.$$

The set $\mathcal{O}$ is open (Theorem H.4(a)), so the integrand is Borel and nonnegative, and Tonelli's theorem exchanges the integrals. The closed segment $[s, x]$ equals $[x, s]$, and $[x, s] \subseteq \mathcal{S}$ forces $s \in \mathcal{S}$; so for $x \in X$ the inner integral over s is $\int \varphi_\sigma(s - x)\mathbf{1}_{V(x)}(s)\, \mathrm{d}s = \alpha(x)$. The proof of $\alpha(b) \geq \alpha_\star$ in Theorem H.4(a) uses only $b \in \mathcal{S}$, so $\alpha(x) \geq \alpha_\star$ for every $x \in X$. Hence $\text{Law}(s_1)(X) \geq \alpha_\star\lambda(X)/(|\mathcal{A}|\lambda(\mathcal{S}))$ and $\nu(X) \geq n_{\min}\lambda(X)$.

Theorem F.25(b), through which Theorem H.4(f) is proved, asks for a Lebesgue density n of ν with $n \geq n_{\min}$ on $\mathcal{S}$; we verify this form. For every $t \geq 0$ and Borel X, $\mathrm{Law}(s_{t+1})(X) = \mathbb{E}\, P(X \mid s_t, a_t) = \mathbb{E} \int_X p(x \mid s_t, a_t)\,\mathrm{d}x = \int_X \mathbb{E}\, p(x \mid s_t, a_t)\,\mathrm{d}x$ by Theorem H.4(c) and Tonelli's theorem, which applies because $(s, a, x) \mapsto p(x \mid s, a)$ is Borel (Theorem H.4(g)) and nonnegative. So ν has the Lebesgue density $n(x) := T_{\mathrm{seg}}^{-1} \sum_{t<T_{\mathrm{seg}}} \mathbb{E}\, p(x \mid s_t, a_t) \leq p_{\max}$, which vanishes off $\mathcal{S}$. The inequality $\nu \geq n_{\min}\lambda$ on $\mathcal{S}$ gives $\int_X (n - n_{\min})\,\mathrm{d}x \geq 0$ for every Borel $X \subseteq \mathcal{S}$, so $n \geq n_{\min}$ almost everywhere on $\mathcal{S}$; redefining n on the null set where this fails changes neither ν nor the conclusions, and then $n \geq n_{\min}$ on all of $\mathcal{S}$. Theorem F.25(b), with λ the Lebesgue measure restricted to $\mathcal{S}$, the density bound $p \leq p_{\max}$ and this n, gives the last claim of (b); ν is a probability measure, hence finite. For the number: $\alpha_\star = \frac{1}{4} - 4.6 \times 10^{-13}$ (Theorem H.4(a)) and $5 \cdot 50 \cdot 40/49 \approx 204.1$.

(c) An empirical measure of N points is atomic; Theorem H.5 shows that an atomic reference measure triggers Theorem F.25(a) at every (s, a) of the continuous maze. □

Consequences. In the finite mazes, every measure on $\mathcal{S}$ is absolutely continuous with respect to ν (Theorem F.25(d)). In the continuous maze, the population replay law satisfies the domination hypothesis of Theorem F.25(b), so every successor measure, the expert's included, has a bounded ν-density; this gives the absolute continuity and square integrability required in (A6) of Theorem G.6 (the joint measurability of a density version is not addressed here); the trained model uses $\hat{\nu}$, for which only the empirical-buffer readings of Theorem F.27(2) apply. No statement here concerns the accuracy of the trained networks.

H.7 Cross-fitted plug-in signals

The f-PG learner (`fpg.crossfit_advantages`) splits the N trajectories $i = 0, \ldots, N-1$ of an iteration into $n_{\mathrm{f}} \geq 2$ deterministic, nonempty folds $\mathcal{I}_1, \ldots, \mathcal{I}_{n_{\mathrm{f}}}$, with $i \in \mathcal{I}_j$ for $j = 1 + (i \bmod n_{\mathrm{f}})$ (`fpg.fold_ids` numbers the folds from 0). The fold index is j and the folds are $\mathcal{I}_j$, so that neither clashes with the time index k or the history σ-algebras $\mathcal{F}_k$ of Sections B and D. For the trajectories of fold j the learner uses a signal $\hat{g}_{-j} : \mathcal{S} \to \mathbb{R}$ and a baseline $\beta_{-j} : \{0, \ldots, H-1\} \to \mathbb{R}$ computed from the other folds and the fixed demonstrations only. The estimator is

$$\hat{\nabla} := \frac{1}{N} \sum_{j=1}^{n_{\mathrm{f}}} \sum_{i \in \mathcal{I}_j} Z_H^{-1} \sum_{m=0}^{H-1} \gamma^m \chi_m^i (\hat{G}_m^i - \beta_{-j}(m)), \qquad \hat{G}_m^i := \sum_{t=m}^{H-1} \gamma^{t-m} \hat{g}_{-j}(s_{t+1}^i) \quad (i \in \mathcal{I}_j).$$

For a bounded measurable $g : \mathcal{S} \to \mathbb{R}$ let $\Gamma_\phi(g) := \nabla_\phi\, Z_H^{-1} \sum_{t<H} \gamma^t \mathbb{E}_\phi\, g(s_{t+1}) = \nabla_\phi \int g\,\mathrm{d}d^+_{\pi_\phi, H}$.

Proposition H.11. [FPG-9 | D] *Assume Theorem D.1(i)–(ii). Let the trajectories be i.i.d. under $\mathbb{P}_\phi$, independent of the demonstrations, and let every $\hat{g}_{-j}$ and β_{-j} be measurable functions of the trajectories outside $\mathcal{I}_j$ and of the demonstrations, bounded by deterministic constants C_g and C_β.*

- *(a) For every j and $i \in \mathcal{I}_j$, conditionally on the data outside $\mathcal{I}_j$, the term of trajectory i has mean $\Gamma_\phi(\hat{g}_{-j})$; the baseline contributes nothing.*
- *(b) If $\hat{g}_{-j}$ is constant on the cells B_b of a finite partition (the states of the finite maze, or the histogram cells), then $\Gamma_\phi(\hat{g}_{-j}) = \sum_b \hat{g}_{-j}(b) \nabla d_\phi(b)$, and the bound of Theorem D.10(a) holds for it.*
- *(c) $\mathbb{E}\hat{\nabla} = N^{-1} \sum_j |\mathcal{I}_j|\, \Gamma_\phi(\bar{g}_j)$ with the pointwise means $\bar{g}_j := \mathbb{E}\, \hat{g}_{-j}$.*
- *(d) The implemented signals satisfy the hypothesis with $\varepsilon \in (0, 1]$: for counts and the histogram, both smoothed masses lie in $[\varepsilon v_{\min}, 1]$; for the cut-and-normalize KDE, both smoothed densities lie in $[\varepsilon/\lambda(\mathcal{S}),\ (1-\varepsilon)/(2\pi h^2 m_\star) + \varepsilon/\lambda(\mathcal{S})]$ with $m_\star := (\Phi(L/(2h)) - \frac{1}{2})^2$. In every case the ratio stays in a compact subset of $(0, \infty)$, where f' is continuous, so $|\hat{g}_{-j}| \leq C_g$ deterministically. The time baseline is a mean of signal-to-go values, so $|\beta_{-j}| \leq C_g/(1-\gamma)$.*

Nothing here makes $\hat{\nabla}$ unbiased for ∇J: the signal is still a plug-in estimate (Theorem D.10(b)). For the KDE signal no exact cell signal exists, and no bound between $\Gamma_\phi(\hat{g}_{-j})$ and a gradient of J is claimed (Theorem D.11).

Proof. (a) Write D_{-j} for the trajectories outside $\mathcal{I}_j$ together with the demonstrations. The trajectory τ^i is independent of D_{-j}, so for a bounded measurable Ψ, $\mathbb{E}[\Psi(\tau^i, D_{-j}) \mid D_{-j}] = \psi(D_{-j})$ with $\psi(\delta) := \mathbb{E}\,\Psi(\tau^i, \delta)$ (Fubini's theorem on the product of the two laws). Fix δ and put $g := \hat{g}_{-j}$ and $\beta := \beta_{-j}$ evaluated at δ: a bounded measurable function and a bounded vector. Theorem D.4 with $h = g(s_{t+1})$, which is bounded, measurable and free of ϕ, gives $\nabla_\phi \mathbb{E}_\phi g(s_{t+1}) = \mathbb{E}_\phi[g(s_{t+1}) \sum_{m<H} \chi_m]$, and Theorem D.5(b) removes the terms $m \geq t+1$. Multiplying by γ^t/Z_H, summing over t and exchanging the finite sums as in the proof of Theorem D.7 gives $\mathbb{E}_\phi[Z_H^{-1} \sum_m \gamma^m \chi_m \hat{G}_m] = \Gamma_\phi(g)$. For the baseline, $\mathbb{E}_\phi[\chi_m]\beta(m) = 0$ by Theorem D.5(a). The integrand is bounded by $HC(C_g/(1-\gamma) + C_\beta)$, with C the score bound of Theorem D.1(ii), so all expectations exist.

(b) For a cell function, $\mathbb{E}_\phi g(s_{t+1}) = \sum_b g(b)\mathbb{P}_\phi(s_{t+1} \in B_b)$, so $\Gamma_\phi(g) = \sum_b g(b)\nabla d_\phi(b)$ by (D.1). The bound is Theorem D.10(a), whose proof uses only this identity and the boundedness of g.

(c) Take expectations in (a) and average over i. By Theorems D.4 and D.5, $\Gamma_\phi(g) = Z_H^{-1} \sum_{t<H} \gamma^t \mathbb{E}_\phi[g(s_{t+1}) \sum_{m \leq t} \chi_m]$, which is linear in g. Since $|\hat{g}_{-j}| \leq C_g$ and the score sums are bounded, Fubini's theorem exchanges the expectation over D_{-j} with $\mathbb{E}_\phi$, which gives $\mathbb{E}\,\Gamma_\phi(\hat{g}_{-j}) = \Gamma_\phi(\bar{g}_j)$.

(d) Smoothing maps a probability vector p to $(1-\varepsilon)p + \varepsilon\upsilon$, whose entries lie in $[\varepsilon\upsilon_{\min}, (1-\varepsilon)+\varepsilon]$, and $\upsilon_{\min} > 0$ (Theorem H.7(c)). For the KDE, every kernel satisfies $\varphi_h(x - x_i)\mathbf{1}_{\mathcal{S}}(x)/m_i \leq (2\pi h^2 m_\star)^{-1}$ by Theorem H.7(b), and the weights sum to one; the smoothed density adds $\varepsilon/\lambda(\mathcal{S})$ to $(1-\varepsilon)\hat{p}$ on $\mathcal{S}$, and the query points lie in $\mathcal{S}$. Both arguments of the ratio lie in a compact interval $[a, b] \subset (0, \infty)$, so the ratio lies in $[a/b, b/a]$. The function f' is continuous on $(0, \infty)$ for the three generators (Theorem C.4), so $(1-\varepsilon)|f'|$ is bounded on $[a/b, b/a]$. A signal-to-go is bounded by $C_g \sum_k \gamma^k \leq C_g/(1-\gamma)$, and so is a mean of such values. □

`tests/test_fpg_gradient.py` checks (a)–(c) for the implemented counts estimator by exhaustive enumeration on a three-state MDP, including folds of unequal size with the time baseline; (d) is not a test but a deterministic bound.

I Method map: from each update to its result and its code

This appendix condenses the project's method map (`METHOD_MAP.md`). For every implemented step it gives the formula, the result that justifies it, its status, and the function that implements it. *Identity*: a definition or exact identity that the code reproduces up to rounding, checked by a test against a closed form. *Proved*: a theorem whose hypotheses the implementation meets by construction. *Conditional*: proved only under hypotheses that the trained system does not meet or cannot verify; an interpretation, never a guarantee. *Heuristic*: no theorem (optimizers, step sizes, target networks, projections, plug-in estimates). The approximation codes in the last column refer to Table 19.

I.1 Data access

Learners receive the simulator (actions in, next observation out), the learner view of the configuration, the demonstration *states* and, for FB, the task-blind replay buffer. They never receive the finite transition matrix, the continuous kernel parameters, the evaluation reward, the goal, the success rule, the expert actions or the held-out evaluation trajectories, and $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ have no environment handle at all. The maze walls are passed to the density estimators only, to normalize the KDE and the histogram reference law on the free space. These rules are enforced statically (`tests/test_no_leakage.py` rejects the forbidden imports and names in the learner modules and training executors) and behaviourally: scrambling the stored expert actions, or changing the reward width, goal radius and success rule, leaves every training output

bit-identical (`tests/test_pipeline.py`). Evaluation transitions are generated by the evaluator with their own random streams, counted, and never written where a learner reads.

I.2 f-PG (`il_memoir/fpg.py`)

step	formula	result	status	code (codes)
generator	f, f' for JS, $\mathrm{KL}(d\|q)$, $\mathrm{KL}(q\|d)$	Theorems C.4 and C.5	identity	`divergences.f`, `fprime`
smoothing	$p_\varepsilon = (1-\varepsilon)p + \varepsilon\upsilon$, $\varepsilon = 0.01$	Theorem C.7	identity (objective becomes J_ε)	`divergences.smooth` (X3)
policy	tabular logits (zero: uniform); MLP, 2 tanh layers of 64, output scaled by 0.01	Theorems D.1 to D.3	identity	`fpg.TabularSoftmaxPolicy`, `MLPPolicy` (N2, N3)
rollouts	$N = 20$ trajectories of $H = 100$ steps from μ_0 per update, all counted	Theorem D.9	identity	`fpg.collect` (S1)
occupancies	weights $\gamma^t/(NZ_H)$ on s_{t+1}; counts, histogram masses or KDE; same for $\hat{q}_E$	Theorem H.7	identity (counts) / heuristic (KDE)	`fpg.SignalEstimator` (S2, S8)
signal	$\hat{g}_{-j} = (1-\varepsilon)f'(\hat{d}_{-j,\varepsilon}/\hat{q}_\varepsilon)$ from the other fold	Theorems D.10 and H.11	heuristic (plug-in bias); cross-fitting proved to remove the same-batch term	`fpg.crossfit_advantages` (S2, S9)
signal-to-go	$G_k = \sum_{t\geq k}\gamma^{t-k}\hat{g}(s_{t+1})$	Theorem D.7	identity	`fpg.rewards_to_go`
baseline	mean signal-to-go of the other fold at step k	Theorems D.8 and H.11	proved (leaves the conditional mean unchanged)	`fpg.baseline` (S4 avoided)
estimate	$\frac{1}{N}\sum_i Z_H^{-1}\sum_k \gamma^k \chi_k (G_k - \beta_k)$	Theorem D.9	proved unbiased with the exact signal	`fpg.surrogate_loss` (S1)
update	Adam descent, lr 0.3 (tabular) or 0.003 (MLP)	Theorem D.9(b) for exact small steps only	heuristic	`FPGLearner.step` (O1)
diagnostic	state–action gradient, $t = k$ term kept	Theorems D.13 and D.14	identity (tests only)	`fpg.gradient_estimate_sa`

I.3 FB pretraining (`il_memoir/fb.py`)

step	formula	result	status	code (codes)
replay	uniform actions, uniform free-space resets, segments of 50; $\hat{\nu}$ = law of stored next states	Theorems F.21 and H.10	identity; domination proved for the population law	`fb.collect_replay` (S5)
latents	z uniform on the sphere $\mathcal{Z}_4$	Theorem F.28	heuristic	`fb.sample_latents` (X1)
networks	$F(s,a,z)$ for all actions; $B(x)$; $d=16$, width 64; LayerNorm and tanh, then SiLU	Theorems F.1 and F.27	heuristic (finite rank)	`fb.ForwardNet`, `BackwardNet` (N1–N3)
policy	$\pi_{z,\tau} = \mathrm{softmax}(F^\top z/\tau)$, $\tau = 1$	(F.1)	identity	`fb.policy_probs` (X2)
target	$\bar{F}^-(s',z)$ under π_z^- from the target copy	(F.3)	identity	`fb.expected_features` (O3)
loss	$\hat{L}$ with an independent target batch and the diagonal term $-F_i^\top B(s_{i+1})$	Theorem F.22, (F.6)	proved unbiased for the buffer loss; residual reading conditional	`fb.successor_loss` (S5)
penalty	orthonormality loss with stop-gradients, $\lambda_{\mathrm{orth}} = 1$	Theorem F.24, (F.8)	proved unbiased gradient; soft weight heuristic	`fb.orth_penalty` (O1)
step	Adam, lr 3×10^{-4}, batch $128 + 128$	Theorem F.27	heuristic	`FBPretrainer.update` (O1)
targets	Polyak, momentum 0.99 after every update	Theorem F.28	heuristic	`fb.polyak_update` (O3)
Gram	$\hat{G}_B = \frac{1}{N}\sum_j B(x_j)B(x_j)^\top$, online copy, float64	Theorem F.20	identity (exact for $\hat{\nu}$)	`fb.gram` (S5, S7)
frozen model	online copy with $\hat{G}_B$, τ, R (decision D1)	Theorem F.28	identity	`fb.FBModel`

I.4 $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ (`il_memoir/fb_il.py`)

step	formula	result	status	code (codes)
z_{ER}	$\frac{1}{N_E}\sum_i \frac{1}{H}\sum_{t=1}^{H} B(s_t^i)$	Theorem G.4, (G.1)	identity	`fb_il.er_fb_z` (S6)
reading	$z_{ER} = z_{r_e}$, optimal for r_e under exactness	Theorem G.4	conditional	—
projection	$z \mapsto Rz/\|z\|$, $R = 4$; raw norm recorded	Theorem G.11	heuristic (changed problem)	`fb_il.project_to_sphere` (X1)
targets	$\psi_t^{(K)} = \sum_{k<K} \gamma^k B(s_{t+k+1})$, $t \in \{0,\dots,40\}$, $K = 60$	(G.2), Theorem G.9	identity	`fb_il.dm_targets` (T1, X4)

step	formula	result	status	code (codes)
loss	$\hat{\ell}^{(K)} = \text{mean}[F^\top \hat{G}_B F - 2F^\top \psi^{(K)}]$, $F(s,z) = \mathbb{E}_{\pi_{z,\tau}} F(s,a,z)$	(G.5)	identity	`fb_il.dm_loss` (S6, S7)
meaning	$\ell_{DM} = \mathcal{L}_{DM} - C_E$	Theorem G.6(a)	conditional (exact model)	—
bias	$\leq 2\gamma^K \beta_B \bar{F}/(1-\gamma) + \|\hat{G}_B - G_B\| \bar{F}^2$ for a fixed z	Theorem G.8	proved	— (T1, S7)
gradient	through $F(s,a,z)$ and $\pi_{z,\tau}$, networks frozen	Theorem G.10, (G.6)	proved (C^1 activations)	`fb_il.dm_grad`
latent step	Adam lr 0.05, projection after each step, 300 steps from projected z_{ER}	Theorem G.11	heuristic	`fb_il.dm_fit` (O1, X1)

I.5 Evaluation (`il_memoir/evaluation.py`)

In the finite mazes the oracle computes, from P, the exact held-out divergence $\text{JS}(d^+_{\pi,H}, q_E)$, the state–action divergence, the H-step return and its gap to $V^\star_H$ (Theorems B.11 and B.17), the exact hit probability and the probability of ending in the goal, and the FB model error of Theorem F.17; 200 rollouts cross-check them. In the continuous maze, 500 rollouts per checkpoint give the success rate, the return, the goal-holding fractions and the plug-in histogram divergence against 2,000 reference rollouts (S10). Evaluation reads each policy through the same function the learner uses, checks every checkpoint file against the hash recorded by its training job, and writes nothing a learner reads.

I.6 Approximation register

Table 19: Approximation register: each code, what it approximates, where, and the result that bounds it or stops applying.

code	approximation	affects	bound or status
S1	Monte Carlo rollouts and minibatches	f-PG gradient; continuous evaluation	unbiased with the exact signal (Theorem D.9)
S2	plug-in densities (counts, histogram, KDE)	f-PG signal	biased in general (Theorem D.10(b)); no bound for the KDE (Theorem D.11)
S3	same-batch signal	f-PG	not used: cross-fitting by default (Theorem H.11)
S4	baseline fitted on the same trajectories	f-PG	not used: the baseline comes from the other fold
S5	finite replay buffer, $\hat{\nu}$ for ν	FB loss, $\hat{G}_B$, z_r	one buffer for all three; unbiased for the buffer loss only; atomic $\hat{\nu}$ in the continuous maze (Theorem H.5)
S6	four demonstrations	$\hat{q}_E$, z_{ER}, DM_{FB} start law	covariance $O(1/N_E)$ for z_{ER} (Theorem G.4)
S7	Gram estimate $\hat{G}_B$	DM_{FB}	Theorem G.8(c)–(d)
S8	Euclidean truncation of KDE kernels	continuous f-PG signal	hidden mass: median 0, mean 1.1%, up to about 31.5% near doorway corners (Theorem H.8)

code	approximation	affects	bound or status
S9	cross-fitting with 2 folds	f-PG	changes the variance of the signal, not the estimator's structure (Theorem H.11)
S10	plug-in histogram JS between two finite samples	continuous evaluation	biased upward; the expert's row measures the floor
T1	$\mathrm{DM}_{\mathrm{FB}}$ truncation at $K = 60$	$\mathrm{DM}_{\mathrm{FB}}$ targets	$\gamma^K/(1-\gamma) \approx 0.92$ against 20 (Theorem G.8(b))
T2	H-step evaluation	returns, occupancies	Theorem B.13, Theorem F.14
N1	finite rank $d = 16$, MLPs	FB model	exactness impossible in the finite mazes (Theorem F.9) and not established in the continuous maze; only Theorem F.17 applies
N2	activations	gradient statements	all activations C^∞ (tanh, SiLU); ReLU would not be covered
N3	float32 networks	all networks	the oracle uses float64
O1	Adam, step sizes, update counts	f-PG, FB, $\mathrm{DM}_{\mathrm{FB}}$	no theorem; only exact small steps are covered
O2	PPO clipping	f-PG	not used (off by default)
O3	target networks (Polyak 0.99)	FB	Theorem F.22(c) assumes equal copies; no convergence
X1	latent on the sphere $\mathcal{Z}_4$	FB, $\mathrm{ER}_{\mathrm{FB}}$, $\mathrm{DM}_{\mathrm{FB}}$	Theorem F.12, Theorem G.11
X2	softmax $\tau = 1$ instead of greedy	FB, $\mathrm{ER}_{\mathrm{FB}}$, $\mathrm{DM}_{\mathrm{FB}}$	Theorem F.12 replaces Theorem F.6
X3	smoothing $\varepsilon = 0.01$	f-PG	J_ε is minimized, not J (Theorem C.7)
X4	$\mathrm{DM}_{\mathrm{FB}}$ start law with $t = 0$	$\mathrm{DM}_{\mathrm{FB}}$	decisions D4–D5 (Theorem G.9); Theorem G.6(e)
K1	floating-point segment test; noise redraw cap	continuous kernel	proved in real arithmetic (Theorem H.4)
K2	CUDA float32 non-determinism	—	not relevant: all final runs on the CPU
E1	evaluation on the CPU	all metrics	metrics independent of the training device

J Source discrepancies and design decisions

This appendix condenses the source audit (`SOURCE_AUDIT.md`), which was made from the supplied PDFs before any code was written. The PDFs were read in full from text extraction, and every ambiguous page was rendered and read visually; page numbers are PDF page indices, which coincide with the printed ones. No later paper version and no author code was consulted. A reference [SA X] in the text points to item X below. A paper is evidence for its own claims only: nothing here, or in the memoir, says that our code reproduces a paper's results.

Table 20: The supplied sources.

ID	work and version evidence	role here
A	O. Cappé, memoir proposal "Approaches for Imitation Learning", July 2025, 2 pp.	research questions
FIRL	Ni et al. [13], CoRL 2020, 23 pp.	theory only
FPG	Agarwal et al. [2], arXiv:2310.06794v1, NeurIPS 2023, 24 pp.	adapted to a fixed target
FB21	Touati and Ollivier [18], arXiv:2103.07945v3, 45 pp.	FB theory and pretraining
FI24	Pirotta et al. [14], ICLR 2024, 40 pp.	$\mathrm{ER}_{\mathrm{FB}}$, $\mathrm{DM}_{\mathrm{FB}}$

J.1 Divergence conventions

D1 FPG Eq. (3), p. 3, writes $\int_{P>0} Pf(Q/P)$, which swaps the arguments relative to FPG's own Table 1 and proofs (Lemma A.2, p. 16; B.1, p. 19) and flips the sign of the $f'(\infty)$ term. We follow the table and the proofs: $D_f(p\|q) = \sum_x q(x) f(p(x)/q(x))$, learner first.

D2 FPG's "fkl" is KL(learner‖target), which FIRL calls "RKL" when the expert is the target. We never write "forward" or "reverse" KL; both directions are named by their arguments.

D3 FPG Table 1 and FIRL Table 2 print JS with the factor $\frac{1}{2}$, but their generator $u \log u - (1+u) \log \frac{1+u}{2}$ gives twice that value. We use the half-weight generator, $0 \le \mathrm{JS} \le \log 2$; FIRL's $h_{\mathrm{JS}} = -\log(1+u)$ corresponds to the doubled generator and drops a constant that a covariance ignores.

D4 FPG Table 1 lists χ^2 with $f = \frac{1}{2}(u-1)^2$ and $f'(u) = u$; the derivative is $u - 1$. Not used.

D5 Perspective map: $D_f(p\|q) = D_{f^\circ}(q\|p)$ with $f^\circ(u) = uf(1/u)$ (Theorem C.5); FIRL's expert-first objective is a learner-first divergence with generator f°.

J.2 f-PG

F1 The sign of FPG's update is consistent: descent with weight $f'(p_\theta/p_g)$ lowers the probability of over-visited states (p. 7).

F2 Thm 4.1 (Eq. 5, p. 4) lacks the factor $1/T$ of the appendix version (Eq. 8, p. 18): a step-size change only.

F3 Index ranges: the trajectory is $(s_0, a_0, \ldots, s_T)$, but the score sums run over $t = 1..T$, including a_T and omitting a_0. In the next-state convention the scores are at $k = 0..H-1$ and the signal at s_{t+1}.

F4 The proof of B.1 (pp. 18–19) treats $\pi_\theta(s_t)$ as a state weight and drops a term without stating that the score has mean zero; we use an independent derivation (Theorem D.6).

F5 Discount weights: FPG Eq. (12), p. 21, discounts score and signal separately, and Eq. (13) discounts twice; neither is the gradient of the discounted-visitation objective. Eq. (10), re-indexed to $s_{t'}$ and used on-policy, is exact. B.3 calls Eq. (10) "biased"; by our derivation its on-policy form is unbiased with the true occupancy, and bias enters through off-policy ratios, the old occupancy inside f', and plug-in estimates. Dropping the γ^k score weight does not give the gradient (Theorem D.7).

F6 The PPO surrogate of Eqs. (6) and (11) is written for maximization while J is minimized; for descent the pessimistic clipping is a maximum. PPO is not used in the runs.

F7 The Dirac-target results (Thm 4.2, Thm 5.1, Lemmas A.1–A.3) rely on $D_f(p\|\delta_g)$ depending on $p(g)$ only, which is false for a non-degenerate expert target. Their replacements are Theorem C.4, Theorem B.11 and Theorem D.18; no convergence claim is made.

F8 Lemma 5.1 (state-MaxEnt identity, p. 6) is correct and transfers to a fixed positive target (Theorem D.16); the inference of Lemma A.4 (p. 17), from an upper bound to the bounded quantity, is invalid and not used.

F9 FPG clips zero target probabilities at ε; we use the renormalized mixture $(1-\varepsilon)q + \varepsilon\upsilon$, which changes the objective but keeps exact derivatives (Theorem C.7).

F10 FPG fits densities by KDE inside the update loop; a plug-in density inside the nonlinear f' biases the gradient estimator (Theorem D.10).

J.3 FB21 and the transfer to our design

T1 Softmax policies in the pretraining targets are source-supported (FB21 Algorithm 1, line 18, $\tau = 200$); using the same softmax family in imitation and evaluation is our adaptation, to which FB21 Thm 2 and the optimality parts of Thms 8–9 do not apply.

T2 Under exactness for the softmax family, $V_r^\star - V_r^\pi \le \tau \log|\mathcal{A}|/(1-\gamma)$, and τ and the scale of z are coupled (Theorem F.12).

T3 Finite exact realizability for our family is possible with $d \ge \#\mathcal{S}$ (one-hot B, softmax fixed points by Brouwer; a sufficient condition) and requires $d \ge \operatorname{rank}\tilde{P}$ (Theorem F.8(c); a necessary condition), which equals $\#\mathcal{S}$ in our finite mazes; with $d = 16 < 40$ the finite-maze models are approximate by construction (Theorems F.8 and F.9); for the continuous maze no such statement is proved.

T4 The policy-evaluation step of the proof of FB21 Thm 8 (Eqs. 102–103, p. 29) does not use greediness and applies to our models (Theorem F.17(a)).

T5 FB21 feeds 21×21 radial-basis features to B in its continuous maze; we use raw (x, y) (decision U-11).

T6 FB21 (Eq. 35, p. 22) and FI24 (p. 3) both define "$\operatorname{Cov} B$" uncentred; so do we (Theorem F.20).

J.4 Timing and successor conventions

R1 FI24 p. 3 defines $Q_r^\pi(s_0, a_0) = \sum_t \gamma^t \mathbb{E}[r(s_t)]$ while its successor measure (Eq. 1) is on s_{t+1} and the same page says that the reward depends on s_{t+1}; its Proposition 1 holds with $r(s_{t+1})$, which we use throughout (Section B.8).

R2 The stationary optimal policy need not be optimal for the H-step return; the oracle therefore also computes the H-step optimum by backward induction, and every gap is measured against $V_H^\star$ (Theorem B.17).

J.5 FI24: the assumptions of the $\mathrm{DM}_{\mathrm{FB}}$ expansion and $\mathrm{ER}_{\mathrm{FB}}$

A1 Exact FB for the policies attached to every z used. Fails here (rank 16); the implemented loss is the model's, not the true one.

A2 $F(s, z)$ averages over the same π_z as in A1: by design, one network copy for both.

A3 $\operatorname{Cov} B$ under the same reference ν as in A1: enforced, one $\hat{\nu}$ for the loss, $\hat{G}_B$ and z_r.

A4 The expert is a stationary Markov policy of the observed state: the finite expert is; the continuous expert was built as a state-feedback controller for this reason.

A5 Infinite continuation after each expert state: fails with $H = 100$; resolved by truncation at $K = 60$ with a bias bound (decisions D4–D5 below).

A6 Square-integrable density of the expert successor measure with respect to ν: holds in the finite mazes with full coverage; under exactness it is automatic.

A7 Greedy π_z: *not* needed by FI24 Thm 3, so the $\mathrm{DM}_{\mathrm{FB}}$ expansion holds for our softmax family (Theorem G.6); $\mathrm{ER}_{\mathrm{FB}}$'s optimality reading does need it.

E1 FI24 Eq. (8) weights expert states uniformly over $s_1..s_H$ with equal trajectory weights; the f-PG target uses γ^t/Z_H weights. The two are different distributions and are never mixed.

E2 Reading z_{ER} as z_r for $r = \rho_e/\nu$ requires $\rho_e \ll \nu$; in the continuous maze the empirical ρ_e is atomic and the reading is formal.

E3 Even under exact greedy FB, $\mathrm{ER}_{\mathrm{FB}}$ maximizes a reward; it carries no matching guarantee (Theorem G.5).

E4 With softmax policies and an exact model, the norm of z acts as an inverse temperature (for a trained network, which also depends on z, this is not guaranteed), so any projection of z_{ER} changes the problem and is stated.

J.6 Design decisions

The design decisions of the theory stage are numbered separately from the divergence items D1–D5 above; the text refers to them as "decision D1" and so on.

decision D1
: After pretraining, the online network copy defines F, B, $\pi_{z,\tau}$, $\hat{G}_B$, z_{ER}, the $\mathrm{DM}_{\mathrm{FB}}$ objective and evaluation; target copies appear only in the pretraining target (Theorem F.28).

decision D2
: Latents on the sphere of radius $R = \sqrt{d} = 4$, sampled uniformly in pretraining, with $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ outputs projected onto it; τ is reported with $\|z_{ER}\|$.

decision D3
: Polyak target networks with momentum 0.99 after every update.

decision D4
: $\mathrm{DM}_{\mathrm{FB}}$ start times $\{0, \ldots, H - K\}$ with truncation $K = 60$; the model bootstrap of the tail was rejected (Theorem G.9).

decision D5
: The $\mathrm{DM}_{\mathrm{FB}}$ start law includes $t = 0$, which gives occupancy control from μ_0 (Theorem G.6(e)); $\mathrm{ER}_{\mathrm{FB}}$ keeps FI24's $s_1..s_H$.

J.7 Items decided during the project

U-1 Supervisor approval of the environments and of state-only matching: no new requirement was received.

U-2 Latent domain and temperature: the sphere of radius 4 and $\tau = 1$, chosen in the pilot (Section 10.4).

U-3 Network copy after pretraining: decision D1.

U-4 $\mathrm{DM}_{\mathrm{FB}}$ tail treatment: decisions D4–D5.

U-5 Continuous kernel: no clipping (moves are segment-checked), a blocked move becomes "stay", and the noise is redrawn until its segment is free; the kernel has no atom and a bounded density (Theorem H.4); the replay law dominates Lebesgue measure with $n_{\min} \approx 1.2 \times 10^{-3}$ (Theorem H.10).

U-6 f-PG signal: cross-fitting with two folds (Theorem H.11).

U-7 The Gibbs hypothesis (5.1) of FIRL is kept as a hypothesis on the law of the learner's state sequences. The source does not show that an exact maximum-entropy policy realizes this law under stochastic dynamics, and no policy in this memoir is claimed to satisfy it.

U-8 The softmax bound and the existence of softmax fixed points are proved (Theorems F.10 and F.12); uniqueness is not claimed.

U-9 Our reading of FPG B.3's "biased" remark is the one of F5.

U-10 Finite slip 0.1 with the executed action uniform over all five actions; $\operatorname{rank} \tilde{P} = \#\mathcal{S}$ (Theorem H.2).

U-11 Raw (x, y) inputs to FB in the continuous maze, kept after the pilot without an alternative being tested.

U-12 tanh activations in the f-PG MLP, so that the gradient results cover it.

U-13 The 356 histogram cells of positive free area partition the free space and are all reachable within H steps (Theorem H.6); the target is smoothed.

U-14 LayerNorm and tanh in the first FB layer, then SiLU, so that the gradient results cover the networks.

U-15 KDE with $h = 0.05$ kept after the pilot; its wall effects are quantified in Theorem H.8.

U-16 Success means a hit of the goal region within H steps; final-state and post-hit goal fractions are reported beside it; the thresholds JS ≤ 0.10 nats and success ≥ 0.80 were frozen as declared.

K Reproducibility and verification record

K.1 What ran, where, and from which code

- **Machine.** The author's laptop: Windows (platform string `Windows-10-10.0.26200`), 12 logical CPUs, 15.35 GiB of RAM, an RTX 3060 Laptop GPU that the final runs did not use. Python 3.11.16, PyTorch 2.14.1 (CUDA 13.2 build), NumPy 2.4.6, SciPy 1.17.1 (`runs/environment_laptop.json`, `requirements-laptop.txt`).
- **Device.** The CPU with 4 PyTorch threads (NumPy and BLAS thread counts were neither set nor recorded), one job at a time. The choice was made by measurement before the final runs: on timing workloads the projected training time of the core suite was 827 s on the CPU against 2163 s on CUDA (`runs/benchmark.json`).
- **Code.** Both suites were launched from commit `db2e6d6` ("freeze core and compact ablation configurations"). The launch record marks the working tree as *dirty*, i.e. it contained uncommitted files; the package hash of the code that ran, `d5fb4a98...`, equals the one recorded in the frozen run plan, and the audit checks that every job's code hash

matches its result and the current code. Which files were uncommitted was not recorded, and the package hash does not cover the command-line scripts in `scripts/`; neither can be reconstructed afterwards. Checksums written when the deliverables were packaged describe the submitted files, not the tree at launch.

- **Configurations.** Resolved suite hashes `acabfe8d...` (core, 144 jobs) and `45499164...` (compact ablations, 39 jobs of which 21 new), equal to the frozen plan; seed manifest `619a8560...`; reference seed 314159 for the continuous evaluation references.
- **Data.** `data/final-v1`: 39 items (demonstrations, replay buffers, continuous references), each with its generation seed and content hash, generated once and shared by all methods.
- **Timeline.** Core suite 2026-10-05, 21:46:55–22:02:35 UTC; ablations 22:20:00–22:22:55 UTC; analysis and audit at 22:23 UTC. All 165 jobs completed at their first attempt; no job failed and none was removed.

K.2 Commands

From the project root (Windows PowerShell, the project's virtual environment):

```
python -m pytest -q
python scripts/prepare_data.py --config configs/core.yaml --check
python scripts/run_suite.py --config configs/core.yaml --device cpu --resume
python scripts/run_suite.py --config configs/compact_ablations.yaml --device cpu --resume
python scripts/analyze.py --runs runs --out results
python scripts/audit_results.py --runs runs --results results
```

The test run on the frozen tree is saved in `audit/prompt4/pytest_laptop_frozen.txt`: 366 tests passed and 1 was skipped, in 155 s; the skipped test sends a real interrupt signal to a child process and runs on POSIX systems only. The report's own scripts (run from the project root) are:

```
python report/verify/verify_results.py   # recompute, compare, tables, claims
python report/figures/make_figures.py    # the PDF figures of the main text
python scripts/build.py --target report  # report/report.pdf, build.log
```

K.3 Provenance audit and a second implementation

Every job directory records the job's configuration identity, the hashes of its code scope, data inputs and parent outputs, the device, the wall time of each phase and the hashes of its outputs. The project's provenance audit (`scripts/audit_results.py`) ran 62 kinds of checks, 31,749 checks in all, on the two final suites and the generated results, with no error and no warning (`results/audit_report.json`): among them, that no planned seed was dropped, that no failed attempt was deleted, that every cited job is intact, that code and dependencies are not mixed across jobs, and that every table row traces back to a job.

The numbers in this memoir were then recomputed by `report/verify/verify_results.py`, a second implementation that uses the Python standard library only and none of the project's analysis code; it was written in the same project as the analysis code, so it is not an independent check. It reads the suite manifests, every job's status, result and evaluation records, and the $\mathrm{DM}_{\mathrm{FB}}$ latent logs, and it recomputes the final rows (last f-PG checkpoint; $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ at the final FB snapshot and the last latent step), means and sample standard deviations, the thresholds, the costs and the FB diagnostics. It compares them with the project's generated tables (`results/tables/*.csv`), records every mismatch, and exits with an error status if there is one. It then writes every table of Sections 10 to 14 and Section L. One of its inputs is not a run file: the coverage of the continuous replay buffers on the evaluation histogram (Section 12) is counted from the stored buffers by `report/verify/replay_hist_coverage.py`, which uses the project's histogram code, and read from its output. The cost conventions of Section 14 and the quantities derived with them are written to `report/verify/costs_closeout.json`.

The numbers in the text are tied to these checks as follows. Every number of the text and captions that is computed from the runs, from the environment and audit check reports or from a closed-form expression is typeset through a macro that takes a claim identifier and the printed literal. `report/verify/claims.py` registers each identifier with its literal and with the value it is recomputed from, and the verifier checks three things: that each registered literal equals the recomputed value at the printed precision; that every occurrence of the macro in the TeX sources names a registered claim and prints exactly its registered literal; and that every claim registered as printed occurs in the text. Configuration settings (such as γ, H, the budgets and the learning rates) are registered as facts and checked against the frozen configuration, but most of their occurrences in the text are not tied to these facts. The numeric tokens of the sources that are typeset without the macro are listed in `report/verify/claims_coverage.txt` for review by hand. The log of the verifier is `report/verify/verification_log.txt`.

K.4 Data-access rules

The rules of Section I were tested before the runs (static and behavioural leakage tests) and hold for the code that ran. In the runs, learners received the demonstration states, the simulator and, for FB, the task-blind replay buffer; evaluation used separate random streams and files; and every evaluation transition is counted separately from the training transitions (Section 14).

K.5 What is not reproducible bit for bit

Results are bitwise reproducible on the same machine and software (resuming a job reproduces the uninterrupted run on the CPU, which the tests check). Another operating system, another PyTorch build or a GPU can change the last bits of floating-point results and hence sampled actions; the run plan therefore copies `data/final-v1` instead of regenerating it on another system.

L Every run: per-seed results and full learning curves

This appendix reports every final run. Table 21 gives the last-checkpoint divergence and success of each seed, including every run that does not meet the operational thresholds. Figure 8 and the following figures show the full learning curves of every core cell, drawn by `scripts/analyze.py` from the evaluation records: f-PG against learner transitions (left), $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ against FB pretraining updates (middle), and the 300 $\mathrm{DM}_{\mathrm{FB}}$ latent steps at the final FB snapshot (right); thin lines are seeds and thick lines means. A run marked "not reached" in Table 10 did not meet the thresholds at any assessed checkpoint within the budget: every evaluated checkpoint for f-PG, and every FB snapshot after completed adaptation for $\mathrm{ER}_{\mathrm{FB}}$ and $\mathrm{DM}_{\mathrm{FB}}$ (for $\mathrm{DM}_{\mathrm{FB}}$, the latent after its 300 steps). The intermediate $\mathrm{DM}_{\mathrm{FB}}$ iterates evaluated at the final snapshot (right-hand panels) are not assessed by this rule; in three runs they entered the threshold region and left it before the final step (Section 11).

Table 21: Every final run: held-out divergence (nats) and success at the last checkpoint for each seed, and the seeds that meet the operational thresholds there (JS ≤ 0.10 and success ≥ 0.80). Finite values are exact; continuous values are Monte Carlo estimates (500 rollouts).

maze	goal	method	JS s0	JS s1	JS s2	succ. s0	succ. s1	succ. s2	met
finite open	(5,1)	f-PG (JS)	0.001	0.003	0.026	1.000	0.999	1.000	s0, s1, s2
		$\mathrm{DM}_{\mathrm{FB}}$	0.490	0.630	0.485	0.147	0.042	0.466	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.423	0.448	0.440	0.952	0.721	0.932	none
finite open	(5,5)	f-PG (JS)	0.020	0.004	0.032	1.000	1.000	1.000	s0, s1, s2
		$\mathrm{DM}_{\mathrm{FB}}$	0.489	0.484	0.398	0.529	0.453	0.368	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.272	0.374	0.230	0.999	0.978	0.999	none

maze	goal	method	JS s0	JS s1	JS s2	succ. s0	succ. s1	succ. s2	met
finite open	(1,5)	f-PG (JS)	0.002	0.002	0.004	0.999	1.000	1.000	s0, s1, s2
		$\mathrm{DM}_{\mathrm{FB}}$	0.450	0.618	0.122	0.291	0.057	0.975	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.497	0.423	0.097	0.471	0.954	1.000	s2
finite four rooms	(5,1)	f-PG (JS)	0.001	0.002	0.002	1.000	0.999	1.000	s0, s1, s2
		$\mathrm{DM}_{\mathrm{FB}}$	0.645	0.559	0.562	0.058	0.068	0.136	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.427	0.448	0.435	0.926	0.932	0.976	none
finite four rooms	(5,5)	f-PG (JS)	0.012	0.021	0.012	1.000	1.000	1.000	s0, s1, s2
		$\mathrm{DM}_{\mathrm{FB}}$	0.244	0.554	0.629	0.950	0.031	0.001	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.226	0.435	0.382	0.998	0.651	0.668	none
finite four rooms	(1,5)	f-PG (JS)	0.004	0.003	0.013	1.000	0.999	0.998	s0, s1, s2
		$\mathrm{DM}_{\mathrm{FB}}$	0.198	0.622	0.275	0.971	0.048	0.865	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.079	0.321	0.434	0.994	0.986	0.828	s0
continuous	(5,1)	f-PG (JS)	0.242	0.267	0.300	0.986	0.974	0.982	none
		$\mathrm{DM}_{\mathrm{FB}}$	0.644	0.355	0.542	0.002	0.884	0.174	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.484	0.500	0.523	0.868	0.934	0.612	none
		expert	0.0008	0.0009	0.0008	1.000	1.000	1.000	s0, s1, s2
continuous	(5,5)	f-PG (JS)	0.359	0.414	0.340	0.974	0.922	0.962	none
		$\mathrm{DM}_{\mathrm{FB}}$	0.555	0.548	0.352	0.584	0.004	0.834	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.659	0.549	0.515	0.434	0.122	0.260	none
		expert	0.0007	0.0013	0.0010	1.000	1.000	1.000	s0, s1, s2
continuous	(1,5)	f-PG (JS)	0.271	0.313	0.322	0.982	0.974	0.982	none
		$\mathrm{DM}_{\mathrm{FB}}$	0.371	0.594	0.693	0.588	0.072	0.000	none
		$\mathrm{ER}_{\mathrm{FB}}$	0.499	0.548	0.479	0.974	0.118	0.994	none
		expert	0.0008	0.0008	0.0005	1.000	1.000	1.000	s0, s1, s2
ablation, four rooms	(5,5)	f-PG ($\mathrm{KL}(d\|q)$)	0.010	0.017	0.009	1.000	1.000	0.998	s0, s1, s2
		f-PG ($\mathrm{KL}(q\|d)$)	0.057	0.071	0.055	0.995	0.998	1.000	s0, s1, s2
		$\mathrm{ER}_{\mathrm{FB}}$, no orth.	0.308	0.387	0.380	0.964	0.982	0.949	none
		$\mathrm{DM}_{\mathrm{FB}}$, no orth.	0.389	0.502	0.508	0.605	0.392	0.433	none

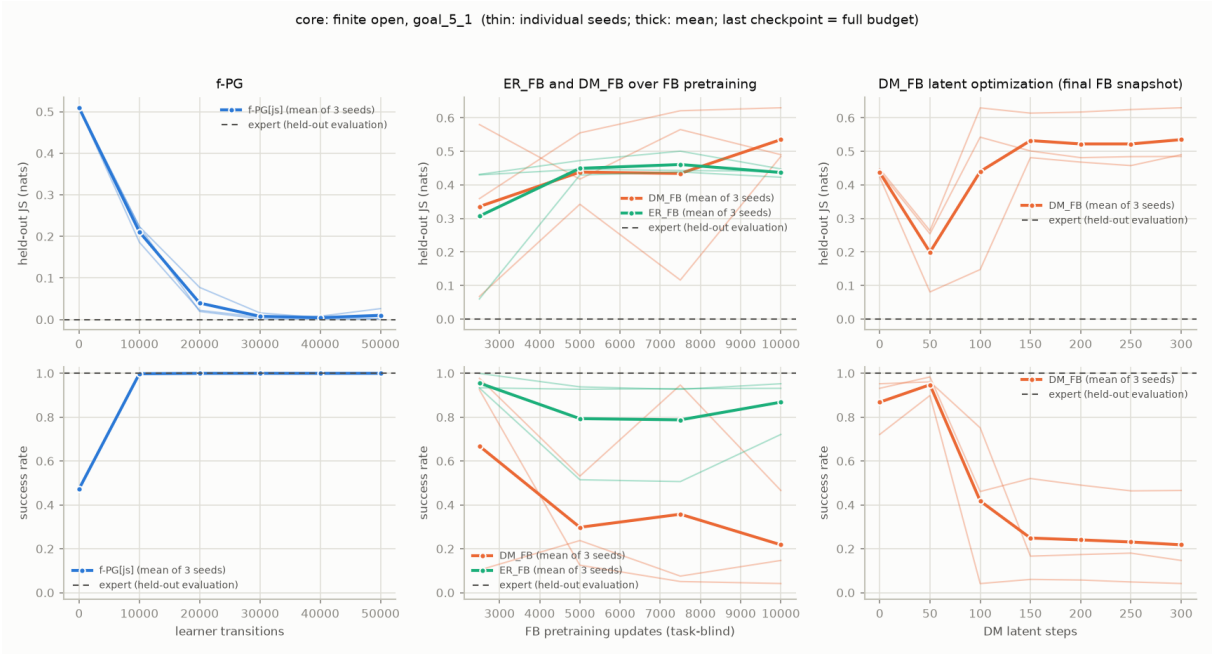


Figure 8: Learning curves, core suite, finite open maze, goal (5,1).

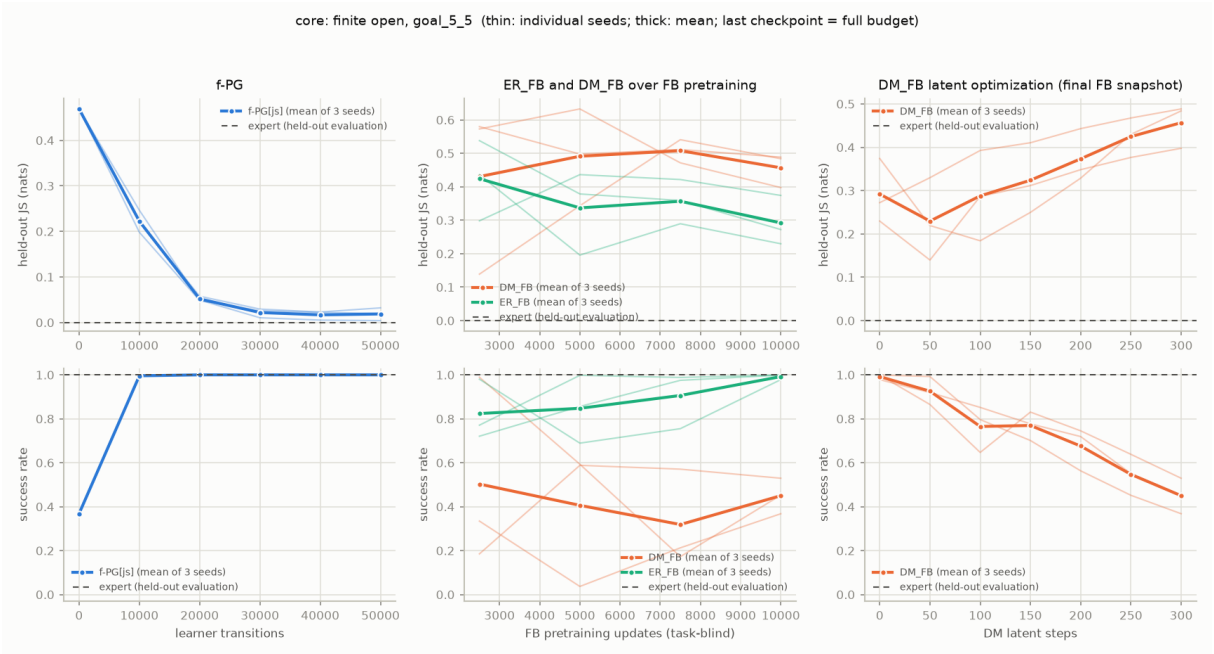


Figure 9: Learning curves, core suite, finite open maze, goal (5,5).

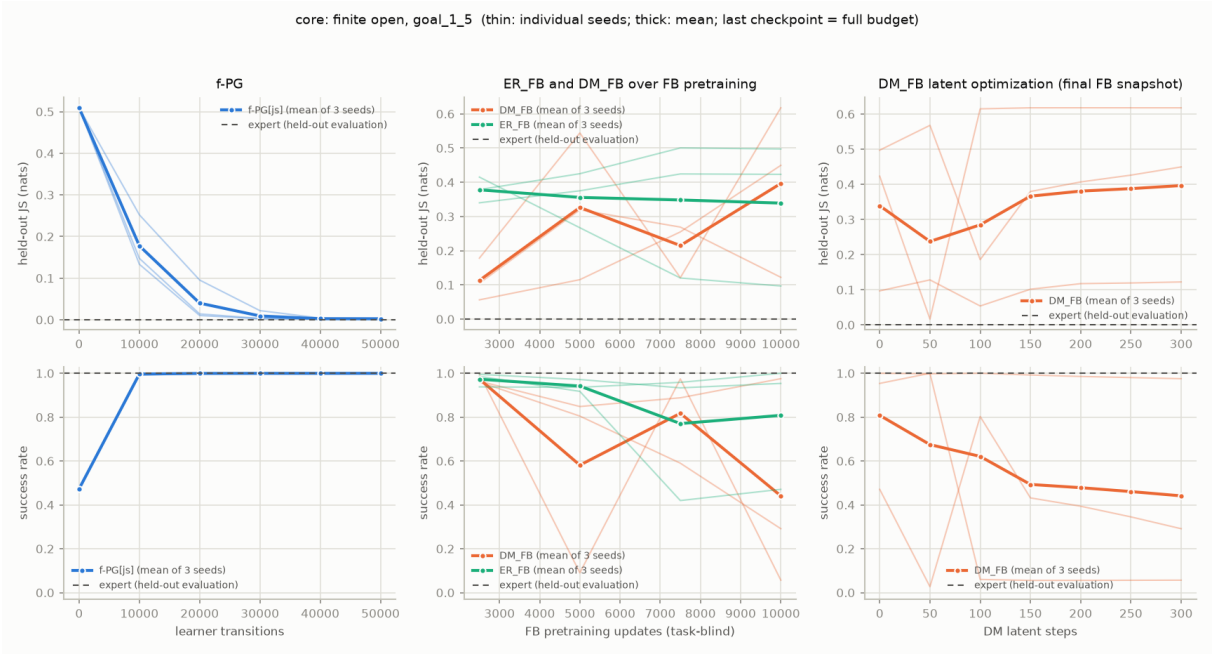


Figure 10: Learning curves, core suite, finite open maze, goal (1,5).

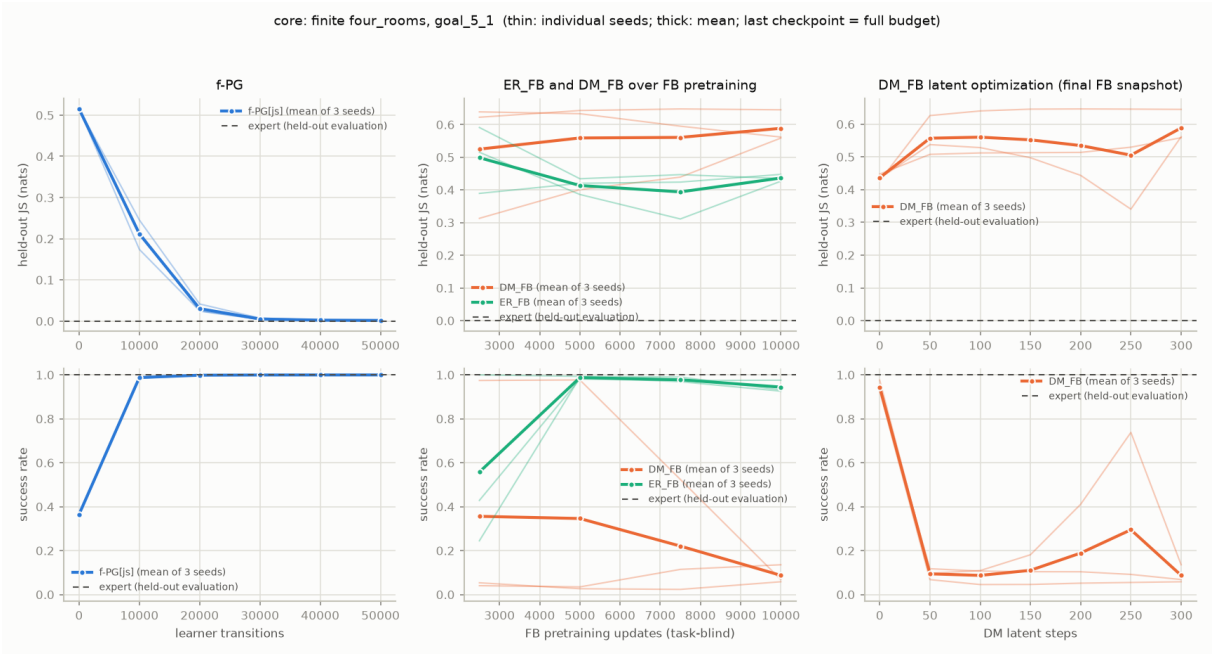


Figure 11: Learning curves, core suite, finite four-rooms maze, goal (5,1).

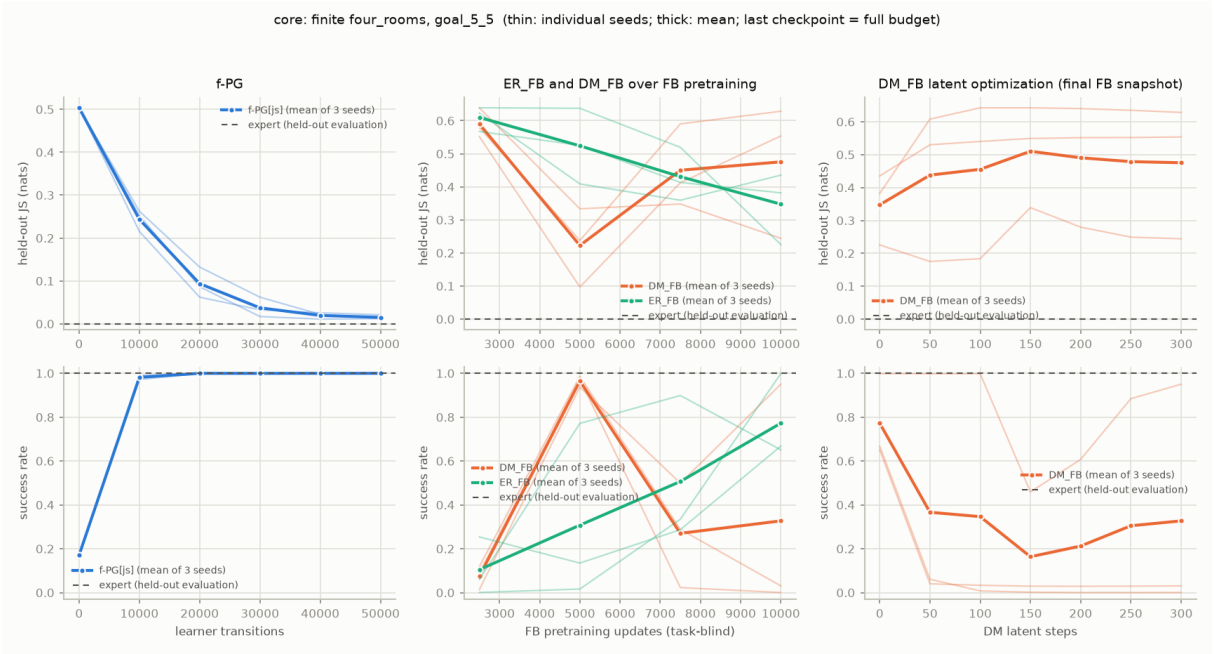


Figure 12: Learning curves, core suite, finite four-rooms maze, goal (5,5).

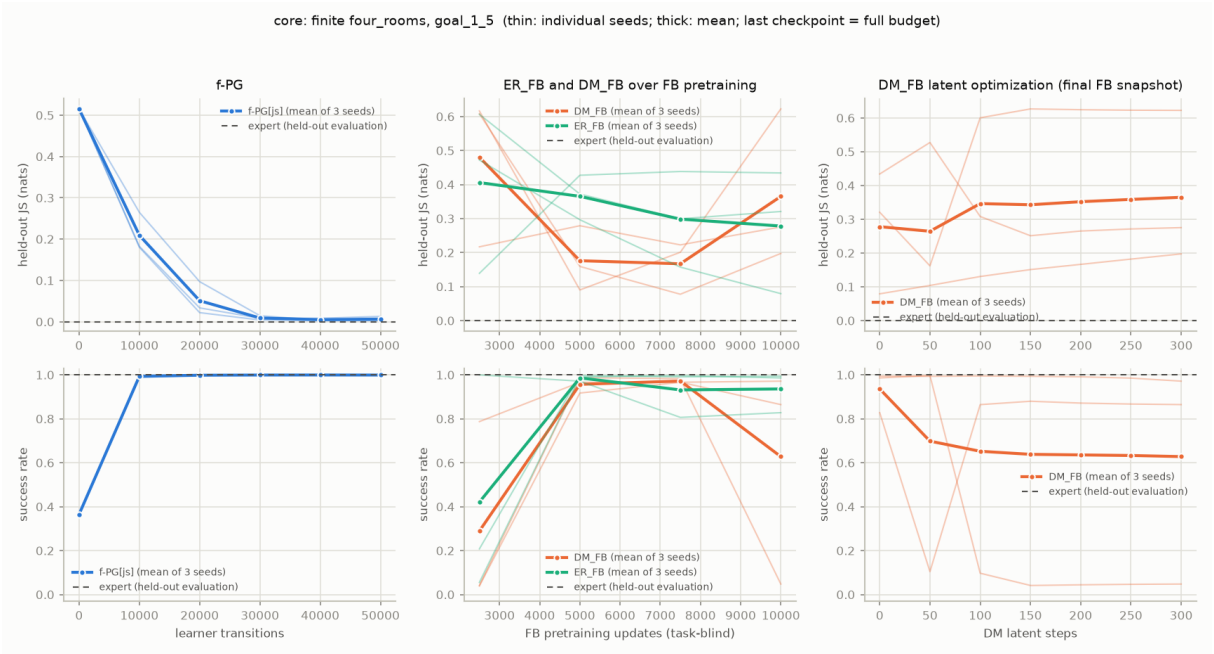


Figure 13: Learning curves, core suite, finite four-rooms maze, goal (1,5).

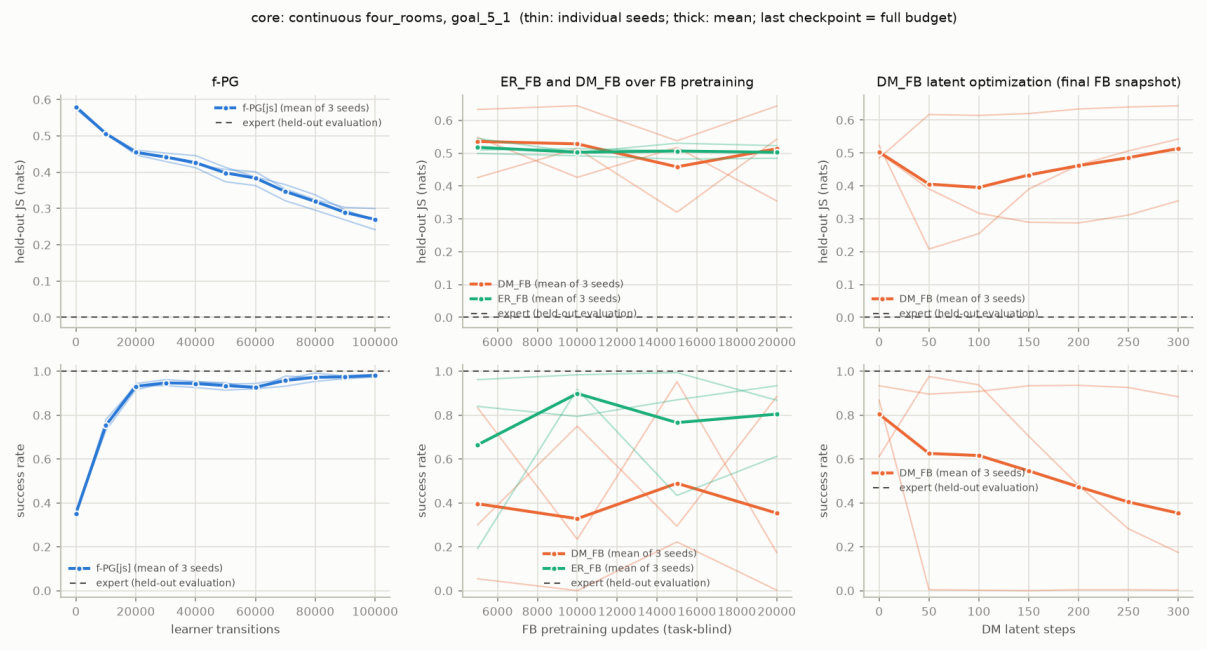


Figure 14: Learning curves, core suite, continuous four-rooms maze, goal (5,1).

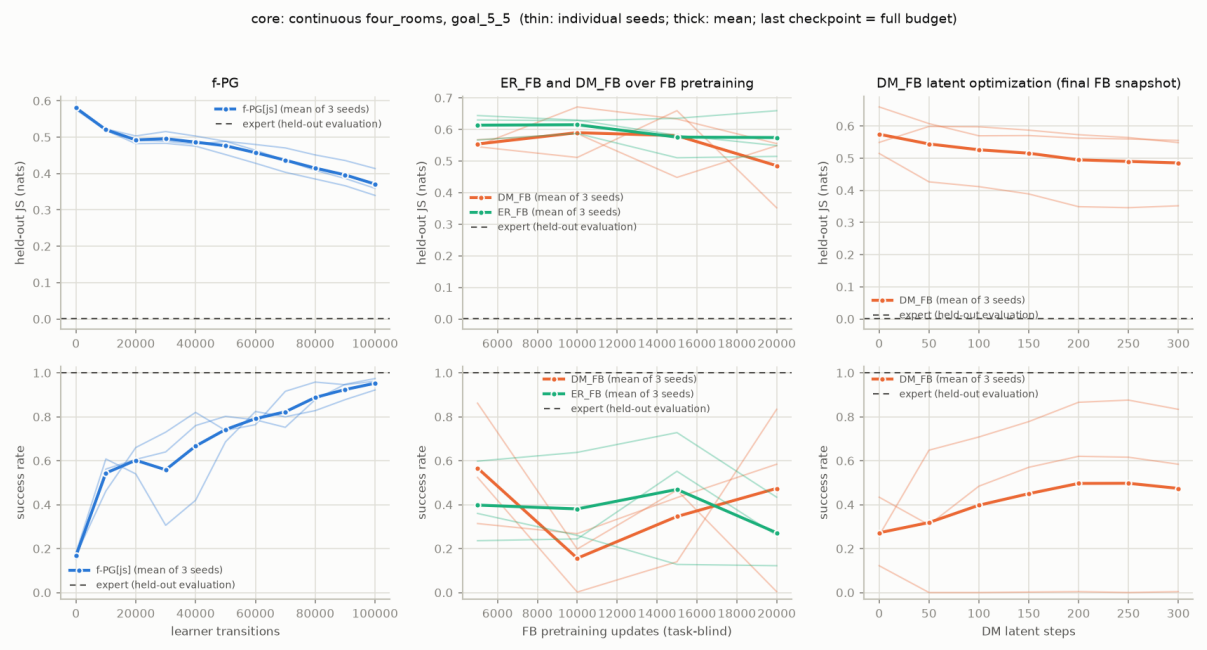


Figure 15: Learning curves, core suite, continuous four-rooms maze, goal (5,5).

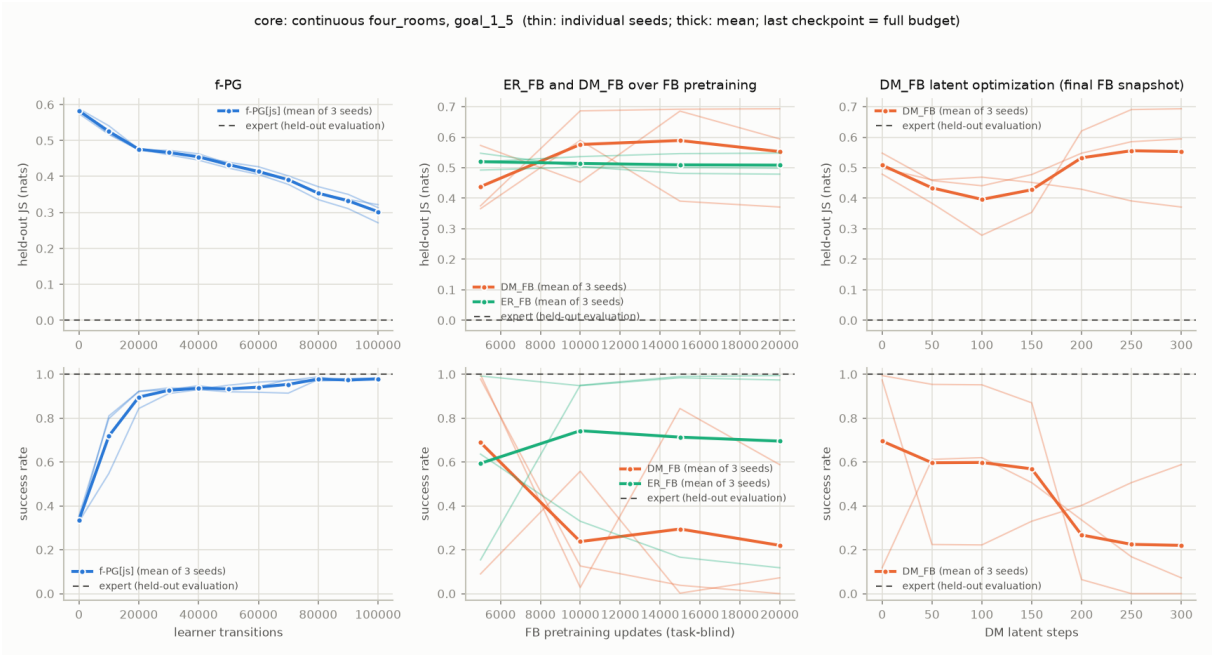


Figure 16: Learning curves, core suite, continuous four-rooms maze, goal (1,5).



Figure 17: Compact ablations (four rooms, goal $(5, 5)$): learning curves of every variant (top) and the final divergence and success of each seed (bottom).